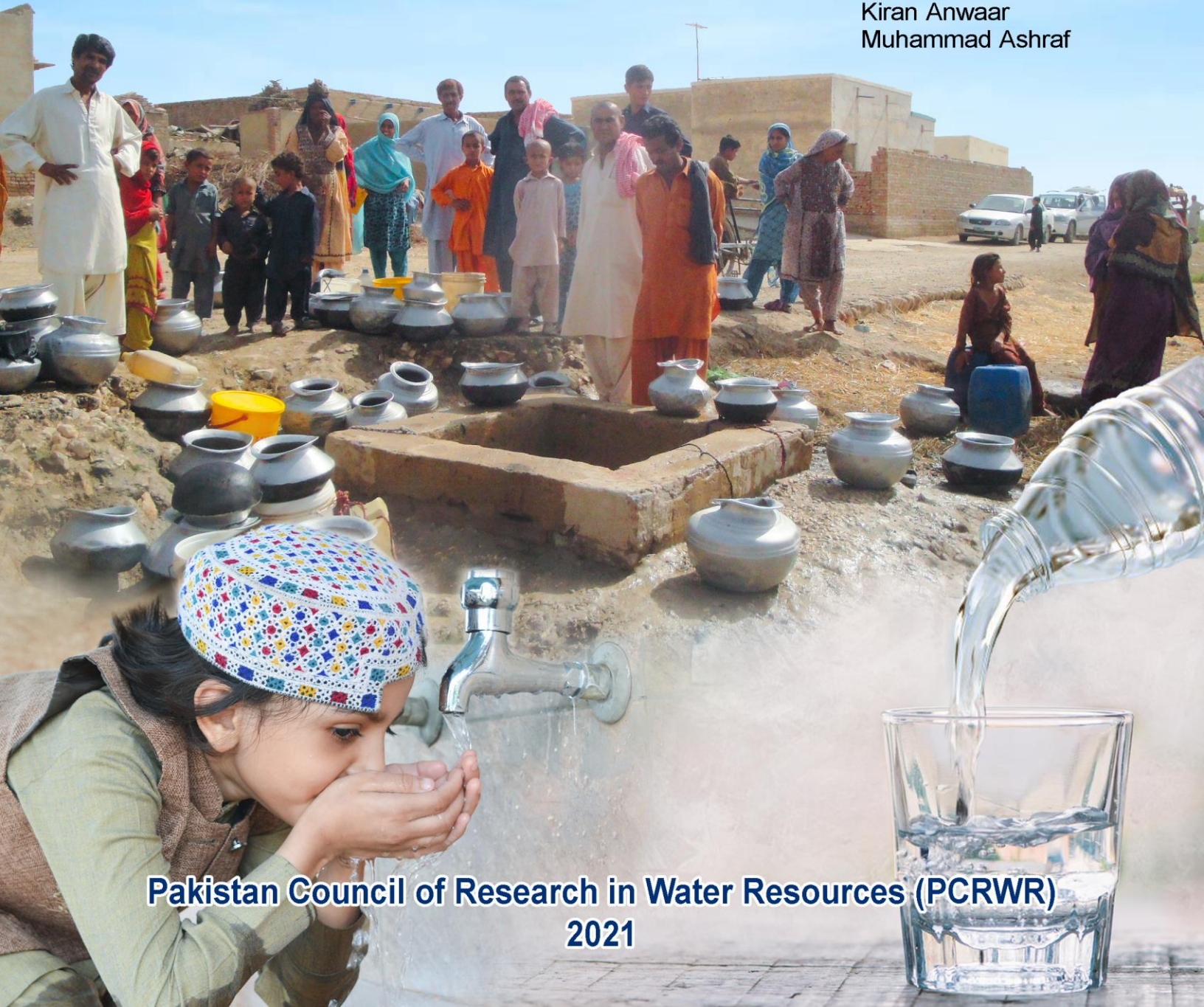


Drinking Water Quality in Pakistan Current Status and Challenges

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Pakistan Council of Research in Water Resources (PCRWR)
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Title Page designed by Mr. Zeeshan Munawar

Top: People waiting for drinking water supply

Bottom Left: A child drinking water from Tap

Bottom Right: Clean drinking water

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Foreword

Access to adequate and safe drinking water is a human right and its provision is the responsibility of state. This responsibility has been conferred through the constitution of Islamic Republic of Pakistan, Vision 2025, national and provincial drinking water policies and international commitments such as Sustainable Development Goals (SDGs).

Monitoring of drinking water quality is a first step towards its management. Pakistan Council of Research in Water Resources (PCRWR) started a nationwide water quality-monitoring program in 2002 focusing on 24 major cities of the country. This has now become a regular program of the Council. This program measures the water quality throughout the country and releases its report after every five years.

The outcome of this program helps raise awareness in the country. It acts as a guideline for the federal, provincial and local governments to take necessary steps towards improvement in water quality. Following this, the respective governments undertook a number of initiatives. However, most of the initiatives could not yield the desired results mainly due to: project approach focusing mostly on infrastructure development, installation of decentralized plants (mostly reverse osmosis), lack of ownership for O&M, lack of professional capacity to address the issue and most importantly lack of service delivery concept. This resulted into slow pace development. For example, from 2002 to 2020, only 19% improvement in safe drinking water supply has been observed (on an average 1% per year). If this situation continues, Pakistan would not be able to meet its national and international obligations.

This is a great challenge and requires a paradigm shift in thinking and approaching the issue. The country needs to declare a water emergency and give due priority to it by allocating adequate resources both in terms of human capacity and finances.

The present report is based on monitoring outcomes of 29 cities. It provides an insight to the current situation of drinking water quality in the country, underlying issues and suggests the way forward to tackle the issues. It would also provide a baseline for setting realistic targets to be achieved by 2030.

Dr. Muhammad Ashraf
Chairman

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Executive Summary

The National Water Quality Monitoring Program (NWQMP) implemented by the PCRWR from 2001 to 2010 generated the first ever annual water quality profile of the country covering 24 cities. This program was continued as a regular activity with monitoring frequency of five years. Following this, with addition of two more cities, it was extended to total 26 cities of Pakistan in 2015-16. This monitoring phase revealed that only 31% of the monitored water sources were safe for drinking purpose. The current water quality monitoring of 2020 is the national program implemented in 29 cities of four provinces, AJK and Gilgit-Baltistan. The water quality data when compared with the National Standards for Drinking Water Quality showed that out of total 435 sources, 168 (39%) sources were safe, whereas 267 (61%) sources were unsafe for drinking.

In the Federal Capital Islamabad, 71% sources were found safe and 29% as unsafe due to bacteriological and Iron contamination. In Bahawalpur city, 76% water sources were found unsafe due to bacteriological (20%), Arsenic (38%), Iron (36%), TDS (24%), Turbidity (20%), Hardness (16%) and Fluoride (4%). In Faisalabad, 59% monitored sources were found unsafe due to high TDS (23%), Iron (23%), Chlorides (23%), Nitrate (18%), Hardness (14%), bacteriological contamination (14%) and Fluoride (9%). In Gujranwala, 50% sources were contaminated because of bacteriological contaminations. In Gujrat, all the monitored sources were found safe. In Kasur, 10% sources were unsafe due to bacteriological contamination and high turbidity.

Lahore, the second largest city of Pakistan has shown 69% of safe and 31% unsafe sources due to bacteriological (12%), Arsenic (6%), Iron (6%) and TDS (6%). The city of Multan has not shown any improvement in water quality i.e. 19% safe in 2015 which decreased to 6% in 2020, bacteriological contamination of 69% and Arsenic (63%). In Rawalpindi, 38% of the monitored sources were contaminated due to bacteriological contamination (38%), Nitrate (15%), Iron (15%) and TDS (8%). In Sargodha city, 83% sources were unsafe for drinking purpose because of bacteriological contamination (21%), Hardness (58%), TDS (58%), Chloride (38%), Nitrate (38%), and Iron (21%). In Sheikhpura, 60% of the monitored sources were supplying unsafe water to the public mainly due to bacteriological contamination (20%) and excessive Nitrate (30%), Hardness (20%), TDS (20%), Arsenic (10%), Turbidity (10%), and Iron (10%). In Sialkot, all the monitored sources were providing safe water to the public.

In KPK, samples were collected from Abbottabad, Mangora, Mardan and Peshawar cities. Overall the water sources were found bacteriologically safe. However, chemically sources were unsafe because of higher level of Iron. The Iron contaminated sources comprised 50% from Peshawar, 55% from Abbottabad, 20% from Mangora, and 45% from Mardan.

In Quetta, the capital of Balochistan, 65% of the monitored sources were unsafe due to bacteriological contaminants (57%), Fluoride (27%), Chlorides (3%), Hardness (3%) and TDS (3%). In Khuzdar, 55% sources were unsafe for drinking because of bacteriological contamination (36%), Turbidity and Nitrate (9%). In Loralai city, 59% sources were supplying contaminated water due to excessive Turbidity and Iron (6%) and bacteriological contamination (59%). In Ziarat city 45% sources were unsafe for drinking purposes due to bacteriological contaminants (45%) and Iron (9%).

In Hyderabad city, 80% of the monitored sources were found contaminated and unsafe for drinking because of bacteriological contamination (73%), Turbidity (60%), Hardness (7%), Chlorides (7%) and TDS (7%). In Karachi, 93% sources were unsafe mainly due to bacteriological contamination (93%), Fluoride (7%), Chlorides and TDS (11%), Turbidity and Hardness (4%). In Sukkar city, 67% of the total sources were found contaminated and unsafe for drinking purposes due to higher concentration of Iron (44%), Chloride and TDS (33%), Hardness, Turbidity, Fluoride and bacteriological contamination (22%) and Arsenic (11%). From Badin city, only one source was found supplying safe drinking water to the community. Major causes of contamination were bacteriological contamination (58%) and chemical contaminations (58%). In Mirpur Khas, 100% of the monitored sources were found unfit for drinking purposes. This was mainly due to Total Coliforms (100%), Chloride (67%), Hardness and TDS (75%), Turbidity (42%) and Iron (17%). In Tando Allahyar city, 57% drinking water sources were found unfit for drinking purpose due to microbiological contamination (36%), Hardness (14%), and TDS (36%). In Shaheed Benazirabad city, 100% of monitored sources were unfit for drinking purposes because of Total Coliforms (100%), Iron (54%), TDS (46%), Chloride (31%), Hardness (23%) and Turbidity (8%).

In Muzaffarabad city, 70% of the monitored sources were found contaminated mainly due to Total Coliforms (60%) and Turbidity (20%). In Gilgit, the monitored sources were found contaminated due to presence of Total Coliforms. Out of these, 50% were also having excessive Turbidity.

Overall analysis of 29 cities has identified 11 major water quality problems in drinking water sources of Pakistan i.e. 41% microbiological, TDS (14%), Iron (14%), Hardness (10 %), Turbidity (9%), Chlorides (8%), Arsenic (5%), Nitrates (4%), Fluoride (4%), and pH (1%).

Comparison of current status of safe water (39% in 2020) with 31% in 2015 reveals slow paced drinking water quality improvement in the country. Consuming 61% of the unsafe waters may pose serious public health risk due to waterborne diseases such as diarrhea, dysentery, typhoid, hepatitis, skeletal and dental fluorosis, methemoglobinemia, and cancer.

The water quality monitoring over the time can help identify the challenges in sustainable access to safe drinking water. The monitoring outcomes of 29 cities make

us realize that all stakeholders such as Federal, Provincial and local Governments, local communities, industries, academia etc. need to play their roles in improving drinking water quality. In this context, establishing a comprehensive drinking water system that integrates water supply, quality and management is essential. This can be accomplished through substantial increase in investment on Water, Sanitation and Hygiene (WASH) sector, enhancement of institutional capacity, rehabilitation of existing water supply infrastructure, public-private partnership arrangements and introducing the concept of service delivery. Otherwise, it would be difficult to achieve the targets set in the Sustainable Development Goals (SDGs) particularly Goal 6.

Chapter 1: Introduction

1.1 Background

Water is a strategic resource and access to affordable and safe drinking water is a fundamental human right of all citizens. Provision of safe drinking water is the responsibility of the State. Pakistan's National Water Policy also envisions safe drinking water for all. However, the earlier studies conducted by Pakistan Council of Research in Water Resources (PCRWR) reported that 31% of the population had access to safe drinking water in the country (Imran et al. 2016).

PCRWR's water quality monitoring reports were shared with the concerned local, provincial and federal authorities for undertaking remedial measures. Consequently, unsafe drinking water became a serious concern for the Water and Sanitation Agencies (WASAs), Public Health Engineering Departments (PHEDs), local governments, NGOs and the general public. Various corrective measures were undertaken by the provinces, which resulted in an improved access to safe water from comparison is from 2010 to 2015 15% in 2010 to 31% in 2015. However, this slow paced improvement despite increased annual spending of Rs. 91 billion financial comparison 2016-17 to 2019-20 (2016-17) to Rs. 157 billion (2019-20) is questionable. This highlights the lack of effective planning and availability of essential human resources to carry out the water sector activities.

Some of the major constraints identified in the supply of safe water were water scarcity, intermittent water supply, improperly designed water supply schemes, cross contamination caused by the closely laid water supply and sewage pipelines, disposal of untreated sewage and industrial effluents and inadequate technical capacity of the service providers. Moreover, unsustainable and unrestricted groundwater abstraction policies, lack of strict enforcement of groundwater laws and poor governance have placed immense stress on the groundwater quality and quantity in the country (Iqbal et al., 2020).

Federal and provincial governments had taken various initiatives in the past to mitigate the above situation such as Clean Drinking Water for All (CDWA) in 2004 and Punjab Saaf Pani Company (PSPC) in 2014. These initiatives failed to provide any breakthrough regarding provision of safe drinking water. The main reasons of their failure were provision of decentralized solutions, lack of ownership, poor water pricing and lack of service delivery concept. Later, in 2015 Pakistan adopted the Sustainable Development Goals (SDGs) which calls for country-wide access to safe drinking water by 2030 through SDG target 6.1. Following this, provincial and local governments expedited their efforts. However, limited progress in achieving this target demands the root cause analysis and to redefine the strategies.

Achieving SDG target 6.1 by 2030 and to correlate the policy actions with performance in the water sector require continuous monitoring and reporting. In this regard, reliable water quality data is very important. The availability of extensive water quality data as provided by the PCRWR in 2015 and as presented in this report have significant importance for the water supply agencies, for WHO/UNICEF Joint Monitoring Programme (JMP), for reporting the progress on the SDG target 6.1, for setting the baselines and targets, and for planning and management of this precious resource.

The current report presents the water quality status of 29 major cities and trend analysis of 26 cities with respect to water quality improvement. The report also outlines gaps for slow paced improvements of drinking water quality to achieve SDG target 6.1 by 2030. The objectives of water quality monitoring programme were:

1. Produce quantitative information on the detection of national drinking water standards violations.
2. Analyze safe water quality improvement trend and assessing changes over time.
3. Identify emerging water quality problems and to support water supply service providers for taking remedial measures.

Chapter 2: Materials and Methods

The general methodological approach used to monitor the drinking water quality at national level comprised establishing the network of monitoring stations for sampling, sampling design and sampling frequency, field and laboratory testing, recording the groundwater level *etc.* The detailed methods implied are discussed as below:

2.1 Water Quality Monitoring Network

The 2020 national level water quality monitoring program is extended from 26 cities in 2015 to 29 main cities encompassing Federal Capital (Islamabad), Punjab (11 cities), Sindh (7 cities), Balochistan (4 cities), Khyber Pakhtunkhwa (4 cities), Gilgit-Baltistan (1 city) and Azad Jammu Kashmir (1 city). The population and area based selection of main cities for this program is aimed to undertake impact monitoring through valid and representative sampling strategy (Figure 1).

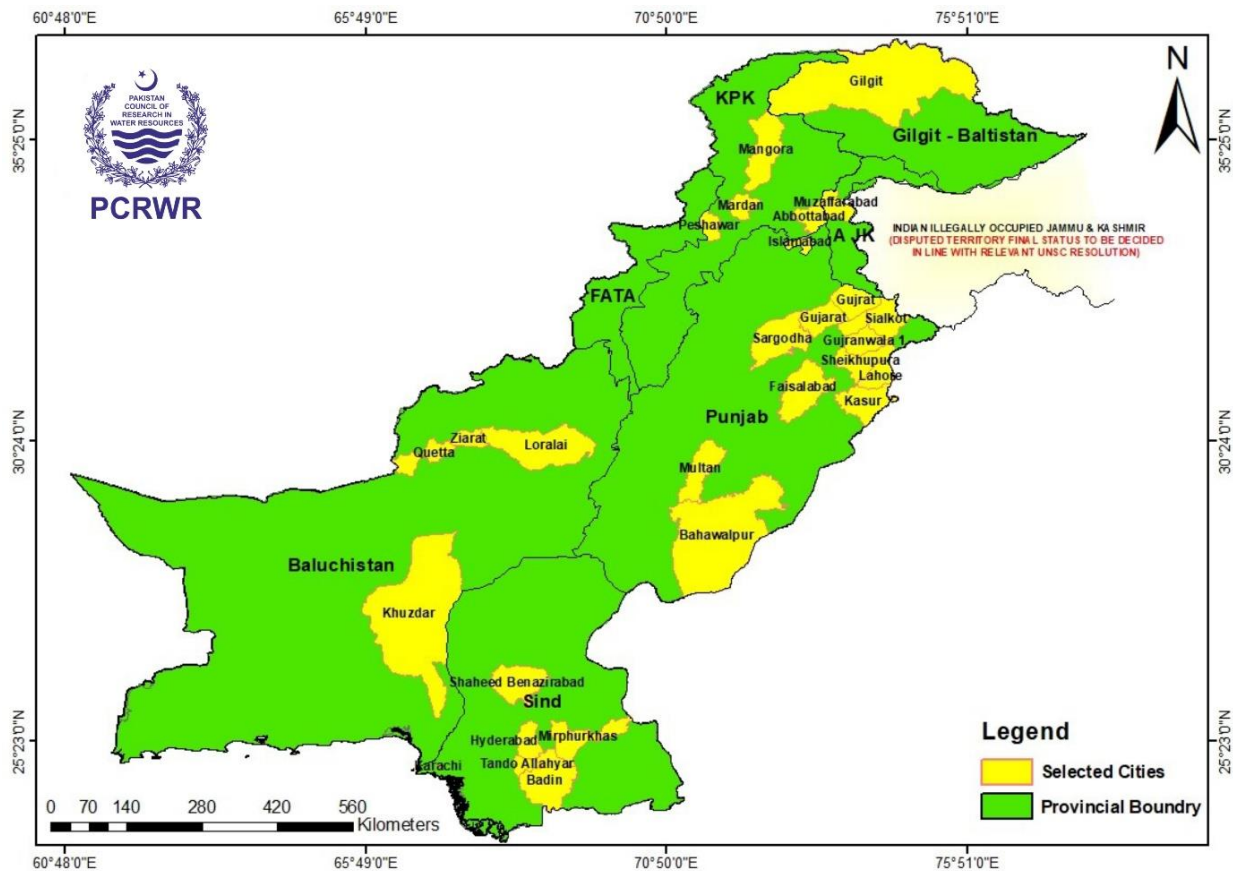


Figure 1: Selected cities for National Water Quality Monitoring Network

As reflected in Figure 1, the country is divided into 7 zones for water quality data collection purposes namely: Federal Capital Territory Area, Punjab, Sindh, Balochistan, Khyber Pakhtunkhwa, Azad Jammu Kashmir and Gilgit-Baltistan. PCRWR field teams

of the regional and district laboratories were assigned the sampling and field data collection tasks in the respective zones. Details of the Monitoring Stations (MS) and their areas of responsibility for collection of water samples are given in Table 1.

Table 1: Detail of monitoring station and cities

Zone	Monitoring Station	Monitoring Cities
Federal Capital Territory Area	NWQL, Islamabad	Rawalpindi and Islamabad
Punjab	Regional Office, Lahore	Lahore, Sheikhpura and Kasur
	Regional Office, Bahawalpur	Bahawalpur
	District Laboratory Sargodha	Sargodha
	District Laboratory Sialkot	Sialkot and Gujrat
	District Laboratory Gujranwala	Gujranwala
	District Laboratory Faisalabad	Faisalabad
	District Laboratory Multan	Multan
Sindh	Drainage Research Center, Tandojam	Hyderabad, Mirpur Khas, Tando Allahyar, Shaheed Benazirabad and Sukkur
	Regional Office, Karachi	Karachi
	District Laboratory Badin	Badin
Khyber Pakhtunkhwa	WRRC, Peshawar	Peshawar, Mardan and Mangora
	District Laboratory Abbottabad	Abbottabad
Balochistan	WRRC, Quetta	Quetta, Loralai and Ziarat
Azad Jammu Kashmir	District Laboratory Muzaffarabad	Muzaffarabad
Gilgit-Baltistan	District Laboratory Gilgit	Gilgit

2.2 Sampling Design: Grid Size and Number of Samples

A uniform and statistically valid criteria was followed for site selection and grid size, i.e. 1 km² (for small cities) 4 and 9 km² (for medium cities) and 16 and 25 km² (for big cities). Samples were taken from locations that are representative of the water source, treatment plant, storage facilities, distribution network, points at which water is delivered to the consumer and points of use. Monitoring was conducted at regular sites ("fixed stations") on a continuous basis. In this regard, fixed public points were given preference for the long term monitoring to gauge changes over time. Geology and depth of aquifers was also considered. The two monitoring points kept apart at minimum distance of 1 km. Site identification mark was done on each city map according to the grid. Sample ID for monitoring purpose was marked on the basis of actual sampling visit sequence of various sites. The details regarding grid size and sampling points (number) are shown in Table 2.

Table 2: Detail of number of samples collected along with grid size

Sr. #	City Name	City Code	Grid Size (km ²)	Total Sample Points	Sr. #	City Name	City Code	Grid Size (km ²)	Total Sample Points
1	Islamabad	ISL	4	26	16	Mangora	MAN	1	12
2	Bahawalpur	BAH	16	25	17	Mardan	MARD	4	10
3	Faisalabad	FAI	4	13	18	Peshawar	PES	16	13
4	Gujranwala	GUJ	4	14	19	Khuzdar	KHU	25	11
5	Gujrat	GUT	1	9	20	Loralai	LOR	1	13
6	Kasur	KAS	1	10	21	Quetta	QUE	4	25
7	Lahore	LAH	16	18	22	Ziarat	ZIA	1	11
8	Multan	MUL	16	16	23	Hyderabad	HYD	4	15
9	Rawalpindi	RAW	9	15	24	Karachi	KAR	25	28
10	Sargodha	SAR	4	24	25	Sukkur	SUK	1	12
11	Sheikhupura	SHE	4	11	26	Badin	BAD	1	29
12	Sialkot	SIA	4	10	27	Mirpur Khas	MK	1	15
13	Muzaffrabad	MUZ	1	10	28	Tando Allahyar	TA	1	14
14	Gilgit	GLG	1	10	29	Shaheed Benazirabad	SB	1	19
15	Abbottabad	ABT	1	11					

2.3 Sampling Pre-Requisites and Procedure

The sampling team well aware of monitoring objectives comprised of a technical officer, a laboratory technician and a driver. Following a uniform methodology in accordance with the American Public Health Association Protocols (APHA, 2017), four types of samples were collected from each site, preserved, labeled and transported to the PCRWR network of water quality laboratories. The details of these samples and preservatives used for each sample are given below in Table 3.

Table 3: Details of samples identification, sampling container, and preservatives

Samples type per site	Purpose	Sites	Sampling container	Preservatives
Type A	Microbiological testing	All sites	Pre-sterilized sampling bottles of 200 ml capacity	Pre-sterilized sampling bottles
Type B	Trace elements	All sites	Polystyrene bottles of 0.5 liter	2 ml/liter Nitric Acid (HNO_3) added in the bottles before going to field
Type C	Nitrate	All sites	-do-	1 ml/100 ml, 1 Molar Boric acid added in the bottles before going to field
Type D	Other physio-chemical parameters	All sites	Polystyrene bottles of 1.0 liter	No preservative

Before collecting samples and adding preservatives, bottles (B, C and D) were washed properly and rinsed thoroughly with deionized water. The samples collected for chemical analysis were transported to the laboratory without ice boxes. Residual chlorine, pH and turbidity were tested immediately after sampling as they could change during storage and transportation.

The water samples for microbiological contamination were collected in clean, sterile plastic bottles (200 ml). Proper care was taken to ensure that no accidental contamination occurs during sampling. Samples were not taken from those taps, which were leaking between the spindle and gland to avoid outside contamination. The type-A samples were transported to the laboratory under controlled temperature (2 and 8 °C) in a properly lightproof and disinfected insulated box. Water samples containing traces of

chlorine were treated with sodium thiosulfate to neutralize any chlorine present, incase chlorine is not neutralized, microbes may be killed during transit and an erroneous result will be obtained. The time between sample collection and analysis did not exceed 6 hours.

All samples were accompanied by an appropriate samples collection form. Field observations and information regarding each sample (such as sample types, sample ID, sample code given to the sample, GPS reading, date and time of sample collection, physical conditions like water-table depth etc.) were recorded on the sample collection proforma. Following procedures and precautionary measures were followed while collecting samples from different water sources.

2.3.1 Tap

Un-rusted taps were selected for collection of water samples. These taps were properly disinfected and allowed to flow for a few minutes before collecting the sample.

2.3.2 Tubewell

The water samples from tube wells were collected after allowing them to flow for at least 10 minutes to get representative sample of the groundwater. The depths of groundwater level were also recorded and the location of the tubewell was properly marked on the topographic survey sheet.

2.3.3 Distribution Network

The water samples from the distribution network were collected from the source of supply (as closely as possible) to minimize the effects of pollution in the distribution system and from point of use (PoU) to evaluate the actual quality of water being used. All water sample containers were filled slowly to avoid turbulence and air bubbles after flushing the system for sufficient time.

2.3.4 Hand Pump/Dug Well

Water samples were collected from hand pumps or dug wells after purging the hand pump or well. The purging was carried out by making one stroke for every foot of depth (A hand pump or dug well having 30 feet of depth, needs 30 strokes for its purging).

2.3.5 Stream

Water samples were collected from the center by standing in the middle of the stream. Care was taken to keep the bottle well above the bed of the stream to avoid unwanted bed material going into the sample.

2.3.6 Spring Water

Water samples were collected directly from the spring in sterilized sampling bottles for microbiology and bottles used with or without preservatives for other water quality parameters.

2.3.7 Dam, River and Lake

It is difficult to obtain a truly representative sample when collecting surface water samples. Sampling point was selected carefully (near to bank in case of river) to avoid any kind of debris in the water. Considerable variations like seasonal stratification, runoff, rainfall and wind were also documented while collecting water sample especially from lake.



Figure 2: Water quality sampling from groundwater sources

2.4 Quality Assurance

For the quality assurance purpose, cross, field blank and replicate samples were also collected.

- Sites for cross samples were selected owing to site number divisible by 10. (10%)
- Sites for field blank and replicates were on the basis of site number divisible by 20 (5%).

Field blank and replicate samples were planned to be analyzed in the same laboratory to see the quality of distilled water and reproducibility of analytical results.

2.5 Distribution of Water Sources

The province wise detail of types and number of sources monitored during the year 2020 is tabulated as Table 4:

Table 4: Distribution of water sources

Sr. No.	Province/Cities	Total No. of Samples	Types and Number of Water Sources Monitored
1	Islamabad	24	Tube well (16), Tap (08)
2	Punjab	168	Tube well (80), Water Supply Schemes (10), Bore (16), Tap (6), Hand Pump (19), Injection Pump (27), Well (1), Filtration Plant (9)
3	KPK	44	Tube well (29), Water Supply Schemes (1), Tap (14)
4	Balochistan	76	Tube well (43), Water Supply Schemes (18), Tap (4), Wind Mill (1), Filtration Plant (1), Dam (1), Bore (1), Well (2), Karez (2), Spring (2), Hand Pump (1)
5	Sindh	103	Water Supply Schemes (9), Tap (48), Hand Pump (37), Tube Well (3), Filtration Plant (1), Rotter(2), Storage Tank (1), Bore (1), Well (1)
6	AJK	10	Water Supply Schemes (3), Spring (5), Filtration Plant (2)
7	Gilgit-Baltistan	10	Surface water (05), Water Supply Schemes (05)

2.6 Laboratory Testing

The water samples were analyzed for aesthetic, physico-chemical and bacteriological parameters by using American Public Health Association (APHA) standard methods [Americana Public Health Association, 23rd Addition]. The details of the parameters and methods used for their testing are listed in Table 5.

Table 5: Water quality parameters and methods used for analysis

Sr.#	Parameters	Analysis Method
1	Alkalinity (mg/l as CaCO ₃)	2320, Standard method (2017)
2	Arsenic (ppb)	AAS Vario 6, Analytik Jena AG (3111B APHA) 2017
3	Bicarbonate (mg/l)	2320, Standard method (2017)
4	Calcium (mg/l)	3500-Ca-D, Standard Method (2017)
5	Carbonate (mg/l)	2320, Standard method (2017)
6	Chloride (mg/l)	Titration (Silver Nitrate), Standard Method (2017)
7	Conductivity (μS/cm)	E.C meter, Hach-44600-00, USA
8	Hardness (mg/l)	EDTA Titration, Standard Method (2017)
9	Magnesium (mg/l)	2340-C, Standard Method (2017)
10	Nitrate as Nitrogen (mg/l)	Cd. Reduction (Hach-8171) by Spectrophotometer
11	pH	pH Meter, Hanna Instrument, Model 8519, Italy
12	Potassium (mg/l)	Flame photometer PFP7, UK
13	Sodium (mg/l)	Flame photometer PFP7, UK
14	Sulfate (mg/l)	SulfaVer4 (Hach-8051) by Spectrophotometer
15	Phosphate (mg/l)	8190 and 8048 Colorimeters (HACH)
16	TDS (mg/l)	2540C, Standard method (2017)
17	Turbidity (NTU)	Turbidity Meter, Lamotte, Model 2008, USA
18	Fluoride (mg/l)	4500-FC.ion-Selective Electrode Method Standard (2017)
19	Total Coliforms	9221-B,C&D, Standard Methods (2017) APHA
20	E-coli	9221-B,C&D, Standard Methods (2017) APHA



Figure 3: Laboratory testing of water samples

All test results were compared with the permissible limits of National Drinking Water Quality Standards of Pakistan (NDWQS) (Table 6) to evaluate the degree of fitness of water sources for drinking purpose.

Table 6: Water quality permissible limits for drinking water

Sr. #	Parameter	Units	Permissible Limits of NDWQS
1.	Alkalinity	mg/l	NGVS
2.	Bicarbonate	mg/l	NGVS
3.	Calcium	mg/l	NGVS
4.	Carbonate	mg/l	NGVS
5.	Chloride	mg/l	250
6.	Colour	TCU	Colourless
7.	Conductivity	μS/cm	NGVS
8.	Fluoride	mg/l	1.5
9.	Hardness	mg/l	500
10.	Iron	mg/l	0.3
11.	Magnesium	mg/l	NGVS
12.	Odour	-	Unobjectionable
13.	Nitrate-N	mg/l	10
14.	pH	-	6.5-8.5
15.	Potassium	mg/l	NGVS
16.	Sodium	mg/l	NGVS
17.	Sulfate	mg/l	NGVS
19.	TDS	mg/l	1000
20.	Turbidity	NTU	<5
22.	Arsenic	μg/l	50
23.	*Total Coliforms	CFU/100ml	0
24.	*E-Coli	CFU/100ml	0

CFU = Colony Forming Unit

TCU = Total Colour Units

WHO= World Health Organization

NGVS= No Guideline Value Set

NDWQS= National Drinking Water Quality Standards

NTU=Turbidity Unit

* For MPN: <2 MPN/100ml which is considered equivalent to "zero"

2.7 Risk Mapping

Water pollution risk assessment is carried out based on potential health impacts of water quality contaminants as well as socio-economic impacts for mitigation of these contaminants. Risk scores ranging from 0 to 1 based on presence of contaminants and their types are assigned to different contaminants (Table 7).

Based on potential health and other socio-economic impacts, risk scores were translated into risk ranges from 0 to 7 (Table 8). Risk score “0” is assigned to the water source, where the parameter under study were marginally safe.

Low Risk: Water samples having either bacterial contamination is high salt concentration only have been considered as low risk contaminants. These contaminants can be removed with less effort and in short time.

Medium Risk: Samples with health hazardous contaminants such as Nitrate, Fluoride, Arsenic is present either solely or in combination with bacterial contamination is considered as medium risk.

High Risk: Water samples having multiple chemical contaminants along with bacteriological contamination are considered as potential high-risk.

Table 7: Ranking method for risk mapping

Contaminants Type	Reasoning for risk determination	Risk Score	Potential Risk
Level of Total Dissolved Solids, Total Coliforms, Fluoride, and Nitrate within permissible limits	Within permissible limits	0	Marginally Safe
Bacterial contamination only (Total Coliforms & Fecal Coliforms)	Bacterial contamination can be removed by simple and low-cost disinfection method	1	Low risk
Total dissolved solids only	Salt concentration without health threatening chemical contamination can be reduced by easy and low-cost methods	2	Low risk
Single chemical contamination	Special ion exchange media would be required	3	Medium Risk

Contaminants Type	Reasoning for risk determination	Risk Score	Potential Risk
Single chemical plus microbial contamination	Health hazardous Double and intensive treatment is required	4	Medium Risk
Double chemical contamination	Health hazardous Intensive chemical treatment is required	5	High Risk
Multiple chemical contamination and microbial contamination	Health hazardous Intensive chemical treatment is required	6	High Risk

Table 8: Risk ranges for potential contamination

Sr. No.	Risk	Risk Range
1	Low	1-2
2	Medium	3-4
3	High	5-6

2.8 Hydro-geochemical Facies

The hydro-chemical data is subjected to various conventional graphical plots to understand and identify the dominant ionic constituents and water types in the aquifer of the monitoring cities. The water quality data of 2015 was assessed by using Piper diagrams to represent the geochemical behavior of water. Piper diagram is a multifaceted plot where in milliequivalents percentage concentrations of major cations (Calcium, Magnesium, Sodium and Potassium) and anions (Bicarbonate, Sulphate, and Chloride) are plotted in two triangular fields, which were then projected further into the central diamond field (Piper, 1944). In contrast, Durov diagram is a composite plot consisting of 2 ternary diagrams where the milliequivalents percentages of the cations of interest were plotted against that of anions of interest; sides form a central rectangular, binary plot of total cations vs. total anions concentrations (Durov, 1948). In this study Durov Diagrams plotted in AquaChem 10.0 also define the hydro-chemical processes involved along with water type.

The water type of a given water source is influenced by the interaction of the water with the minerals that it has been in contact with. These types of interactions include precipitation, dissolution, ion exchange, geological structure and the mineralogy of watershed/aquifer. In AquaChem 10.0, the water type is determined by calculating the percent contribution of each cation and anion to the total concentration of ions in solution on equivalents per liter basis. The water type can be defined as a "short" water type or a "long" water type. The "short" water type is calculated by concatenating the cation and anion with the highest equivalent concentrations. The "long" water type is calculated by concatenating cations and anions with concentrations above a specified percent contribution threshold. The threshold for determining if an ion is "major" is normally taken as 10%. In some countries a threshold of 12.5% is more common. The higher the threshold, the smaller number of parameters may exceed this threshold level. Usually, only major ions (Sodium, Calcium, Magnesium, Bicarbonate, Sulphate, and Chloride) will have a concentration that allows them to exceed the defined threshold (Waterloo Hydrogeologic, 2020).

Chapter 3: Results and Discussion

This chapter pertains to the water quality results and related situation analysis of 29 major cities of Pakistan, region and city wise description of potential contaminants exceeding the National Drinking Water Quality Standards (NDWQS), trend analysis over the time (2002-20), Risk maps showing health impacts of hazardous contaminants and Durov diagrams indicating groundwater type. Location wise detailed water quality profile of 435 monitored sources can be seen at Annexures 1 to 29.

3.1 Islamabad Capital Territory (ICT)

Water supplies of Islamabad are mostly supplemented by groundwater through more than 290 tubewells. Twenty four fixed locations were monitored for the water quality of city. In total, (29%) sources were found contaminated and unfit for the drinking purpose mainly due to bacteria and Iron (Fe) as shown in Table 9.

Table 9: The level of major parameters exceeding the NDWQS in Islamabad City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	24	04	17
2	Total Coliforms	CFU/100 ml	24	06	25
3	E. coli	CFU/100 ml	24	03	13

In Islamabad, the main sources of domestic water supply are, the reservoirs built at Simly and Khanpur, tube wells and few springs located at Saidpur, Nurpur and Shahdra-hills. About 1.12 million inhabitants of the city receive some or all of their drinking water from these water sources. The conventional treatment methods such as coagulation, flocculation, sedimentation, filtration and chlorination are used for the treatment of Simly and Khanpur reservoir's water to make it potable. The level of microbial contamination (25%) in city raises the questions on the sustainable supply of safe drinking water from these filtration plants. Further, the presence of fecal contamination indicated by the presence of *E-coli* in 13% of the sources also signposts the unsafe disposal of domestic sewage.

Time trend analysis by comparing the 2020 data with previous monitoring studies (2002-2015) reveals an improvement in safe water access from 32% to 71% indicating significant improvement over the time (Figure 4).

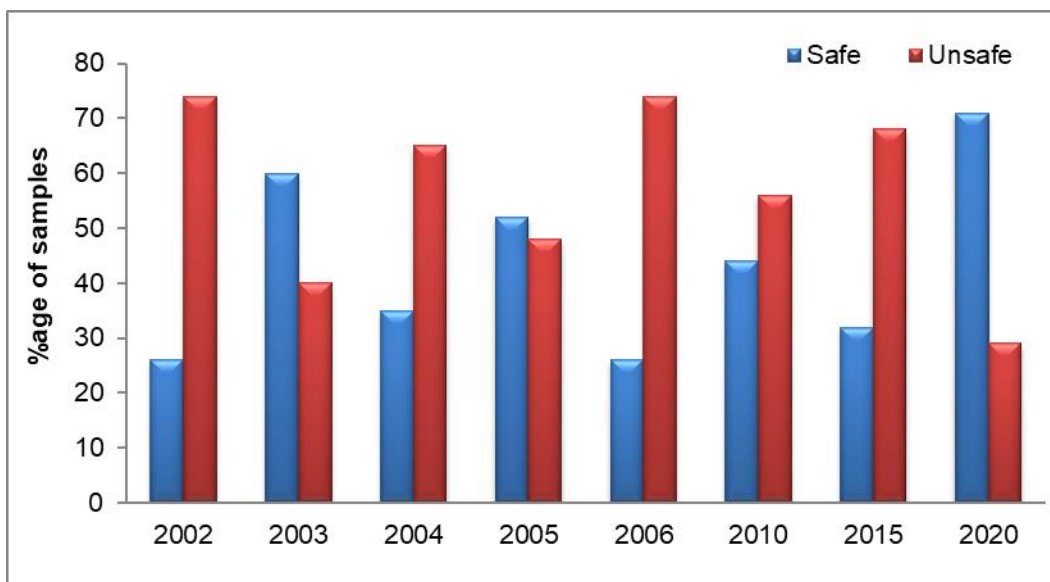


Figure 4: Year wise Drinking Water Quality Status of Islamabad City

The possible reason for the improvement of safe drinking water quality may be attributed to bringing the chlorinators at water supply tubewells into functional state. Further improvement requires sustainable and continuous disinfection of ICT water supply as well as disposal of sewage after proper treatment.

The hydro-chemical facies shown as Durov Diagram (Figure 5) indicate the water rock interaction in terms of Calcium-Magnesium-Bicarbonate (46%), Magnesium-Calcium-Bicarbonate (29%) and Calcium-Sodium-Magnesium-Bicarbonate (8%). Therefore, it can be concluded that main type of hardness in the water supplies is temporary.

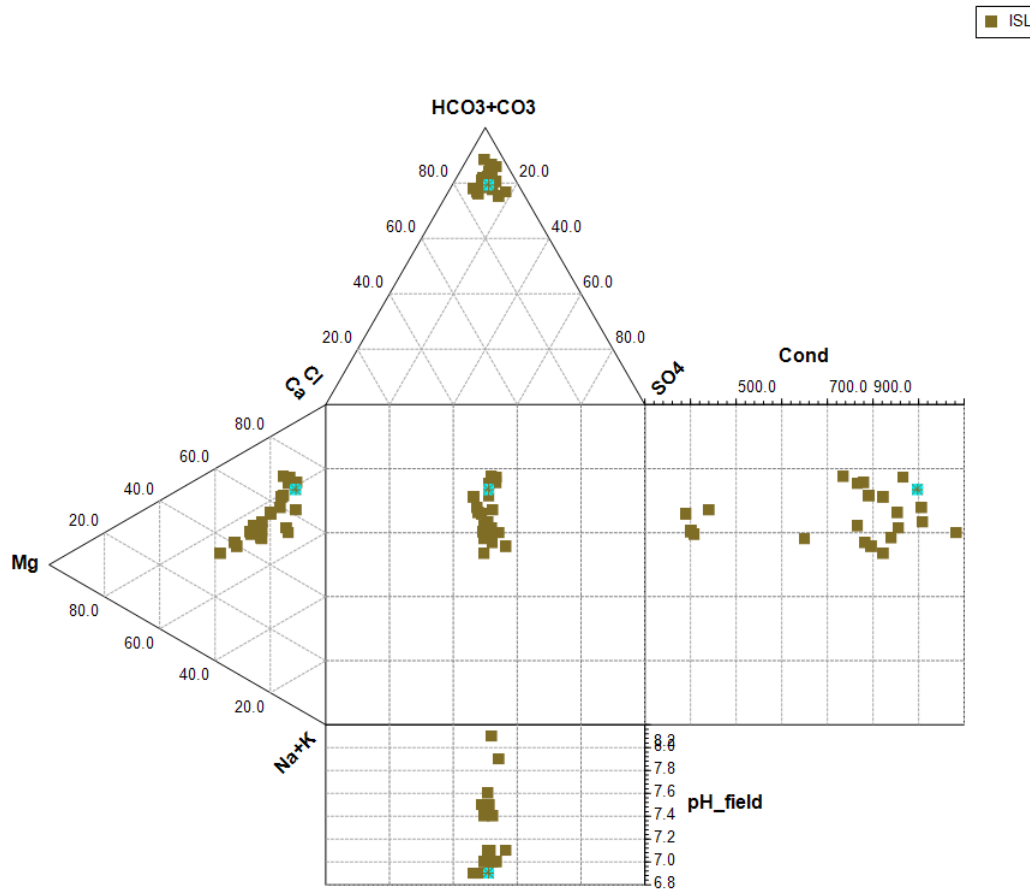


Figure 5: Durov diagram of Islamabad City

3.2 Punjab Province

Punjab occupies an area of about 205,344 km². It is located between 27°44'N 69°20'E and 34°15'N 75°20'E at the northwestern edge of the geologic Indian plate in South Asia. Nearly 60% of Pakistan's population lives in the province.

3.2.1 Bahawalpur

Bahawalpur is the 11th largest city of Pakistan with a population of 0.76 million (Population Census Report, 2017). In Bahawalpur city, 10-15% of the population consumes piped water supply, while remaining depends on the domestic groundwater (Grayman, et al., 2003). Altogether, 25 water samples were collected from city. The data when compared with the NDWQS revealed that 76% of the water samples were unsafe due to microbial contamination, excessive Arsenic, Fluoride, Hardness, Iron, Turbidity and TDS (Table 10).

Table 10: The level of major parameters exceeding the NDWQS in Bahawalpur City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Arsenic (As)	µg/l	25	7	28
2	Fluoride (F)	mg/l	25	1	4
3	Hardness	mg/l	25	4	16
4	Iron (Fe)	mg/l	25	9	36
5	Turbidity	(NTU)	25	5	20
6	TDS	(mg/l)	25	6	24
7	Total Coliforms	(MPN/100 ml)	25	5	20
8	E-coli	(MPN/100 ml)	25	4	16

A comparison of the present data with previous monitoring studies (2015) shows an improvement from 4% to 28% in the water quality of the city (Figure 6.)

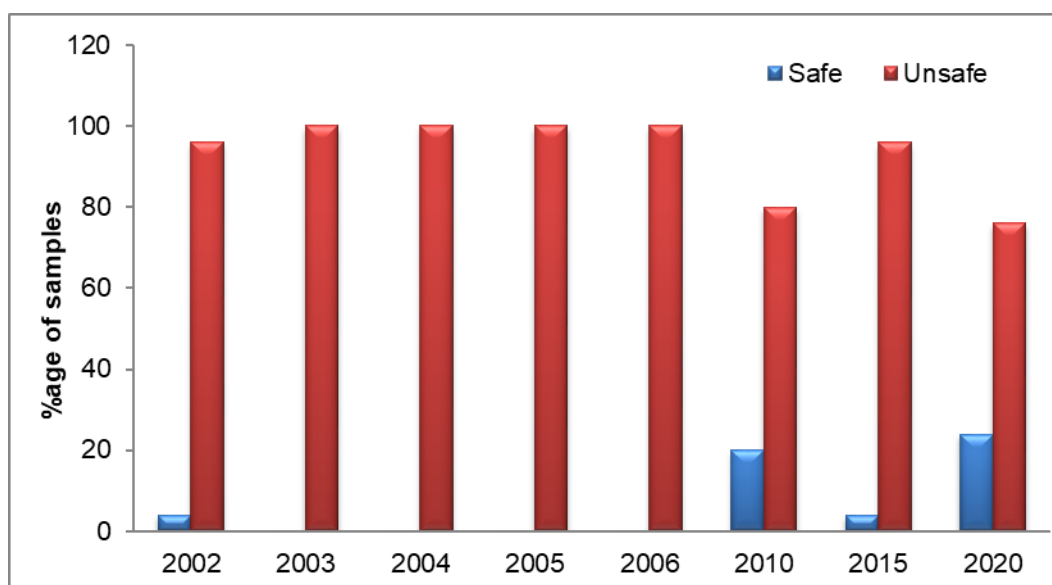


Figure 6: Year wise Drinking Water Quality Status of Bahawalpur City

3.2.2 Faisalabad

Faisalabad is the third most populous city in Pakistan, and the second-largest in Punjab. The total population of the city is around 3.2 million (Population Census Report, 2017). Faisalabad is located in Rechna Doab, the land between Chenab and Ravi rivers. The water supply of the city is dependent on groundwater and surface water sources.

In total, 22 locations were monitored for the water quality of the city in 2020. Out of these, 12 sites are fixed sites and being monitored since 2002, whereas 10 new locations were added recently in the 2020 monitoring plan. The data of 2020 showed 59% sources as unfit for drinking purposes. The main causes were microbial contamination, Chloride, Fluoride, Hardness, and TDS (Table 11). The groundwater samples of the city belong to Sodium-Bicarbonate (36%), Magnesium-Bicarbonate (18%), Sodium-Chloride (18%), Calcium-Bicarbonate (14%) and Sodium-Sulphate (9%). This suggests that groundwater has both temporary and permanent type of hardness at different sites (Figure 7).

Table 11: The level of major parameters exceeding the NDWQS in Faisalabad City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Chloride (Cl)	mg/l	22	5	23
2	Flouride (F)	mg/l	22	2	9
3	Hardness	mg/l	22	3	14
4	Iron (Fe)	mg/l	22	5	23
5	Nitrate (NO ₃)	mg/l	22	4	18
6	TDS	mg/l	22	5	23
7	Total Coliforms	MPN/100 ml	22	3	14
8	E-coli	MPN/100 ml	22	3	14

A visible improvement can be seen in the water quality of the city from 2010 (20%) to 2020 (40%) (Figure 7).

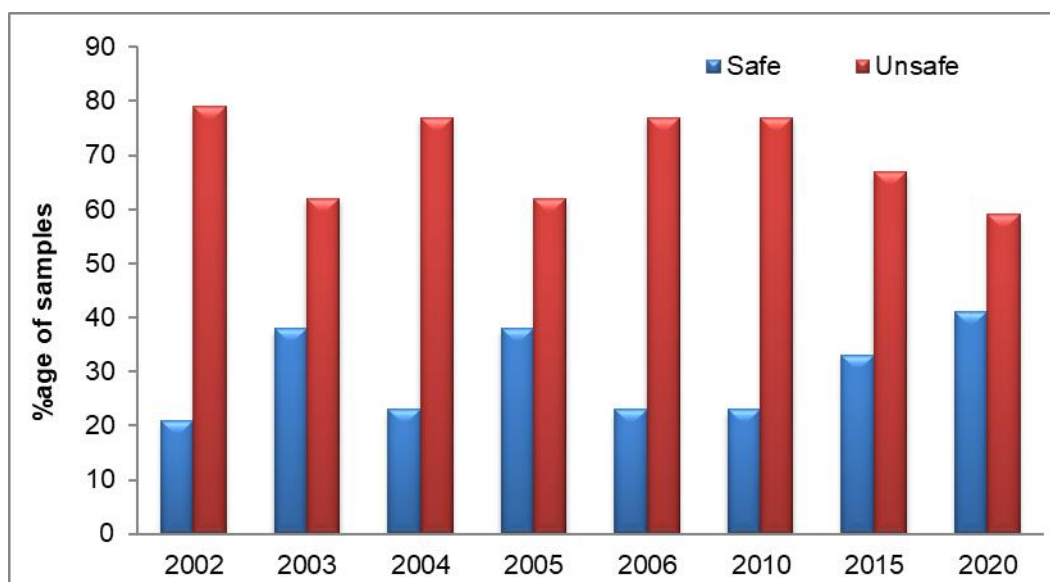


Figure 7: Year wise Drinking Water Quality Status of Faisalabad City

3.2.3 Gujranwala

The Gujranwala is the 5th most populous metropolitan city. Groundwater as the main drinking water sources is available at a depth of 17 m, while the depths of tubewells vary from 137 to 198 m.

The results of samples collected from 14 fixed sites showed that 50% of the sources were unsafe for drinking purposes only due to microbial contamination. Fortunately, no chemical contamination was detected (Table 12). The microbial contamination in the city is worrisome and depicts the unsafe disposal of domestic sewage, source protection and inadequate disinfection practices.

Table 12: The level of major parameters exceeding the NDWQS in Gujranwala City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Coliforms	MPN/100 ml	14	7	50
2	E-coli	MPN/100 ml	14	7	50

A water quality trend analysis over the years (2002-2006) indicates a decline in water quality from 35% safe to 7%. However, significant improvement in the water quality of the city observed from 2010-2020 (10% to 50%) (Figure 8). Local water supply agencies still require undertaking all possible measures to improve the remaining 50% of the unsafe sources.

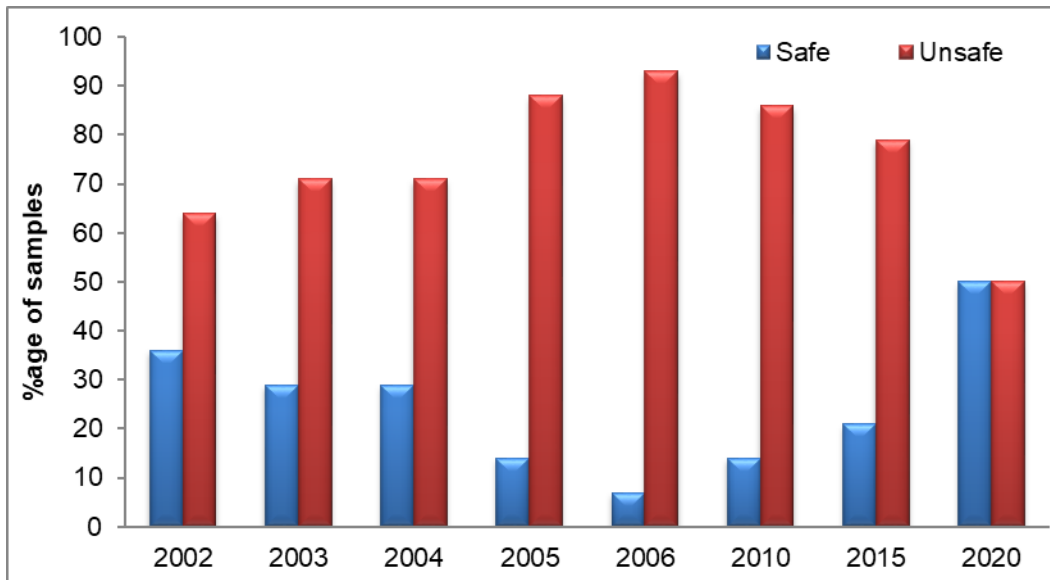


Figure 8: Year wise Drinking Water Quality Status of Gujranwala City

3.2.4 Gujrat

Gujrat is the 20th largest city of Pakistan by population and is located between river Jhelum and Chenab (Population Census Report, 2017). In total, 9 fixed sites within the city are monitored regularly. All the sources were found safe.

Compared to 2015, an abrupt water quality improvement trend is observed as the safe water sources increased from 67 to 100% (Figure 9).

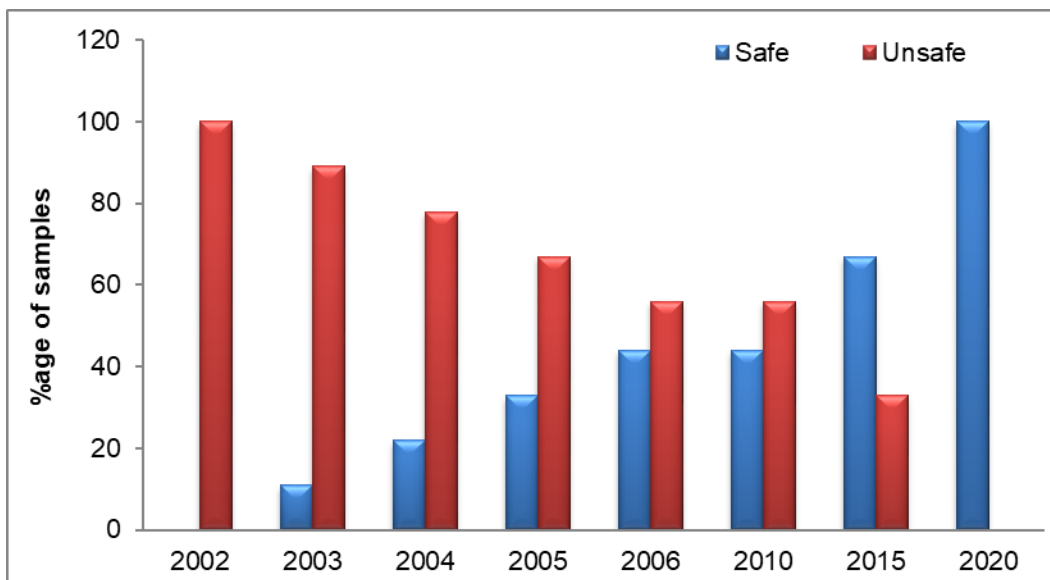


Figure 9: Year wise Drinking Water Quality Status of Gujrat City

3.2.5 Kasur

Kasur, located between Ravi and Sutlej rivers, is an industrial city. Ten water samples were collected from the pre-selected sources of the city. Table 13 shows that 90% of the samples were found safe. Remaining 10% samples were unsafe for drinking due to microbial contamination. The monitored sources belong to Sodium-Bicarbonate type hydro-chemical facies (Figure 10) which shows fresh water recharge.

Table 13: The level of major parameters exceeding the NDWQS in Kasur City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Turbidity	NTU	10	1	10
2	Total Coliforms	MPN/100 ml	10	1	10
3	E-coli	MPN/100 ml	10	1	10

Figure 10 shows an improved water quality situation in the year 2020.

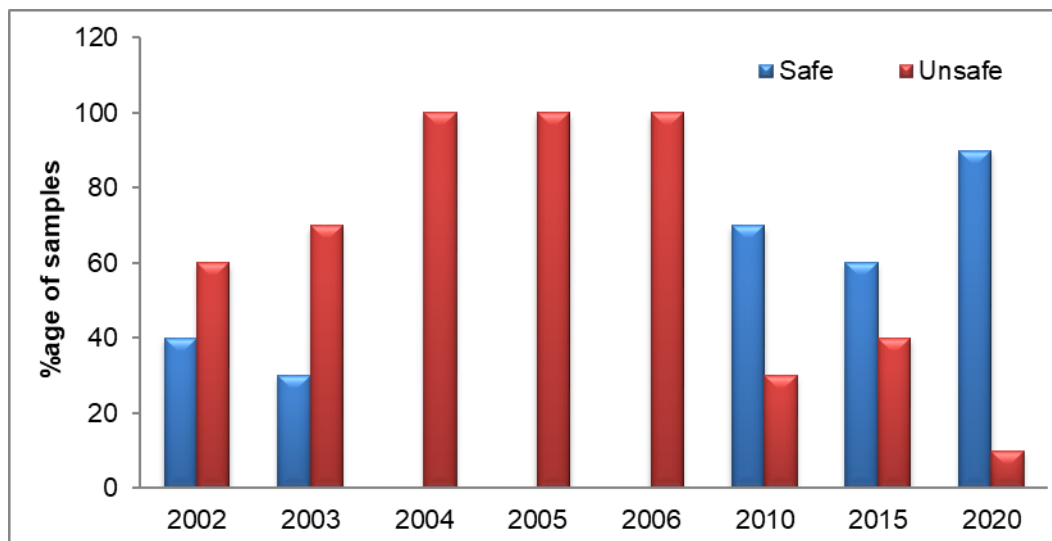


Figure 10: Year wise Drinking Water Quality Status of Kasur City

3.2.6 Lahore

Lahore is the 2nd largest populous city of Pakistan. Altogether, 16 water samples were collected from fixed locations. 31% of the monitored sources were found unsafe due to high level of Arsenic, Turbidity, Iron, TDS and bacteria (Table 14). Remaining 69% sources were found safe for drinking purposes.

Table 14: The level of major parameters exceeding the NDWQS in Lahore City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1.	Arsenic (As)	µg/l	16	1	6
2.	Iron (Fe)	mg/l	16	1	6
3.	TDS	mg/l	16	1	6
4.	Total Coliforms	MPN/100 ml	16	2	12
5.	E-coli	MPN/100 ml	16	2	12

Figure 11 shows an improvement in the quality of drinking water in Lahore over the time. However, in 2020 there is slight decline in water quality and situation seems same as it was observed in 2010.

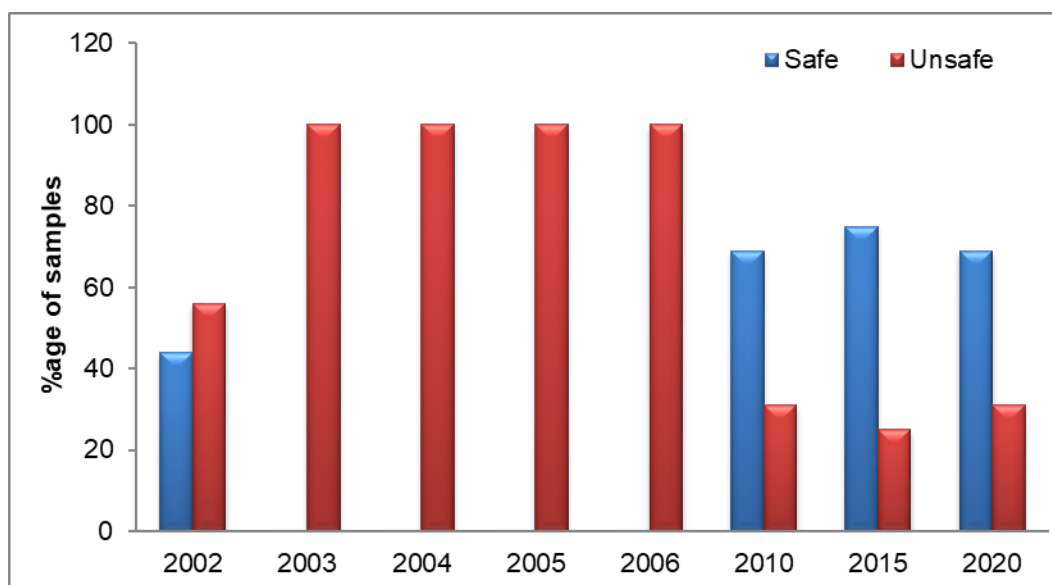


Figure 11: Year wise Drinking Water Quality Status of Lahore City

3.2.7. Multan

Multan is the main industrial city of South Punjab having population of 1.87 million as per 2017 Population Census Report. Chenab river is the main source of groundwater recharge. Sixteen sources from fixed locations were monitored. Overall, 94% of the sources were unsafe for drinking purposes due to microbial contamination (69%) and Arsenic (63%) as given in Table 15.

Untreated municipal waste and its mixing with the safe water supplies is possibly the major cause of microbial contamination; whereas, Arsenic is geogenically present in the mineral composition of groundwater of the southern Punjab.

Table 15: The level of major parameters exceeding the NDWQS in Multan City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Arsenic (As)	µg/l	16	10	63
2	Total Coliforms	MPN/100 ml	16	11	69

The trend analysis with previous monitoring studies indicates further deterioration in the water quality situation from 19% in 2015 to 6% in 2020 (Figure 12). This is an alarming situation for the local water supply agencies.

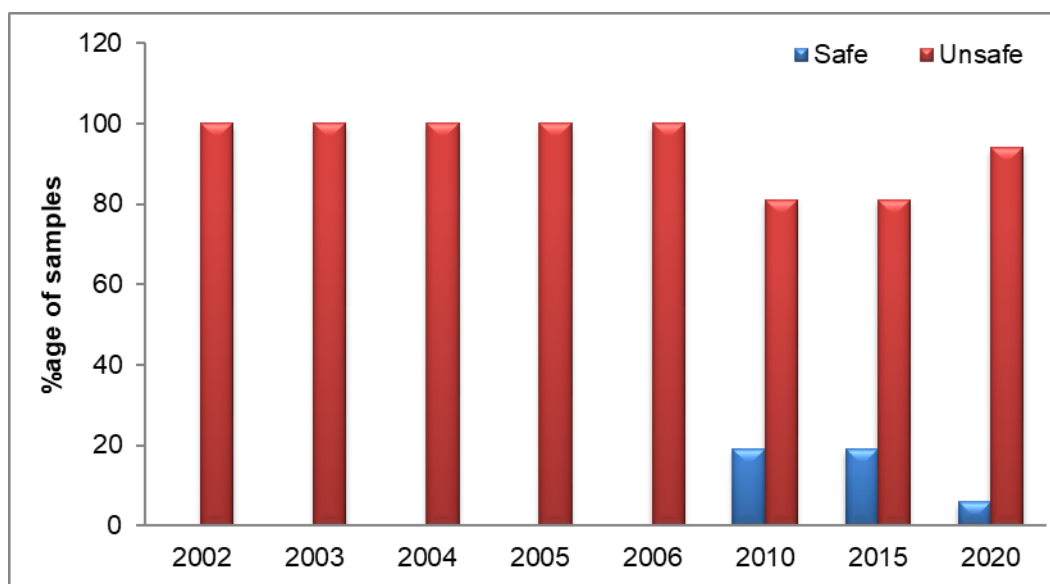


Figure 12: Year wise Drinking Water Quality Status of Multan City

3.2.8 Rawalpindi

Rawalpindi is the 4th largest city in Pakistan by population. Khanpur and Rawal Dams supply water to the city.

The analytical results show 38% of the sources as unsafe. The main reasons were microbial contamination (38%), Nitrate (31%), Iron (23%) and excessive level of TDS (8%) (Table 16). Presence of fecal contamination represented by the E-coli in three water sources is important and requires immediate attention.

Table 16: The level of major parameters exceeding the NDWQS in Rawalpindi City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	TDS	mg/l	13	1	8
2	Nitrate (NO ₃)	mg/l	13	2	15
3	Iron (Fe)	mg/l	13	2	15
4	Total Coliforms	MPN/100 ml	13	5	38
5	E-coli	MPN/100 ml	13	3	23

A comparison of the analytical data of 2020 with previous monitoring studies (Figure 13) indicates slight decline after 2010.

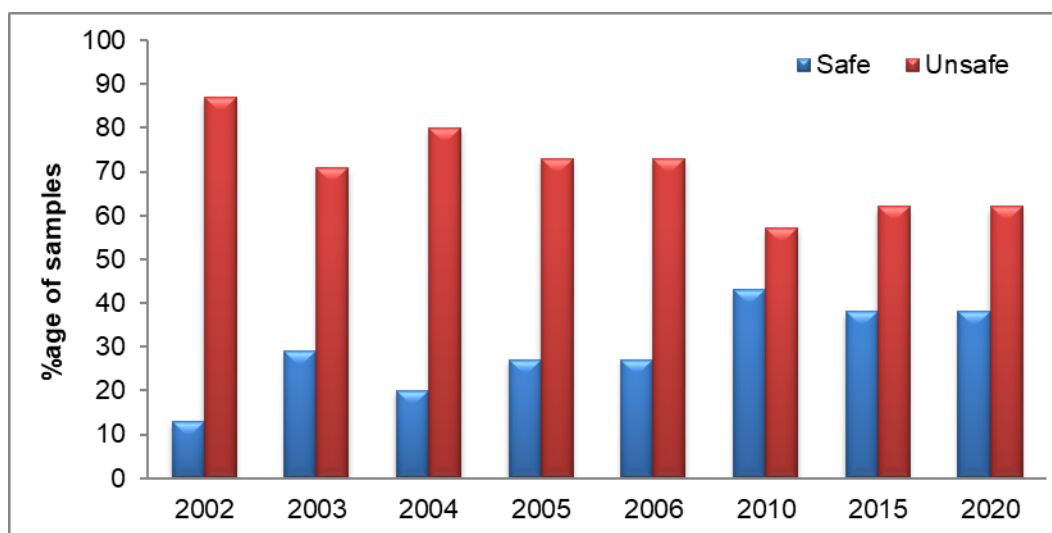


Figure 13: Year wise Drinking Water Quality Status of Rawalpindi City

3.2.9 Sargodha

Sargodha is a densely populated and agricultural city of Punjab Province. Twenty four sources from fixed locations of the city were monitored. The test data reflected 83% of water sources as unsafe for drinking purposes (Figure 14). The major contaminants are microbial contamination, excessive Chloride, Nitrate, Hardness, Iron, and TDS (Table 17). Brackish water is usually found in the study area due to major contribution from Sodium and Chloride ions. Furthermore, the type of hardness is of permanent nature in most of the water supplies of the city (Figure 14).

Table 17: The level of major parameters exceeding the NDWQS in Sargodha City

Sr. No.	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Hardness	mg/l	24	14	58
2	Iron (Fe)	mg/l	24	5	21
3	TDS	mg/l	24	14	58
4	Chloride (Cl)	mg/l	24	9	38
5	Nitrate (NO ₃)	mg/l	24	9	38
6	Total Coliforms	MPN/100 ml	24	5	21

Figure 14 shows that the safe drinking water quality has declined from 25% in 2010 to 17% in 2020.

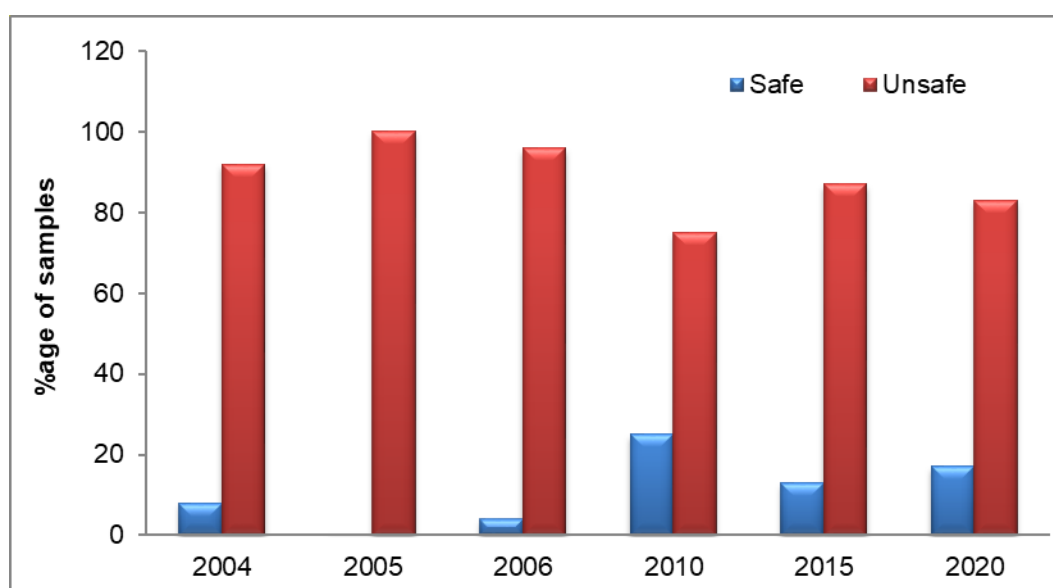


Figure 14: Year wise Drinking Water Quality Status of Sargodha City

3.2.10 Sheikhpura

Sheikhpura is located in the Rechna Doab, the area between Ravi and Chenab river. Ten water samples were collected from the major areas of the city as per the sampling design. It was found that 60% sources were unsafe for drinking purposes mainly due to microbial contamination, excessive levels of Arsenic, Hardness, Iron, TDS, Turbidity, and Nitrate (Table 18). This is a quite alarming situation indicating diversity of chemical contaminants along with presence of fecal contamination.

Table 18: The level of major parameters exceeding the NDWQS in Sheikhpura City

Sr. No.	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Arsenic (As)	µg/l	10	1	10
2	Hardness	mg/l	10	2	20
3	Iron (Fe)	mg/l	10	1	10
4	TDS	mg/l	10	2	20
5	Turbidity	NTU	10	1	10
6	Nitrate (NO ₃)	mg/l	10	3	30
7	Total Coliforms	MPN/100 ml	10	2	20
8	E-coli	MPN/100 ml	10	2	20

Over the years, the quality of water has declined (73% safe in 2010) to 55% in 2015 and is almost the same in 2020 (Figure 15).

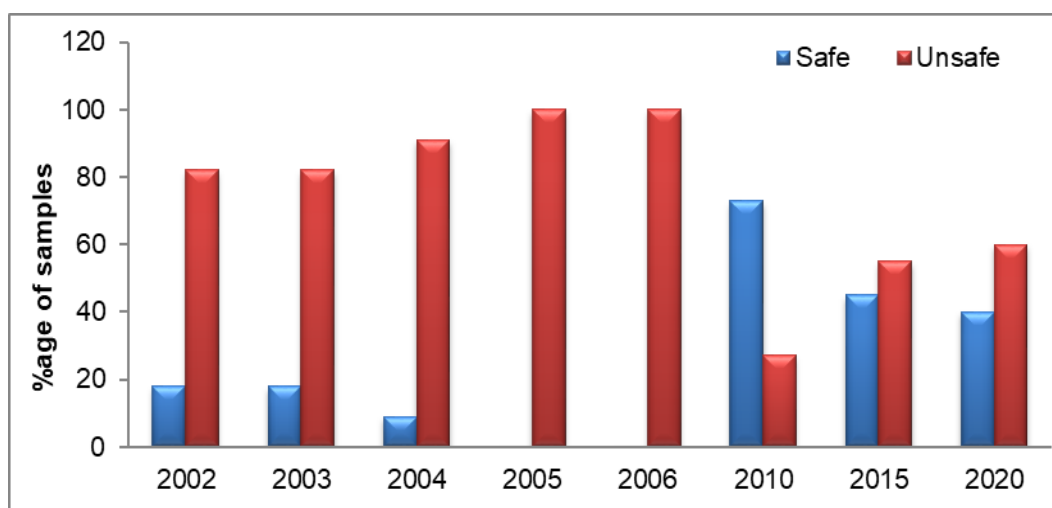


Figure 15: Year wise Drinking Water Quality Status of Sheikhpura City

3.2.11 Sialkot

Sialkot has been known as Pakistan's largest sports city. It is known as a city full of industries such as leather, textile, metallurgical and pharmaceutical. Ten drinking water sources from fixed locations of the city were monitored. Fortunately, 100% sources were safe for drinking purposes. A gradual improvement is evident in the city's drinking water supplies i.e. 90% safe in 2010-2015 and 100% safe in the year 2020 (Figure 16).

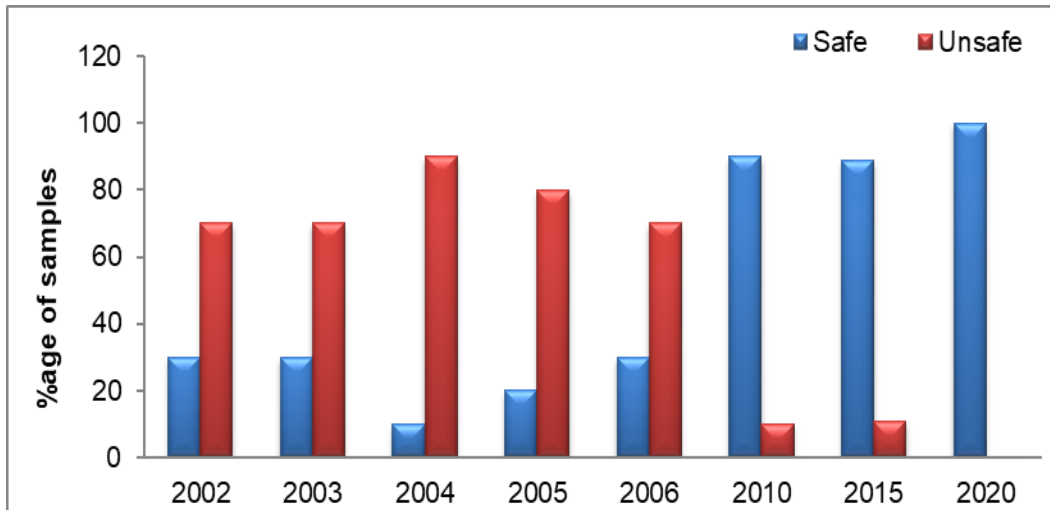


Figure 16: Year wise Drinking Water Quality Status of Sialkot City

3.2.12 Risk Map of Punjab

A risk map indicates the health impact of different contaminants in the study area. For the mapping, quality of groundwater is divided into four categories as marginally safe, low, medium and high-risk. Figure 17 shows that groundwater of Gujrat and Sialkot cities are marginally safe. The groundwater in Faisalabad, Kasur and some areas of Sargodha and Gujranwala cities are in low-risk category. However, the drinking water in Multan, Bahawalpur, Lahore, Sheikhupura, Sargodha and in certain areas of Gujranwala and Faisalabad are categorized as medium-risk region. Similarly, drinking water at certain areas of Sargodha, Bahawalpur, Faisalabad and Sheikhupura is placed at high-risk (Figure 17).

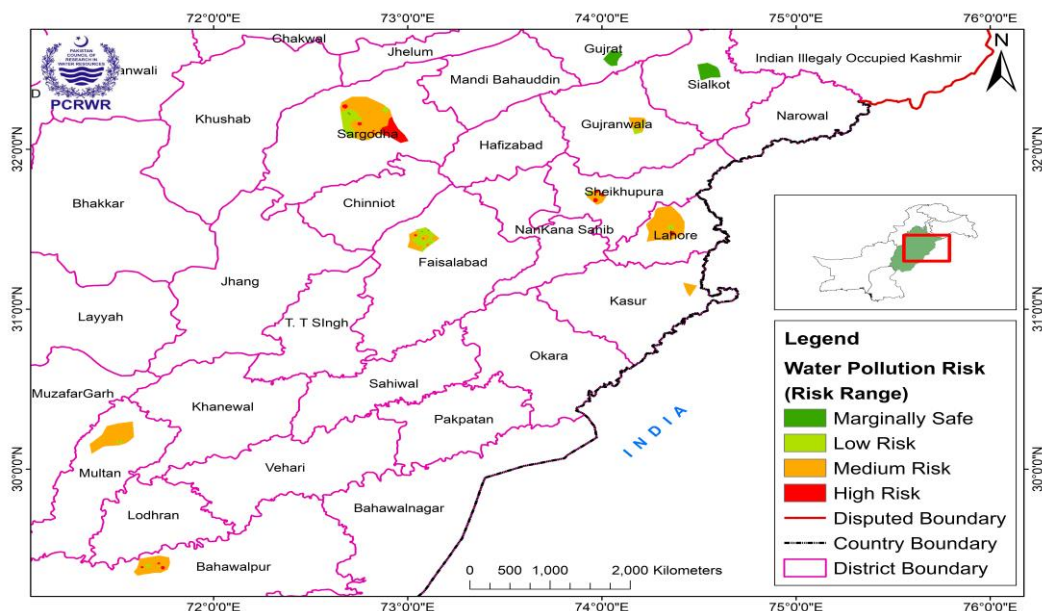


Figure 17: GIS risk mapping of Punjab province

Immediate measures are required in the medium and high-risk regions. Detail of hotspot areas of Punjab province according to contaminants types are depicted in Annexure (30-33).

3.2.13 Overall Hydro-chemical Facies of Punjab

Durov diagram (Figure 18) was generated to describe the hydro-chemical characteristics and type of groundwater. The pH part of the plot reveals that groundwater in Punjab is neutral to alkaline which is preferred for drinking. The Electrical Conductivity of most of groundwater samples lies in the range of drinking water standards except some samples from Sargodha city that have higher EC (310-22140 $\mu\text{S}/\text{cm}$) due to brackish water. The hydro-chemical facies shown as Durov Diagram indicates the water rock interaction in terms of Sodium-Bicarbonate (45% of the total samples), Calcium-Bicarbonate (17%), Magnesium-Bicarbonate (14%), Sodium Carbonate (7%) and Sodium Sulphate (5%). The dominant Sodium-Bicarbonate type water designates weathering of silicate rocks or ion exchange during water infiltration through the subsurface soil. Moreover, Calcium-Magnesium-Bicarbonate type not only suggests fresh recharge to the groundwater but also specifies the dissolution of calcite or dolomite minerals (Mondal et al., 2012). Overall, main type of hardness in the groundwater of Punjab province is of temporary nature.

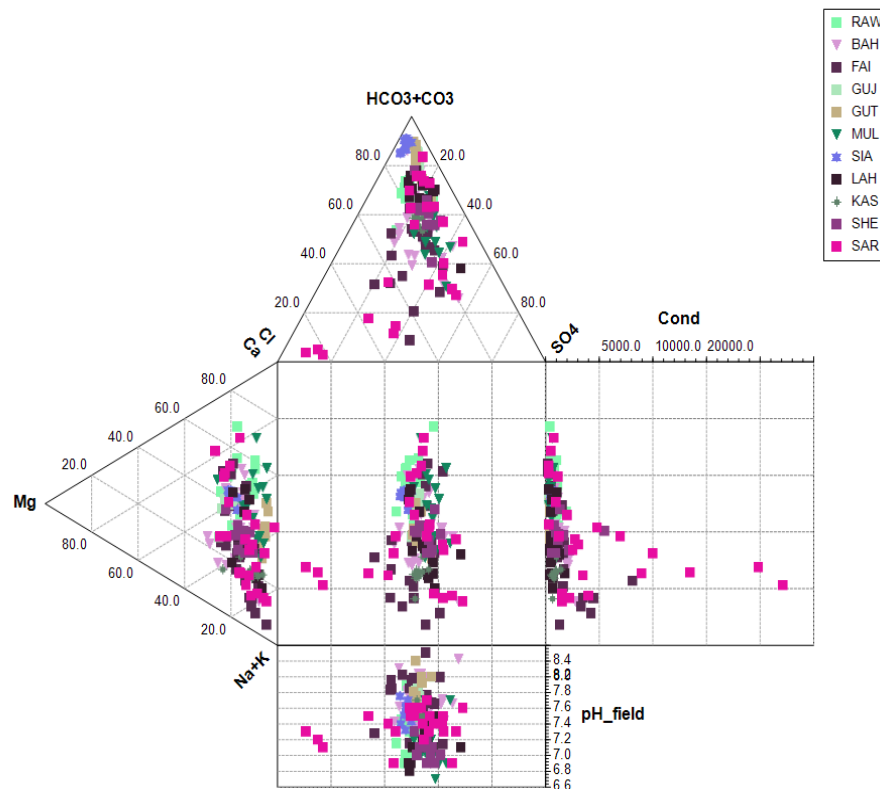


Figure 18: Durov Diagram depicting geo-chemical processes in groundwater of Punjab

3.3 Khyber Pakhtunkhwa Province

Chemical and microbial aspects of water samples were collected from four cities of KPK, i.e. Abbottabad, Mardan, Peshawar and Mangora.

3.3.1 Abbottabad

Water quality samples were collected for the monitoring purpose from 11 different sources of Abbottabad. In 2015, four sampling locations were non-functional. To manage this situation, alternative four sites were chosen for samples collection in 2020.

Microbiologically all samples were found safe. However, 55% of the sources contained Iron (Fe) beyond the NDWQS level of 0.03 mg/l (Table 19).

Table 19: The level of major parameters exceeding the NDWQS in Abbottabad City

Sr.#	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	11	6	55

Figure 19 indicates gradual drinking water quality improvement from 43% in 2015 to 45% in 2020.

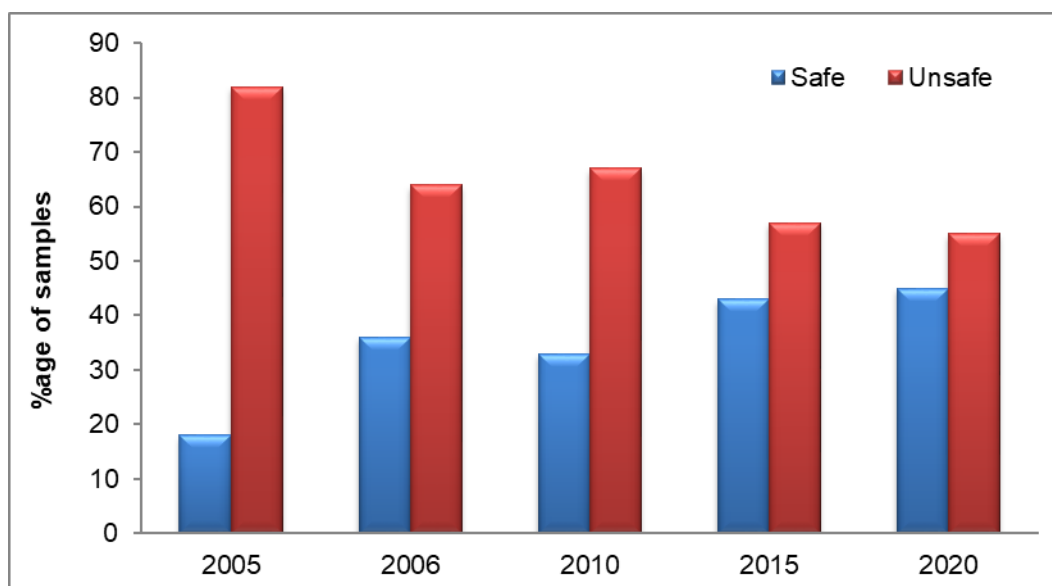


Figure 19: Year wise Drinking Water Quality Status of Abbottabad City

3.3.2 Mangora

Ten pre-determined locations of drinking water were monitored for water quality in 2020. One location was closed; therefore, alternative site was chosen for the monitoring purpose. Major contaminant found was Iron (Fe) in 20% of the sources. All other chemical and microbiological parameters were within the safe permissible limits (Table 20).

Table 20: The level of major parameters exceeding the NDWQS in Mangora City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	10	2	20

The water quality of the Mangora shows significant improvement possibly due to improvement of sanitation practices and increased public awareness (Figure 20).

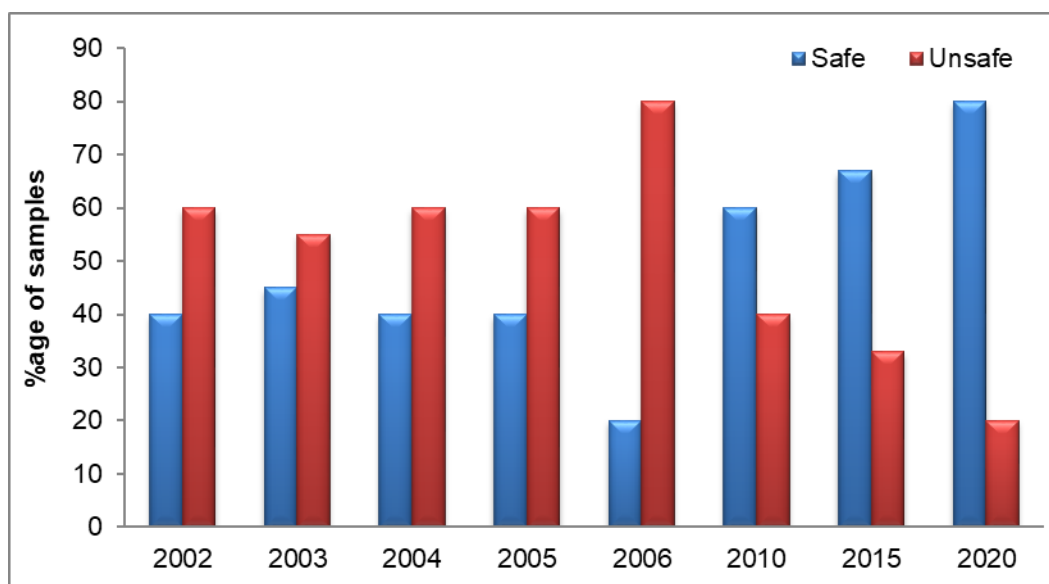


Figure 20: Year wise Drinking Water Quality Status of Mangora City

3.3.3 Mardan

In total, 11 fixed sites were monitored for water quality in the year 2020. Iron (Fe) as a major contaminant was found in 45% of monitored water sources, whereas, no other microbial or chemical contamination was detected in any water source (Table 21).

Table 21: The level of major parameters exceeding the NDWQS in Mardan City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	11	5	45

Figure 21 shows a significant improvement in water quality of Mardan city from 17% to 55% (from 2002 to 2020).

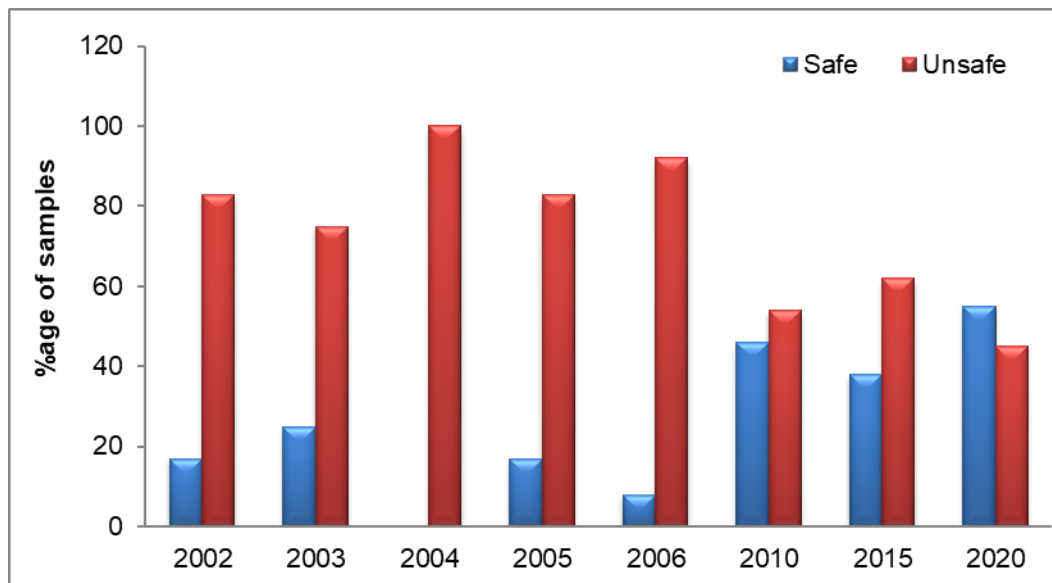


Figure 21: Year wise Drinking Water Quality Status of Mardan City

3.3.4 Peshawar

In total, 12 water samples were collected covering the major localities of the Peshawar city as per established sampling design. Analytical findings shows that 50% of water samples collected was unfit due to presence of Iron (Fe). Microbial and chemical contaminations were not found (Table 22).

Table 22: The level of major parameters exceeding the NDWQS in Peshawar City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	12	6	50

A year wise trend analysis of water quality of consecutive years shows a decline in safe water percentage of city from 60% in 2010 to 50% in 2020 (Figure 22).

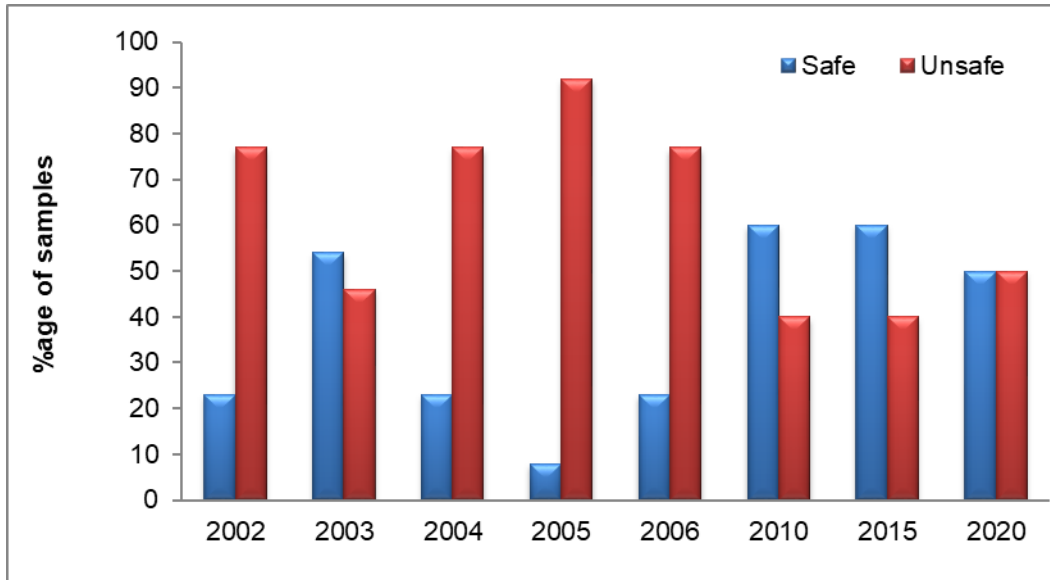


Figure 22: Year wise Drinking Water Quality Status of Peshawar City

3.3.5 Risk Map of KPK

The risk map of monitored cities in KPK is shown as Figure 23. It can be seen that groundwater of Peshawar, Abbottabad, and Swat cities are marginally safe. However, groundwater in some areas of Mardan city is at Low-risk. Details of hotspot areas of KPK according to contaminants type is given as Annexure 34-35.

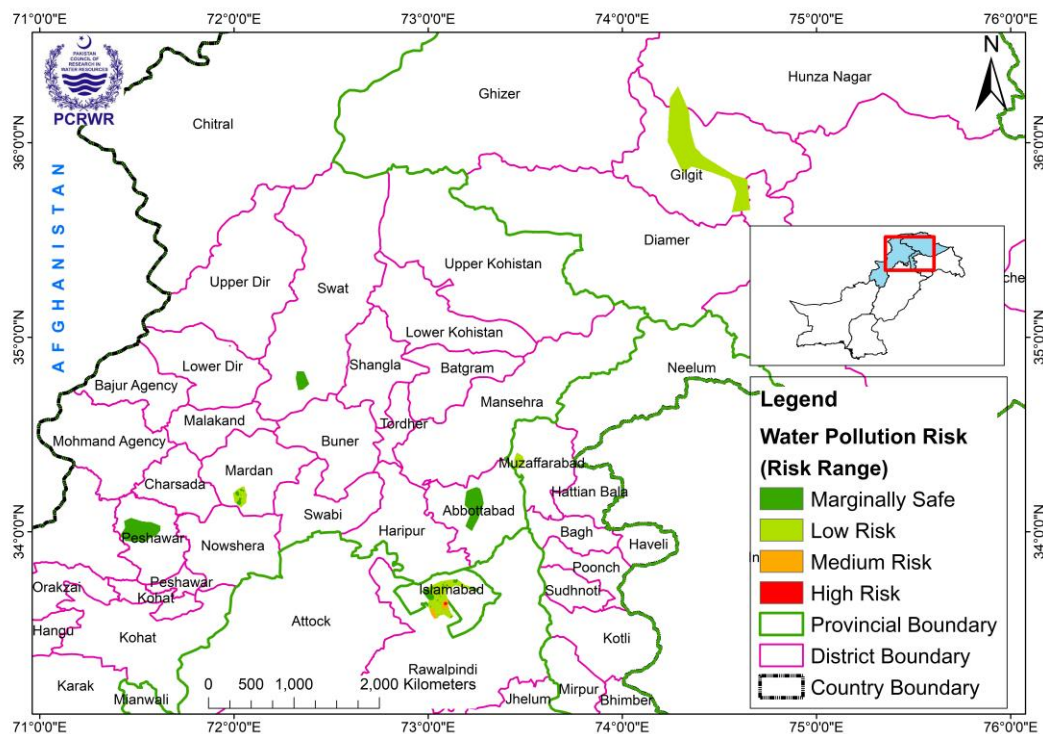


Figure 23: GIS risk mapping of KPK province

3.3.6 Overall Hydro-chemical Facies of KPK

Plotting the 44 samples of KPK as Durov diagram (Figure 24) represents the hydro-geochemical behavior of the Province. This shows that the groundwater mainly belongs to rocks that have dominance of Calcium-Bicarbonate (43%) followed by Magnesium-Bicarbonate (36%) and Sodium-Bicarbonate (20%). No facies of cation with Sulphate and Chloride are observed. It can therefore, be said that groundwater in the study area is mainly of temporary hardness.

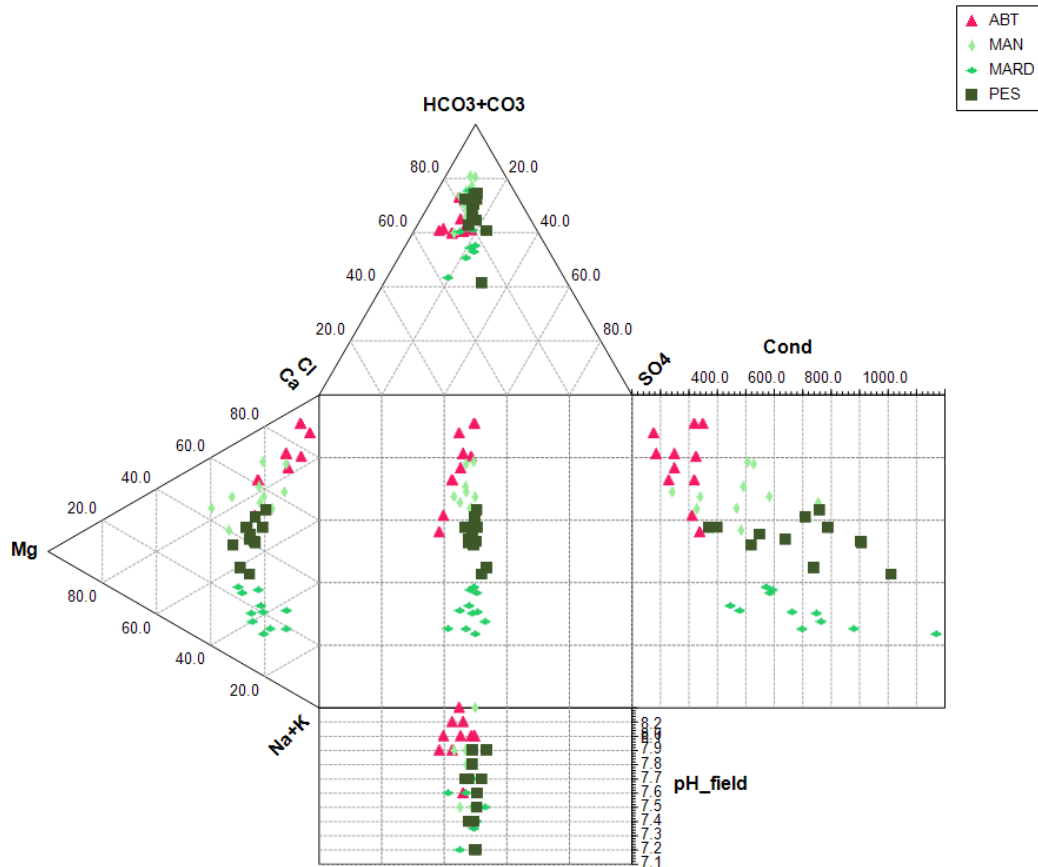


Figure 24: Durov diagram depicting geochemical processes in groundwater of KPK

3.4 Balochistan Province

Balochistan is an arid to semi-arid region characterized by low rainfall and frequent dry spells and persistent droughts. The scarcity of water is one of the most critical issues of the province. The main sources of drinking water are springs, streams, rivers and karezes. In most of the area, karezes were persistent and sufficient water was available to meet requirements. With the widespread introduction of tubewell technology, many karezes dried because of overexploitation. The increase in population and demand for economic development has resulted in indiscriminate abstraction of groundwater, which

is only 9% of total water resource available in Balochistan. Now the deep groundwater is being used through drilling of tubewells up to a depth of 300 m and water table in some areas is lowering at a rate of about 5 meters per annum (Ashraf and Ashfaq, 2017). A city wise description of current water quality situation is given as below:

3.4.1 Khuzdar

Khuzdar is the 3rd largest city of Balochistan province. Due to law and order situation, samples could not be collected in 2010 and 2015, but in 2020, samples from 11 sites were obtained. In total, 55% of these water sources were found unsafe mainly due to microbiological contamination (36%), Nitrates and Turbidity (9%) (Table 23). Moreover, presence of fecal contamination (as E-coli) is an alarming situation and demands for immediate attention.

Table 23: The level of major parameters exceeding the NDWQS in Khuzdar City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Turbidity	NTU	11	01	9
2	Nitrate (NO ₃)	mg/l	11	01	9
3	Total Coliforms	MPN/100 ml	11	4	36
4	E-coli	MPN/100 ml	11	2	18

Figure 25 shows that there is decline in the year 2020 compared to water quality situation in 2006.

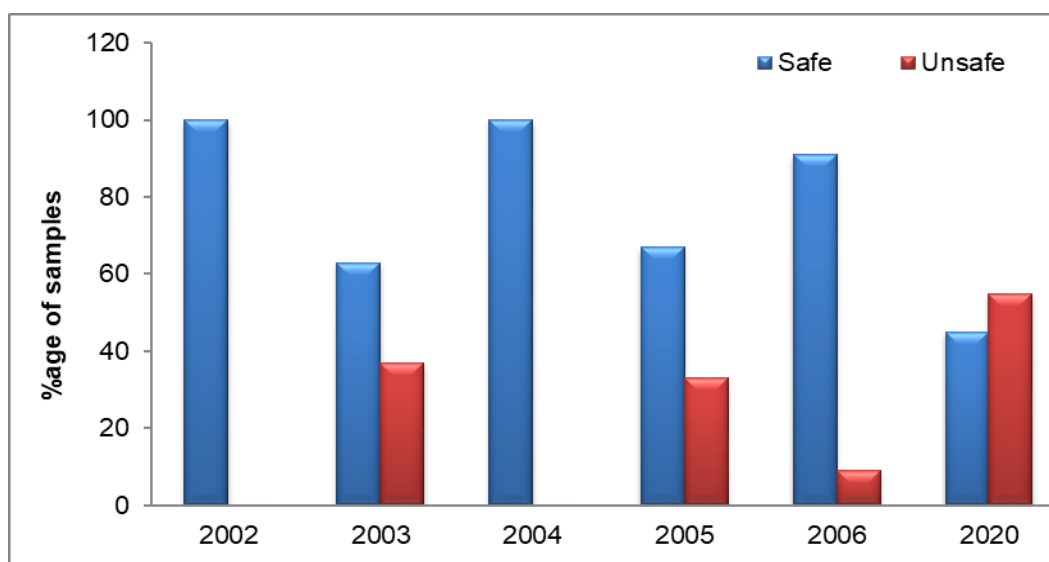


Figure 25: Year wise Drinking Water Quality Status of Khuzdar City

3.4.2 Loralai

In addition to 13 fixed locations of previous years, four new sampling sites were also selected for sampling in 2020. Overall 59% of the samples were found unsafe mainly due to microbiological contamination including fecal contamination and excessive Turbidity (Table 24).

Table 24: The level of major parameters exceeding the NDWQS in Loralai City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Turbidity	NTU	17	01	6
2	Iron (Fe)	mg/l	17	01	6
3	Total Coliforms	MPN/100 ml	17	10	59
4	E-coli	MPN/100 ml	17	04	24

Water quality shows a gradual improvement from 2002 to 2004, a declined trend during 2005 to 2006, and then improvement from 2010 to 2020 (Figure 26).

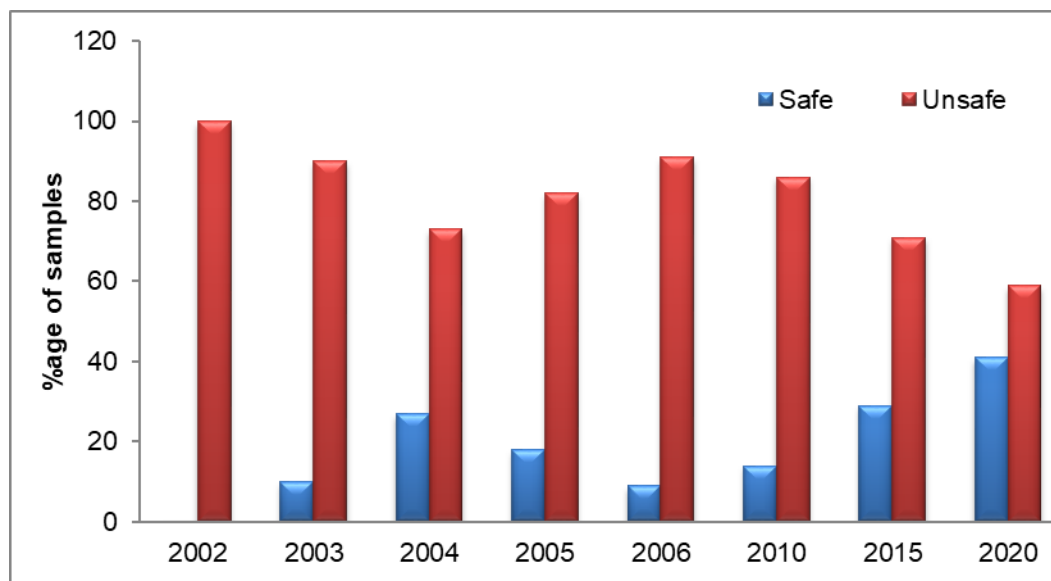


Figure 26: Year wise Drinking Water Quality Status of Loralai City

3.4.3 Quetta

Thirty eight sites were selected for the sampling purpose. In 2015, water samples were collected from 31 sources instead of 38 sites as seven sources were found dried due to

depleted water table. In 2020, those seven locations were replaced with alternative sources.

Overall 57% of the water sources were found contaminated mainly with microbial contamination followed by the presence of excessive Fluoride (26%), TDS, Hardness and Chlorides (9%) (Table 25).

Table 25: The level of major parameters exceeding the NDWQS in Quetta City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	TDS	mg/l	37	01	3
2	Hardness	mg/l	37	01	3
3	Chloride (Cl)	mg/l	37	01	3
4	Fluoride (F)	mg/l	37	10	27
5	Total Coliforms	MPN/100 ml	37	21	57
6	E-coli	MPN/100 ml	37	04	11

The water quality trend of Quetta city is similar as of Loralai city showing a gradual improvement from 2002 to 2004, from 2005 to 2010 there was a decline in water quality and again improvement during 2015 and 2020 as shown (Figure 27).

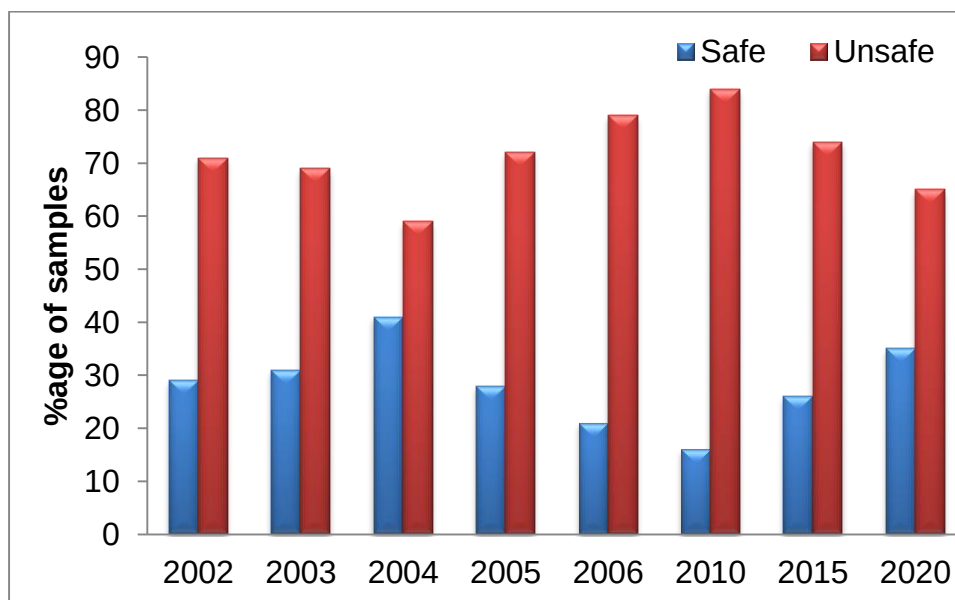


Figure 27: Year wise Drinking Water Quality Status of Quetta City

3.4.4 Ziarat

In total, 11 water sources were monitored during 2020 obtaining samples from 9 fixed sites and two new sites replacing the older dried out water sources. Overall 45% of the sources were found unsafe for drinking mainly due to microbial contamination indicated by the total Coliforms and E-coli (Table 26).

Table 26: The level of major parameters exceeding the NDWQS in Ziarat City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Iron (Fe)	mg/l	11	1	9
2	Total Coliforms	CFU/100 ml	11	5	45
3	E-coli	CFU/100 ml	11	2	18

The grave situation of drinking water quality with 100% unsafe sources (2002-06) was improved gradually, and in the year 2020 significant improvement can be seen (Figure 28).

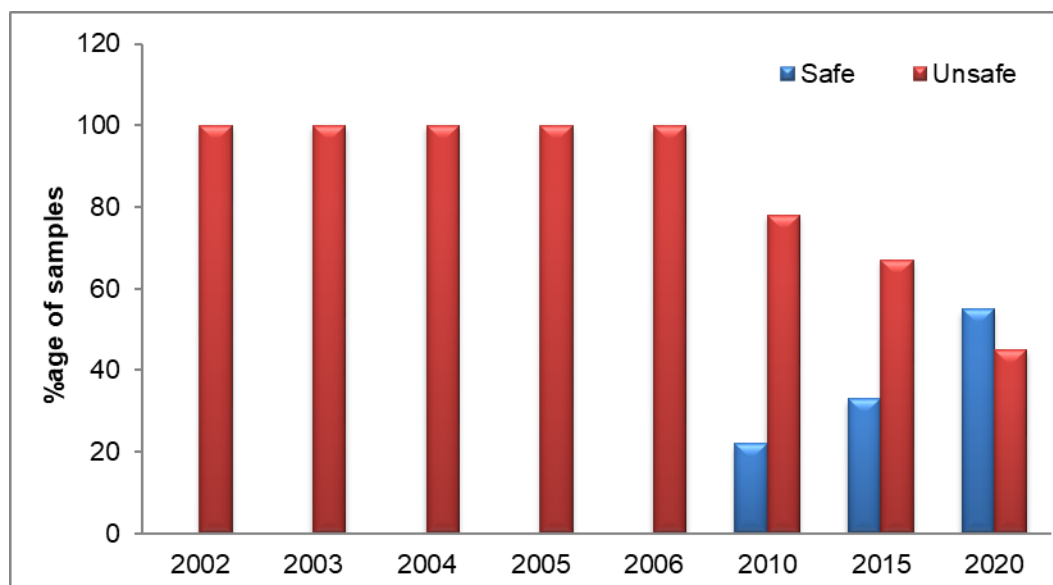


Figure 28: Year wise Drinking Water Quality Status of Ziarat City

3.4.5 Risk Map of Balochistan

Risk map of Balochistan province (Figure 29) shows that groundwater of the monitored cities falls within low risk category. However, groundwater at certain locations in Quetta and Ziarat cities are in medium to high-risk. The hotspot areas of monitored cities with respect to contaminants type is given in Annexure 36-38.

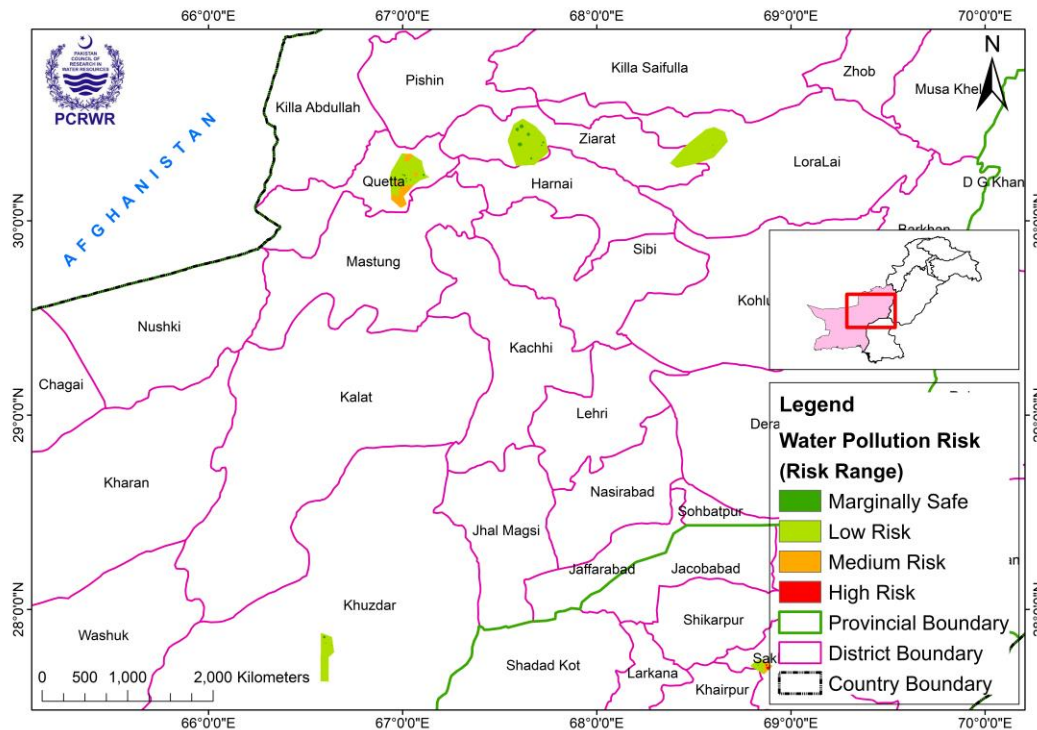


Figure 29: GIS risk mapping of Balochistan province

3.4.6 Overall Hydro-chemical Facies of Balochistan

The Durov diagram of Balochistan province (Figure 30) shows that 28% of water sources belong to Sodium-Chloride facies, Sodium-Sulphate (21%), Calcium-Bicarbonate (20%) and Calcium-Chloride (15%). Accordingly, water supplies of Balochistan have dominance of the major ions such as Sodium, Calcium, Chloride, Sulphate and Bicarbonate. Therefore, the groundwater in the study area is considered mainly of permanent hardness type. Furthermore, the percentages of these ions show that the freshwater in this region mainly belongs to the Calcium-Bicarbonate facies (20%), and seawater belongs to the Sodium-Chloride facies (28%). However, Sulphate in the groundwater might originate from several sources, potentially including (i) natural and artificial sulphate in rainwater, (ii) dissolution of sulphate-bearing evaporates (e.g. gypsum and anhydrite), (iii) seawater, and (iv) anthropogenic pollutants (e.g. domestic sewage, detergent and agricultural fertilizers) (Dongmei Han et al., 2016).

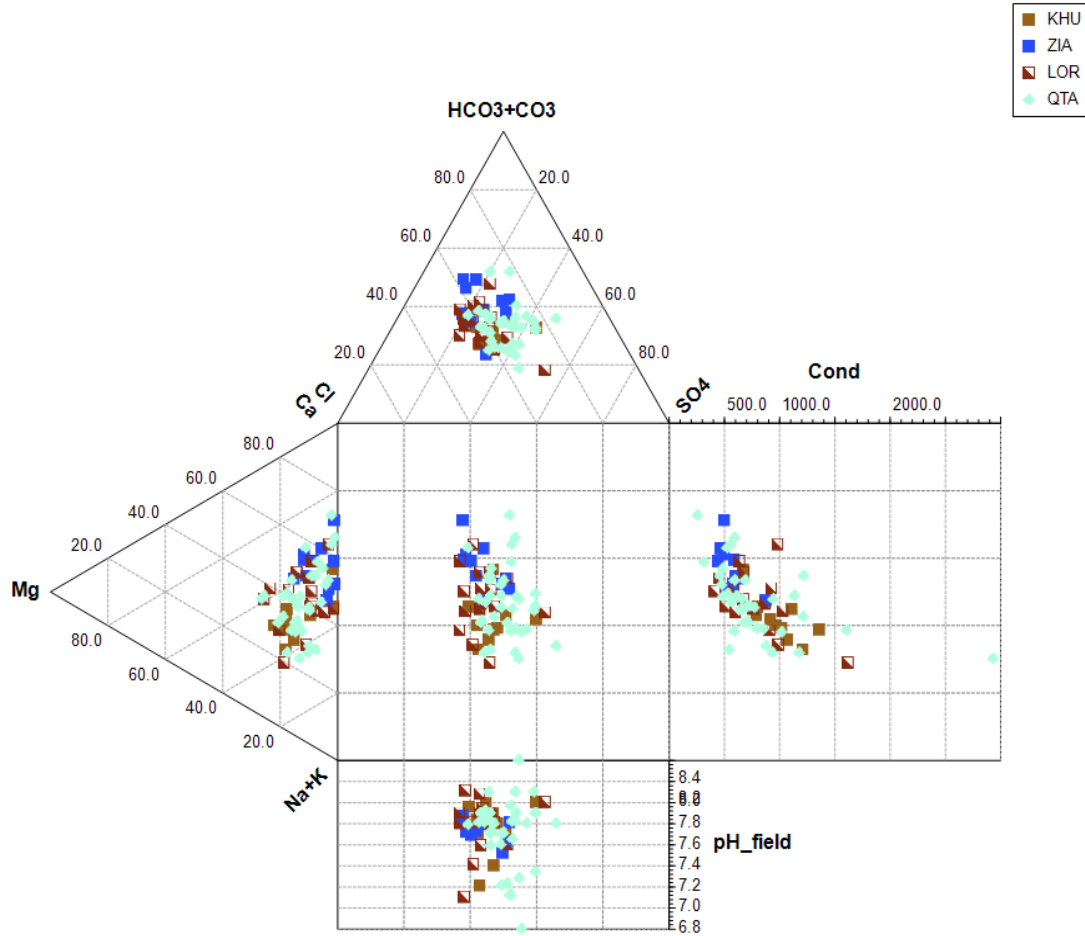


Figure 30: Durov Diagram depicting geochemical processes in groundwater of Balochistan

3.5 Sindh Province

Sindh, the second most populated province with population of around 30.44 million is located in south-eastern part of Pakistan. It stretches from 24°4'N 66°8' E to 28°7'N 71°1E', and covers about 140,914 km² area. Province is bounded by the Thar Desert to East, the Kirthar Mountains to the West and the Arabian Sea in the South. It is divided into six divisions namely Karachi, Hyderabad, Sukkur, Shaheed Benazirabad, Mirpurkhas and Larkana.

3.5.1 Hyderabad

Hyderabad is the 2nd largest city in Sindh and 8th largest in Pakistan (Population Census Report, 2017). Fifteen samples were collected from fixed locations for drinking water monitoring. Table 27 shows that 80% of the total sources were unsafe for drinking. Major contamination causes were microbial, followed by excessive Turbidity, Hardness, Chlorides and TDS.

Table 27: The level of major parameters exceeding the NDWQS in Hyderabad City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Chloride (Cl)	mg/l	15	1	7
2	Hardness	mg/l	15	1	7
3	Turbidity	NTU	15	9	60
4	TDS	mg/l	15	1	7
5	Total Coliforms	MPN/100 ml	15	11	73
6	E-coli	MPN/100 ml	15	7	47

Figure 31 shows a visible improvement (20% safe) in the year 2020 than in 2015 (7% safe).

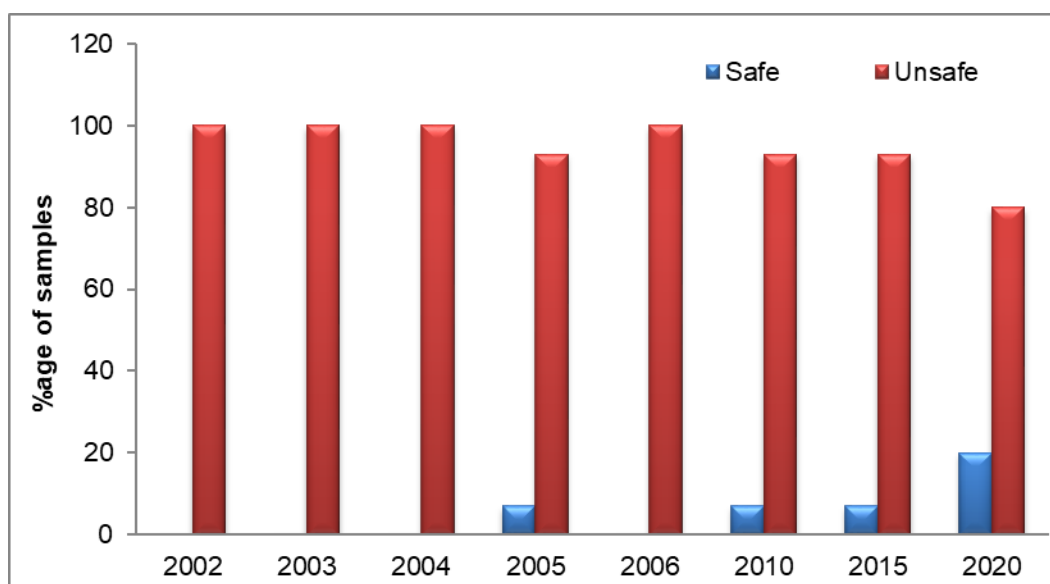


Figure 31: Year wise Drinking Water Quality Status of Hyderabad City

3.5.2 Karachi

Karachi, inhabits approximately 16.025 million (Population Census Report, 2017) people and houses about 60% of the industries of the country (UN Census, 2021). The water supply of Karachi is dependent on surface water and groundwater sources. Surface water sources include Haleji Lake, Keenjhar Lake and Hub Dam, while

groundwater source includes Dumlottee well-fields. Altogether, 28 fixed sources were monitored for drinking water quality. Out of these sources, 93% were found as unsafe for drinking (Table 28). The groundwater of the city belongs to mainly Sodium-Chloride (92%) and a minor involvement from Sodium-Bicarbonate (8%). This indicates the presence of seawater belonging to the Sodium-Chloride facies and permanent type water hardness.

Table 28: The level of major parameters exceeding the NDWQS in Karachi City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Chloride (Cl)	mg/l	28	3	11
2	Flouride (F)	mg/l	28	2	7
3	Hardness	mg/l	28	1	4
4	Turbidity	NTU	28	1	4
5	TDS	mg/l	28	3	11
6	Total Coliforms	MPN/100 ml	28	26	93
7	E-coli	MPN/100 ml	28	10	36

The situation of water quality in Karachi continues to worsen since 2002 (23%) and no remarkable improvement could be seen up to the recent study of 2020 (7%) (Figure 32). The water supply pipelines and sewerage pipelines are corroded and often lie parallel to each other causing cross contamination. Resultantly, majority of citizens in Karachi do not receive safe water.

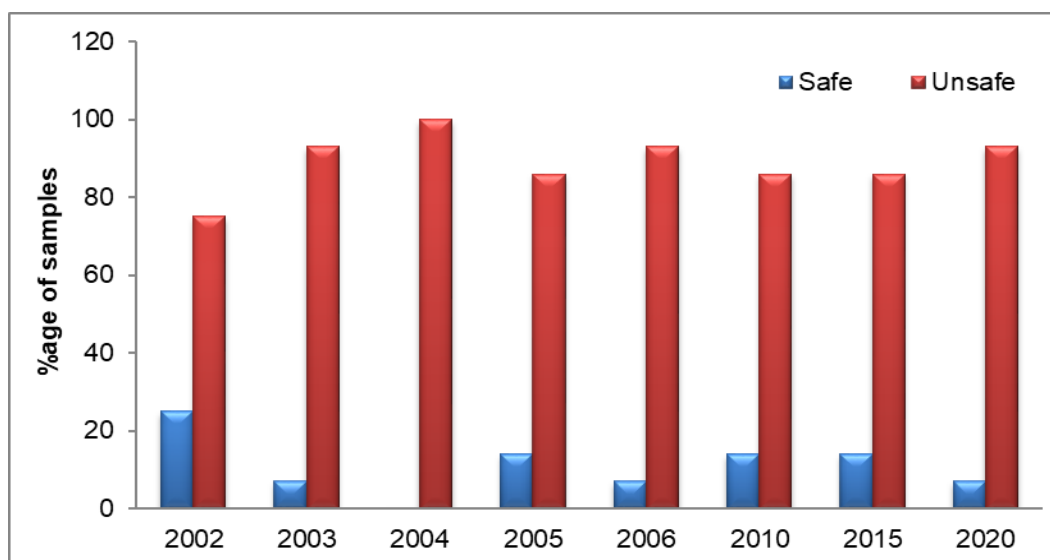


Figure 32: Year wise Drinking Water Quality Status of Karachi City

3.5.3 Sukkur

Sukkur city is located along the western bank of the Indus river. Sukkur is the third largest city in Sindh after Karachi and Hyderabad. 9 samples were collected from pre-determined locations of drinking water. It was found that 67% of the sources are unsafe for drinking purposes. The major parameters found above the permissible limits include microbial contamination, Arsenic, Chloride, Fluoride, Hardness, Turbidity, TDS and Iron (Table 29).

Table 29: The level of major parameters exceeding the NDWQS in Sukkur City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Arsenic	µg/l	9	1	11
2	Chloride (Cl)	mg/l	9	3	33
3	Flouride (F)	mg/l	9	2	22
4	Hardness	mg/l	9	2	22
5	Iron	mg/l	9	4	44
6	Turbidity	NTU	9	2	22
7	TDS	mg/l	9	3	33
8	Total Coliforms	MPN/100 ml	9	2	22

The water quality comparison of all monitoring years is shown in Figure 33. Compared to our previous monitoring in the year 2015 (100% unsafe), there seems some improvement of drinking water quality in 2020 (67% unsafe).

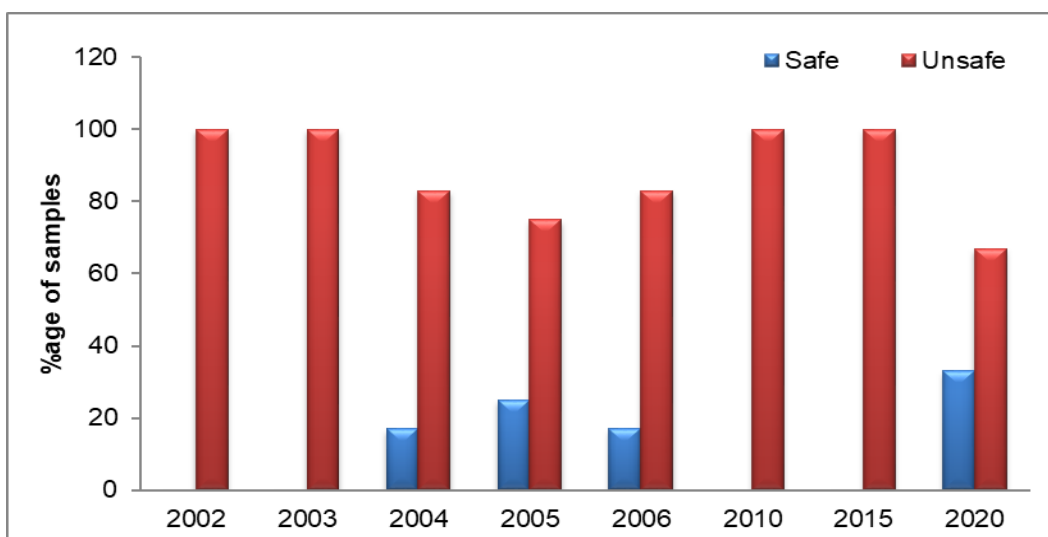


Figure 33: Year wise Drinking Water Quality Status of Sukkur City

3.5.4 Badin

Badin city was included in the NWQMP in 2010. Altogether, 12 sources were chosen for water sampling. It was found that 92% of the sources are unsafe for drinking. The analytical parameters that were found beyond the permissible limits included microbial contamination (Total Coliforms and E-coli), pH, Chloride, Hardness, Turbidity, and TDS, (Table 30). Brackish water is usually found in the study area due to major influence from Sodium and Chloride ions. The pH of some samples were high showing high alkalinity, thus unfit for drinking purposes.

Table 30: The level of major parameters exceeding the NDWQS in Badin City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	pH	mg/l	12	4	33
2	Chloride (Cl)	mg/l	12	1	8
3	Hardness	mg/l	12	1	8
4	Turbidity	NTU	12	3	25
5	TDS	mg/l	12	2	17
6	Total Coliforms	MPN/100 ml	12	7	58
7	E-coli	MPN/100 ml	12	5	42

Figure 34 shows deteriorated water quality with 62% unsafe sources in 2015 which further worsened in 2020 indicating 92% unsafe sources for drinking purposes.

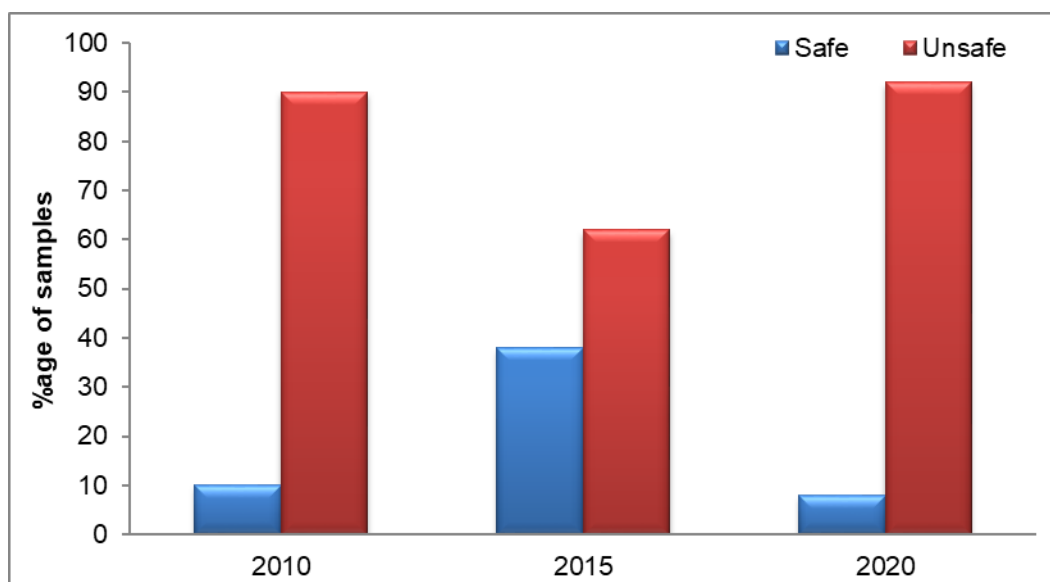


Figure 34: Year wise Drinking Water Quality Status of Badin City

3.5.5 Mirpur Khas

Mirpur Khas is the 7th largest city in Sindh province inhabiting 0.23 million population (Population Census Report, 2017). Mirpur Khas is recently included in the National Water Quality Program (2020). Fifteen samples were collected from the city. Table 31 indicates that 100% of the monitored sources were unfit for drinking purposes. The analytical parameters that were beyond the permissible limits included microbial contamination represented by the total Coliforms and E-coli, Chloride, Hardness, Turbidity, and TDS. This is an alarming situation and requires immediate attention of the concerned authorities.

Table 31: The level of major parameters exceeding the NDWQS in Mirpur Khas City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Chloride (Cl)	mg/l	12	8	67
2	Hardness	mg/l	12	9	75
3	Turbidity	NTU	12	5	42
4	TDS	mg/l	12	9	75
5	Iron (Fe)	mg/l	12	2	17
6	Total Coliforms	MPN/100 ml	12	12	100
7	E-coli	MPN/100 ml	12	2	17

3.5.6 Tando Allahyar

The Arabian Sea is about 200 km south-west of Tando Allahyar. The source of groundwater for human consumption in the area is a shallow aquifer created and recharged partially by the rainfall and mostly by seepage of water from the unlined canals. In the absence of treated piped water, the residents switch to the groundwater for drinking. Out of 14 monitored sites, 57% of the sources were unsafe due to microbiological contamination, Hardness, and TDS (Table 32).

Table 32: The level of major parameters exceeding the NDWQS in Tando Allahyar City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Hardness	mg/l	14	2	14
2	TDS	mg/l	14	5	36
3	Total Coliforms	MPN/100 ml	14	5	36

3.5.7 Shaheed Benazirabad

Thirteen water samples were collected from different locations of Shaheed Benazirabad. Table 33 shows that all sources were unsafe for drinking purposes. The major contaminations were microbial (Total Coliforms and E-coli), Chloride, Hardness, Iron, Turbidity, and TDS (Table 33)

Table 33: The level of major parameters exceeding the NDWQS in Shaheed Benazirabad City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Chloride (Cl)	mg/l	13	4	31
2	Hardness	mg/l	13	3	23
3	Iron (Fe)	mg/l	13	7	54
4	Turbidity	NTU	13	1	8
5	TDS	mg/l	13	6	46
6	Total Coliforms	MPN/100 ml	13	13	100
7	E-coli	MPN/100 ml	13	2	15

3.5.8 Risk Map of Sindh

Risk mapping of Sindh province (Figure 35) depicts that majority of the monitored water sources in Karachi, Hyderabad, Badin, Shaheed Benazirabad and Sukkur were within low-risk category. The Medium-risk water quality is seen in Tando Allahyar and some areas of Shaheed Benazirabad. Details of hotspot areas are given in Annexure 39-41.

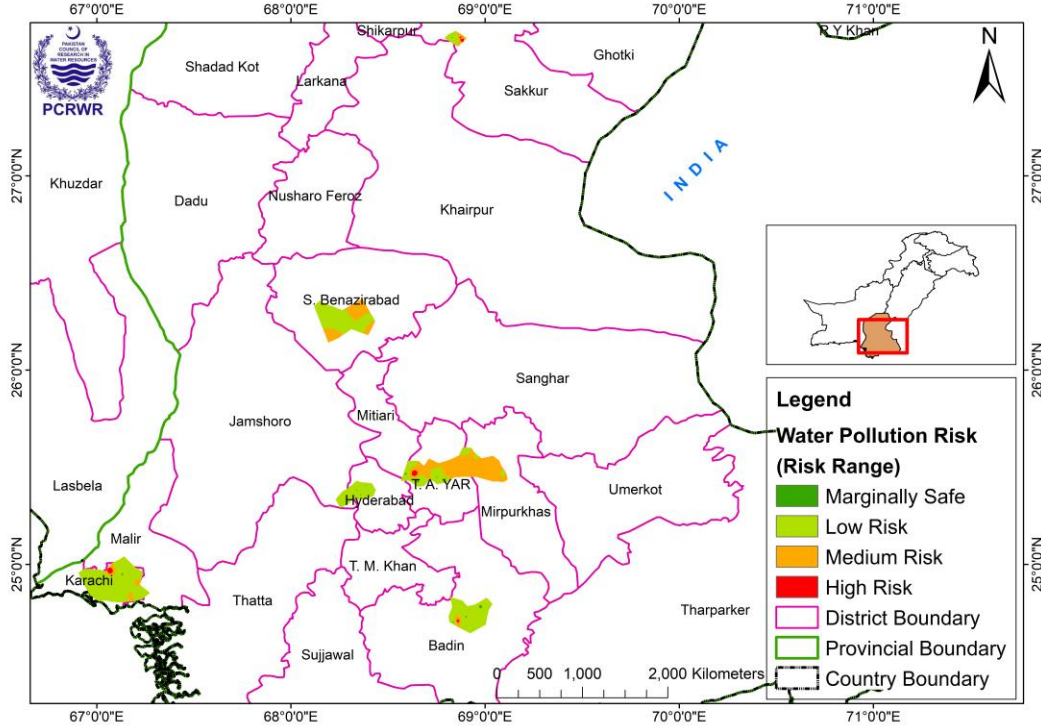


Figure 35: GIS risk mapping of Sindh province

3.5.9 Overall Hydro-chemical Facies of Sindh

The Durov diagram (Figure 36) of Sindh province based on 103 samples indicates that most of the monitored sources were classified as Sodium type of cation facies and Chloride and Bicarbonate type of anion facies. The diagram shows an increase in concentration of chloride ions because of natural source of chloride in the study area. Altogether, 46% of samples show Sodium-Chloride type hydro-chemical facies, which indicates that the rock–water interaction significantly interprets hydrochemical behavior, anthropogenic activities, ion exchange and reverse ion exchange (Talib et al., 2019). Whereas, 35% samples show Sodium and Magnesium mixed type cations facies with Bicarbonate type of anion facies representing the fresh water recharge and carbonate weathering.

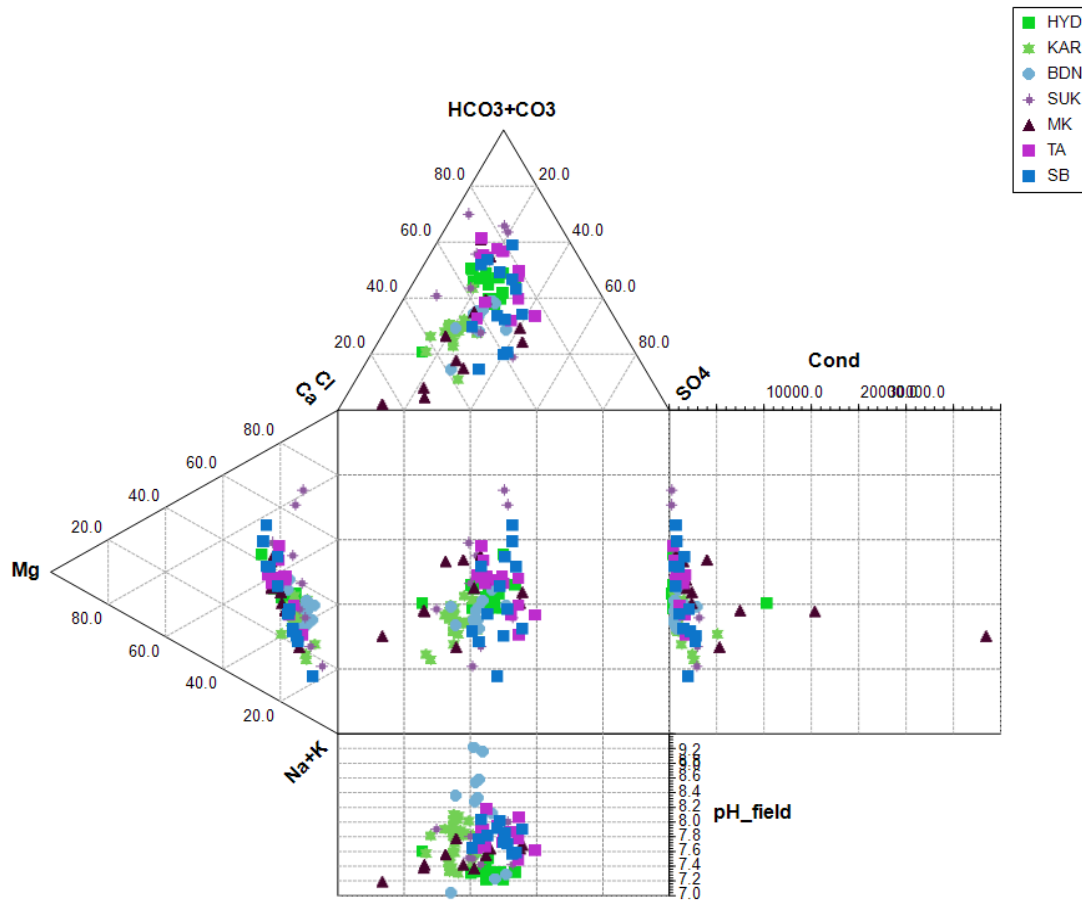


Figure 36: Durov diagram depicting geochemical processes in groundwater of Sindh

3.6 Azad Jammu and Kashmir

The administrative territory of Azad Jammu and Kashmir (AJK) covers a total area of 13,297 km² and has a total population of 4,045,366 (Population Census Report, 2017). AJK is divided into two divisions – Muzaffarabad and Mirpur. Rivers and springs are the main drinking water sources in the northern part of the AJK. On the other hand wells are the major source of drinking water in the southern part. In the urban areas of AJK, 81% of the population has access to piped water supply mostly built on springs, while 55% of the population of rural areas has been provided with piped water (The World Bank, 2016).

3.6.1 Muzaffarabad

The city of Muzaffarabad is located at the confluence of Jhelum and Neelum rivers. Muzaffarabad was included in the NWQMP in 2015. In total, 10 water samples were collected from Muzaffarabad. It was found that 70% sources were unsafe due to microbial contamination and Turbidity (Table 34).

Table 34: The level of major parameters exceeding the NDWQS in Muzaffarabad City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Turbidity	mg/l	10	2	20
2	Total Coliforms	MPN/100 ml	10	6	60
3	E-coli	MPN/100 ml	10	5	50

A comparison of water quality in 2020 with that of 2015 indicates that percentage of unsafe sources increased from 40 to 70% (Figure 37).

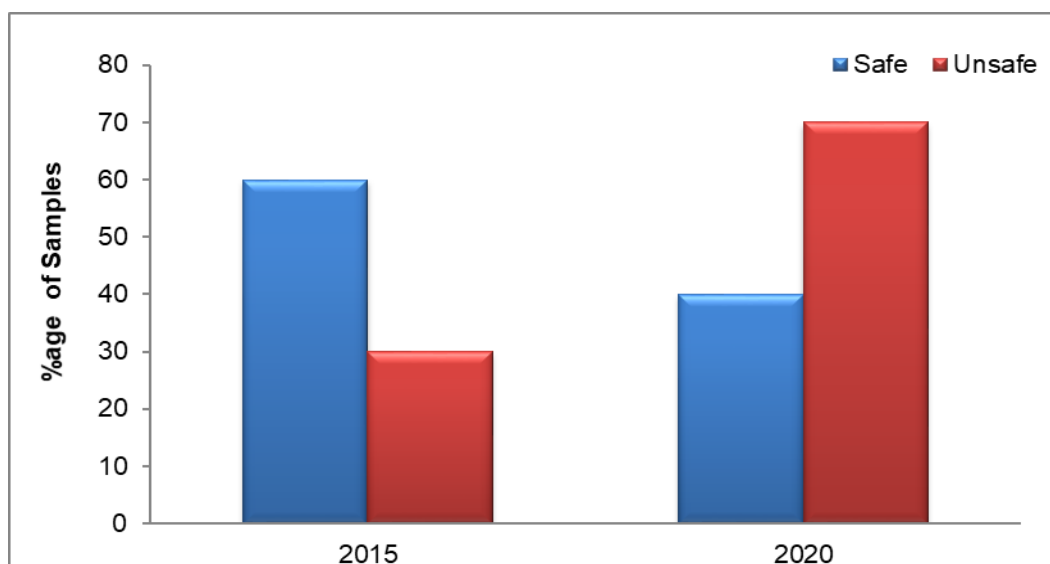


Figure 37: Year wise Drinking Water Quality Status of Muzaffarabad City

3.6.2 Overall Hydro-chemical Facies of AJK

The Durov plot of Muzaffarabad (Figure 38) shows that groundwater samples have Calcium and Magnesium dominance for the cations and bicarbonate for the anions. Two types of water can be identified, 50% samples with Calcium-Bicarbonate type and remaining 50% with Magnesium-Bicarbonate type showing that the freshwater in this area mainly belongs to the Calcium-Magnesium-Bicarbonate facies.

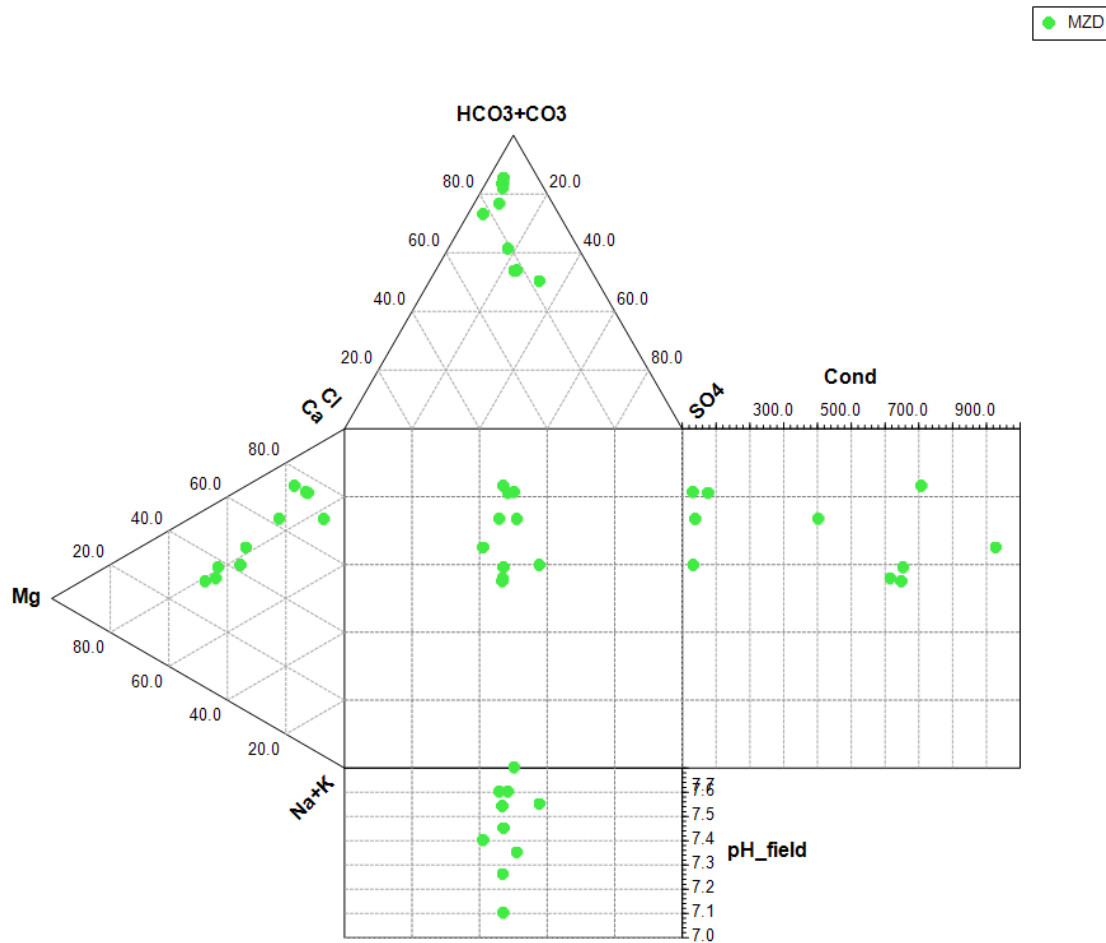


Figure 38: Durov diagram depicting geochemical processes in groundwater of AJK

3.7 Gilgit-Baltistan

Gilgit-Baltistan covers an area of over 72,971 km² and is highly mountainous. It has an estimated population of 1.249 million (Population Census Report, 2017). Glaciers and snow deposits are the principal water source in Gilgit-Baltistan.

3.7.1 Gilgit

Ten samples were collected from different sources. These included direct surface water and water supply schemes (based on surface water). 100 percent samples were found unsafe due to microbial contamination and excessive Turbidity (Table 35).

Table 35: The level of major parameters exceeding the NDWQS in Gilgit City

Sr. #	Water Quality Parameter	Unit	Total No. of Samples Analyzed	Number of Contaminated Samples	%age of Contaminated Samples
1	Turbidity	NTU	10	5	50
2	Total Coliforms	CFU/100 ml	10	10	100
3	E-coli	CFU/100 ml	10	5	50

A comparison of water quality in 2020 with that of 2015 indicates that there is no improvement in the drinking water quality and all the selected sources are unsafe (Figure 39). However, another study by PCRWR showed unsafe water quality in district Astor (100%), followed by district Skardu (80%) and Hunza (80%), Gizer (74%), Gilgit (70%) and district Diamer (67%) as reported by (Ahsan et al. 2020).

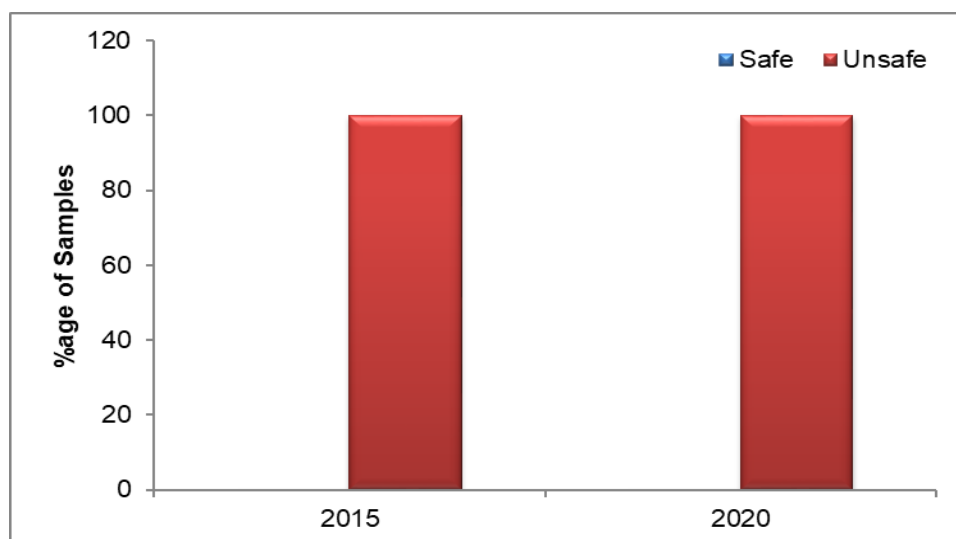


Figure 39: Year wise Drinking Water Quality Status of Gilgit City

3.7.2 Risk Map of Federal Capital, AJK and Gilgit

Figure 40 shows that the majority of monitored sources in Islamabad are at low risk. Similarly, all the sources of Gilgit and Muzaffarabad fall in low-risk category and may be used after some basic treatments. The hotspot areas of Gilgit city according to contaminants type is given as Annexure 42 and 43.

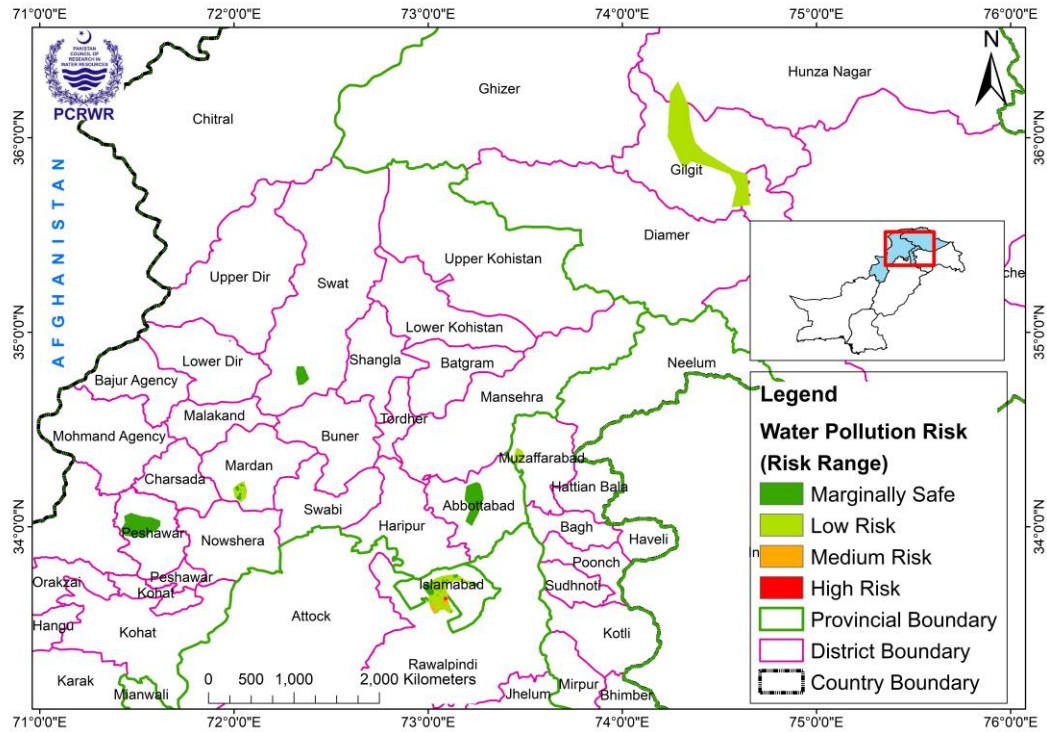


Figure 40: GIS risk mapping of Federal Capital, AJK and Gilgit

3.7.3 Overall Hydro-chemical Facies of Gilgit

The hydro-chemical facies identified from Figure 41 are; Calcium-Bicarbonate facies (50%), Calcium-Sulphate facies (30%), Magnesium-Bicarbonate facies (10%) and Magnesium-Sulphate facies (10%). This suggests presence of temporary hardness of 60% and permanent hardness of 40%. The groundwater in the area is mainly made of alkaline earth metals and predominantly Bicarbonate and Sulphate water. This indicates that fresh water belongs to the Calcium-Bicarbonate facies.

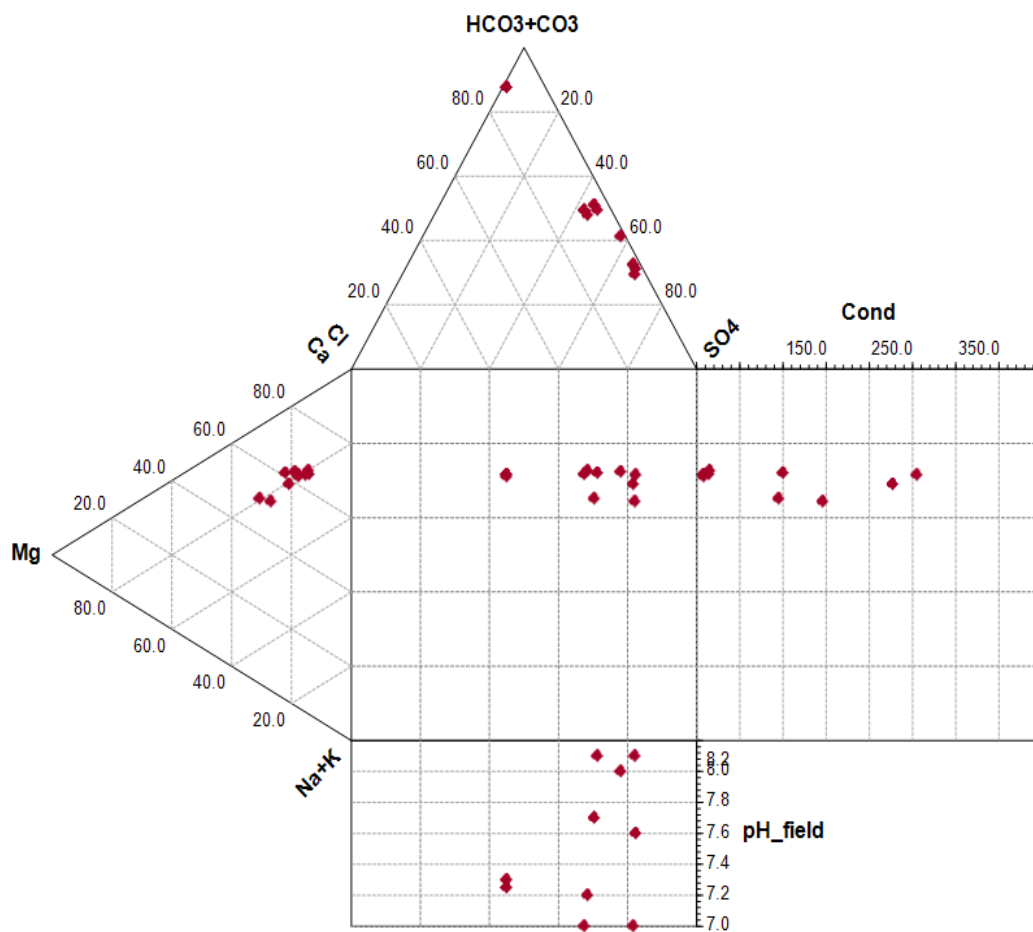


Figure 41: Durov diagram depicting geochemical processes in groundwater of Gilgit

3.8 Overall Water Quality Situation in Pakistan

The water quality monitoring in the year 2020 comprised monitoring of 15 types of water supply sources by collecting 435 samples from 29 major cities of Pakistan. The details regarding types and number of water sources are given in Table 36.

Table 36: Types of water sources and number of samples collected for monitoring

Source	Number	Source	Number
Tube well	171	Karez	02
Water Supply	46	Spring	07
Surface Water	05	Well	04
Bore	18	Hand Pump	57
Tap	80	Injection Pump	27
Reotter	02	Filtration Plant	14
Water Storage Tank	01	Dam	01
Total			435

Like earlier monitoring outcomes (2002-15), bacteriological contamination is wide spread and the main reason of unsafe drinking water supply across the country. Number and percentage of safe and unsafe water sources (microbiologically or chemically) are given in Table 37. Depending on the geological diversity, water supplies were found to have some commonly occurring contaminants such as Hardness, excessive TDS and Iron, whereas, others such as Arsenic, Nitrate and Fluoride were also found in some cities (Table 37).

Table 37: Overall water quality situation of 29 cities and causes of contamination

Sr. #	City	Total No. of Samples	Safe Samples		Unsafe Samples		Type of Contamination
			No.	%age	No.	%age	
1.	Islamabad	24	17	71	07	29	Iron, Bacteriological
2.	Bahawalpur	25	06	24	19	76	Turbidity, Hardness, Fluoride, TDS, Iron, Arsenic, Bacteriological
3.	Faisalabad	22	09	41	13	59	Hardness, Chloride, TDS, Iron, Fluoride, Bacteriological
4.	Gujranwala	14	07	50	07	50	Bacteriological
5.	Gujrat	09	09	100	0	0	-
6.	Kasur	10	09	90	01	10	Turbidity, Bacteriological
7.	Lahore	16	11	69	05	31	Arsenic, TDS, Iron, Turbidity, Bacteriological
8.	Multan	16	01	06	15	94	Arsenic, Bacteriological
9.	Rawalpindi	13	08	62	05	38	Bacteriological , TDS
10.	Sargodha	24	04	17	20	83	Hardness, Chloride, Fluoride, TDS, Iron, Nitrate, Bacteriological
11.	Sheikhupura	10	04	40	06	60	Arsenic, Turbidity, Hardness, TDS, Iron, Nitrate, Bacteriological
12.	Sialkot	09	09	100	0	0	-
	Sub-Total	192	94	49	98	51	-
13.	Abbottabad	11	05	45	06	55	Iron
14.	Mangora	10	08	80	02	20	Iron
15.	Mardan	11	06	55	05	45	Iron
16.	Peshawar	12	06	50	06	50	Iron
	Sub-Total	44	25	57	19	43	-
17.	Khuzdar	11	05	45	06	55	Turbidity, Nitrate, Bacteriological

18.	Loralai	17	07	41	10	59	Turbidity, Bacteriological, Iron
19.	Quetta	37	13	35	24	65	Turbidity, Bacteriological, Fluoride, Hardness, Chlorides, TDS.
20.	Ziarat	11	06	55	05	45	Bacteriological, Iron
	Sub-Total	76	31	41	45	59	-
21.	Hyderabad	15	03	20	12	80	Turbidity, Hardness, Chloride, TDS, Bacteriological
22.	Karachi	28	02	07	26	93	Turbidity, Hardness, Chloride, TDS, Fluoride, Bacteriological
23.	Sukkur	09	03	33	06	67	Arsenic, Turbidity, Hardness, Chloride, TDS, Iron, Fluoride, Bacteriological
24.	Badin	12	01	08	11	92	pH, Turbidity, Hardness, Chloride, TDS, Bacteriological
25.	Mirpur Khas	12	0	0	12	100	Turbidity, Hardness, Chloride, TDS, Bacteriological
26.	Tando Allahyar	14	06	43	08	57	Hardness, TDS, Bacteriological
27.	Shaheed Benazirabad	13	0	0	13	100	Turbidity, Hardness, Chloride, TDS, Iron, Bacteriological
	Sub-Total	103	15	15	88	85	-
28.	Muzaffarabad	10	03	30	07	70	Turbidity, Bacteriological
29.	Gilgit	10	0	0	10	100	Turbidity, Bacteriological
GRAND TOTAL		435	168	39	267	61	-

Overall water quality status with respect to water quality parameters as shown in Table 38 indicates a situation not very different from the past monitoring outcomes (2002-15).

Table 38: Overall water quality parameters found beyond permissible limits

Sr. #	Parameters	Number of Samples Beyond Permissible Limit	%age
1	Turbidity	38	9
2	Hardness	43	10
3	pH	4	1
4	Chloride	35	8
5	Nitrate	19	4
6	TDS	59	14
7	Arsenic	22	5
8	Iron	61	14
9	Fluoride	17	4
10	Total Coliforms	179	41
11	E. Coli	85	20

Followed by the bacteriological contamination (41%), the major health concerned parameters found are Hardness, Arsenic, Nitrate and Fluoride. The consumers may get long term exposure due to their tasteless and odorless characteristics even at higher concentrations of these contaminants.

3.9 Water Quality Trend in Provinces (2002-20)

The province wise water quality trend is discussed as under.

3.9.1 Punjab Province

The overall water quality situation of the Punjab province is shown in Figure 42. The monitoring outcomes of the year 2020 with 51% of unsafe water sources was comparatively lowest than the previous monitoring years (2002-06, 2010-15). The major problems, identified in province, were Bacterial contaminants, Arsenic, Fluorides, TDS and Nitrates present in the drinking water. These problems need to be addressed immediately; otherwise waterborne diseases will increase and effect the public health.

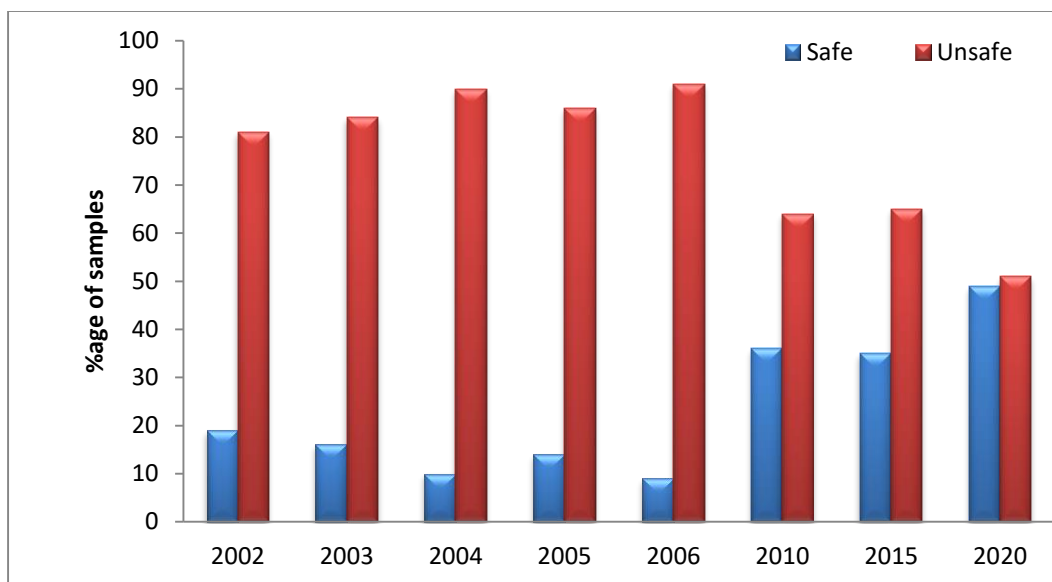


Figure 42: Water Quality Trend in Punjab Province

3.9.2 Khyber Pakhtunkhwa (KPK)

Different spikes of safe drinking water quality were noted in the KPK (Figure 43). From 2002 to 2006, it ranged from 26% to 48%, which decreased to 35% in 2010. This level of safe water supplies gradually decreased in 2015 and 2020. Iron is main contaminant in the province during monitoring year 2015 and 2020. It may be due to nearby high amount of parent material having iron rich minerals.

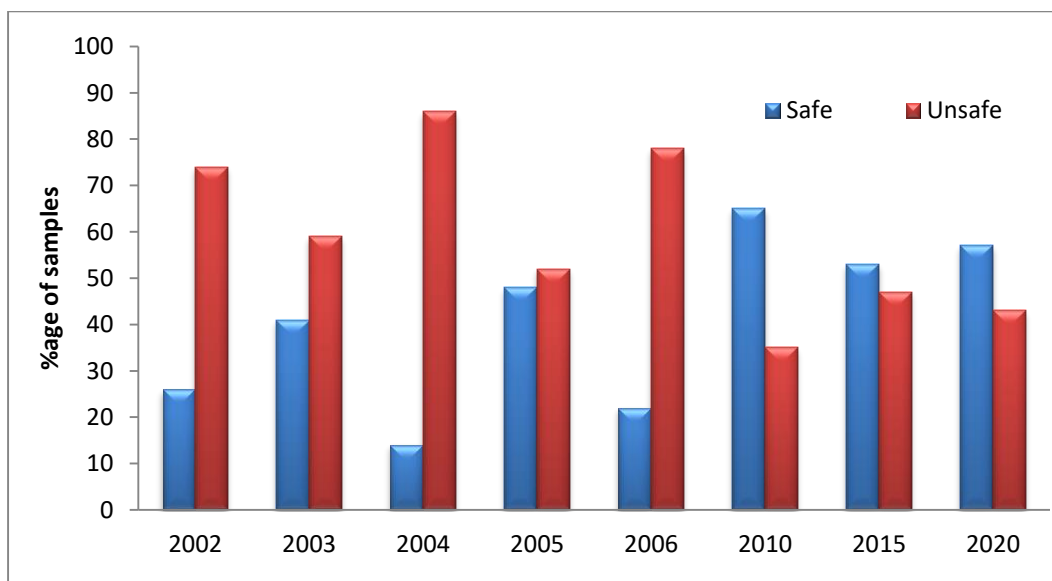


Figure 43: Water Quality Trend in KPK

3.9.3 Balochistan Province

In Balochistan, safe drinking water sources were in the range of 14 to 26% during the five years monitoring period 2002-2006. Later on, there seems gradual improvement from 17% to 41% during 2010 to 2020 (Figure 44). The level of unsafe samples decreased gradually with notable water quality improvement in the year 2020. Despite this improvement, the unsafe water supplies of 59% in the year 2020 may pose a serious health risk and requires urgent attention of the concerned authorities.

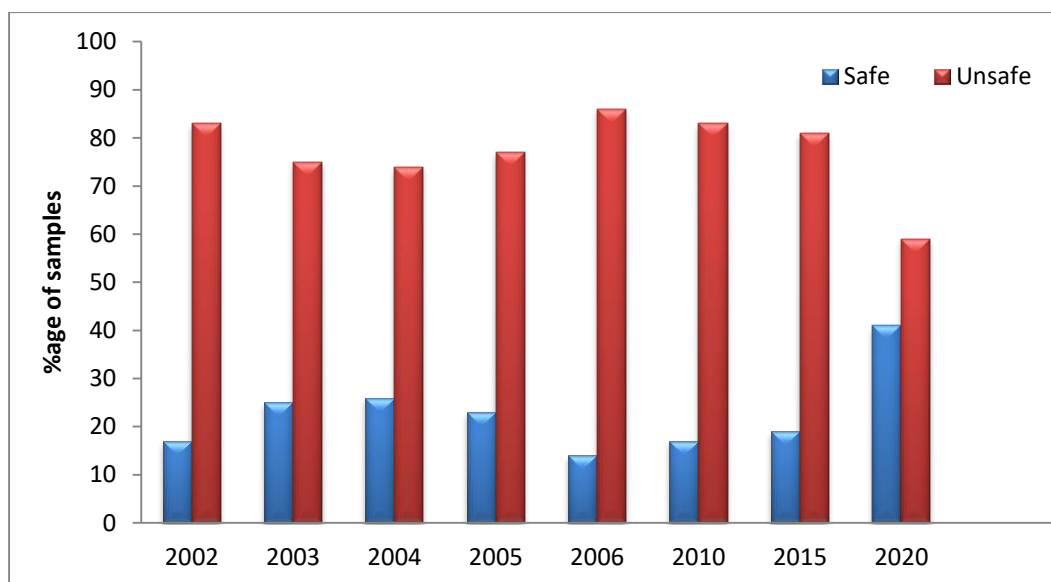


Figure 44: Water Quality Trend in Balochistan Province

3.9.4 Sindh Province

An overall quality of water supplies in Sindh (2002-2006) remained unsatisfactory as 91% of the water samples were found unsafe for drinking. This situation was not very different later during 2010 and 2015. In 2020, the percentage of unsafe water samples increased to 85%. The Honorable Court has taken several actions from time to time to expedite the provision of safe drinking water in the province. A judicial commission was appointed by the Honorable Court through an order dated January 14, 2018 to probe the detailed investigation on unsafe water supply in the province. The recommendations of this commission focused on the supply of clean drinking water to the end-users/consumers. Despite this, no considerable improvement could be seen (Figure 45).

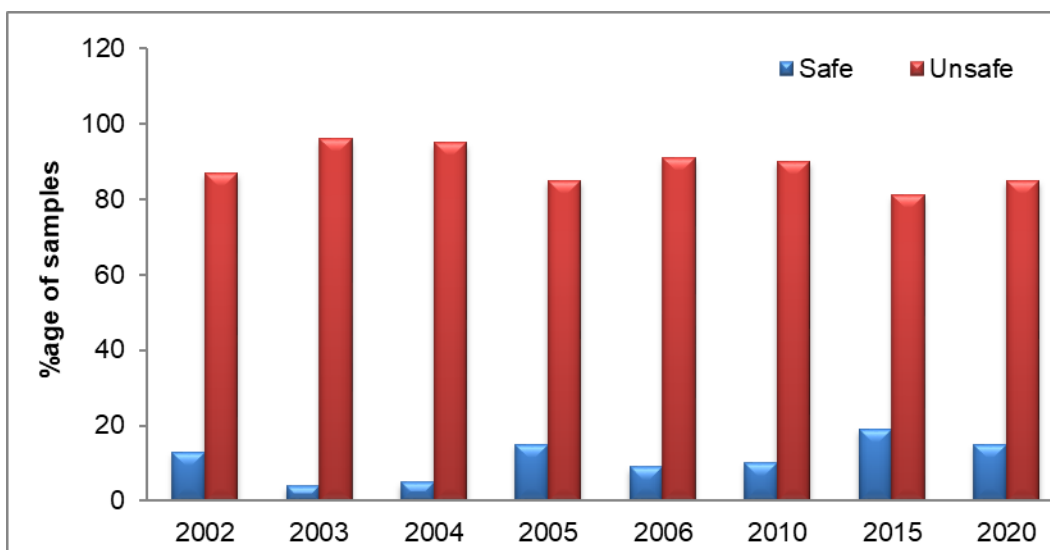


Figure 45: Water Quality Trend in Sindh Province

3.9.5 Azad Jammu and Kashmir (AJK)

The Muzaffarabad City was selected from AJK for water quality monitoring in 2015. The percentage of unsafe samples was 40% and 70% in 2015 and 2020, respectively (Figure 46). The percentage of unsafe sources increased which is alarming and required immediate action to mitigate this situation.

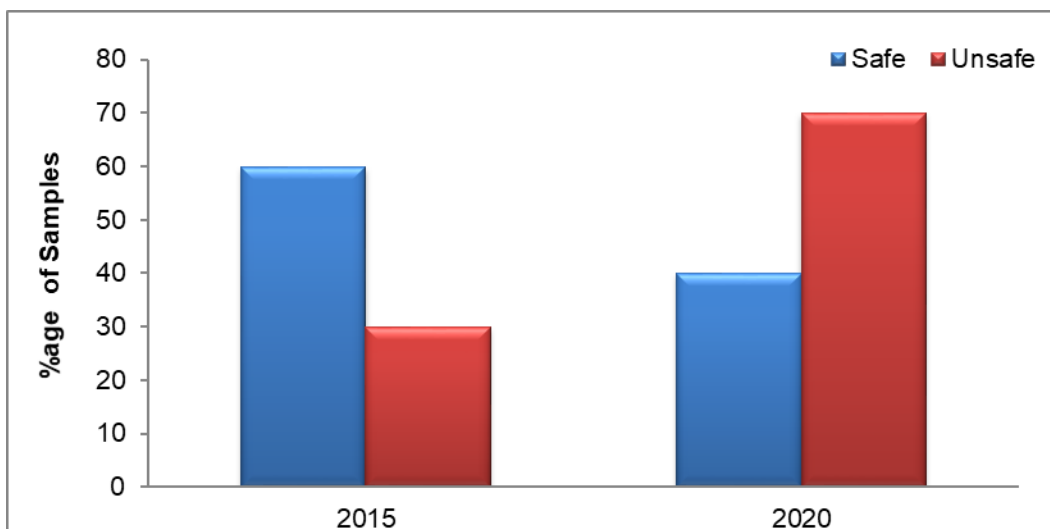


Figure 46: Water Quality Trend in AJK

3.9.6 Overall Water Quality Trend of Pakistan (2002-20)

An overall comparison of all provinces/regions (Figure 47) reveals that the highest percentage of unsafe water quality was found in Gilgit (100%) followed by Sindh (85%), AJK (70%), Punjab (51%) and KPK (43%). Therefore, urgent measures are required to mitigate the unsafe water supplies along with regular monitoring of water quality.

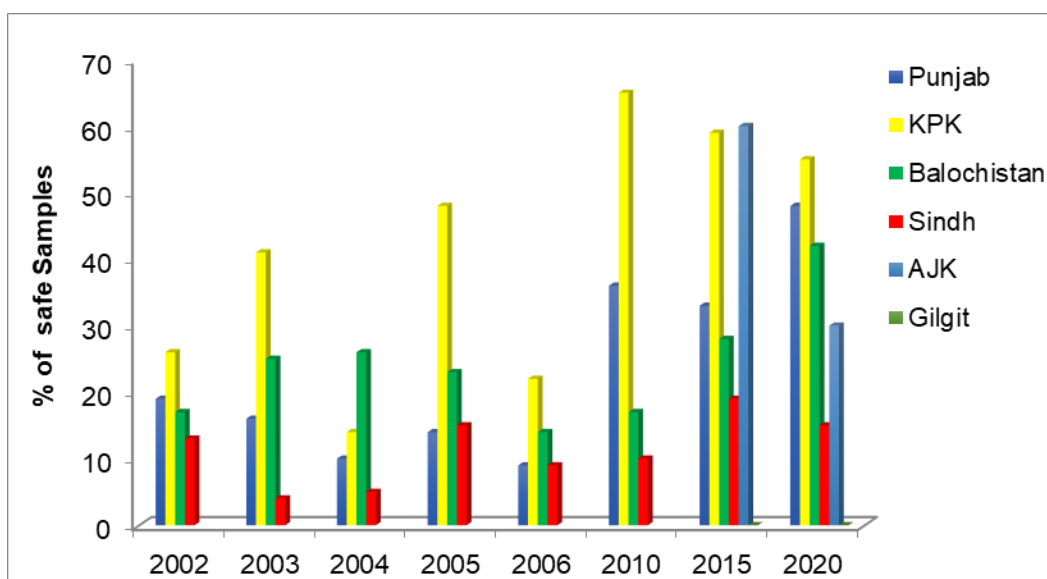


Figure 47: Comparison of Water Quality Trend in Provinces/Regions

Table 39 indicates gradual improvement in water supplies of the country over a period of two decades. The distribution of unsafe water samples proportion was found within the higher range of 61 to 88% during all monitoring years, whereas safe water sources were in the range of 12-39% (Figure 48).

Table 39: Year wise summary of water quality status of Pakistan

Sr. No.	Year	No. of Samples			%age of Samples	
		Total	Safe	Unsafe	Safe	Unsafe
1	2002	295	54	241	18	82
2	2003	287	53	234	18	82
3	2004	319	44	275	14	86
4	2005	326	48	278	15	85
5	2006	330	39	291	12	88
6	2010	333	100	233	30	70
7	2015	344	108	236	31	69
8	2020	435	166	267	39	61
Total		2669	612	2055	23	77

There is considerable improvement during the period from 2016 to 2020 mainly due to awareness through electronic and print media, SDGs obligations and the actions taken up by the Honorable courts from time to time. The role of public-private partnerships in overcoming these challenges would be more effective.

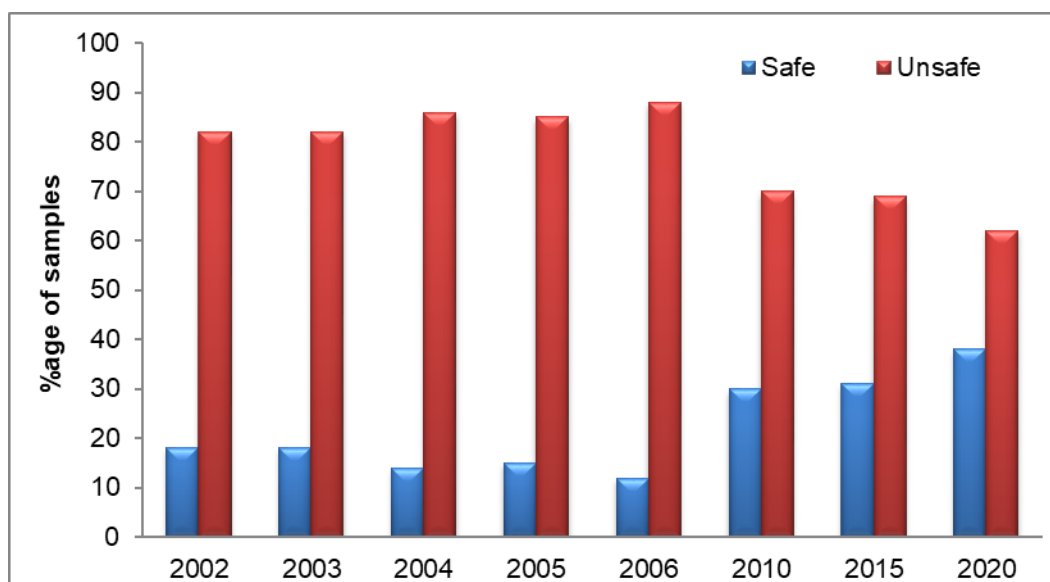


Figure 48: Overall Water Quality Status of Pakistan

The nature and type of water quality problems in the water supplies of Pakistan during last two decades are almost the same. However, presence of microbial contamination, arsenic, nitrate and fluoride are of considerable importance due to their health hazards and related socio-economic implications. The unsafe supplies due to microbial contamination are mainly attributed to the dried out pipelines caused by intermittent water supply. In such pipes, a single *E. coli* bacterium can multiply into trillions in just a week's time. Kumpel and Nelson (2016) have also found that 0.7% of tap water samples were positive with continuous water supply compared to 31.7% of intermittent water supply (IWS). Likewise, groundwater sources located near wastewater conduits are also on the risk of microbial contamination. The burden of diarrheal disease associated with IWS likely exceeds the WHO health-based normative guideline for drinking water.

3.10 Potential Contaminants: Health Effects and Required Treatments

Some drinking water contaminants may be harmful if consumed at certain levels while others may be less harmful depending on the immunity levels of the the consumers. Generally, public water supply systems use water treatment and monitoring protocols to protect consumers from those contaminants. Whereas, private wells do not receive the same services. In this context, a reference table (Table 40) listing potential contaminants, their sources, possible health impacts and general treatment methods is given.

Table 40: Health effects, sources and possible treatment of potential contaminants

Contaminants	Hot spots identified in 2020-21	Potential Health Effects from Ingestion of Water	Sources of Contaminant in Drinking Water	*Possible treatment
Microbiological contamination	Islamabad, Bahawalpur, Faisalabad, Gujranwala, Kasur, Lahore, Multan, Rawalpindi, Sheikhupura, Sargodha, Khuzdar, Loralai, Quetta, Ziarat, Hyderabad, Karachi, Sukkur, Badin, Mirpur Khas, Shaheed Benazirabad, Tando Allahyar Muzaffarabad, and Gilgit	Cholera Diarrhea Typhoid Dysentery Gastroenteritis Hepatitis A & E	- Coliforms are naturally present in the environment; as well as feces; - Fecal coliforms and <i>E. coli</i> only come from human and animal fecal waste	- Chemical disinfectants (chlorine, chlorine dioxide, and ozone) - Physical disinfectants (ultraviolet light, and boiling) - Filtration
Arsenic	Bahawalpur, Lahore, Multan, and Sheikhupura	Skin hyperpigmentation Oxidative stress Peripheral neuropathy Cancer of bladder, skin, liver, lung, and lymphatic cancer	Geologically present however mobilized into groundwater through weathering of rocks	- Arsenic removal media - Reverse osmosis
Nitrate	Sargodha, Sheikhupura and Khuzdar	Blue baby syndrome in infants: Symptoms include shortness of breath	- Nitrogen fertilizer - Industrial wastewater discharges - Urban domestic sewage - Septic systems - Animal feedlots - native soil organic matter	- Reverse osmosis - Distillation - Ion exchange resin
Fluoride	Bahawalpur, Faisalabad, Sargodha, Quetta, Loralai, Karachi and Sukkur,	Skeletal Fluorosis: Bone disease (pain and tenderness of the bones); Dental Fluorosis: Children may get mottled teeth	- Water additive which promotes strong teeth; - Erosion of natural deposits; - Discharge from fertilizer and aluminum factories	- Precipitation/coagulation - Membrane-based processes, - Ion-exchange methods - Adsorption methods
Turbidity	Bahawalpur, Kasur, Lahore, Sheikhupura, Khuzdar, Loralai, Quetta, Hyderabad, Karachi,	Higher turbidity levels are often associated with higher levels of disease-	Soil runoff	Surface water: Coagulation-flocculation aided with filtration

Contaminants	Hot spots identified in 2020-21	Potential Health Effects from Ingestion of Water	Sources of Contaminant in Drinking Water	*Possible treatment
	Sukkur, Badin, Mirpur Khas, Shaheed Benazirabad Muzaffarabad, and Gilgit	causing microorganisms such as viruses, parasites and some bacteria		Groundwater: Filtration
Hardness	Bahawalpur, Faisalabad Sheikhpura, Sargodha, Khuzdar, Loralai, Quetta, Hyderabad, Karachi, Sukkur, Badin, Mirpur Khas, Tando Allahyar and Shaheed Benazirabad	Hardness is not a health concern, but it can cause mineral buildup in plumbing, fixtures, and water heaters, and poor performance of soaps and detergents.	Calcium and magnesium present in many sedimentary rocks (i.e. limestone and chalk)	- Water Softeners - Reverse Osmosis
TDS	Bahawalpur, Faisalabad, Lahore, Rawalpindi, Sheikhpura, Sargodha, Quetta, Hyderabad, Karachi, Sukkur, Badin, Mirpur Khas, Tando Allahyar and Shaheed Benazirabad	High TDS level alters the taste of water and makes it salty, bitter, or metallic. High TDS levels also indicate the presence of health hazardous toxic minerals	TDS is made up of inorganic salts, as well as a small amount of organic matter	- Reverse Osmosis - Distillation - Deionisation (DI)
Chlorides	Faisalabad, Sargodha, Quetta, Hyderabad, Karachi, Sukkur, Badin, Mirpur Khas, Shaheed Benazirabad	Chloride in drinking water is not harmful, however, the sodium part of table salt has been linked to heart and kidney disease	- Rocks containing chlorides - Agricultural runoff - Industrial wastewater - Oil well wastes - Effluent wastewater from treatment plants	- Reverse Osmosis - Ion exchange resins
Iron	Islamabad, Bahawalpur, Faisalabad, Lahore, Sheikhpura, Sargodha, Abbottabad, Mangora, Mardan, Peshawar, Sukkur and Shaheed Benazirabad	Iron is not hazardous to health, but it is considered a secondary or aesthetic contaminant	- Aeration of iron-containing layers in the soil - Rusting of water supply infrastructures such as pipes	- Sediment filters - Water Softening (Ion Exchange) - Oxidizing Filters - Shock chlorination (for bacterial Iron)

*Selection of treatment method/technology depends on the available financial and human resources, scale of treatment required and level of potential contaminant.

Chapter 4: Constraints in Slow Progress towards Access to Safe Drinking Water

4.1 Constraints and Challenges

4.1.1 Slow Progress towards Safe Water Supply

Pakistan along with many developing countries failed to achieve the Millennium Development Goals (MDGs) mainly due to: lack of its understanding, lack of institutional setup and policy, lack of capacity (infrastructural, institutional and human resources), lack of mechanism for tracking the progress (both inputs and outputs) and transparency issues in the use of resources. One of the major constraints was the lack of baseline data and aspirational targets. The same is almost true for the SDGs, particularly Goal 6. So far the Government of Pakistan has not been able to develop baselines for all the indicators of Goal 6.

The current study reveals slow paced improvements in safe water quality i.e. between 2015 and 2020 quality improved only by 7% (Figure 49). Therefore, the access to safe drinking water to 100% of the population by 2030 seems to be a huge challenge for Pakistan.

The NWQMP report of 2015 (Imran et al., 2016) predicted that with safe water improvement of 31% in 2015, Pakistan would reach hardly up to 53% access to safe drinking water by 2030. Based on this, achieving SDG target 6.1 by 2030 would require speeding up the improvement by three times than that of 2015. Now with the current safe water access of 39% in 2020, only 50% of the population is projected to have safe water access by 2030 (Figure 49). To achieve 100% targeted coverage by 2030, it will require speeding up the efforts by four times the current pace. Thus, it is important to critically analyze the root causes for this slow paced improvement, redefine the priorities for the current decade and rigorously implement these redefined priorities to achieve SDG target/goal 6, to which Pakistan has committed.

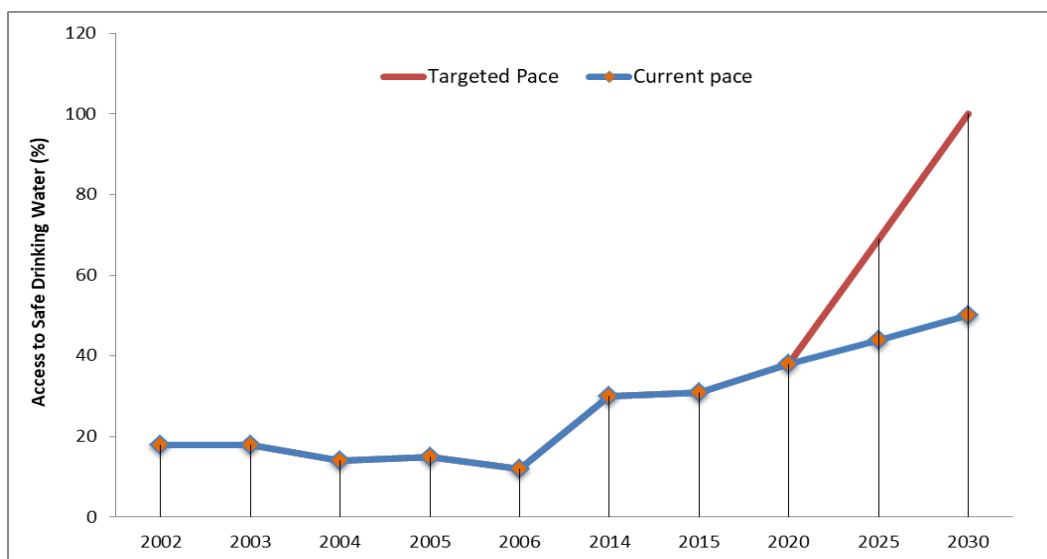


Figure 49: Safe drinking water access based on the current pace and the projected

4.2 Water Shortage

Sustainable domestic water supplies depend on the availability of reliable water resources that can be easily developed. Over abstraction of groundwater leading to groundwater depletion is a growing concern, exacerbated by climate change, population growth and urbanization. As a result, groundwater sources across the country are increasingly being polluted through intensive agriculture, industry and poor sanitation. This is clearly evident by the lower level of improvement in microbial and chemical contaminations as shown in this report.

4.3 Ineffectiveness of Existing Water Supply Model

The provision of safe drinking water is the responsibility of the state as per the constitution of Islamic Republic of Pakistan. However, after 18th amendment in the constitution, provision of safe drinking water falls within the domain of the provincial governments. Thus, the responsibility sets on the district government organizations such as Tehsil Municipal Authority (TMA), Water and Sanitation Agency (WASA), Public Health Engineering Department (PHED) etc.

Provincial and Local governments have struggled to provide safe and reliable drinking water through traditional, typically centralized and decentralized solutions. The ideal solution is safe and affordable drinking water piped into people's homes. However, setting aside the huge water supply infrastructure in the country, past investments were mainly made on the decentralized solutions in the form of constructing new water supply schemes, unregulated private tankers and water filtration plants without strengthening the human capacity for sustainable operation and maintenance of already existing

infrastructure. Consequently, much of Pakistan's water supply infrastructure is found in poor condition, due to a practice of "build-neglect-rebuild". Following this, the required investment in piped distribution systems and its maintenance seems to be the dominant cost of achieving SDG 6.1. However, the revenues recovered from the consumers of improved water supplies are mostly insufficient to meet the operation and maintenance costs of water supply network. If this practice continues, there is a real risk that existing issues of sustainability and ownership as well as appropriate technical considerations within the setup of water supply will become critical.

The lack of consumer's confidence on water supply along with unsafe water quality proved very favorable and advantageous for the growth of improperly regulated bottle water industry in Pakistan. On the other hand, regular monitoring of bottle water brands by PCRWR since 2005 also reveals the unsafe quality of many brands.

4.4 Low and Inefficient Financial Investment

Safe and affordable drinking water for all by 2030 requires effective investment in water supply infrastructure. The World Bank has estimated that achieving a higher level of "safely managed" water services would require additional financing in the order of three times the current spending (Hutton and Varughese, 2016). The budgetary allocations for Water, Sanitation and Hygiene in 2019-2020 (including federal and provincial budgets) were PKR 157.258 billion and overall PKR 739 per capita. However, the per capita was highest in Balochistan with PKR 1,872 and the lowest in Punjab with PKR 371 capita (UNICEF, 2020).

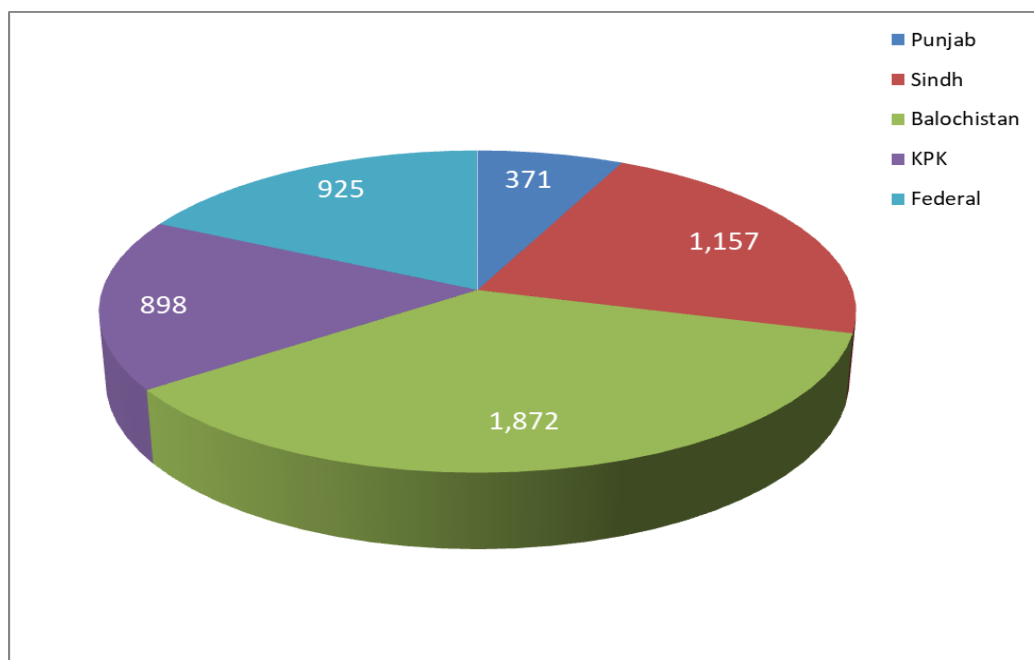


Figure 50: 2019-20 per capita WASH allocation
(Data source: UNICEF, 2020)

The WHO (2014) reported that for every US\$ invested in Water and Sanitation, there is a return of US\$ 4.3 in the form of reduced health care costs for individuals and economics returns for the country (WHO, 2014). Based on SDG costing tool, it is estimated that Pakistan needs PKR 450 billion annually to meet SDG targets by 2030 (Pakistan Economic Survey, 2019). However, without effective planning and consideration of sustainability and service delivery concept within the existing model of water supply services, increased investment on water sector may result in failures like Clean Drinking water for All (CDWA) and Punjab Saaf Paani Company Project (PSPC).

4.5 Management of Water Supply Technologies

Main cause of surface or groundwater pollution is the addition of untreated municipal sewage and industrial wastewater at different points of the water distribution network. Drinking water supply lines and sewage lines run in parallel, which causes leakages and intermixing result in deterioration of water quality. This has increased the costs of providing access to safe water using expensive water treatment. The past experience with CDWA and PSPC shows that inappropriate technology selection like reverse osmosis, ultrafiltration units etc. may result in wastage of capital cost without achieving the objective of improved access to safe drinking water. These technologies are only manageable if the right skills, resources, support structure and incentives exist. The planning should prefer to invest on improving safe tap water at point of use rather than improperly managed decentralized solutions.

Water supply models on the basis of public-private partnership in different residential and commercial sectors of major cities should be introduced. A classic example for such a model is the Changa Pani Program at Bhalwal, district Sargodha. This program provides metered safe water almost around the clock at a reasonable cost. For more than 10 years, the Program has relieved the TMA from any repair and maintenance cost (Rasheed et al., 2021). The rationalized water prices for volumetric use of water by different social categories will result in revenue generation which can ultimately be used to improve water supply and treatment infrastructure. However, complete rehabilitation of main and distribution pipe network is very important for uninterrupted and good quality water supply.

4.6 Cost of Not Providing Safe Water

An inadequate water supply whether as a result of poor access or quality, low reliability, high cost, or difficulty of management is associated with significant health risks. The bacteriological and chemical contaminations identified in this study may result in certain waterborne health implications such as diarrhea, cholera, dysentery, typhoid, and polio. The JMP Report (UNICEF and WHO, 2018) reported that 27,000 children die each year from diarrhea related diseases in Pakistan. Thus, unsafe drinking water quality imposes

both direct and indirect costs that are experienced through multiple pathways reflected in Figure 51.

Pakistan's Economic Survey (2011) reported that WASH related diseases cost Pakistan's economy about PKR 112 billion per year, over PKR 300 million a day in terms of health cost and lost earnings. Out of this, the cost associated with stomach-related diseases alone has been estimated at around PKR 55-80 billion per year. These costs include those borne by the government and households and the wider economy like healthcare. As mentioned earlier, a major national financing gap exist between current annual WASH spending of PKR 80 billion and requirement of PKR 450 billion annually to meet SDG targets by 2030.

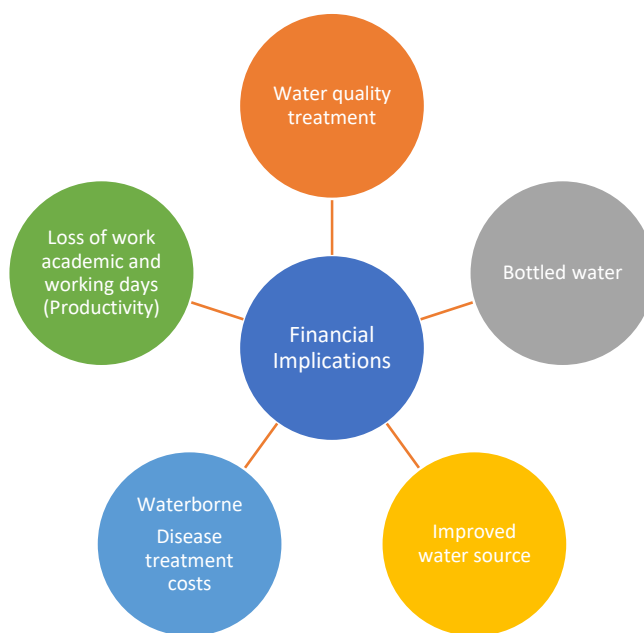


Figure 51: Financial implications of unsafe water quality

4.7 Implementation of Policies and Regulatory Framework

The National Water Policy (NWP) approved in 2018 outlines a framework for interventions by the Federal and Provincial governments to address various issues aggravating declining supply and deteriorating quality of water. The Section 12 of NWP also shows the commitment to National Drinking Water and Sanitation Policies and Sustainable Development Goals (NWP Clause 12.8). However, the current state of safe water access highlights the need to analyze the policy implementation framework to identify gaps and weaknesses responsible for slow paced safe water access of 31% in 2015 and 39% in 2020. The NWP is a national framework within which the provinces are trying to develop their own policies, regulations and master plans for sustainable management of water resources. PCRWR on the direction of the Federal government is

also contributing in developing a five-year National Groundwater Management Plan to provide a framework for coordination among groundwater stakeholders across Pakistan. Approval and implementation of this plan would also help to control the factors responsible for increased groundwater contamination.

4.7.1 National Drinking Water Policy, 2009

The National Drinking Water Policy (NDWP) was approved in 2009. According to the NDWP, provision of safe drinking water is a fundamental human right and the responsibility of the state. The main objective of the policy is to provide adequate quantity of safe water to the entire population at an affordable cost and in an equitable, efficient and sustainable manner by 2025. The policy has also provided guidelines for the protection and conservation of water resources, appropriate water treatment technologies and standardization, capacity building, public awareness, legislation etc.

4.7.2 Punjab Water Policy (2018)

The overall objective of Punjab's Water Policy is to provide clear policy directions to the Government of Punjab on the sustainable management and development of water sector. The Chapter 4 (Clause 4.1: Drinking Water) of the policy emphasizes that drinking water is the top most priority of this policy with specific objective to improve water quality of surface and groundwater through control of pollutants discharge from agriculture, industrial, municipal wastes and over abstraction of groundwater near saline groundwater zones. In this connection, Punjab Water Act (2019) is also instrumental to ensure the implementation of Provincial Water Policy. An improvement of 14% in the quality of water supply (49% safe water in 2020 vs. 35% in 2015) depicts the implementation status of Punjab Water Policy. The Government of Punjab needs to further strengthen its efforts following the policy guidelines and through enforcement of clauses of Punjab Water Act across the province. Punjab is also developing a provincial Groundwater Management Plan which is also expected to be instrumental in controlling the groundwater contamination.

Sustainable Development Goals 2015–2030

Goal 6: Ensure the availability and sustainable management of water and sanitation for all.

Target 6.1: By 2030, achieve universal and equitable access to safe and affordable drinking-water for all.

Priority indicator: Percentage of population using safely managed drinking-water services.

4.7.3 Khyber Pakhtunkhwa Drinking Water Policy (2015) and Water Act (2020)

The overall goal of the KPK's Drinking Water Policy is to streamline the sector and ensure that the entire population of KPK has access to adequate quantity of potable water at affordable cost through equitable, efficient and sustainable services by 2025. Political commitment of the province to SDG 6 is also reflected in the policy as Objective No. 1 (Section 5.3). One of the important features of the policy is the allocation of at least 10% of the provincial ADP (Annual Development Plan) for the achievement of its goal and objectives. Likewise, Chapter 5 of the Khyber Pakhtunkhwa Water Act (2020) is focused on water supply and includes responsibilities, liabilities and penalties of water supply service providers, citizens and all of the relevant stakeholders (Clauses 19 to 27). Both of the policy instruments provide clear directions for complete surveillance of drinking water quality in the province and hence to ensure improved access to the safe water supply. Though KPK has made visible and high ranked improvement in the quality of its water supplies after 2010, however the current status of 43% of the unsafe water supplies in four monitored cities indicates the need to further transform the policy directives into actions.

4.7.4 Sindh Drinking Water Policy (2017)

Sindh Drinking Water Policy was approved on 3rd May, 2017. The goal of the policy is to improve the quality of life of the people in Sindh by reducing morbidity and mortality caused by water-borne diseases through provision of safely managed and potable drinking water to the entire population. The objective 4 of this policy aims to enhance the coverage of safely managed drinking water supply in the province to achieve the SDGs targets of universal access. However, the situation of unsafe water supplies in the province as indicated in this study (85%) shows weaker enforcement of the ambitious commitment stated in the policy. Moreover, a draft water policy of Sindh is also underway to provide policy directions for conjunctive management of surface water and groundwater. A stronger enforcement of both drinking water policy and water policy (once approved) in the province may improve the safe water coverage.

4.7.5 Gilgit-Baltistan Drinking Water Policy (2019)

The draft Gilgit-Baltistan drinking water policy aims to provide an inclusive policy framework and principles to the GB government, local government and all stakeholders to address the key-issues and challenges in provision of safely managed drinking water to its citizens. The Objective 6 of the draft policy ensures that all future drinking water supplies are constructed and the existing drinking water supply systems are rehabilitated in line with the targets and indicators of Sustainable Development Goal 6. The current state of unsatisfactory improvement of water quality as revealed in this study demands taking appropriate and immediate actions in order to achieve the policy objectives.

Although some progress has been made in Punjab and KPK however, concrete actions are required in all the provinces. The current situation of 59% unsafe water supplies in the Balochistan province also demands the Provincial government for institutional strengthening, restructuring and capacity building of the water supply stakeholders as a part of the thrust areas of Balochistan Integrated Water Resources Management Policy approved by the Cabinet in March, 2006.

The results of this country-wide monitoring also confirmed that the interaction between policy and water supply sub-sectors would play an important role for the successful implementation of the policy actions. In this process, provinces also need strong institutions, effective legal frameworks and sufficient human and financial resources. Furthermore, encouraging the provinces to reform their water supply systems to bring these into sustainable state would be in compliance to the NWP and provincial water policies.

Chapter 5: Conclusion and Recommendations

5.1 Conclusions

- a. Pakistan Council of Research in Water Resources (PCRWR) in compliance to the Clause 21.4 (Section 21) of the National Water Policy 2018 has monitored the water quality across the country through its well-established monitoring network. The study outcomes are straight aligned with study objectives to identify problem areas and the causes of contamination for preventive and corrective actions.
- b. The microbiological contamination (41%) is the primary challenge for water supply agencies. At the same time, it is relatively easy and cost effective to treat such kind of contaminant if the source of contaminant is identified and appropriate technologies are used to treat it.
- c. The country-wide chemical contamination of water supplies due to Turbidity (9%), Iron (14%), Hardness (10%), Chloride (8%), TDS (14%), Fluoride (4%), Nitrates (4%), Arsenic (5%) and pH (1%) also needs to be considered seriously by the water supply agencies.
- d. In Balochistan, water quality improvement is observed with safe water supplies of 41% in 2020 compared to 19% in 2015.
- e. The overall water quality situation of the Punjab province also shows an improvement in water quality from 35% in 2015 to 49% in 2020.
- f. In Sindh, there is a decline in water quality from 19% safe water supplies in 2015 to 15% in 2020.
- g. Though Khyber Pakhtunkhwa has shown highest ranking in safe water supplies with (57%), however, a decline occurred when compared with situation in 2015 (59% safe). The main issue is of excessive level of Iron which can be very easily addressed by replacing the rusted pipes and components.
- h. The situation in Gilgit (GB) is not good as 100% of surface water supplies were found contaminated due to bacteria and turbidity. In Muzaffarabad (AJK), 70% of the monitored sources were found contaminated.
- i. With the current pace of water quality improvement, only 50% of the population is projected to have safe water access by 2030. Achieving SDG target 6.1 by 2030 requires speeding up the current slow paced quality improvement by four times.

5.2 Recommendations

The National Water Policy, Pakistan Vision 2025 and UN Sustainable Development Goals (SDG's) 2030 impose obligations to ensure access to clean drinking water for all. To achieve this goal following recommendations are given in the light of this monitoring outcome:

- a. The provincial governments should strengthen water supply institutions; implement policy frameworks and arrange sufficient human and financial resources.
- b. Pakistan Water Vision 2025 and National Water Policy support an architecture that leads to 100% safe water supply. However, water sector is highly subsidized, and any discussion of tariff and groundwater abstraction reforms is encountered with political opposition. Therefore, achieving SDG 6.1 by 2030 in Pakistan requires substantial reform of water services and controlled groundwater abstraction.
- c. A proper groundwater monitoring network is needed in all areas of the country. This would help in making informed decisions thus helping to use the available resources optimally.
- d. Disposal of untreated wastewater results in the contamination of water supplies. The approach of 4R (refuse, recycling, reduce and reuse) of wastewater would result in safe and environmentally sustainable way of wastewater utilization. In all cities of the country, domestic and industrial effluents should be discharged after proper treatment preferably following zero liquid discharge policy (ZLD) and in conformity to the National Environmental Quality Standards (NEQS, 1999). In addition, proper sewerage system including sewage collection should also be placed in all the urban and rural areas. Proper distance should be maintained and water supply pipelines should not allow passing across the sewerage lines.
- e. New water supply schemes should not be approved unless feasibility study of water quality, quantity and its sustainability is carried out. Currently service providers are monitoring their water supplies by themselves. To improve this process and to ensure safe water quality, PCRWR network of water quality laboratories can provide technical support to all the local water supply agencies.
- f. Alternate sources of water should be identified in areas where the groundwater sources are contaminated due to arsenic, TDS, nitrate and fluoride etc.
- g. Education, training and awareness from the community level to policy makers on improving access to safe drinking water by removing all barriers will play an important role in achieving SDG 6.
- h. Strengthening the technical capacity of local, provincial and national water supply authorities is essential for managing and regulating water supply systems across the country.

- i. Water supply agencies should switch to the application of smart technologies that can improve management and service delivery. This may include smart water meters, online monitoring of water supplies, web based and mobile applications etc.
- j. Water supply agencies need to track and resolve the possible causes of microbial contamination i.e. leakage in water supply pipes, unhygienic physical situation at source, inadequate chlorination etc. In this case, leakage repair as well as use of water purification and disinfection methods (e.g. chlorinators) either at source-to-supply network or at point of use should be regulated.
- k. Compliance to the permissible limits set in the National Drinking Water Quality Standards of Pakistan must be ensured by the Federal, Provincial and Local water supply agencies at all levels from source to consumer end.
- l. The concept of service delivery and Public-Private Partnerships needs to be introduced. Such models exists in the country. However, it needs a paradigm shift in thinking and strong personal, institutional and political comitments.

References

- Ahsan, M., Hifza, R., Ashraf, M., and Kiran, A., (2020). Assessment of Water Quality Status in Gilgit-Baltistan. Pakistan Council of Research in Water Resources (PCRWR), pp. 43.
- APHA, (2017). Standard Method for Examination of Water and wastewater. 23rd Edition, American Public Health Association and Water Pollution Control Federation, New York, Washington DC.
- Ashraf, M., and Sheikh, A.A., (2017). Sustainable Groundwater Management in Balochistan. Pakistan Council of Research in Water Resources, pp. 34.
- Dongmei, H., Xianfang, S., and Matthew, J. C (2016). Identification of anthropogenic and natural inputs of sulfate into a karstic coastal groundwater system in northeast China. Hydrology and Earth System Sciences, Vol. 20, pp.1983–1999.
- Durov, S. A. (1948). Natural waters and graphic representation of their composition. In Dokl Akad Nauk SSSR, Vol. 59, No. 3, pp. 87-90.
- Grayman, W.M., Day, H.J. and Luken, R. (2003). Regional water quality management for the Dong Nai river basin, Vietnam. *Water Science and Technology*, 48(10), pp.17-23. World Health Organization, (2014). Global status report on non-communicable diseases 2014 (No. WHO/NMH/NVI/15.1). World Health Organization.
- Hutton, G., and Varughese, M. (2016). The costs of meeting the 2030 sustainable development goal targets on drinking water, sanitation, and hygiene. The World Bank.
- Imran, S., Anwaar, K., Bukhari, L. N., and Ashraf, M. (2016). Water Quality Status of Major Cities of Pakistan 2015-16. Pakistan Council of Research in Water Resources, (PCRWR), pp. 93.
- Iqbal N., M. Ashraf, M. Imran, Salam H. A., Hassan F. U., Khan, A. D (2020). Groundwater Investigations and Mapping in the Lower Indus Plain. Pakistan Council of Research in Water Resources (PCRWR), Islamabad, pp. 70.
- Kumpel, E. and Nelson, K. L. (2016). Intermittent water supply: prevalence, practice, and microbial water quality. *Environmental science & technology*, 50, pp. 542-553.
- Ministry of Environment (MOE) Government of Pakistan Statutory Notification S.R.O 1063 (1), (2010). National Standards for Drinking Water Quality., The gazette of Pakistan, Part II pp 3207 http://environment.gov.pk/images/rules/SRO2010_NEQS_Air_Water_Noise.pdf accessed on October 10th, 2020.

- Ministry of Finance (2011). Pakistan Economic Survey. Chapter 16: Environment. (www.finance.gov.pk). Accessed: May 18, 2021.
- Ministry of Finance (2017). Pakistan Economic Survey. Chapter 11: Health and Nutrition. (www.finance.gov.pk). Accessed: March 16, 2021.
- Ministry of Finance (2019). Pakistan Economic Survey. Chapter 16: Climate Change. (www.finance.gov.pk). Accessed: March 16, 2021.
- Mondal, N. C., and Singh, V. P. (2012). Chloride migration in groundwater for a tannery belt in Southern India. Environmental Monitoring and Assessment, 184(5), pp. 2857-2879.
- Pakistan Census Reports (2017) Statistical Division, Government of Pakistan. Final Results Census-2017 | Pakistan Bureau of Statistics (pbs.gov.pk). Accessed: July 10, 2021.
- Piper, A. M. (1944). A graphic procedure in the geochemical interpretation of water-analyses. Eos, Transactions American Geophysical Union, 25(6), pp. 914-928.
- Rasheed H., Badshah S., Ashraf M., Ahmed M., and Rehman S. (2021). Comparative Assessment of Changa Pani and Traditional Water Supply Schemes (Bhalwal, District Sargodha, Punjab). Pakistan Council of Research in Water Resources (PCRWR), Islamabad, pp. 44.
- Waterloo Hydrogeologic (2020). Hydro GeoAnalyst 9.0. https://wwaterloohydrogeologic.com/help/hga/index.html?sample_water_type.htm. Accessed: April 12, 2021.
- Talib, M. A., Tang, Z., Shahab, A., Siddique, J., Faheem, M., and Fatima, M. (2019). Hydrogeochemical characterization and suitability assessment of groundwater: a case study in Central Sindh, Pakistan. International Journal of Environmental Research and Public Health, 16(5), pp. 886.
- UNICEF (2018). Drinking water, sanitation and hygiene in schools: global baseline report 2018.
- UN Census (2021). Population Stat, 2017-2021. Data source: World Bank, United Nations, Census, GeoNames. Accessed: March 10, 2021.

Annexures

Annexure-01

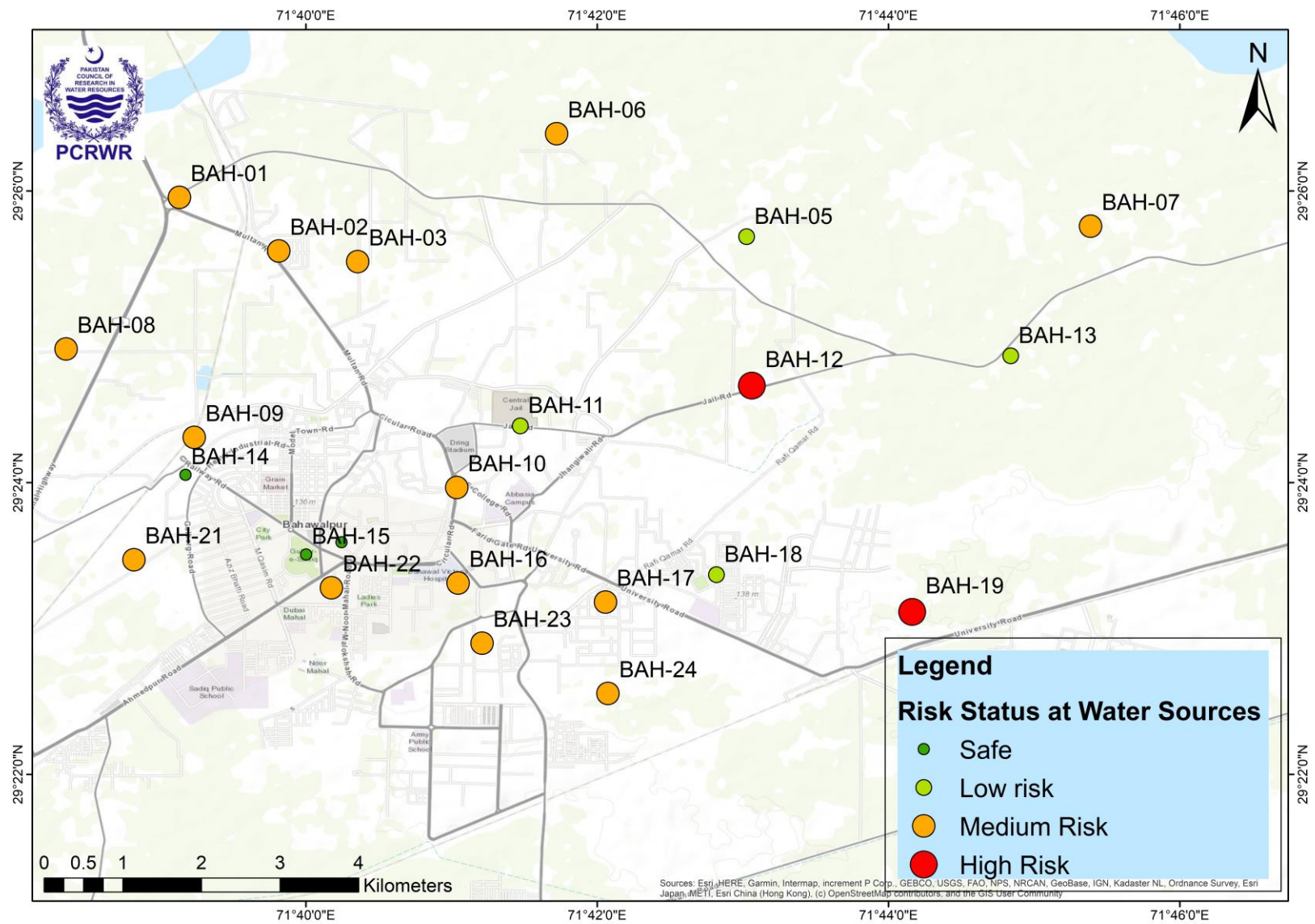
Detailed Water Quality Results of Islamabad

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	Safe/Unsafe
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	
1	ISL-01	T.Well No.193,F-6 NEFDEC Cinema	T.Well	898	6.90	0.30	350	0	108	19	350	24	33	0.9	40	5	493	0.45	0.26	0.12	0	0	Unsafe
2	ISL-02	Quaid-e-Azam University	Tap.	400	7.50	4.60	170	0	32	19	160	12	16	2.6	16	1	220	0.69	0.13	0.12	0	0	Unsafe
3	ISL-03	Noor Pur Shahan(Simly Dam)	Tap.	402	7.50	3.20	170	0	32	19	160	12	16	2.6	15	1	221	0.30	0.14	0.60	0	0	Safe
4	ISL-04	T.Well No.37,G-5	T.Well	441	7.40	0.30	180	0	52	10	170	12	23	2.5	25	1	243	0.02	0.09	0.70	0	0	Safe
5	ISL-05	IMCG F-7/4 Boring	Tap.	866	7.00	0.30	370	0	108	22	360	9	20	2.8	36	4	476	0.57	0.35	0.12	26	0	Unsafe
6	ISL-06	T.Well Polyclinic Hostel	T.Well	797	7.10	0.30	360	0	44	53	330	14	30	0.5	67	4	438	0.02	0.43	0.12	0	0	Safe
7	ISL-07	T.Well No.61 G-7/3-2	Tap.	735	7.00	0.30	310	0	96	24	340	20	15	1.5	39	4	404	0.19	0.33	0.12	0	0	Safe
8	ISL-08	T.Well PIMS Near Storage	Tap.	766	7.50	0.30	340	0	100	22	340	16	23	1	30	4	421	0.02	0.44	0.50	8	0	Unsafe
9	ISL-09	T.Well No.64,St#37,F-8/1(New Site)		Non Functional																			
10	ISL-10	T.Well#2,E-8 GE Navy, MES Off		Non Functional																			
11	ISL-11	T.Well No.200 F-9,Fatima Jinnah Park	T.Well	856	8.10	0.30	370	0	80	24	200	12	48	1.8	30	3	470	0.27	0.44	0.12	0	0	Safe
12	ISL-12	T.Well No.105,F-10/2	T.Well	840	7.10	0.30	380	0	72	46	250	14	46	1.1	30	5	462	0.02	0.42	0.60	0	0	Safe
13	ISL-13	T.Well No.41,G-9/3	T.Well	792	7.00	0.30	330	0	88	24	320	17	26	1.4	30	4	436	0.02	0.46	0.12	0	0	Safe
14	ISL-14	T.Well No.100,G-11/2	T.Well	983	7.90	0.30	380	0	92	27	340	24	61	1.7	67	5	540	0.02	0.39	0.12	0	0	Safe
15	ISL-15	T.Well No.103,F-11/3	T.Well	780	7.00	0.30	300	0	104	17	330	14	27	1.3	39	5	429	0.02	0.44	0.12	0	0	Safe
16	ISL-16	T.well No.194,Lunda Mustan,H-11	T.Well	766	7.00	0.30	400	0	56	39	210	16	0.9	41	20	3	421	0.02	0.45	0.12	0	0	Safe
17	ISL-17	T.well No.151,G-10/2		Non Functional																			
18	ISL-18	T.Well No.10/48,PCSIR Lab H-9	T.Well	823	7.00	0.30	350	0	48	93	340	24	36	1.1	30	4	453	0.02	0.36	0.12	0	0	Safe
19	ISL-19	National Inst. of Science & Technology Edu	T.Well	909	7.10	0.30	400	0	80	44	380	23	35	1.2	38	6	500	0.02	0.43	0.12	0	0	Safe
20	ISL-20	T.Well No.118,1-8/2 Deep Turbine	T.Well	823	6.90	0.30	440	0	100	29	370	48	30	1.2	34	5	463	0.47	0.38	1.00	0	0	Unsafe
21	ISL-21	T.well No.139,1-9/4,Pindora	T.Well	854	6.90	0.30	350	0	84	36	360	38	32	1.1	36	3	470	0.22	0.42	0.80	34	0	Unsafe
22	ISL-22	T.well No.137,1-10/4	T.Well	907	6.90	0.30	370	0	100	32	380	40	38	1.2	35	5	499	0.02	0.34	0.60	0	0	Safe
23	ISL-23	7MGR,F-5/2	T.Well	408	7.60	0.30	160	0	28	22	160	12	14	2.4	19	1	224	0.02	0.08	0.90	0	0	Safe
24	ISL-24	Tap Water, Most,D-Block	Tap.	400	7.50	0.30	170	0	28	22	160	12	13	2.2	14	1	220	0.02	0.10	0.12	10	6	Unsafe
25	ISL-25	TapWater,H-2,St#15,F-6/3	Tap.	390	7.50	0.30	180	0	40	17	170	14	15	2.2	15	1	214	0.02	0.13	0.12	0	0	Safe
26	ISL-26	H-63,Gomal Rd,E-7	Tap.	650	7.40	0.30	270	0	30	19	280	20	19	1.5	25	4	376	0.28	0.28	0.12	56	16	Unsafe
27	ISL-27	H-16,St 83,G-6/4	T.Well	783	7.00	0.30	340	0	44	53	330	16	26	0.6	36	4	430	0.23	0.41	0.12	16	6	Unsafe
Total No. of samples				24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
%age of Samples Exceeding Permissible Limits					0%	0%	0%		0%	0%	0%	0%	0%	0%	0%	0%	0%	17%	0%	0%	25%	13%	38%
No. of Samples Exceeding Permissible Limits					0	0			0	0						0	0	4	0	0	6	3	9
Maximum Concentration				983.00	8.10	5.00	440	0	108	93	500	250	61	41.0	67	10	1000	0.69	1.50	50.00	56	16	
Minimum Concentration				390.00	6.90	0.30	160	0	28	10	160	9	1	0.5	14	1	214	0.02	0.08	0.12	0	0	

Annexure-02

Detailed Water Quality Results of Bahawalpur

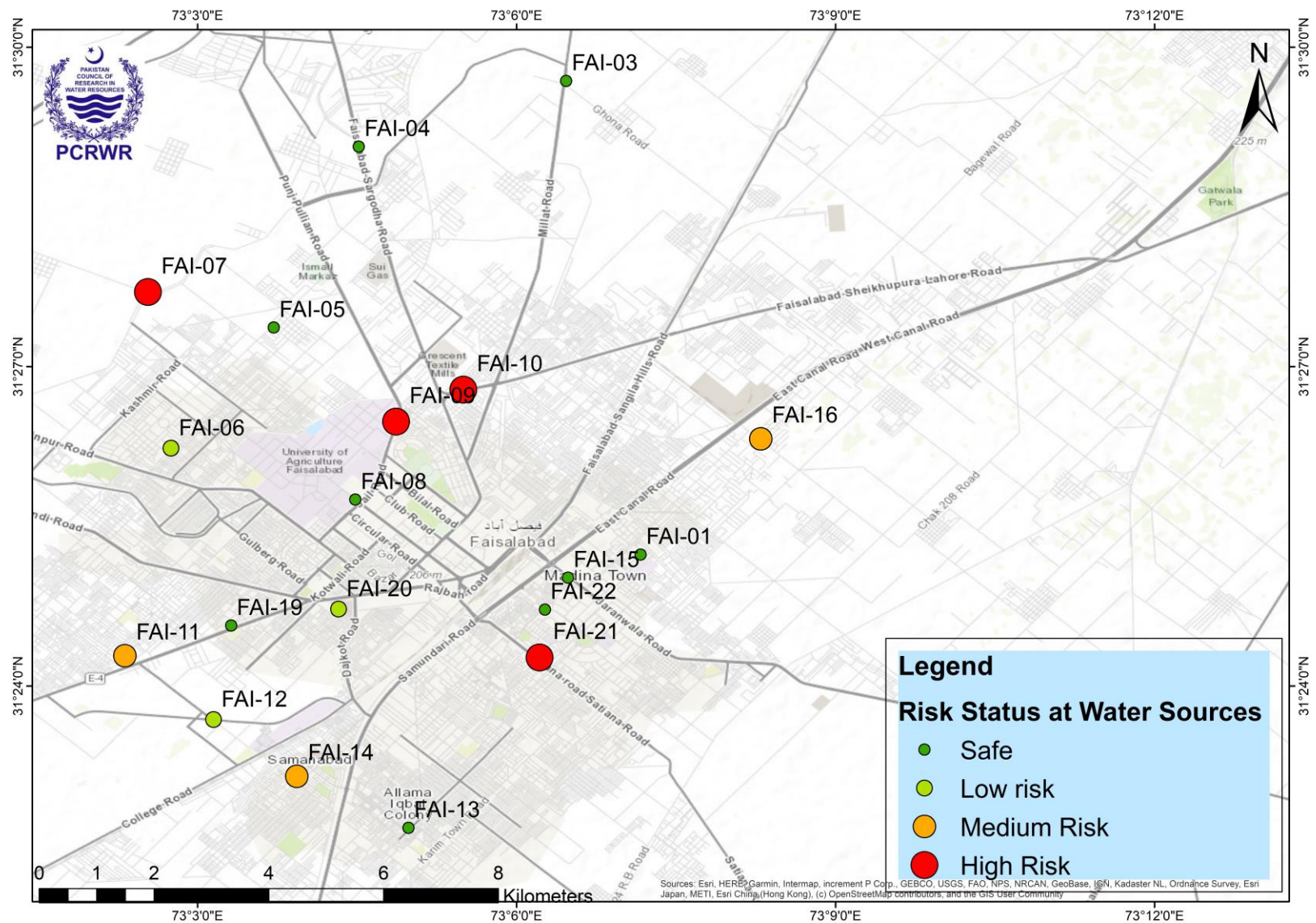
Sr. No.	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk.	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
			Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
			Maximum Permissible Limits NDWQS		NGVS	6.5-8.5	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10	1000	0.3	1.50	50.00	0/100 ml	0/100 ml	
1	BAH-01	1	Sutlej Hotel (New bore installed)	I. Pump	870	7.58	6.20	270	0	68	44	350	38	52	6.1	104	1.371	557	0.30	1.50	104.5	0	0	Unsafe
2	BAH-02	2	Water Supply Small Industrial State	I. Pump	715	7.82	1.19	240	0	72	22	270	53	45	5.2	62	0.055	458	1.35	0.44	34.00	0	0	Unsafe
3	BAH-03	3	Madrasa Abdullah Bin Abbas, Abbasi Town	I. Pump	518	7.86	2.82	200	0	52	17	200	14	27	4.9	31	0.455	332	0.16	0.41	76.50	7	4	Unsafe
4	BAH-04	4	Jamia Masjid Kot Noora	I. Pump	1148	7.42	0.16	250	0	24	70	350	96	99	7.3	79	2.528	735	0.54	0.28	0.60	0	0	Unsafe
5	BAH-05	5	Govt. High School Sahlan Basti	H. Pump	1798	7.39	2.34	410	0	100	56	480	134	189	12.5	290	0.483	1151	0.03	0.40	4.00	0	0	Unsafe
6	BAH-06	6	Basti Nayan (New Bore)	I. Pump	1190	7.43	17.74	310	0	88	44	400	82	86	6.5	158	0.865	762	0.21	0.52	85.00	0	0	Unsafe
7	BAH-07	7	Govt. High School Jhangi Wala (Opp. New Bore)	I. Pump	719	7.74	21.38	260	0	52	24	230	34	58	7.6	54	1.91	460	1.25	0.46	69.00	52	25	Unsafe
8	BAH-08	8	Al-Haq Cotton Factory	I. Pump	568	7.79	0.9	250	0	44	17	180	14.4	43	4.2	13	0.526	363	4.75	0.28	66.00	0	0	Unsafe
9	BAH-09	9	Railway Station (WSS)	T. Well	580	8.03	0.32	230	0	44	12	160	15	58	5.4	38	0.211	371	0.14	0.75	42.00	0	0	Safe
10	BAH-10	10	Municipal Corp. Old Jumma Bazar	F. Plant	955	7.61	0.38	250	0	48	31	250	43	80	6.7	113	0.269	611	0.20	0.40	55.00	0	0	Unsafe
11	BAH-11	11	Opp. Central Jail Masjid (New Bore)	I. Pump	2190	7.36	0.21	560	0	68	90	540	206	281	12.3	305	5.258	1402	0.40	0.43	0.12	0	0	Unsafe
12	BAH-12	12	Govt. Primary School, Akhtar Abad	I. Pump	2120	7.58	23.35	490	0	40	99	510	220	256	13.4	304	0.522	1357	0.10	0.62	34.00	0	0	Unsafe
13	BAH-13	13	Govt. Girls Primary School Ferozwala	I. Pump	1967	7.44	0.12	490	0	28	109	520	192	223	10.4	242	4.391	1259	2.35	1.02	0.12	0	0	Unsafe
14	BAH-14	14	Cooperative Office (WSS)	T. Well	543	7.9	0.11	250	0	50	22	190	14	51	4.4	37	0.05	348	0.04	1.01	0.12	0	0	Safe
15	BAH-15	15	Gulzar-e- Saddique Ground	T. Well	1050	7.66	1.67	270	0	44	46	300	37	104	7.1	190	0.342	672	0.10	0.61	2.20	0	0	Safe
16	BAH-16	16	Pump-1, BWP Victoria Hospital (WSS)	T. Well	1160	7.71	2.49	280	0	68	41	340	66	122	9	210	0.202	742	0.33	0.46	54.90	0	0	Unsafe
17	BAH-17	17	One Unit Colony (WSS)	T. Well	1269	7.65	3.09	310	0	72	51	390	75	131	10.3	230	0.424	812	0.18	0.47	45.50	0	0	Safe
18	BAH-18	18	Commercial Area, Satellite Town (WSS)	T. Well	530	7.62	19.69	160	0	36	12	140	47	60	6	43	0.983	339	0.27	0.58	2.60	6	0	Unsafe
19	BAH-19	19	Govt. High School Hamaitiyan	I. Pump	2820	8.22	0.09	420	0	48	46	310	184	483	33.6	702	1.598	1805	0.10	0.15	10.50	0	0	Unsafe
20	BAH-20	20	Fisheries Department	T. Well	848	8.03	0.31	270	0	24	36	210	61	113	8.1	105	0.158	543	0.12	6.05	36.00	0	0	Unsafe
21	BAH-21	21	B.I.S.E. Bahawalpur (WSS)	T. Well	730	7.99	0.92	250	0	40	22	190	33	93	5.6	80	0.637	467	0.20	0.47	37.00	0	0	Safe
22	BAH-22	22	Officer Colony Near High Court (New Bore)	T. Well	768	7.59	3.75	270	0	64	24	260	33	56	7.1	91	0.488	492	0.14	0.38	35.00	0	0	Safe
23	BAH-23	23	Abbasia Campus Cantt	I. Pump	1250	7.69	4.81	380	0	56	39	300	70.5	160	8.9	165	0.266	800	0.72	0.33	20.00	25	16	Unsafe
24	BAH-24	24	Green Town (Behind Fauji Basti) (New Source)	T. Well	540	8.11	0.33	160	0	32	17	150	52	65	6.1	48	0.526	346	0.85	0.70	3.20	18	12	Unsafe
25	BAH-25	25	Masjid Hasnain Rame Fatak	I. Pump	1690	7.44	0.21	310	0	44	112	570	130	138	13.4	370	2.475	1082	0.08	0.29	0.90	0	0	Unsafe
Total No. of Sample Collected					25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
%age of Samples beyond permissilbe limit						0%	20%					16%	0%				0%	24%	36%	4%	28%	20%	16%	76%
No. of Samples beyond permissilbe limit						0	5					4	0				0	6	9	1	7	5	4	19
Maximum Concentration					2820	8.22	23.35	560	0	100	112	570	220	483	33.6	702	5.258	1805	4.75	6.05	104.5	52	25	0
Minimum Concentration					518	7.36	0.09	160	0	24	12	140	14	27	4.2	13	0.05	332	0.03	0.15	0.12	0	0	0



Water quality risk map of Bahawalpur

Detailed Water Quality Results of Faisalabad

Sr. No.	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk.	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms (MPN/100 ml)	E. coli (MPN/100 ml)	Remarks
Units -					μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	0/100 ml	0/100 ml	
Maximum Permissible Limits PSQCA/NDWQS					NGVS	6.5-8.5	5	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10	1000	0.3	1.5	50	0/100 ml	0/100 ml	
1	FAI-01	5	Nishatabad opposite Post Office (New Bore)	I. Pump	1698	8.00	0.58	430	0	36	58	330	151	200	17	106	4	934	0.10	0.99	8.87	0	0	Safe
2	FAI-02	24	Kalama Wali Tanki Sheikupura Road	W. Supply.	692	7.90	1.12	200	0	40	34	240	46	52	5	77	0	381	0.66	0.30	3.45	0	0	Unsafe
3	FAI-03	2	Treatment Plant Millat Town	T.Well	208	7.10	1.37	60	0	20	12	100	17	9	2	52	0	114	0.11	0.10	3.94	0	0	Safe
4	FAI-04	1	Adda Chak 7JB Sargodha Road	I. Pump	1668	7.80	1.45	480	0	44	27	220	120	260	12	174	6	917	0.40	0.79	4.50	0	0	Unsafe
5	FAI-05	3	Baghdadi Chowk Sidupura Near Masjid Noor	W. Supply.	954	7.80	0.30	200	0	40	32	230	96	96	6	77	2	525	0.14	0.34	4.00	0	0	Safe
6	FAI-06	6	Awan Chok Sadar Bazar	W. Supply.	617	7.80	0.30	180	0	40	24	206	36	46	4	80	0	340	11.50	0.29	3.30	10	6	Unsafe
7	FAI-07	21	Jamia Sulaimania 100 Chak Kakkawala	I. Pump	3320	8.00	0.30	620	0	28	80	400	372	600	16	410	8	1826	0.14	1.50	13.58	0	0	Unsafe
8	FAI-08	7	Agriculture University, Faisalabad	I. Pump	1367	8.00	0.30	430	0	60	46	340	105	130	35	145	7	752	0.02	0.20	0.12	0	0	Safe
9	FAI-09	22	M Shakeel Kareem City Housing Socity Daewoo Road	I. Pump	3530	7.80	0.30	600	0	40	80	420	456	520	16	387	0	1941	1.77	1.60	3.03	0	0	Unsafe
10	FAI-10	23	Al- Khidmad Markaz Gov. Punjab Opposite Moun Trust Hospital	I. Pump	1568	7.60	0.30	410	0	92	73	530	96	100	20	144	24	862	0.22	0.80	0.28	0	0	Unsafe
11	FAI-11	9	Shadab Colony, Shadab Park FSD	W. Supply.	585	7.90	0.30	180	0	40	29	220	33	41	4	41	14	322	0.02	0.25	3.60	0	0	Unsafe
12	FAI-12	13	Ghusia General Store Near Goal Chock	I. Pump	8120	7.20	0.30	400	0	160	209	1260	1248	1050	86	1652	1	4466	1.11	1.49	2.71	0	0	Unsafe
13	FAI-13	12	M.C Girl School	I. Pump	530	7.80	1.02	180	0	60	29	270	42	35	2	80	1	345	0.11	0.50	2.92	0	0	Safe
14	FAI-14	15	Darbar Naiaz Hussain Haidar Main Jarnwala Road Faisalabad	I. Pump	4230	8.00	0.30	650	0	24	78	430	336	724	16	842	3	2327	0.22	1.40	4.88	10	7	Unsafe
15	FAI-15	8	Tube Well 3 Canal Bank	T. Well	1668	7.70	0.30	400	0	60	63	410	96	200	13	251	10	917	0.12	0.80	2.43	0	0	Safe
16	FAI-16	16	Sadqain Masjid Zia Town Opposite United Hospital	I Pump	1761	7.70	0.30	400	0	52	87	490	96	160	13	233	21	978	0.02	1.10	2.88	0	0	Unsafe
17	FAI-17	18	New Gulshane Fatima Colony, Abdula Pull Service Road	I. Pump	1245	7.60	0.30	350	0	40	58	340	75	125	11	148	7	685	0.02	1.00	1.88	0	0	Safe
18	FAI-18	19	New Truck Stand Masjid Muhammad Jhang Road	I. Pump	1310	8.30	0.30	400	0	8	12	70	52	250	9	125	5	720	0.02	3.90	5.87	0	0	Unsafe
19	FAI-19	20	H# 97 Near Amman Aysha Masjid MS Traders	I. Pump	521	7.80	0.30	140	0	48	15	180	36	38	4	74	1	287	0.02	0.30	3.30	0	0	Safe
20	FAI-20	10	Makhdoom Road Khalilabad	W. Supply.	1120	7.30	0.30	190	0	36	36	240	168	130	7	98	3	616	0.13	0.30	0.53	12	8	Unsafe
21	FAI-21	11	National College of Computer Sciences	I. Pump	4500	8.00	0.30	500	0	60	95	540	552	700	24	786	23	2475	0.02	1.20	5.70	0	0	Unsafe
22	FAI-22	17	Main WASA Storage Tank Cottan Wala Chok	T. Well	237	7.70	0.30	70	0	24	10	100	14	10	2	32	1	130	0.23	0.20	1.35	0	0	Safe
23	FAI-23	4	Allied Hospital Punjab Medical College	Closed due to Covid																				
Total No. of Sample Collected					22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22
%age of Samples beyond permissilbe limit						0%	0%					14%	23%				18%	23%	23%	9%	0%	14%	14%	59%
No. of Samples beyond permissilbe limit						0	0					3	5				4	5	5	2	0	3	3	13
Maximum Concentration					8120	8.30	1.45	650	0	160	209	1260	1248	1050	86	1652	24	4466	11.50	3.90	13.58			
Minimum Concentration					208	7.10	0.30	60	0	8	10	70	14	9	2	32	0	114	0.02	0.10	0.12			



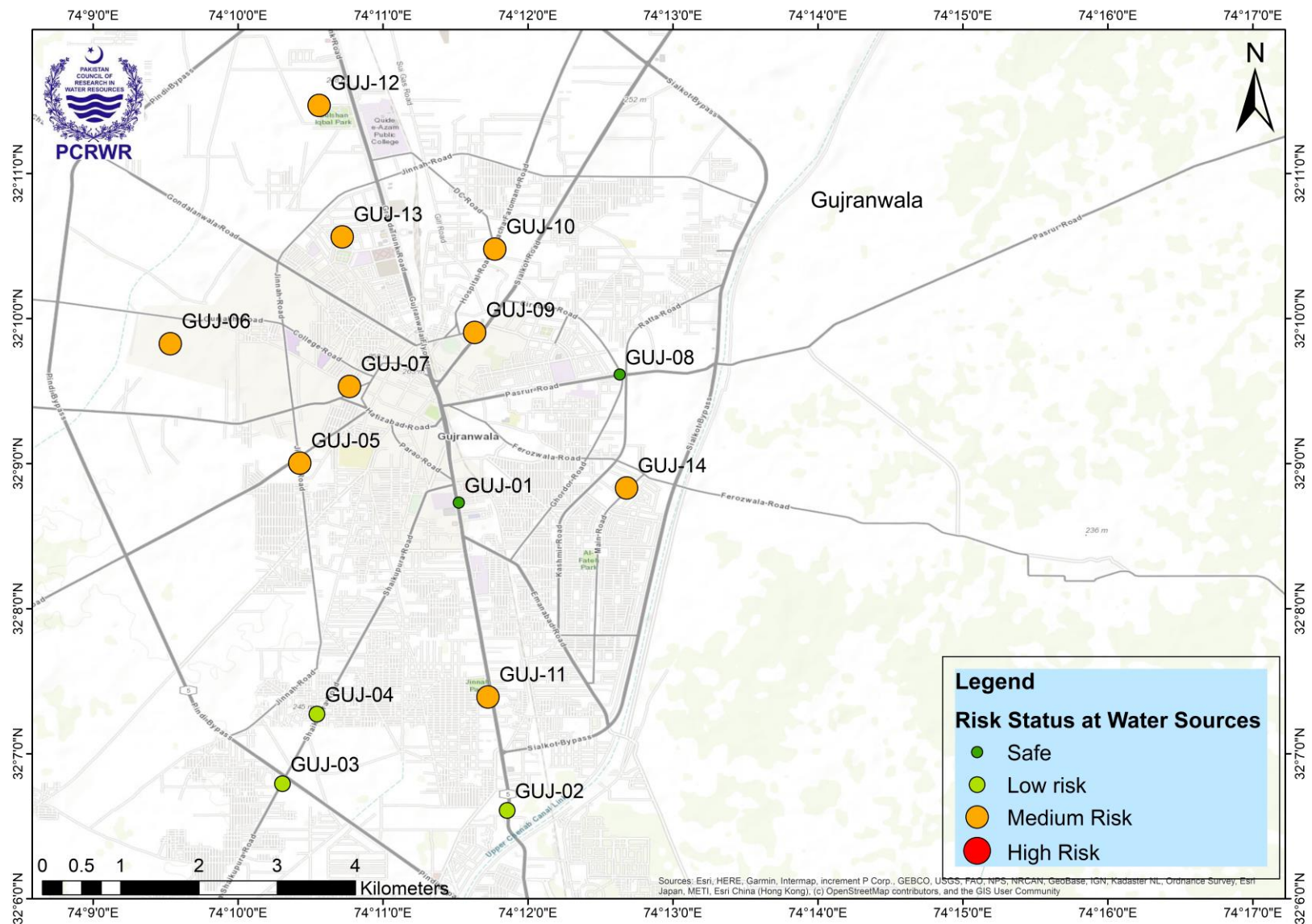
Water quality risk map of Faisalabad

Annexure-04

Detailed Water Quality Results of Gujranwala

Sr. No.	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
			Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/ 100 ml)	(MPN/ 100 ml)	
			Maximum Permissible Limits PSQCA/NDWQS		NGVS	6.5-8.5	5	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10	1000	0.3	1.5	50	0/100 ml	0/100 ml	
1	GUJ-01	1	Sheikhupura chowk G.T road	T.Well	506	7.80	0.8	220	0	36	22	180	11	32	3	28	0.19	265	0.02	0.03	3.10	0	0	Safe
2	GUJ-02	2	Super Asia M/D Sons Factory	T.Well	320	7.80	0.5	140	0	24	17	130	10	24	2	20	0.12	168	0.02	0.03	8.50	10	8	Unsafe
3	GUJ-03	3	Govt. High School Khiali By Pass	F. Plant	477	7.80	0.3	230	0	32	22	170	9	32	2	22	0.15	253	0.02	0.03	6.70	16	20	Unsafe
4	GUJ-04	4	T.Well Tanki, Shahpur Khiali	T.Well	483	7.70	0.5	240	0	32	22	170	9	32	3	23	0.17	259	0.02	0.13	2.60	10	12	Unsafe
5	GUJ-05	5	Saghir Park Badri Chowk	F. Plant	470	7.80	0.3	230	0	30	22	170	8	30	2	21	0.19	250	0.02	0.03	31.80	0	0	Safe
6	GUJ-06	6	Rehmanpura 400 Meter from Girjakh	F. Plant	490	7.80	0.2	240	0	32	24	180	11	35	3	22	0.15	261	0.02	0.03	24.10	8	6	Unsafe
7	GUJ-07	7	Janat Bibi Park Water Works WASA	F. Plant	473	7.80	0.3	230	0	30	22	170	8	30	2	21	0.15	251	0.02	0.03	26.60	12	18	Unsafe
8	GUJ-08	8	Water Filtration Plant, Satellite Town Chowk Farid Twon (New Source)	F. Plant	750	7.60	0.9	260	0	24	39	220	40	65	5	55	0.36	401	0.02	0.04	0.60	0	0	Safe
9	GUJ-09	9	Hamilton Water works near SP house	F. Plant	507	7.80	0.3	250	0	36	20	180	12	32	3	28	0.22	269	0.02	0.03	22.00	0	0	Safe
10	GUJ-10	10	Commissioner road	T.Well	470	7.80	0.5	230	0	30	22	170	8	30	2	20	0.17	250	0.02	0.03	15.30	0	0	Safe
11	GUJ-11	11	Govt. Girls High School Sarfraz Colony	F. Plant	450	7.20	0.3	220	0	24	22	150	9	30	2	20	0.12	240	0.02	0.32	19.20	16	10	Unsafe
12	GUJ-12	12	Sheikh Chowk main bazaar Shaheen abad	T.Well	540	7.70	0.5	260	0	40	22	190	15	40	3	28	0.23	286	0.02	0.03	14.80	8	6	Unsafe
13	GUJ-13	13	Block-A Tank Model Town	F. Plant	516	7.80	0.2	250	0	36	22	180	11	40	3	28	0.23	274	0.02	0.03	13.00	0	0	Safe
14	GUJ-14	14	Peoples Colony, Faqeer Pura	T.Well	495	7.80	0.3	240	0	32	22	170	10	36	3	24	0.16	263	0.02	0.66	23.20	0	0	Safe

Total No. of Sample Collected	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
%age of Samples beyond permissilbe limit		0%	0%						0%	0%							0%	0%	0%	0%	0%	50%	50%	50%
No. of Samples beyond permissilbe limit		0	0						0	0							0	0	0	0	0	7	7	7
Maximum Concentration		750	7.80	0.9	260	0	40	39	220	40	65	5	55	0.36	401	0.02	0.66	31.80						
Minimum Concentration		320	7.20	0.2	140	0	24	17	130	8	24	2	20	0.12	168	0.02	0.03	0.60						



Water quality risk map of Gujranwala

Detailed Water Quality Results of Gujrat

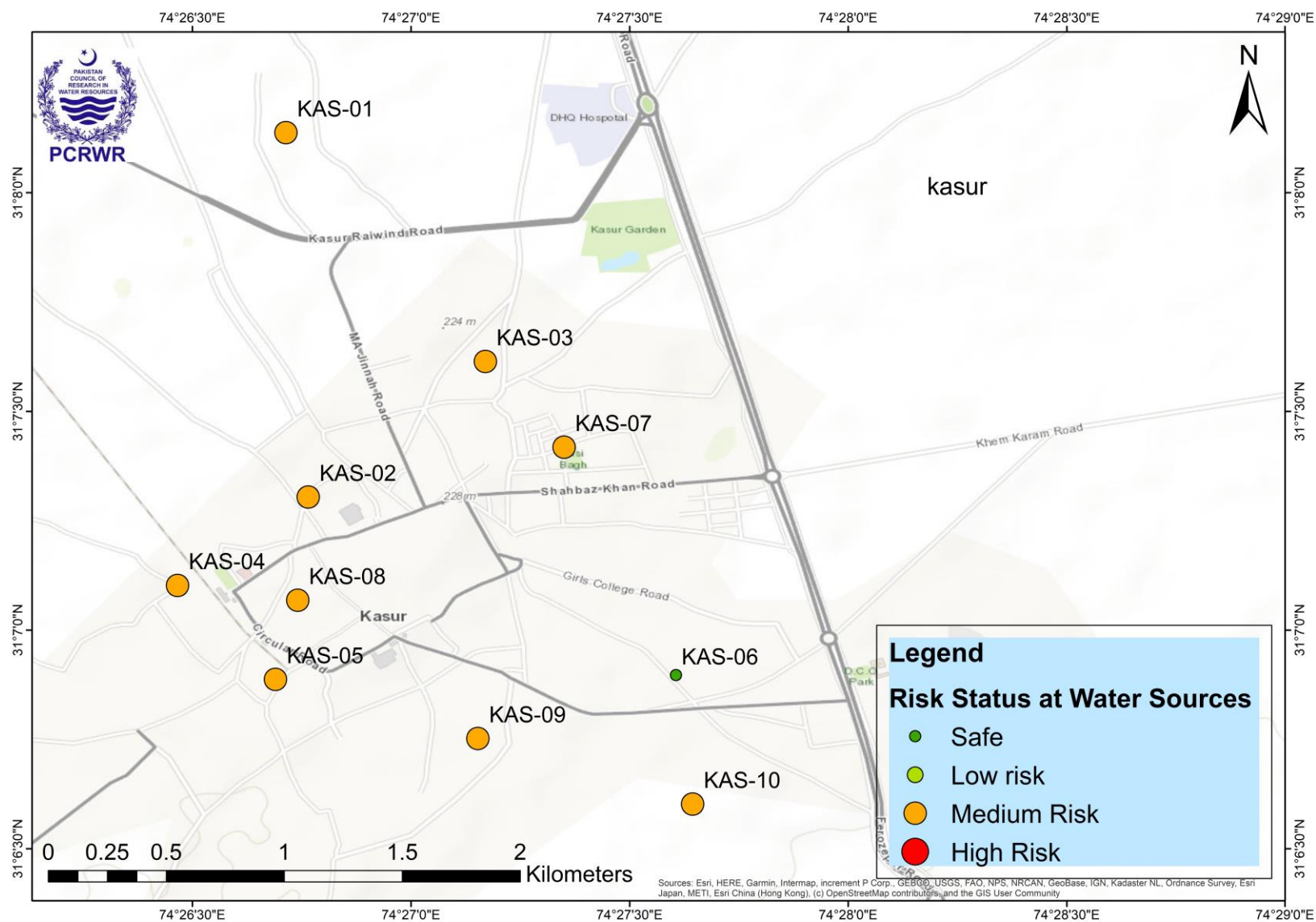
Sr. No.	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk.	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
			Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
			Maximum Permissible Limits PSQCA/NDWQS		NGVS	6.8-8.5	5	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10	1000	0.3	1.5	50	0/100 ml	0/100 ml	
1	Guj-01	1	Mohallah Faizaabad Dera Gulab Shah	T. well	375	7.80	0.7	285	0	40	7	130	8	68	1.2	15	0.05	265	0.08	0.12	2.40	0	0	Safe
2	Guj-02	2	Mohallah Meena abad Sargodha Road	T. well	461	7.92	0.6	290	0	48	7	150	20	78	1.7	52	0.05	168	0.02	0.08	1.39	0	0	Safe
3	Guj-03	3	Village Haria Near Elementary school(New sample)	Tap	408	8.00	0.9	305	0	52	5	150	10	66	2.5	35	0.43	253	0.12	0.15	3.39	0	0	Safe
4	Guj-04	4	Faiz Madina Masjid Bus stand	Tap	370	8.20	0.9	280	0	48	5	140	10	54	1.5	22	0.03	259	0.07	0.25	1.35	0	0	Safe
5	Guj-05	5	Girls stadium behind old jail	T. well	386	7.60	1.2	300	0	36	10	130	12	80	1.5	27	0.05	250	0.04	0.20	0.46	0	0	Safe
6	Guj-06	6	Municipal Corporation	Tap	441	7.75	0.7	310	0	44	7	140	16	84	2.1	32	0.20	261	0.08	0.13	5.20	0	0	Safe
7	Guj-07	7	Mahnda More Bhimber road Gujrat	T. well	383	7.60	1.5	290	0	36	7	120	10	68	1.0	17	0.02	251	0.05	0.09	3.17	0	0	Safe
8	Guj-08	8	Marghzar Colony Gujrat	T. well	375	7.70	0.8	290	0	36	7	120	8	70	1.2	20	0.04	401	0.03	0.30	0.38	0	0	Safe
9	Guj-09	9	Jamia School Baghdad Colony	Tap	580	8.00	1.5	305	0	36	10	130	25	105	1.4	88	0.30	269	0.13	0.09	1.93	0	0	Safe

Total No. of Sample Collected	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
%age of Samples beyond permissilbe limit		0%	0%							0%	0%					0%	0%	0%	0%	0%	0%	0%	0%
No. of Samples beyond permissilbe limit		0	0							0	0					0	0	0	0	0	0	0	0
Maximum Concentration	580	8.20	1.5	310	0	52	10	150	25	105	3	88	0.43	401	0.13	0.30	5.20						
Minimum Concentration	370	7.60	0.6	280	0	36	5	120	8	54	1	15	0.02	168	0.02	0.08	0.38						

Detailed Water Quality Results of Kasur

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ / Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms (MPN/ 100 ml)	E. coli (MPN/ 100 ml)	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l			
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/ Unsafe
1	KAS-01	Govt. Primary School Mohalla Qadirabad	T. well	1010	7.40	2.04	350	0	36	34	230	57	130	0.3	118	0.00	586	0.02	0.03	14.00	0	0	Safe
2	KAS-02	Nafees Colony Chandni Chowk	T. well	951	7.50	5.8	300	0	30	23	170	62	136	0.2	100	0.00	532	0.02	0.03	13.00	38	12	Unsafe
3	KAS-03	Kot Azam Khan Near Islamia College (New bore) Nimman Wala School	T. well	807	7.50	2.01	290	0	24	19	140	48	130	0.2	78	0.00	474	0.02	0.03	14.00	0	0	Safe
4	KAS-04	Chaki Piyarolal Opp. Railway Station (New bore)	Bore	1436	7.10	1.94	540	0	20	77	370	67	190	0.2	180	0.00	859	0.02	0.03	11.00	0	0	Safe
5	KAS-05	Chowk Shahedan Kot Murad Khan (New bore)	T. well	927	7.50	1.61	300	0	32	19	160	62	138	0.2	108	0.00	540	0.02	0.03	13.00	0	0	Safe
6	KAS-06	Capt. Naseem Kot Ghulam Muhammad	T. well	802	7.60	1.69	270	0	28	15	130	53	124	0.2	79	0.00	461	0.02	0.03	8.00	0	0	Safe
7	KAS-07	MCB Middle School Kot Munir Shaheed Colony	T. well	732	7.60	1.95	240	0	26	13	120	43	120	0.2	77	0.00	424	0.02	0.03	13.00	0	0	Safe
8	KAS-08	Kot Osman Khan near MCP School	T. well	1017	7.50	1.93	320	0	32	24	180	67	148	0.3	130	0.00	594	0.02	0.03	11.00	0	0	Safe
9	KAS-09	Kot Mulvi Abdul Qadeer	T. well	658	7.60	2.33	200	0	16	7	70	43	122	0.2	66	0.00	375	0.02	0.03	15.00	0	0	Safe
10	KAS-10	Basti Barat Shah, Ansari Mohalla	T. well	600	7.70	1.77	220	0	28	10	110	38	116	0.1	64	0.00	388	0.02	0.03	15.00	0	0	Safe

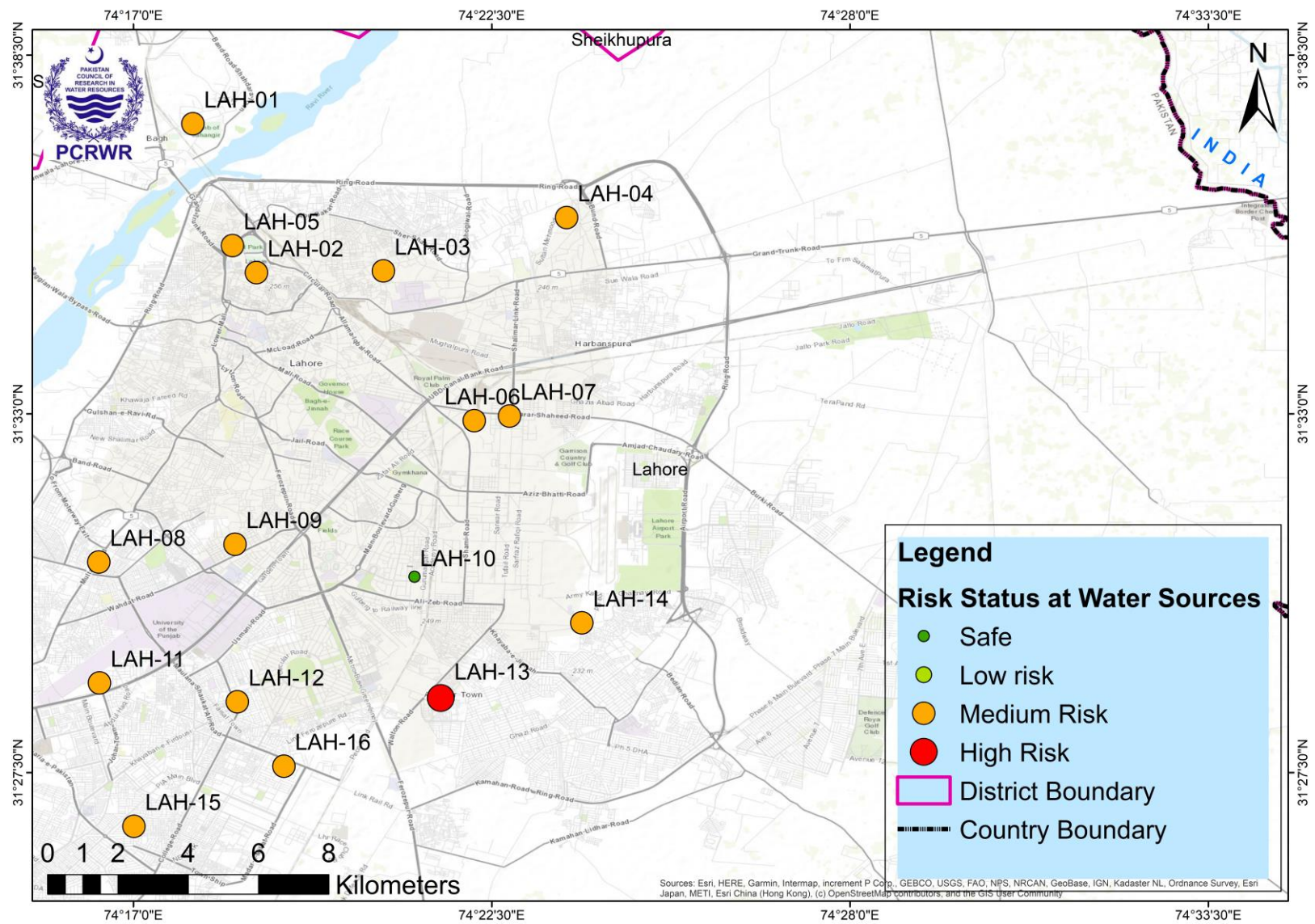
Total No. of Sample Collected	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
%age of Samples beyond permissilbe limit	0%	10%						0%	0%				0%	0%	0%	0%	0%	10%	10%	10%
No. of Samples beyond permissilbe limit	0	1						0	0				0	0	0.00	0.00	0.00	1	1	1
Maximum Concentration	1436	7.70	5.80	540	0	36	77	370	67	190	0.3	180	0.00	859	0.02	0.03	15.00	38	12	
Minimum Concentration	600	7.10	1.61	200	0	16	7	70	38	116	0.1	64	0.00	375	0.02	0.03	8.00	0	0	



Water quality risk map of Kasur

Detailed Water Quality Results of Lahore

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50	0	0	Safe/Unsafe
1	LAH-01	Old Shahdara Town Centre Tubewell	T.Well	332	6.90	1.9	150	0	36	12	140	14	32	0.0	19	0.11	204	0.02	0.03	33.25	0	0	Safe
2	LAH-02	Ali Park Tubewell-1 Fort Road	T.Well	580	6.90	1.6	250	0	56	17	210	29	60	0.2	34	0.09	347	0.02	0.03	54.78	0	0	Unsafe
3	LAH-03	Sultanpura TW Near Chah Meeran Shah	T.Well	526	7.10	2.3	200	0	40	17	170	24	54	0.1	47	0.05	302	0.02	0.03	24.82	0	0	Safe
4	LAH-04	Goharabad Tubewell Shalimar Town	T.Well	507	7.10	2.0	280	0	32	24	180	19	73	0.1	37	0.05	354	0.02	0.03	28.88	0	0	Safe
5	LAH-05	Tubewell Riwarz Garden	T.Well	550	7.10	2.1	240	0	16	27	150	14	81	0.1	64	0.05	346	0.02	0.03	20.54	0	0	Safe
6	LAH-06	Tubewell Federal Lodge, Chamba House	T.Well	1080	6.80	5.0	450	0	88	47	415	67	80	0.4	84	0.13	637	1.05	0.03	32.25	0	0	Unsafe
7	LAH-07	Tubewell Tufail Road Saddar Bazar	T.Well	528	7.20	1.9	210	0	28	17	140	14	72	0.2	45	0.05	303	0.02	0.03	24.95	0	0	Safe
8	LAH-08	TW-12, Ravi Block, Allama Iqbal Town	T.Well	447	7.10	2.2	200	0	24	15	120	24	70	0.1	34	0.05	286	0.02	0.03	21.72	53	21	Unsafe
9	LAH-09	Goal Bagh Tubewell Wahdat Colony	T.Well	444	7.30	1.7	160	0	24	10	100	17	62	0.1	42	0.05	251	0.02	0.03	25.85	0	0	Safe
10	LAH-10	Guromanget Tubewell Gulberg-III	T.Well	1308	7.10	1.9	260	0	48	44	300	50	164	0.4	266	0.55	731	0.02	0.03	2.11	0	0	Safe
11	LAH-11	PCSIR Housing Society, Canal Bank Rd	T.Well	386	7.30	1.9	160	0	16	7	70	9	70	0.1	34	0.05	233	0.02	0.03	24.11	0	0	Safe
12	LAH-12	LDA Flats Opp. Faisal Town, Ghosia Masjid	T.Well	720	7.40	2.4	300	0	24	15	120	17	120	0.2	65	0.05	421	0.02	0.03	28.92	17	7	Unsafe
13	LAH-13	Farooq Colony, Walton Road, Police Line	T.Well	1809	7.20	2.2	730	0	60	44	330	72	284	0.4	205	0.03	1103	0.02	0.10	27.87	0	0	Unsafe
14	LAH-14	Tubewell Cantonment Boad Asghari Flats	T.Well	1089	7.20	2.7	470	0	54	30	260	38	140	0.4	94	0.05	639	0.02	0.03	21.77	0	0	Safe
15	LAH-15	Punjab Govt. Co.Operative Housing Society	T.Well	693	7.50	2.4	280	0	20	10	90	14	120	0.1	75	0.05	407	0.02	0.03	24.25	0	0	Safe
16	LAH-16	Govt. Housing Scheme Township A-1	T.Well	592	7.50	1.8	240	0	20	12	100	14	91	0.2	64	0.05	346	0.03	0.03	23.25	0	0	Safe
Total No. of Sample Collected				16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
%age of Samples beyond permissilbe limit					0%	-	-	-	-	-	0%	0%	-	-	-	0%	6%	6%	0%	6%	13%	13%	31%
No. of Samples beyond permissilbe limit					0	0	-	-	-	-	0	0	-	-	-	0	1	1	0	1	2	2	5
Maximum Concentration				1809	7.50	5.0	730	0	88	47	415	72	284	0.4	266	0.55	1103.3	1.05	0.10	54.78	53	21	
Minimum Concentration				332	6.80	1.6	150	0	16	7.29	70	9	32	0	19	0.03	204.03	0.02	0.03	2.11	0	0	

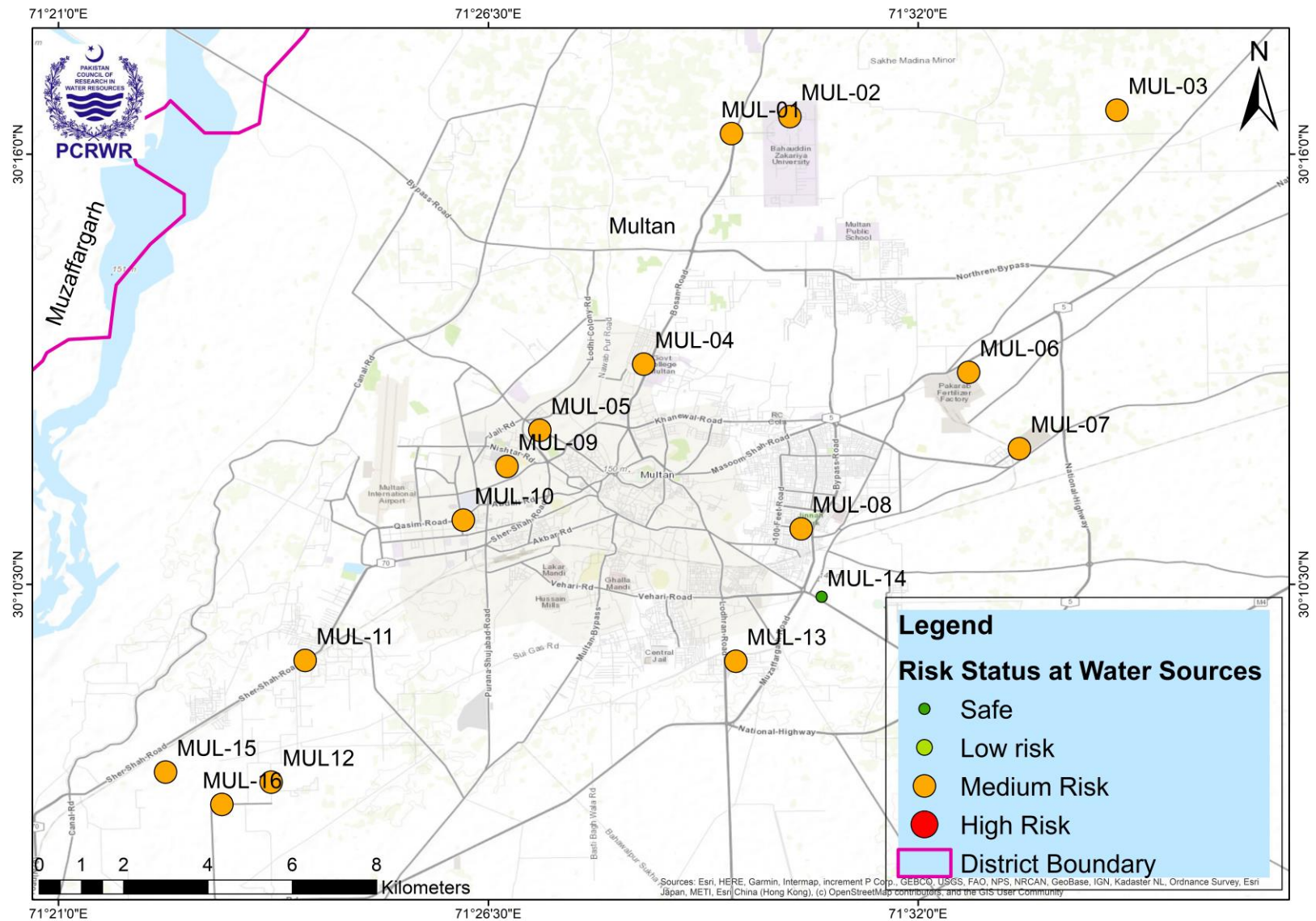


Water quality risk map of Lahore

Detailed Water Quality Results of Multan

Sr. #	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	K	Na	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	MUL-01	1	Chah Keemay Wala Opp. Zakaria University	H. Pump	631	7.00	0.5	210	0	84	12	260	31	5.9	25	61	0.20	347	0.02	0.24	39.33	4	0	Unsafe
2	MUL-02	2	Bahauddin Zakariya University	T.Well	632	7.20	0.6	215	0	46	18	190	22	7.5	54	70	0.12	348	0.02	0.26	53.43	0	0	Unsafe
3	MUL-03	3	Punjab Police Line, Malital Road	T.Well	578	7.10	1.1	185	0	42	13	195	22	7.4	58	69	0.12	318	0.02	0.20	54.23	6	0	Unsafe
4	MUL-04	4	Jamia Qasim ul Uloom Gul Gusht Colony	T.Well	540	7.20	1.2	155	0	55	7	160	40	6.5	45	59	0.16	297	0.02	0.29	45.88	10	0	Unsafe
5	MUL-05	5	Jamia Masjid Bilal, Tariqabad	H. Pump	392	7.50	1.5	155	0	42	9	140	10	6.6	25	32	0.12	216	0.02	0.23	52.52	0	0	Unsafe
6	MUL-06	6	Well -9, C-20, Pak Arab Fertilizer Corp	T.Well	740	7.70	0.9	190	0	46	9	150	29	8.4	96	133	0.06	407	0.02	0.37	43.14	5	0	Unsafe
7	MUL-07	7	WAPDA (NGPS) Piran Ghaib	Well	751	7.20	0.2	200	0	40	10	140	49	8.2	103	101	0.17	413	0.20	0.30	56.94	12	0	Unsafe
8	MUL-08	8	Shah Rukan Alam Colony, G Block	W. Supply	765	7.40	1.2	230	0	34	22	175	32	8.7	92	106	0.15	421	0.02	0.38	41.78	8	0	Unsafe
9	MUL-09	9	Nishter Hospital, Multan	T.Well	520	7.40	0.9	150	0	30	9	110	27	7.1	60	62	0.20	286	0.02	0.47	89.20	6	0	Unsafe
10	MUL-10	10	Well -1, Cantonment Board Metro Plaza	T.Well	346	7.00	1.6	130	0	26	11	110	8	6.9	27	27	0.13	190	0.02	0.22	86.50	0	0	Unsafe
11	MUL-11	11	Ch. Medical Store Basti Khudadad	H. Pump	937	6.90	2.0	230	0	90	9	260	53	7.0	93	157	0.26	515	0.02	0.41	83.30	7	0	Unsafe
12	MUL12	12	Munir Hotel Oppsite, Solkex Factory	H. Pump	413	6.70	0.5	110	0	40	7	130	22	6.8	32	62	0.44	227	0.02	0.60	78.50	8	0	Unsafe
13	MUL-13	13	WAPDA Colony, Qasim Pur	T.Well	755	6.90	0.3	130	0	88	7	250	54	7.5	54	161	0.30	415	0.02	0.60	28.20	6	0	Unsafe
14	MUL-14	14	132 KV Grid Station, Vehari Road	T.Well	634	7.00	0.1	200	0	40	10	140	24	9.8	74	77	0.22	349	0.02	0.33	9.49	0	0	Safe
15	MUL-15	15	Ismail Textile Mills (Pvt.) Ltd	H. Pump	318	7.40	1.3	120	0	24	18	135	10	4.8	12	32	0.08	175	0.02	0.31	95.60	0	0	Unsafe
16	MUL-16	16	Lucky Linker Pesticide Company	H. Pump	988	7.70	0.5	240	0	54	16	131	74	6.7	154	145	0.10	543	0.02	0.14	85.70	7	0	Unsafe

Total No. of Sample Collected	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
%age of Samples beyond permissilbe limit	-	0%	0%	-	-	-	-	0%	0%	-	-	-	0%	0%	0%	0%	63%	69%	0%	94%
No. of Samples beyond permissilbe limit	-	0	0	-	-	-	-	0	0	-	-	-	0	0	0	0	10	11	0	15
Maximum Concentration	988	7.70	2.0	240	0	90	22	260	74	9.8	154	161	0.44	543	0.20	0.60	95.60	12	0	
Minimum Concentration	318	6.70	0.1	110	0	24	7	110	8	4.8	12	27	0.06	175	0.02	0.14	9.49	0	0	

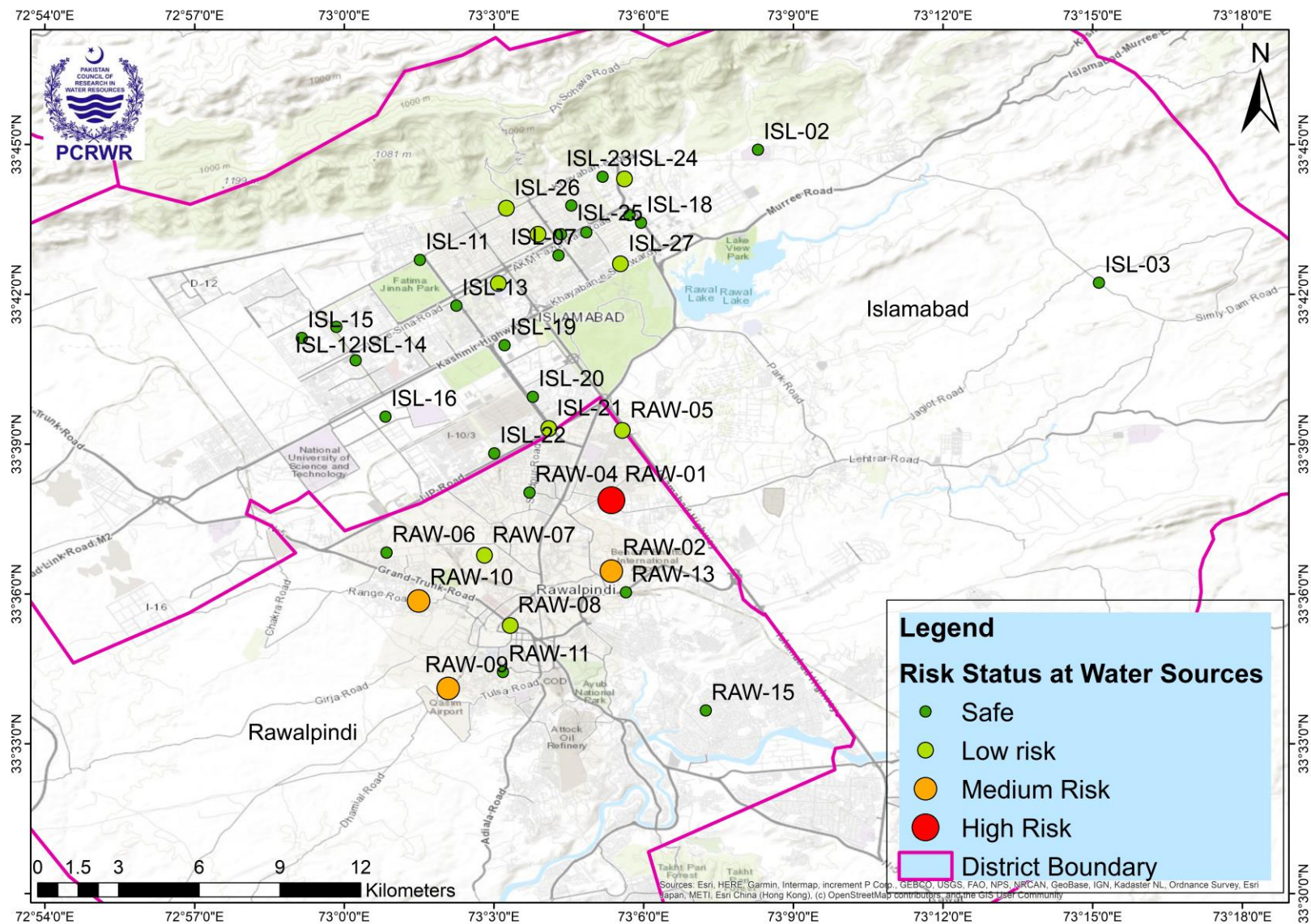


Water quality risk map of Multan

Detailed Water Quality Results of Rawalpindi

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	K	Na	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	RAW-01	T. Well-30, Muslim Town Haji Chowk	T. well	1942	7.15	0.3	460	0	128	61	500	144	2.8	170	119	29.00	1068	0.24	0.56	0.22	18	6	Unsafe
2	RAW-02	T. Well-4, PAF Base, Minhas Camp (N/F Security) Raja Bazar, 64 TMA Office (Alt.)	T. well	1034	7.30	0.3	340	0	108	19	350	36	2.6	56	45	11.00	569	0.93	0.46	0.12	8	4	Unsafe
3	RAW-03	T. Well-148 Banni Thana	T. well																				
4	RAW-04	Chitti Tanki, Saidpur Rd.	W.Supply	1001	6.90	0.3	340	0	60	46	330	48	3.3	50	46	4.90	556	0	0.58	0.50	0	0	Safe
5	RAW-05	T. Well-17 Dk. Kala Khan	T. well	1191	7.20	0.3	400	0	60	49	350	60	3.6	110	90	3.40	657	0.01	0.54	0.20	10	5	Unsafe
6	RAW-06	T. Well-42 Football Grd. Westridge	T. well	478	7.45	0.3	200	0	48	17	190	12	2.8	19	27	1.50	262	0.66	0.27	0.12	0	0	Safe
7	RAW-07	T. Well-25, Military Hospital	T. well	423	8.00	0.3	150	0	52	15	190	12	1.4	7	46	1.30	233	0.01	0.37	0.12	8	0	Unsafe
8	RAW-08	T. Well-7 Hockey Stadium, Saddar	T. well	1077	7.00	0.3	380	0	68	36	320	48	1.8	96	68	10.00	592	0.03	0.32	0.12	12	0	Unsafe
9	RAW-09	Rajgan Masjid, Dhamial Rd.	Bore	1131	6.90	0.3	360	0	108	29	390	60	1.7	88	62	10.00	622	0.02	0.42	0.21	0	0	Safe
10	RAW-10	T. Well-53, Afshan Colony, Qasim Rd.	T. well	847	7.00	0.3	300	0	92	17	300	36	1.8	57	36	10.00	466	0.23	0.39	0.21	0	0	Safe
11	RAW-11	T. Well-41 Dheri Hasanabad	T. well	1045	7.88	0.3	380	0	100	24	350	57	2.5	96	62	2.10	575	0.13	0.39	0.24	0	0	Safe
12	RAW-12	T. Well-67 Civil Line Club, Jhanda	T. well	784	7.45	0.3	290	0	72	39	340	48	2.6	43	40	6.10	431	0.01	0.44	0.12	0	0	Safe
13	RAW-13	T. Well Chaklala Railway Station	T. well	481	7.36	0.3	180	0	52	12	180	24	2.4	40	48	9.00	265	0.02	0.47	0.12	0	0	Safe
14	RAW-14	T. Well Swan Nullah Near High Court	T. well																				
15	RAW-15	Bahria Town, Phase-II, St.49	T. well	545	7.40	0.3	200	0	58	26	200	28	1.6	29	30	4.80	299	0.01	0.36	0.12	0	0	Safe

Total No. of Sample Collected	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
%age of Samples beyond permissilbe limit	-	0%	0%	-	-	-	-	0%	0%	-	-	-	15%	8%	15%	0%	0%	38%	23%	38%			
No. of Samples beyond permissilbe limit	-	0	0	-	-	-	-	0	0	-	-	-	2	1	2	0	0	5	3	5			
Maximum Concentration	1942	8	0.3	460	0	128	61	500	144	3.6	170	119	29	1068	0.93	0.58	0.5	18	6				
Average Concentration	921.46	7.31	0.30	306.15	0.00	77.38	30.00	306.92	47.15	2.38	66.23	55.31	7.93	507.31	0.18	0.43	0.19	4.31	1.15				
Minimum Concentration	423	6.90	0.3	150	0	48	12	180	12	1.4	7	27	1.3	233	0	0.27	0.12	0	0				

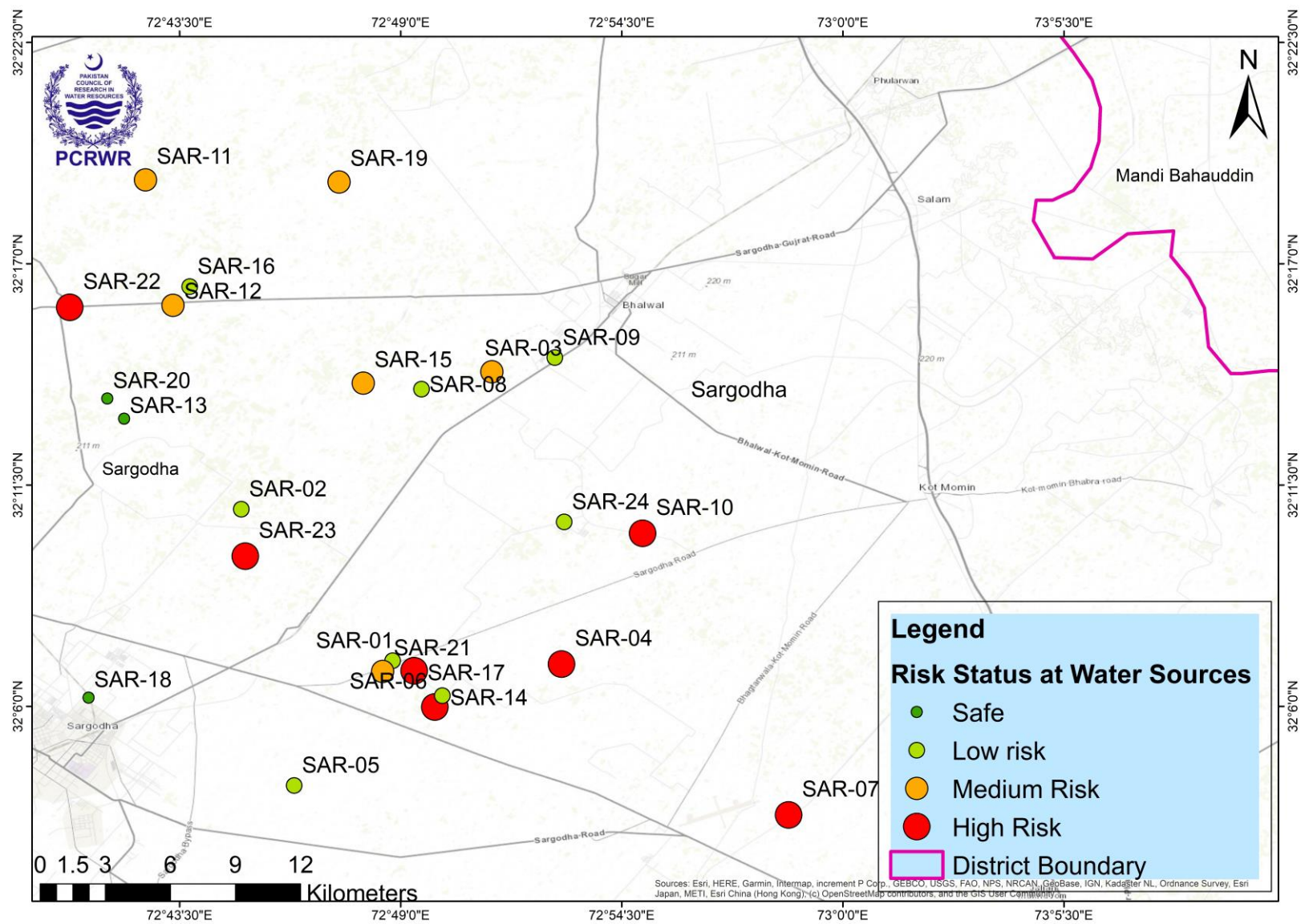


Water quality risk map of Rawalpindi

Annexure-10

Detailed Water Quality Results of Sargodha

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
		Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
		Maximum Permissible Limits NDWQS	-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	SAR-01	Haji Nazar Hussain Awan, 341, St.8, Farooq Colony	Bore	19830	7.30	0.90	565.00	0.00	820	401	3700	7630	3100	14	1050	1.30	13361	0.98	1.40	0.93	0.00	0	Unsafe
2	SAR-02	Nalka Chowk (N/F) Civil Line Tank, Canal Colony, Near Chief Engineer Office (Alt.)	W.Supply	310	7.60	1.10	140	0.00	28	16	135	11	15	1	21	0.60	178	0.15	0.30	3.38	10.00	0	Unsafe
3	SAR-03	Govt. Poly Technic Institute	Bore	3490	7.40	1.40	575	0.00	90	81.4	560	438	470	6	350	2.60	1791	0.22	0.40	2.60	30.00	0	Unsafe
4	SAR-04	Govt. Technical Training Institute Chak No.47 NB	Bore	4030	6.90	1.30	710	0.00	62	96	550	280	670	7	960	12.30	2556	0.08	1.00	1.40	0.00	0	Unsafe
5	SAR-05	Sargodha Sultan Textile Mill PAF Road	Bore	6960	7.30	0.90	650	0.00	280	328	2050	1260	700	8	1300	7.50	4300	0.13	1.00	0.13	0.00	0	Unsafe
6	SAR-06	Muhammad Sultan Boota, Chak No.50 NB	H.P	10010	6.90	1.10	780	0.00	600	122	2000	1995	1540	44	2010	12.70	6835	1.28	0.80	1.20	0.00	0	Unsafe
7	SAR-07	Gulzar Madina Darulabom Gosia Mosque	Bore	4800	7.30	1.00	815	0.00	400	12	950	420	630	35	840	12.80	2859	0.08	0.80	0.48	0.00	0	Unsafe
8	SAR-08	Khawar Nadeem Warraich/Mr. Khalid, Kabary Bazar	Bore	2030	7.50	1.30	610	0.00	150	60	620	133	210	7	200	8.90	1665	0.26	0.40	0.85	0.00	0	Unsafe
9	SAR-09	New Sabzi Mandi, Muhallah Abdul Aziz	W.Supply	310	7.60	1.40	140	0.00	32	15	140	9	15	1	23	0.80	182	0.63	0.03	1.70	60.00	0	Unsafe
10	SAR-10	Muhamad Munir/Muhammad Anwar, Sultan Town	H.Pump	8970	7.50	1.20	1120	0.00	240	340	2000	2100	1400	67	1260	12.90	6137	0.27	1.10	1.70	80.00	0	Unsafe
11	SAR-11	Small Imam Bargah, Fatima Jinnah Colony	H.Pump	1580	7.40	1.10	505	0.00	36	30	215	46	280	9	170	11.90	927	0.25	0.90	1.60	0.00	0	Unsafe
12	SAR-12	Park, Major Aziz Bhatti Town	WSS	340	7.50	1.40	130	0.00	24	7	90	7	39	1	29	1.00	189	0.06	0.03	15.80	0.00	0	Safe
13	SAR-13	IBEX Mart, Khushab Road, Sargodha.	H.Pump	990	7.60	1.20	370	0.00	66	45	350	49	72	3	60	8.80	556	0.00	0.03	0.20	0.00	0	Safe
14	SAR-14	Govt. Central Model School (High) New S. Town	Bore	22140	7.10	1.00	420	0.00	440	656	3800	7035	3560	19	1830	2.60	13804	0.07	1.20	1.54	0.00	0	Unsafe
15	SAR-15	Madrsa Zia-ul-Quran Factory Area	Bore	13480	7.20	0.70	460	0.00	340	498	2900	4445	2020	16	930	6.10	8553	0.08	0.50	2.03	24.00	0	Unsafe
16	SAR-16	Abdul Majeed, Chak No.78 NB	H.Pump	2250	7.50	1.40	690	0.00	52	28	245	68	370	63	320	8.00	1352	0.82	1.00	1.20	0.00	0	Unsafe
17	SAR-17	Haji Zafar Chak No.86 NB	Bore	1570	7.60	1.10	430	0.00	36	15	150	33	290	2	310	5.50	970	0.08	1.50	12.78	0.00	0	Unsafe
18	SAR-18	Mosque Chak No.87, Nikki	H.Pump	1290	7.70	1.20	460	0.00	44	68	390	54	101	33	140	4.20	736	0.30	0.80	1.20	0.00	0	Safe
19	SAR-19	Mazhar Hussain, St.1, Amir Colony, Istaqlalabad	H.Pump	1180	7.50	1.00	385	0.00	102	62	510	68	58	6	90	11.90	670	3.66	0.03	0.49	0.00	0	Unsafe
20	SAR-20	M. Akram Gujjar, 89 NB	Bore	520	7.40	1.30	250	0.00	52	32	260	7	10	2	30	1.60	290	0.26	0.30	1.70	0.00	0	Safe
21	SAR-21	M. Mumtaz/M. Safdar, Chak No.9, Jhang Road	H.Pump	790	7.20	1.00	310	0.00	98	28	360	21	18	11	60	11.20	472	0.30	0.10	0.12	0.00	0	Unsafe
22	SAR-22	Haji Barkat Raza Town Silanwali Road	H.Pump	2540	7.30	1.30	580	0.00	116	62	545	151	310	20	480	12.80	1543	0.08	0.20	3.60	0.00	0	Unsafe
23	SAR-23	Haji Saifullah, Chak No.43 NB	H.Pump	3080	7.40	1.10	620	0.00	158	67	670	213	375	10	610	10.20	1851	0.10	0.20	0.21	0.00	0	Unsafe
24	SAR-24	D-Block, S. Town, Jamia Ulmia Madarsatulbanat	Bore	2770	7.30	1.40	440	0.00	142	80	685	186	315	18	680	4.90	1707	0.22	0.40	2.24	0.00	0	Unsafe
		Total No. of Sample Collected		24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24	24
		%age of Samples Exceeding Permissible Limits	-		0%						58%	38%				38%	58%	21%	0%	0%	21%	0%	83%
		No. of Samples Exceeding Permissible Limits			0	0					14	9				9	14	5	0	0	5	0	20
		Maximum Concentration	-	310	6.90	0.70	130	0.00	24	7	90	7	10	1	21	0.60	178	0.00	0.03	0.12	0	0	
		Minimum Concentration	-	22140	7.70	1.40	1120	0.00	820	656	3800	7630	3560	67	2010	12.90	13804	3.66	1.50	15.80	80	0	

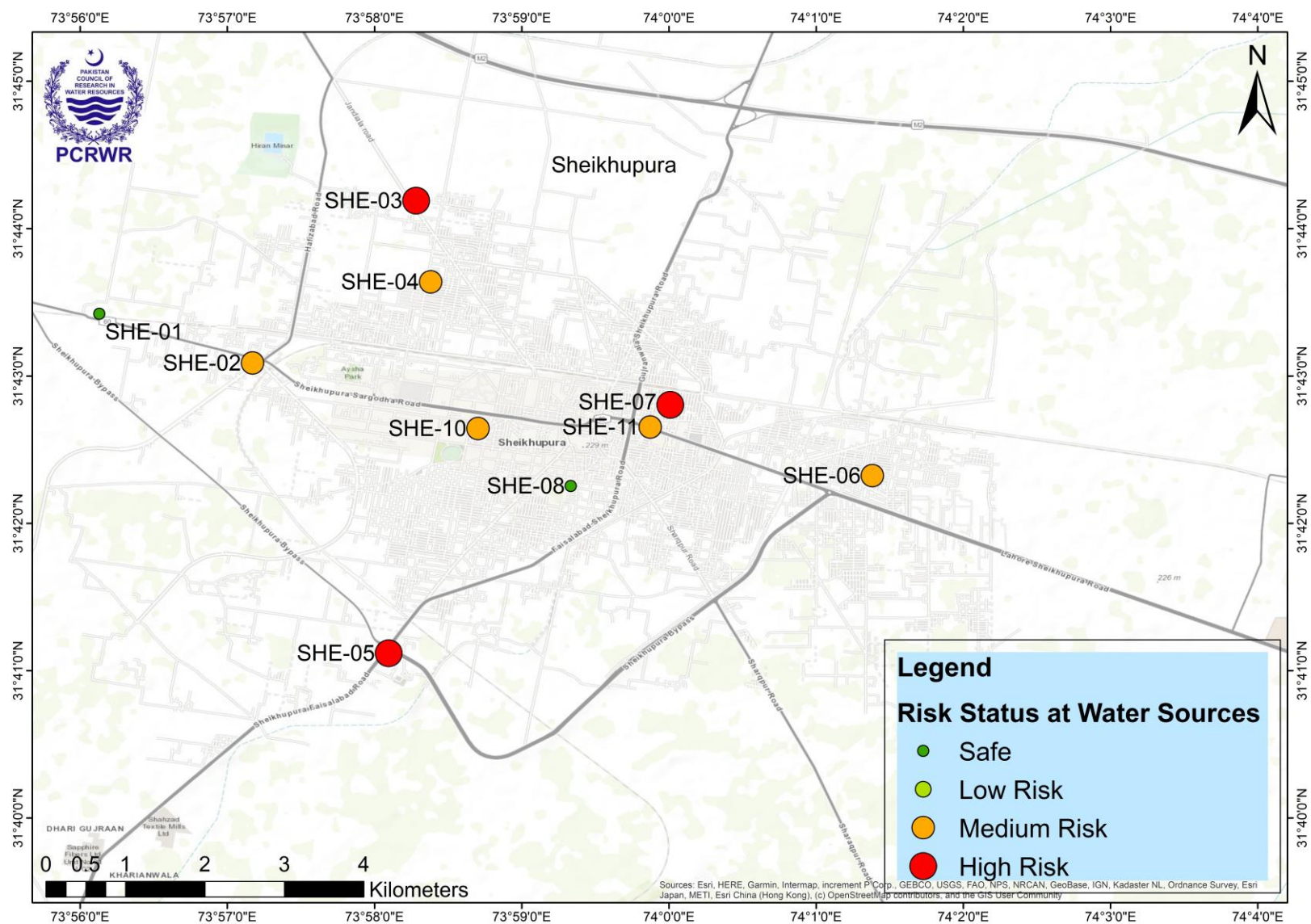


Water quality risk map of Sargodha

Detailed Water Quality Results of Sheikhpura

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	SHE-01	Ghazi Manara Near Railway Crossing	H.Pump	1230	7.00	18.76	450	0	68	46	360	48	146	0.80	200	0.23	779	2.02	0.03	7.14	0	0	Unsafe
2	SHE-02	Nati Pura near Yamin Madran Tubewell-12	T.Well	947	7.00	1.85	360	0	48	22	210	38	136	0.60	118	0.11	578	0.11	0.03	33.56	0	0	Safe
3	SHE-03	Bhadroo Minara Chowk Jhandik	H.Pump	1368	6.90	1.92	410	0	84	68	490	58	110	0.80	180	26.32	863	0.09	0.06	55.10	0	0	Unsafe
4	SHE-04	T.W-14, Qadirabad Near Shah Colony	T.Well	628	7.20	1.64	240	0	28	19	150	24	86	0.40	66	0.11	368	0.02	0.03	18.81	8	3	Unsafe
5	SHE-05	Sheikhpura Filling Station F.abad Road	Bore	2240	7.10	4.64	560	0	90	84	570	178	282	2.80	410	13.05	1440	0.02	0.03	5.12	0	0	Unsafe
6	SHE-06	TW-18, Housing & Physical Colony	T.Well	5521	7.30	2.17	260	0	36	19	170	19	70	0.20	32	0.11	333	0.02	0.03	33.65	0	0	Safe
7	SHE-07	Kohistan Rehman Hotel near G.M Rice Mill	Bore	1704	6.90	1.49	585	0	88	73	520	101	200	1.00	230	21.08	1137	0.02	0.03	12.98	0	0	Unsafe
8	SHE-08	M.C Primary School, Shabistan Cinema	Tap	984	7.00	1.92	375	0	64	27	270	58	124	0.40	98	0.60	598	0.02	0.03	0.97	0	0	Safe
9	SHE-09	T.W-15, Bhikhi Road	T.Well	TW Non-Operational Since 2018																			
10	SHE-10	Distt. Headquarter Hospital	T.Well	1180	7.10	1.93	450	0	64	39	320	58	150	0.60	155	0.09	736	0.02	0.03	35.58	0	0	Safe
11	SHE-11	Live Stock Training Services Centre	Bore	821	7.10	1.91	320	0	52	34	270	48	94	0.20	92	0.72	515	0.02	0.03	29.59	77	30	Unsafe

Total No. of Sample Collected	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
%age of Samples Exceeding Permissible Limits		0%	10%					20%	0%				30%	20%	10%	0%	10%	20%	20%	60%
No. of Samples Exceeding Permissible Limits		0	1					2	0				3	2	1	0	1	2	2	6
Maximum Concentration	628	6.90	1.49	240	0	28	19.4	150	19.2	70	0.2	32	0.093	333	0.02	0.03	0.97	0	0	
Minimum Concentration	5521	7.30	18.76	585	0	90	83.8	570	178	282	2.8	410	26.32	1440	2.02	0.06	55.1	77	30	



Water quality risk map of Sheikhpura

Annexure-12

Detailed Water Quality Results of Sialkot

Sr. No.	Sample Code	Site ID	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk.	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
			Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
			Maximum Permissible Limits PSQCA/NDWQS	NGVS		6.50-8.50	5.0	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10	1000	0.3	1.5	50	0/100 ml	0/100 ml	
1	SIA-01	1	MES Services Club Nr F.G School Ghazi road	T.Well	516	7.63	0.8	300	0	48	29	240	12	43	3.5	7	1	310	0.03	0.12	6.74	0	0	Safe
2	SIA-02	2	Jco family accomodation sarwar line MES-No.19	T.Well	500	7.51	0.7	290	0	48	29	240	14	39	3.4	6	1	300	0.17	0.08	1.69	0	0	Safe
3	SIA-03	3	Kotli Behram Wazirabad road Sian Mana Darbar TMA	T.Well	683	7.43	1.5	355	0	64	36	310	12	53	4.3	17	2.3	410	0.17	0.15	6.24	0	0	Safe
4	SIA-04	4	Nala Behr Paris Road(Poly technical college)TMA	T.Well	705	7.40	2.1	370	0	68	39	330	28	59	4.3	12	2.2	423	0.13	0.25	5.28	0	0	Safe
5	SIA-05	5	Allama Iqbal Memorial Hospital T.M.A	T.Well	791	7.75	1.5	435	0	72	45	365	35	68	4.9	14	2.3	475	0.05	0.2	5.02	0	0	Safe
6	SIA-06	7	Masjid Kabturan,Nr Ahmedia School	T.Well	525	7.61	1.8	290	0	52	29	245	14	47	3.9	9	0.8	315	0.30	0.13	4.84	0	0	Safe
7	SIA-07 New	9	TMA No-70 behind Govt Boys Primary School Fateh Garh	T.Well																				
8	SIA-08	6	Tubewell No-70 roras road Nr lunda railway crossing graveyard	T.Well	533	7.52	1.2	280	0	52	33	265	16	48	5.1	12	0.8	320	0.04	0.09	0.7	0	0	Safe
9	SIA-09	8	Jamia Masjid Shah Khasi Wali (Rangpura)	Tap	566	7.32	2.0	300	0	56	27	250	18	55	3.7	13	1.3	340	0.10	0.3	5.5	0	0	Safe
10	SIA-10	10	Gulshane -Iqbal PARK	T.Well	525	7.7	1.3	310	0	44	24	210	12	56	2.9	10	0.6	315	0.04	0.09	2.28	0	0	Safe

Total No. of Sample Collected	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
%age of Samples beyond permissilbe limit		0%	0%						0%	0%						0%	0%	0%	0%	0%	0%	0%	0%	0%
No. of Samples beyond permissilbe limit		0	0						0	0						0	0	0	0	0	0	0	0	0
Maximum Concentration	500	7.32	1	280	0	44	24	210	12	39	3	6	1	300	0	0	1	0	0					
Minimum Concentration	791	7.75	2	435	0	72	45	365	35	68	5	17	2	475	0	0	7	0	0					

Annexure-13

Detailed Water Quality Results of Abbottabad

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	ABT-01	Town Labriary Nawansher (Alt.)	Tap	286	7.60	2.90	100	0	84	15	150	23	10	2.60	21	2.90	199	1.42	0.37	0.00	0	0	Unsafe
2	ABT-02	Kunj Ground Near G.D.College Abbottabad	Tap	420	8.00	3.10	180	0	88	7.29	250	20	3	1.70	26	1.80	262	0.02	0.07	0.02	0	0	Safe
3	ABT-03	Takia Camp Near Scout Ground	Tap	330	7.90	3.80	130	0	44	19.4	170	35	5	4.10	22	2.20	212	0.01	0.01	0.01	0	0	Safe
4	ABT-04	Sheen Technical Vocational Centre	Tap	350	8.00	4.10	140	0	62	11	200	28	12	4.30	22	3.60	239	1.79	0.02	0.00	0	0	Unsafe
5	ABT-05	CMH Abbottabad	T.Well	426	8.00	2.30	160	0	80	7.29	230	32	16	2.30	38	1.90	280	0.77	0.79	0.02	0	0	Unsafe
6	ABT-06	Near Ilyasi Masjid Chonakari (Alt.)	Tap	277	8.2	3.80	120	0	60	2.4	160	18	6	2.70	11	1.20	177	0.01	0.20	0.05	0	0	Safe
7	ABT-07	Banda Dilazak, Lamideri (Alt.)	T.Well	412	8.00	3.10	160	0	40	24	200	45	19	2.40	19	1.50	252	0.86	0.36	0.02	0	0	Unsafe
8	ABT-08	Jahngi Qazian Sadian (Alt.)	T.Well	439	7.90	3.50	150	0	36	28	200	45	25	2.40	16	4.30	260	0.03	0.12	0.01	0	0	Safe
9	ABT-09	TW#13 GE(A), Habib Ullah Colony (Alt.)	T.Well	350	8.10	3.20	100	0	84	15	150	23	10	2.60	21	2.90	199	0.49	0.39	0.00	0	0	Unsafe
10	ABT-10	Ayub Medical College (T.W# 6)	T.Well	450	8.00	3.90	180	0	88	7.29	250	20	3	1.70	26	1.80	262	0.8	0.41	0.02	0	0	Unsafe
11	ABT-11	Water Supply Scheme Mirpur	W.Supply	420	8.10	4.10	130	0	44	19.4	170	35	5	4.10	22	2.20	212	0.01	0.47	0.00	0	0	Safe

Total No. of samples	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
%age of Samples Exceeding Permissible Limits			0%							0%	0%				0%	0%	55%	0%	0%	0%	0%	0%	55%
No. of Samples Exceeding Permissible Limits		0	0							0	0				0	0	6	0	0	0	0	0	6
Maximum Concentration	450	8.20	4.10	180	0	88	28	250	45	25	4.30	38	4.30	280	1.79	0.79	0.05	0	0				
Minimum Concentration	277	7.60	2.30	100	0	36	2.4	150	18	3	1.70	11	1.20	177	0.01	0.01	0.00	0	0				

Detailed Water Quality Results of Mangora

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	Safe/Unsafe
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	
1	MAN-01	Shaghi, Saidu Sharif	T.Well	428	7.50	2.1	190	0	52	19.40	210	18	24	4.10	20	4.30	271	0.2	0.1	0.00	0	0	Safe
2	MAN-02	TW, DHDC Hospital, Saidu Sharif	T.Well	584	7.70	1.5	270	0	28	35.00	300	22	15	3.00	21	4.00	339	0.39	0	0.01	0	0	Unsafe
3	MAN-03	WAPDA Office, Saidu Sharif (N/F)	T.Well	855	7.50	1.9	350	0	86	42.00	390	51	30	5.00	32	3.30	471	0.04	0.12	1.43	0	0	Safe
4	MAN-04	Makand Bagh, Gulshan Chowk	T.Well	568	7.70	2.2	250	0	36	51.00	300	30	1	3.00	32	1.90	315	0.61	0.67	0.01	0	0	Unsafe
5	MAN-05	TW#1, Amankot, Chinar Colony	T.Well	607	7.40	2.3	300	0	84	30.00	360	30	5	1.00	37	3.10	391	0.03	0.4	0.58	0	0	Safe
6	MAN-06	Sharifabad Bridge, Haji Baba Road,	T.Well	592	7.90	1.1	290	0	80	36.00	350	45	17	2.00	40	2.50	406	0.08	0.32	0.05	0	0	Safe
7	MAN-07	Ward-2, Usmanabad, Nawa Killi	T.Well	683	7.90	2.8	240	0	82	35.00	350	63	26	3.60	42	3.30	411	0.05	0.38	0.79	0	0	Safe
8	MAN-08	TW#2, Nawa Killi, Takhtabad Road	T.Well	342	7.80	3.2	140	0	52	12.00	180	25	20	1.00	24	2.10	228	0.06	0.01	0.00	0	0	Safe
9	MAN-09	Sector-C, Kanju Township Chowk Mingora	T.Well	440	8.20	1.5	210	0	40	34.00	240	15	5	1.00	20	3.00	254	0.24	0.12	1.70	0	0	Safe
10	MAN-10	TW-1, Amankot, Zahidabad, near Bus Adda	T.Well	628	7.90	1.8	250	0	80	15.00	320	30	15	1.50	25	4.30	350	0.03	0.06	0.00	0	0	Safe

Total No. of samples	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
%age of Samples Exceeding Permissible Limits			0%					0%	0%				0%	0%	20%	0%	0%	0%	0%	0%	20%
No. of Samples Exceeding Permissible Limits		0	0					0	0				0	0	2	0	0	0	0	0	2
Maximum Concentration	855	8.20	3.20	350	0	86	51.00	390	63	30	5.00	42	4.30	471	0.61	0.67	1.70	0	0		
Minimum Concentration	342	7.40	1.10	140	0	28	12.00	180	15	1	1.00	20	1.90	228	0.03	0.00	0.00	0	0		

Detailed Water Quality Results of Mardan

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	Safe/Unsafe
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	
1	01-Mar	MDA TW Sheikh Maltoon Colony	T.Well	798	7.60	2.10	260	0	28	24	190	83	136	7.50	88	3.20	555	0.01	0.33	6.02	0	0	Safe
2	02-Mar	Govt. High School Lab. (N/F)	Tap	1270	7.51	1.80	400	0	16	58	280	95	176	11.10	128	2.90	740	0.50	0.44	1.71	0	0	Unsafe
3	03-Mar	Govt. College No.1	T.Well	694	7.71	1.50	240	0	32	41	250	63	82	7.30	74	2.00	479	0.01	0.34	1.46	0	0	Safe
4	04-Mar	PHED Office, Shamsi Road (Alt.)	T.Well	579	7.20	2.50	220	0	28	17	140	53	88	5.30	43	1.20	373	0.29	0.51	10.79	0	0	Safe
5	05-Mar	Dang Baba, TW#12	T.Well	763	7.60	1.10	300	0	24	41	230	45	108	6.60	66	3.40	486	0.02	0.3	4.54	0	0	Safe
6	06-Mar	Fire Brigade Office, Shamsi Road (Alt.)	Tap	546	7.71	1.90	300	0	20	32	180	33	76	6.50	32	1.60	362	1.22	0.24	5.22	0	0	Unsafe
7	07-Mar	Hujra Haji Zareen Khan (N/F)	T.Well																				
8	08-Mar	Janabad Nisha road Mardan (Alt.)	Tap	980	7.60	2.20	240	0	20	39	220	120	135	9.00	86	3.10	565	1.05	0.67	0.64	0	0	Unsafe
9	09-Mar	Nawi Road Kaskoruna, TW#18	T.Well	849	7.70	3.50	270	0	16	49	250	53	97	11.00	65	4.60	484	0.85	0.32	1.67	0	0	Unsafe
10	10-Mar	Baghdada, TW-19	T.Well	685	7.40	1.60	250	0	20	49	250	38	70	5.60	54	2.60	407	0.02	0.08	0.78	0	0	Safe
11	11-Mar	Pirano park Mardan T.W#15 Mardan (Alt.)	T.Well	672	7.35	1.70	200	0	20	49	250	53	62	3.60	69	3.20	390	0.79	0.09	1.68	0	0	Unsafe
12	12-Mar	U.E.T University Mardan (Alt.)	T.Well	865	7.50	2.10	270	0	12	49	270	43	109	5.20	80	4.00	486	0.02	0.07	0.85	0	0	Safe
Total No. of samples				11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
%age of Samples Exceeding Permissible Limits						0%					0%	0%				0%	0%	45%	0%	0%	0%	0%	45%
No. of Samples Exceeding Permissible Limits					0	0					0	0				0	0	5	0	0	0	0	5
Maximum Concentration				1270	7.71	3.50	400	0	32	58.00	280	120	176	11.10	128	4.60	740	1.22	0.67	10.79	0	0	
Minimum Concentration				546	7.20	1.10	200	0	12	17.00	140	33	62	3.60	32	1.20	362	0.01	0.07	0.64	0	0	

Annexure-16

Detailed Water Quality Results of Peshawer

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	PES-01	Children Hospital Near Haji Camp	Tap	1111	7.70	1.5	260	0	52	68	410	100	97	8.5	156	4.1	656	1.11	0.00	2.20	0	0	Unsafe
2	PES-02	Khatak Chowk	Tap	840	7.90	1.9	280	0	36	56	320	43	59	8.4	85	5.3	486	2.07	0.12	0.23	0	0	Unsafe
3	PES-03	Soraizai Payan Inqilab Road																					
4	PES-04	(site is reactivated on Solar System)	Tap	860	7.50	2.5	350	0	88	39	380	38	39	7.90	55	3.50	500	0.54	0.01	0.00	0	0	Unsafe
5	PES-05	Ahmad Khail	Tap	740	7.70	3.1	250	0	52	44	310	48	42	5.80	52	4.60	414	0.09	0.02	0.72	0	0	Safe
6	PES-06	Dabgary Garden Colony	T.Well	1006	7.20	1.1	380	0	76	58	460	60	68	3.80	84	4.20	597	0.02	0.01	0.11	0	0	Safe
7	PES-07	Afghan Colony, Madina Masjid, Latifabad UC-3, (TW No.15)	T.Well	1007	7.40	2.3	370	0	76	58	470	72	70	5.00	76	3.30	595	0.75	0.01	1.00	0	0	Unsafe
8	PES-08	Irrigation Colony, Warsak Road	Tap	810	7.20	2.5	320	0	72	44	360	33	35	5.50	44	6.50	454	0.01	0.03	0.01	0	0	Safe
9	PES-09	Nothia Road, Gulberg No.1	T.Well	500	7.80	1.7	180	0	40	32	230	25	23	4.40	30	5.10	292	0.07	0.02	0.75	0	0	Safe
10	PES-10	University Town, Gul Mehr Road	T.Well	620	7.40	1.8	260	0	36	49	290	33	32	4.50	42	6.00	379	1.40	0.03	0.39	0	0	Unsafe
11	PES-11	Agriculture University Campus	T.Well	470	7.60	2.1	210	0	44	24	210	20	29	3.70	30	3.70	287	0.16	0.07	1.40	0	0	Safe
12	PES-12	(T.W#2)	Tap	890	7.70	2.2	360	0	68	56	400	50	41	4.40	42	3.20	492	0.59	0.02	0.15	0	0	Unsafe
13	PES-13	Hamid Khan Hujra (N/F)	T.Well	650	7.90	3.5	250	0	52	41	300	40	38	3.60	49	4.10	392	0.17	0.01	0.82	0	0	Safe
Total No. of samples				12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12
%age of Samples Exceeding Permissible Limits				0%	0%					0%	0%					0%	0%	50%	0%	0%	0%	0%	50%
No. of Samples Exceeding Permissible Limits				0	0					0	0					0	0	6	0	0	0	0	6
Maximum Concentration				1111	7.90	3.50	380	0	88	68.00	470	100	97	8.50	156	6.50	656	2.07	0.12	2.20	0	0	
Minimum Concentration				470	7.20	1.10	180	0	36	24.00	210	20	23	3.60	30	3.20	287	0.01	0.00	0.00	0	0	

Detailed Water Quality Results of Khuzdar

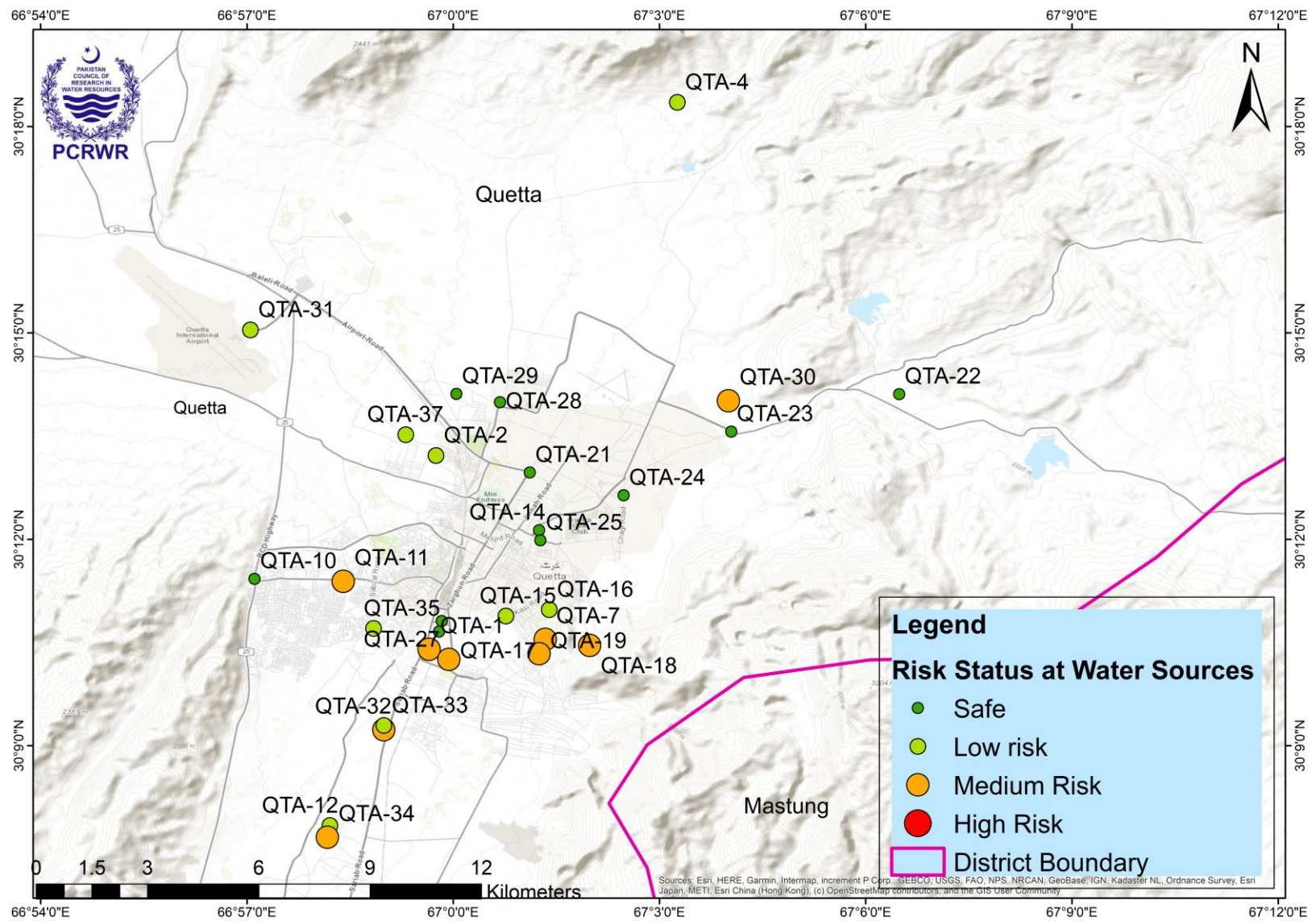
Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	KHU-01	Dy. Commissioner Office	T.Well	789	7.70	1.80	170	0	60	17	220	90	96	1.00	128	0.20	495	0.04	0.36	0.00	0	0	Safe
2	KHU-02	Irrigation Colony	T.Well	968	7.83	11.60	190	0	44	48.6	310	140	98	1.00	118	7.30	596	0.05	0.41	0.02	0	0	Unsafe
3	KHU-03	Park Near Public School	T.Well	1209	7.71	1.70	190	0	46	49.8	320	180	156	1.00	160	14.00	769	0.04	0.26	0.01	0	0	Unsafe
4	KHU-04	Civil Hospital Faqirabad	T.Well	1360	7.40	0.90	230	0	66	57.1	400	190	150	3.00	220	2.10	833	0.01	0.29	0.24	17	6	Unsafe
5	KHU-05	Madrisa Jama-e-Rizvia Khalilia Khund (New source)	T.Well	1070	7.83	3.51	185	0	52	36.6	280	155	135	1.00	165	0.80	659	0.1	0.34	0.01	0	0	Safe
6	KHU-06	Filtration Plant Khuzdar Cantt.	W.Supply	802	7.96	0.70	180	0	76	2.4	200	120	104	2.00	83	0.10	496	0.02	0.18	0.39	9	0	Unsafe
7	KHU-07	Madrisa Tajweed-ul-Quran Katan	Well	1111	7.21	2.80	190	0	73	45.5	370	180	112	2.00	164	0.60	693	0.05	0.16	0.12	0	0	Safe
8	KHU-08	Pir Umar Mazar	T.Well	874	7.99	0.40	130	0	84	7.2	240	140	107	3.00	144	1.00	568	0.08	0.24	0.00	13	4	Unsafe
9	KHU-09	Super Shashan Hotel Karachi Road (New Site)	Tap	1018	7.80	1.10	170	0	52	36.4	280	130	110	2.00	160	0.80	596	0.03	0.18	0.02	18	0	Unsafe
10	KHU-10	Engineering University Khuzdar (New Site)	T.Well	918	8.00	0.16	170	0	48	34	260	72	88	2.00	180	1.60	533	0.06	0.24	0.01	0	0	Safe
11	KHU-11	Rahim Abad Near BRC College (New Site)	W.Supply	674	7.90	0.70	130	0	80	2.4	210	90	68	3.00	102	0.30	425	0.01	0.29	0.01	0	0	Safe
Total No. of samples				11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
%age of Samples Exceeding Permissible Limits					0%	9%					0%	0%				9%	0%	0%	0%	0%	36%	18%	55%
No. of Samples Exceeding Permissible Limits					0	1					0	0				1	0	0	0	0	4	2	6
Maximum Concentration				1360	8.00	11.60	230	0	84	57.1	400	190	156	3.00	220	14.00	833	0.1	0.41	0.39	18	6	
Minimum Concentration				674	7.21	0.16	130	0	44	2.4	200	72	68	1.00	83	0.10	425	0.01	0.16	0.00	0	0	

Detailed Water Quality Results of Loralai

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	LOR-1	Army Water Purification Plant	F.Plant	416	7.10	0.12	90	0	38	9	130	65	43	1	41	0.8	254	0.01	0.1	0.01	0	0	Safe
2	LOR-2	Commissioner Office	T.Well	765	8.10	0.6	130	0	72	2	190	120	101	2	109	1.1	489	0.01	0.2	1.40	0	0	Safe
3	LOR-3	Rodlin Village Quetta Road - House (Alt.)	W.Supply	676	7.90	0.16	190	0	64	22	250	70	53	2.00	70	3.00	408	0.03	0.2	0.01	8	2	Unsafe
4	LOR-4	Tank Quetta Road	W.Supply	600	8.00	0.21	60	0	47	5	140	55	70	1.00	141	3.10	369	0.02	0.6	1.10	11	0	Unsafe
5	LOR-5	PHED Pathan Kot	T.Well	638	7.80	0.27	145	0	71	13	230	95	52	2.00	52	3.50	387	0.03	0.1	0.02	0	0	Safe
6	LOR-6	Pathan Kot	Spring	676	7.90	0.16	165	0	64	22	250	85	53	2.00	70	3.00	408	0.05	0.4	0.01	0	0	Safe
7	LOR-7	OHSR Baloch Colony	W.Supply	910	7.90	0.27	170	0	42	43	280	160	100	1.00	98	1.10	550	0.04	0.7	0.01	12	2	Unsafe
8	LOR-8	Zangiwal Near Oregi River	T.Well	924	7.60	0.16	170	0	62	50	360	140	66	1.00	128	1.10	554	0.01	0.1	0.00	14	0	Unsafe
9	LOR-9	Torkhezai Delay Action Dam	Dam	512	7.80	41.8	80	0	42	11	150	75	59	1.00	88	0.70	327	0.90	1.2	0.02	21	0	Unsafe
10	LOR-10	T & T Centre Near Awami Centre	W.Supply	674	7.90	0.12	160	0	38	40	260	90	51	1.00	81	0.80	401	0.02	0.2	0.50	0	0	Safe
11	LOR-11	Zangiwal Village	T.Well	982	7.40	1.65	230	0	125	7	340	130	77	2.00	94	0.80	576	0.01	0.2	1.00	2	6	Unsafe
12	LOR-12	Torkhezai Road Zangiwal (New Site)	T.Well	991	7.80	1.25	210	0	52	24	230	145	136	1.00	110	0.70	597	0.02	0.3	0.01	0	0	Safe
13	LOR-13	Main Bazar Near Police Line (New Site)	W.Supply	1620	7.80	1.74	330	0	44	68	390	220	217	4.00	233	1.00	988	0.04	0.1	1.00	8	4	Unsafe
14	LOR-14	Govt School Zangiwal Village	H.Pump	869	7.90	2.1	165	0	76	18	265	135	103	1.00	124	1.70	564	0.02	0.2	0.01	8	0	Unsafe
15	LOR-15	UNICEF Water Purification Plant	T.Well	454	7.60	0.45	80	0	44	10	150	55	41	1.00	80	0.30	280	0.01	0.74	0.00	0	0	Safe
16	LOR-16	Civil Hospital	Tap	586	7.90	2.7	130	0	46	23	210	75	51	1.00	81	1.10	360	0.02	0.3	0.02	12	0	Unsafe
17	LOR-17	Baloch Colony	Tap	1030	8.10	3.4	210	0	80	19	280	165	120	1.00	110	1.20	627	0.01	0.1	0.01	6	0	Unsafe
Total No. of samples				17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17	17
%age of Samples Exceeding Permissible Limits					0%	6%					0%	0%				0%	0%	6%	0%	0%	59%	24%	59%
No. of Samples Exceeding Permissible Limits					0	1					0	0				0	0	1	0	0	10	4	10
Maximum Concentration				1620	8.11	41.80	330	0	125	68	390	220	217	4.00	233	3.50	988	0.90	1.20	1.40	21	6	
Minimum Concentration				416	7.10	0.12	60	0	38	2	130	55	41	1.00	41	0.30	254	0.01	0.10	0.00	0	0	

Detailed Water Quality Results of Quetta

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	QTA-1	Agri. Department. Rani Bagh	T.Well	1180	8.10	2.1	160	0	40	44	280	140	144	2.00	230	2.10	705	0.01	3.62	0.23	13	0	Unsafe
2	QTA-2	WRRC Quetta Tajik Abad	T.Well	616	7.20	1.8	130	0	44	27	220	75	54	1.00	99	0.33	379	0.00	0.90	0.01	17	15	Unsafe
3	QTA-3	Railway Housing Society	T.Well	561	7.80	0.5	110	0	72	2	190	60	45	1.00	95	0.20	342	0.01	0.62	0.21	0	0	Safe
4	QTA-4	Nasir Abad Maidani, Mariabad	T.Well	479	7.60	1.6	75	0	49	4	140	65	51	1.00	85	0.20	301	0.06	0.42	0.48	14	0	Unsafe
5	QTA-5	Pumping Station Ayub Stadium	W.Supply	596	7.70	2.4	130	0	52	22	220	70	49	1.00	96	0.20	369	0.03	0.19	0.49	19	0	Unsafe
6	QTA-6	WASA Chaman Phatak	T.Well	705	7.90	2.1	140	0	44	22	200	100	80	1.00	92	1.10	428	0.04	0.92	0.02	18	0	Unsafe
7	QTA-7	Haji Ghaibi Road	T.Well	880	7.80	1.6	190	0	48	19	200	50	123	3.00	202	1.10	564	0.02	2.18	0.01	0	0	Unsafe
8	QTA-8	Kharot Abad-1, Samugly Road	T.Well	595	7.80	1.8	120	0	78	1	200	55	45	1.00	94	1.60	354	0.04	0.19	0.01	13	0	Unsafe
9	QTA-9	Killi Khaizy, Samugly Road	T.Well	774	7.70	1.7	160	0	60	17	220	90	86	1.00	118	0.80	472	0.05	0.74	0.05	0	0	Safe
10	QTA-10	Met Office Brewery	T.Well	528	7.80	2.2	120	0	62	13	210	80	36	1.00	54	0.30	320	0.01	0.62	0.04	0	0	Safe
11	QTA-11	Bolan Medical Complex	W.Supply	657	8.10	2.1	140	0	52	15	190	55	76	4.00	134	1.40	426	0.06	2.20	0.09	21	0	Unsafe
12	QTA-12	Gurgai Karez Near Kechi Beg	Bore	1610	7.80	3.7	230	0	79	54	420	225	174	2.00	242	0.90	918	0.01	0.72	0.19	18	0	Unsafe
13	QTA-13	Mosque Near Helper Hospital	W.Supply	692	7.80	1.7	150	0	76	5	210	90	75	1.00	95	0.50	434	0.04	0.98	0.23	0	0	Safe
14	QTA-14	Lourds Hotel	T.Well	490	7.70	0.8	85	0	55	6	160	70	46	1.00	78	0.00	306	0.02	0.42	0.03	0	0	Safe
15	QTA-15	Gawalmandi Chowk	T.Well	534	7.30	0.1	100	0	38	21	180	45	44	2.00	110	0.30	321	0.04	0.41	0.05	16	0	Unsafe
16	QTA-16	Khanwari Road Near Children Hospital	T.Well	685	7.90	1.2	140	0	42	23	200	100	79	1.00	90	1.10	424	0.04	0.62	0.01	18	5	Unsafe
17	QTA-17	Block-5, Satellite Town	T.Well	463	7.60	1.4	105	0	48	10	160	60	43	2.00	64	0.00	290	0.01	4.65	0.19	21	0	Unsafe
18	QTA-18	Taro Chowk Reservoir Pashtoon Abad	W.Supply	316	7.10	2.1	65	0	36	5	110	35	27	1.00	55	0.00	198	0.01	2.49	0.80	13	0	Unsafe
19	QTA-19	Rehmat Colony Gali No. 1, Sirki Road	W.Supply	547	7.90	2.2	175	0	32	10	120	55	84	2.00	54	1.40	348	0.06	3.65	0.12	18	0	Unsafe
20	QTA-20	Civil Hospital Near Abdul Malik Ward	T.Well	704	7.90	1.3	140	0	52	22	220	60	74	3.00	148	1.80	451	0.03	2.79	0.02	24	13	Unsafe
21	QTA-21	Army Purification Plant Ordinance Chowk	W.Supply	1216	7.80	0.8	190	0	68	51	380	170	121	2.00	200	1.80	735	0.02	1.25	0.04	0	0	Safe
22	QTA-22	Staff College Purification Plant	T.Well	1137	7.70	0.6	215	0	65	65	430	155	81	1.00	168	0.00	664	0.03	0.85	0.03	0	0	Safe
23	QTA-23	Staff College Urak Road	T.Well	797	8.10	0.7	160	0	48	24	220	85	98	1.00	144	2.20	506	0.01	0.67	0.02	0	0	Safe
24	QTA-24	Chiltan Booster Near Cavalry Zamzama Road	W.Supply	594	8.00	1.1	95	0	36	17	160	75	76	1.00	115	1.90	385	0.01	1.31	0.01	0	0	Safe
25	QTA-25	Govt. Girls College Cantt	T.Well	1220	8.10	1.8	220	0	120	19	380	160	110	2.00	169	1.20	718	0.04	1.16	0.03	0	0	Safe
26	QTA-26	Forest Park, Sabzal Raod	T.Well	656	7.70	2.6	110	0	52	15	190	80	74	4.00	127	0.90	422	0.04	0.41	0.17	16	6	Unsafe
27	QTA-27	Mosque Hajji Ghaffar , Killi Ahmed Zai	T.Well	856	7.80	1.1	190	0	48	29	240	75	102	1.00	160	1.80	537	0.01	1.24	0.21	0	0	Safe
28	QTA-28	Jamia Masjid Nawan Killi	T.Well	660	7.80	1.2	155	0	36	40	255	90	48	1.00	75	0.80	387	0.03	0.65	0.29	0	0	Safe
29	QTA-29	Killi Kotwal	T.Well	489	7.90	1.8	120	0	40	15	160	45	48	3.00	80	1.20	308	0.01	0.45	0.69	0	0	Safe
30	QTA-30	Killi Gull Muhammad	T.Well	2935	8.40	1.9	350	0	116	85	640	390	411	6.00	668	2.10	1901	0.03	0.80	0.42	14	0	Unsafe
31	QTA-31	Air Port	T.Well	939	7.80	0.8	190	0	48	22	210	135	141	1.00	130	0.70	594	0.03	1.24	0.20	19	0	Unsafe
32	QTA-32	Irrigation Colony	T.Well	1022	6.80	3.3	190	0	52	32	260	95	115	2.00	180	1.10	594	0.01	6.21	0.05	0	0	Unsafe
33	QTA-33	D.G WAPDA Office	W.Supply	260	7.10	1.6	80	0	40	1	105	20	16	1.00	32	0.00	158	0.03	0.34	0.51	19	0	Unsafe
34	QTA-34	Killi Habib	T.Well	571	7.90	1.2	130	0	52	7	160	75	68	1.00	72	0.30	355	0.02	4.51	0.21	0	0	Unsafe
35	QTA-35	Muhammad Masjid Podgali Chowk	T.Well	719	7.20	1.9	110	0	36	34	230	95	74	1.00	138	1.40	450	0.01	1.24	0.24	118	0	Unsafe
36	QTA-36	Pepsi Cola Factory Industrial Area	T.Well	945	7.30	1.7	160	0	72	34	320	110	93	1.00	195	1.80	609	0.02	3.92	0.03	20	0	Unsafe
37	QTA-37	Killi Paind Khan	T.Well	728	7.90	2.6	140	0	52	22	220	60	74	1.00	148	1.30	447	0.01	1.20	0.01	24	0	Unsafe
Total No. of samples				37	37	37	37	37	37	37	37	37	37	37	37	37	37	37	37	37	37	37	37
%age of Samples Exceeding Permissible Limits						0%					3%	3%				0%	3%	0%	27%	0%	57%	11%	65%
No. of Samples Exceeding Permissible Limits					0	0				1	1	30				0	1	0	10	0	21	4	24
Maximum Concentration				2935	8.40	3.7	350	0	120	85	640	390	411	6.00	668	2.20	1901	0.06	6.21	0.80	118	15	
Minimum Concentration				260	6.80	0.1	65	0	32	1	105	20	16	1.00	32	0.00	158	0.00	0.19	0.01	0	0	



Water quality risk map of Quetta

Annexure-20

Detailed Water Quality Results of Ziarat

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units				-	µS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	µg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS				-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	ZIA-1	Pechi Karez Ziarat Road - House (Alt.)	W.Supply	874	7.79	0.1	130	0	84	7.29	240	140	107	1.00	144	1.10	566	0.40	0.1	0.00	8	0	Unsafe
2	ZIA-2	P.H.E Water Supply Scheme	Spring	440	7.72	0.26	80	0	52	1.22	135	55	40	2.00	36	8.40	271	0.05	0.12	0.01	0	0	Safe
3	ZIA-3	P.H.E Water Supply Near Jinnah Top	W.Supply	441	7.69	0.08	90	0	50	8.51	160	65	37	0.50	48	2.90	275	0.01	0.18	0.01	0	0	Safe
4	ZIA-4	Jamia Masjid Ziarat Town	Well	623	7.62	0.6	140	0	56	21.87	230	65	51	2.00	93	4.6	393	0.01	0.11	0.02	0	0	Safe
5	ZIA-5	Shalimar Hotel	Tap	499	7.87	0.19	110	0	78	1.20	190	75	35	1.00	45	1.8	305	0.07	0.1	0.01	12	0	Unsafe
6	ZIA-6	Cheena Karez Ziarat Road	Karez	588	7.83	0.24	170	0	64	15.80	225	75	45	1.00	36	3.2	352	0.02	0.21	0.03	8	6	Unsafe
7	ZIA-7	Zandra Karez	Karez	498	7.72	0.26	135	0	56	13.37	195	65	36	1.00	36	3.10	302	0.03	0.18	0.00	15	7	Unsafe
8	ZIA-8	Kwas water Supply Scheme	T.Well	576	7.52	0.32	135	0	62	1.22	160	55	64	2.00	74	3.60	355	0.01	0.19	0.01	0	0	Safe
9	ZIA-9	Kashki Near Kharwari Baba (N/F)	Wind Mill	526	7.81	0.18	135	0	56	3.65	155	50	61	2.00	77	1.60	337	0.01	0.32	0.03	8	0	Unsafe
10	ZIA-10	P.H.E Tubewell Ziarat Town	T.Well	597	7.73	0.42	175	0	60	14.58	210	70	54	1.00	49	1.30	359	0.03	0.12	0.02	0	0	Safe
11	ZIA-11	Faran Tangi W. Sply. Scheme killi Pechi	W.Supply	471	7.87	0.21	110	0	60	6.08	175	60	38	1.00	55	2.30	296	0.02	0.18	0.01	0	0	Safe

Total No. of samples	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	
%age of Samples Exceeding Permissible Limits			0%					0%	0%					0%	0%	9%	0%	0%		45%	18%	45%
No. of Samples Exceeding Permissible Limits		0	0					0	0					0	0	1	0	0		5	2	5
Maximum Concentration	874	7.87	0.60	175	0	84	21.87	240	140	107	2.00	144	8.40	566	0.4	0.32	0.03		15	7		
Minimum Concentration	440	7.52	0.08	80	0	50	1.20	135	50	35	0.50	36	1.10	271	0.01	0.10	0.00		0	0		

Annexure-21

Detailed Water Quality Results of Hyderabad

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	HYD-1	Sapphair Cotton Mill, Khurshid Colony	T.Well	10331	7.60	0.4	1250	Nil	600	243	2500	2250	1127	12.00	720	2.34	6611	0.18	0.30	2.90	0	0	Unsafe
2	HYD-2	City Water Supply Tank Kotri	W. Supply	390	7.30	125	80	Nil	20	12	100	42	39	2.30	48	0.78	250	0.07	0.22	2.64	50	12	Unsafe
3	HYD-3	Hussainabad Pacca Tank HAD-8	W. Supply	400	7.50	45	90	Nil	20	12	100	46	42	1.90	46	0.45	256	0.05	0.12	2.69	50	8	Unsafe
4	HYD-4	B.I.S.E Hyderabad	Tap	409	7.32	22	100	Nil	20	15	110	42	40	2.00	33	0.59	262	0.03	0.10	4.91	60	23	Unsafe
5	HYD-5	Tayyab Masjid Unit Latifabad	Tap	445	7.31	3.1	110	Nil	24	17	130	32	34	2.20	57	0.36	285	0.07	0.09	3.59	0	0	Safe
6	HYD-6	New Wahdat Colony, Qasimabad	Tap	380	7.23	2.5	85	Nil	20	10	90	35	40	2.70	47	0.33	243	0.08	0.07	3.03	0	0	Safe
7	HYD-7	Al-Mustaf Town	Tap	390	7.21	4.5	100	Nil	20	16	115	39	36	2.40	36	0.41	250	0.09	0.08	1.76	32	0	Unsafe
8	HYD-8	Near Poultry Market, Quetta Hotel, Jail Road	Tap	440	7.21	2.1	120	Nil	28	24	140	37	20	2.80	50	0.54	282	0.02	0.07	1.89	40	0	Unsafe
9	HYD-9	Haji Shah, Cantonment Area	Tap	360	7.32	33	90	Nil	20	13	105	31	33	2.00	38	0.32	230	0.01	0.10	2.77	70	21	Unsafe
10	HYD-10	Pritabad Govt. Hospital	Tap	380	7.32	2.2	85	Nil	20	10	90	39	40	2.90	50	0.29	243	0.04	0.11	1.71	0	0	Safe
11	HYD-11	Near Bachal Bhatti Goth Ghanghra Mori	Tap	365	7.32	22	90	Nil	20	10	105	38	32	2.10	37	0.32	233	0.04	0.13	2.76	50	11	Unsafe
12	HYD-12	Custom Post Office Pocket Colony	Tap	401	7.40	12	105	Nil	20	16	115	39	36	2.10	35	0.39	257	0.06	0.15	1.90	88	4	Unsafe
13	HYD-13	Pump Station Behind Karachi Wah Site Kotri	Tap	393	7.58	88	95	Nil	20	18	125	44	31	2.30	30	0.66	252	0.04	0.14	3.90	90	5	Unsafe
14	HYD-14	Naseem Nagar Chowk Tahir Hotel	Tap	419	7.30	12	115	Nil	24	16	125	46	34	2.40	27	0.51	268	0.02	0.13	1.95	65	0	Unsafe
15	HYD-15	Tando Jahania, Defernce Society	Tap	436	7.32	10	120	Nil	24	11	135	48	49	1.30	30	0.71	279	0.05	0.11	1.11	74	0	Unsafe

Total No. of samples	15	15	15	15	0	15	15	15	15	15	15	15	15	15	15	15	15	15	15	15	
%age of Samples Exceeding Permissible Limits		0%	60%					7%	7%					0%	7%	0%	0%	0%	73%	47%	80%
No. of Samples Exceeding Permissible Limits		0	9					1	1					0	1	0	0	0	11	7	12
Maximum Concentration	10331	7.60	125	1250	0	600	243	2500	2250	1127	12	720	2.34	6611	0.18	0.3	4.91	90	23		
Minimum Concentration	360	7.21	0.4	80	0	20	10	90	31	20	1.3	27	0.29	230	0.01	0.07	1.11	0	0		

Detailed Water Quality Results of Karachi

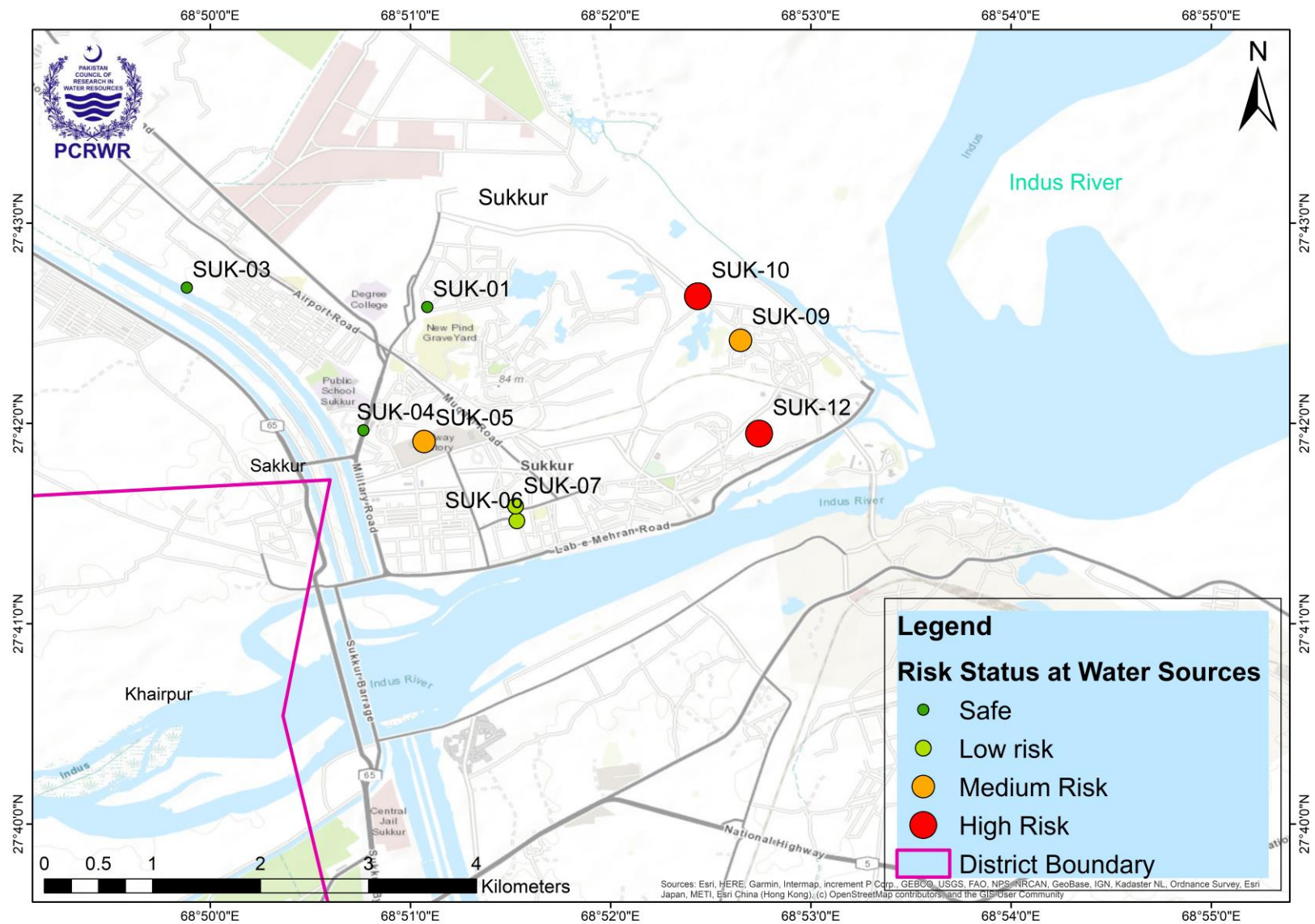
Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
		Units	-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
		Maximum Permissible Limits NDWQS	-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	KAR-01	Darweshabad Hotel, Yousaf Goth	Tap	864	7.35	1.69	140	0	44	29	230	147	88	5.60	77	0.58	477	0.05	0.45	1.30	0	0	Safe
2	KAR-02	Mamoor Masjid, Z-6, Gulshin-e-Maymar	Tap	819	7.90	0.66	130	0	40	24	200	145	92	7.40	68	0.60	457	0.01	0.08	1.50	54	0	Unsafe
3	KAR-03	Faizan-ul-Tegia, Sec.11/2, Orangi Town	Tap	610	7.85	0.96	110	0	36	19	170	91	59	5.40	60	0.58	339	0.04	0.34	0.70	50	0	Unsafe
4	KAR-04	Masjid Rabani, Mango Pir Road	Tap	1155	7.58	0.35	180	0	48	29	240	170	144	7.60	148	0.54	657	0.02	0.61	0.12	21	0	Unsafe
5	KAR-05	Jamai Masjid Saddique-Akbar Churangi	Tap	868	7.32	2.48	150	0	44	21	195	145	105	6.90	71	0.66	452	0.05	0.36	1.10	48	17	Unsafe
6	KAR-06	Hamidia Rizvi Univ. Near PCSIR Lab	Tap	808	7.53	2.46	130	0	40	22	190	140	95	6.20	72	0.55	456	0.10	0.44	1.50	0	0	Safe
7	KAR-07	KNS Poultry Farm, Safura Goth	W. Supply	840	7.44	1.26	140	0	44	22	200	143	97	6.00	70	0.40	468	0.02	0.51	1.50	23	0	Unsafe
8	KAR-08	MP Check Post-4, Malir Cantt.	Tap	792	7.80	2.88	110	0	32	22	170	140	98	6.20	82	0.52	448	0.07	0.40	1.50	43	0	Unsafe
9	KAR-09	Police Station Saeedabad	W. Supply	848	7.43	1.38	130	0	44	24	210	148	95	4.60	78	0.62	474	0.05	0.47	0.60	87	2	Unsafe
10	KAR-10	Masjid Usmania, Opp. MCB Market Poposh	W. Supply	2490	7.57	0.63	290	0	74	57	420	510	370	8.40	180	9.25	1414	0.03	1.55	0.12	35	0	Unsafe
11	KAR-11	KBCA Civic Centre	Tap	826	7.94	1.68	140	0	40	24	200	135	94	5.90	73	0.63	459	0.05	0.49	1.40	44	3	Unsafe
12	KAR-12	Muslim Masjid, C-16, Near Safari Park	Tap	816	8.03	0.71	130	0	40	22	190	137	95	6.10	76	0.62	457	0.07	0.45	1.20	70	0	Unsafe
13	KAR-13	Bachelor Flat A-7, CDA Colony Airport	Tap	883	7.58	0.89	140	0	48	24	220	147	96	6.50	84	0.67	492	0.10	0.41	1.40	67	0	Unsafe
14	KAR-14	Muhammadi Masjid, H Area, Khokhrapar	Tap	2690	7.81	2.01	400	0	72	61	400	520	415	4.50	180	3.56	1508	0.02	1.35	0.12	57	0	Unsafe
15	KAR-15	Kafi Nasir, Gulbai Chowk	Tap	818	7.38	2.98	130	0	40	22	190	139	95	6.20	75	0.52	457	0.05	0.46	1.00	20	2	Unsafe
16	KAR-16	Markazi Jamia Masjid, Jacob Line, Jinnah Road	Tap	1350	8.10	0.97	210	0	56	22	230	223	198	7.90	126	4.21	777	0.07	1.19	0.12	50	0	Unsafe
17	KAR-17	Ahmed General Hospital, Tipu Sultan Road	Tap	914	7.81	0.68	150	0	44	22	200	141	114	7.50	93	2.94	524	0.02	0.46	1.70	60	0	Unsafe
18	KAR-18	Shoukat U.M Hospital, Shah Faisal Colony-3	Tap	991	7.59	0.49	128	0	40	27	210	177	128	3.90	105	0.82	563	0.06	0.45	0.60	54	0	Unsafe
19	KAR-19	Seraj-Al-Masajid, Opp. P.S, Malir City	Tap	1620	7.82	1.07	380	0	104	36	410	191	180	7.00	132	6.04	905	0.04	0.51	0.12	70	13	Unsafe
20	KAR-20	Rabani Masjid, Quaidabad Chowk	W. Supply	1042	8.01	15.10	180	0	52	24	230	153	128	8.20	112	2.02	594	0.04	0.46	1.00	31	0	Unsafe
21	KAR-21	Manzal Petrol Pump, Highway	Tank	816	8.08	2.35	130	0	36	24	190	133	96	6.30	82	0.55	458	0.08	0.40	5.60	80	1	Unsafe
22	KAR-22	Masjid Aqsa, Block-5, Clifton	Tap	833	7.89	2.16	130	0	40	22	190	141	101	5.70	80	0.60	470	0.12	0.42	1.40	11	0	Unsafe
23	KAR-23	Kafe Gulshin, Mani Stop, Akhtar Colony	Tap	832	7.71	0.89	130	0	40	22	190	140	100	5.80	81	0.56	469	0.04	0.45	1.50	50	5	Unsafe
24	KAR-24	Sindh Govt. Dispensary Korangi	W. Supply	804	7.85	0.90	130	0	40	22	190	135	94	6.00	74	0.90	453	0.05	0.57	1.70	48	0	Unsafe
25	KAR-25	Madina Furniture, Main Market Landhi-5	Bore	5140	7.30	0.77	330	0	148	216	1150	1025	630	16.00	740	5.18	2870	0.03	2.35	0.12	62	22	Unsafe
26	KAR-26	Usmania Masjid Gul Ahmed, Bilal Colony	Tap	962	7.44	0.80	170	0	40	32	230	137	110	7.70	106	1.25	540	0.08	0.39	1.40	50	8	Unsafe
27	KAR-27	Pak. Machine Tool Factory, Landhi	Tap	846	7.88	4.25	140	0	40	22	190	133	103	6.20	84	1.10	477	0.11	0.43	0.12	54	0	Unsafe
28	KAR-28	DHA, SKBZ College, Def. Housing Society	Tap	835	8.03	0.74	140	0	36	24	190	135	98	6.20	80	0.65	466	0.01	0.40	1.20	70	0	Unsafe
Total No. of samples				28	28	28	28		28	28	28	28	28	28	28	28	28	28	28	28	28	28	28
%age of Samples Exceeding Permissible Limits					0%	4%	0%		0%	0%	4%	11%	0%	0%	0%	0%	11%	0%	7%	0%	93%	32%	93%
No. of Samples Exceeding Permissible Limits					0	1					1	3				0	3	0	2	0	26	9	26
Maximum Concentration				5140	8.08	15	400	0	148	216	1150	1025	630	16	740	9	2870	0	2	6	87	22	0
Minimum Concentration				610	7.30	0	110	0	32	19	170	91	59	4	60	0	339	0	0	0	0	0	0

Annexure-23

Detailed Water Quality Results of Sukkur

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	SUK-01-GW	Maka Goth, Near Sheraz Hospital	H.Pump	741	7.80	1.3	180	Nil	52	19.00	210	92	70	2.80	60	0.23	474	0.2	0.87	3.30	0	0	Safe
2	SUK-03-GW	Mastoi Hotel Opp. Al-Faisal Med. Centre	Tap	1058	7.90	0.8	240	Nil	56	29.00	260	171	115	7.50	44	0.63	677	0.1	0.67	2.00	0	0	Safe
3	SUK-04-GW	Water Supply Scheme Barrage Colony	Tap	600	7.80	0.5	180	Nil	48	19.00	200	57	40	3.00	36	0.30	384	0.16	0.3	1.00	0	0	Safe
4	SUK-05-GW	Rahmania Masjid, Pathan Colony	Filer Plant	287	8.00	0.7	100	Nil	36	7.00	120	15	8	2.00	21	0.62	184	0.41	0.28	40.40	0	0	Unsafe
5	SUK-06-GW	Jamai Masjid, Police Line SRTC Bus Stop	H.Pump	2930	7.50	1.3	480	Nil	92	32.00	360	435	460	46.10	345	9.82	1875	0.08	1.42	1.20	0	0	Unsafe
6	SUK-07-GW	Khalil Plaza Near Al-Habib Hotel	Tap	625	7.51	1.1	230	Nil	48	29.00	240	49	28	3.50	12	0.35	400	0.69	0.57	1.00	5	0	Unsafe
7	SUK-09-GW	Humayoon Gymkhana	Tap	267	8.00	19	90	Nil	30	8.00	110	14	9	1.50	22	0.49	171	2.9	0.19	86.00	0	0	Unsafe
8	SUK-10-GW	Hotel Sukkur Inn, Clock Tower	Tap	3180	7.42	1.9	350	Nil	160	73.00	700	408	351	55.80	635	3.55	2035	0.33	1.77	3.00	0	0	Unsafe
9	SUK-12-GW	Mosque Near Mukhtiar Kar Officer	Tap	2978	7.42	14	470	Nil	108	57.00	520	427	400	50.10	395	1.53	1906	0.12	1.87	1.60	25	0	Unsafe

Total No. of samples	9	9	9	9	0	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	
%age of Samples Exceeding Permissible Limits			22%					22%	33%					0%	33%	44%	22%	11%	22%	0%	67%
No. of Samples Exceeding Permissible Limits		0	2					2	3					0	3	4	2	1	2	0	6
Maximum Concentration	3180	8.00	19.00	480	0	160	73.00	700	435	460	55.80	635	9.82	2035	2.9	1.87	86.00	25	0		
Minimum Concentration	267	7.42	0.50	90	0	30	7.00	110	14	8	1.50	12	0.23	171	0.08	0.19	1.00	0	0		

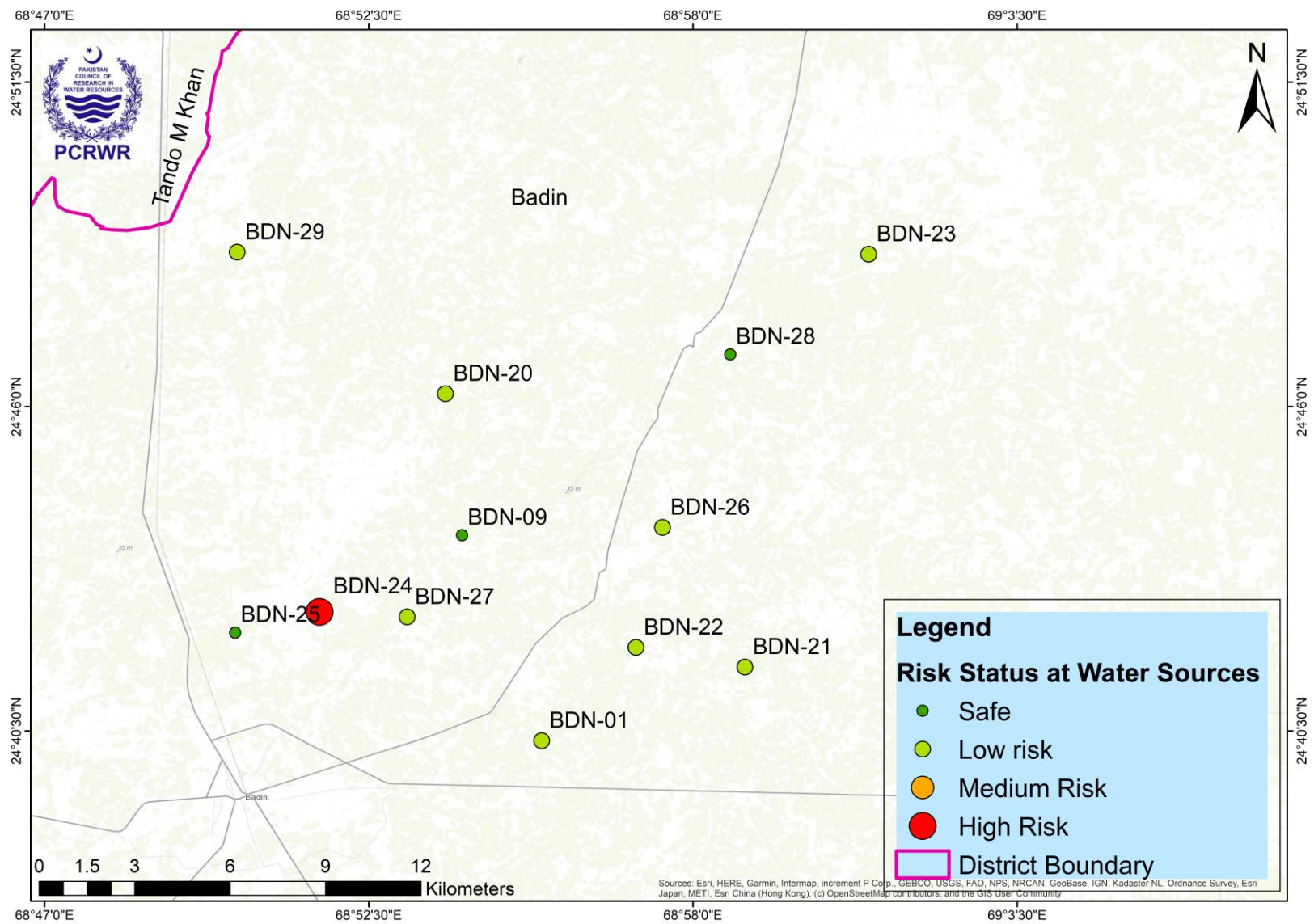


Water quality risk map of Sukkur

Detailed Water Quality Results of Badin

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	BDN-01	Water Supply Sche-Phase-11, Pond	W. Supply	625	8.53	16.54	120	10	36	14	150	84	74	5.00	68	0.72	400	0.02	0.25	3.51	24	2	Unsafe
2	BDN-09	Water Supply Sche-,Phase-111,Pond	W.Supply	593	9.01	2.77	115	10	32	12	130	82	75	5.00	64	0.54	380	0.01	0.37	5.79	0	0	Unsafe
3	BDN-20	Laghari Village, HP	H.Pump	2960	7.03	0.87	250	Nil	152	96	775	597	312	19.00	375	0.80	1894	0.01	0.87	1.62	0	0	Unsafe
4	BDN-21	Chandia Nangar Mohalla	Tap	641	8.27	0.71	120	Nil	32	14	140	89	80	6.00	72	0.48	410	0.02	0.38	4.70	16	0	Unsafe
5	BDN-22	Poly Technical College	Tap	696	7.64	56.00	130	Nil	40	17	170	93	78	7.00	80	1.38	445	0.01	0.26	6.85	70	50	Unsafe
6	BDN-23	Virtual University Campus, Badin (PSO Training Center)	Tap	1453	7.22	0.62	310	Nil	96	51	450	159	119	14.00	185	0.54	930	0.03	0.41	0.96	18	0	Unsafe
7	BDN-24	Shahnawaz Chowk	H.Pump	1723	7.28	0.46	275	Nil	112	29	400	196	198	23.00	280	4.93	1103	BDL	0.25	24.31	0	0	Unsafe
8	BDN-25	Marvi Girls College, Badin	Tap	849	8.57	6.37	135	10	36	23	185	122	104	11.00	110	1.00	543	BDL	0.37	0.66	0	0	Unsafe
9	BDN-26	Abid Town	Tap	666	8.35	0.68	125	10	32	15	140	125	84	6.00	72	0.72	426	0.01	0.31	5.11	48	1	Unsafe
10	BDN-27	Bilawal Park, Filter Plant (GBPS & GGPS Iqra Gharibabad)	Tap	632	8.95	0.97	125	15	40	14	160	79	69	5.00	73	0.88	404	0.02	0.29	3.61	6	4	Unsafe
11	BDN-28	Army Sugar Mill, Badin (Vil: Ghulam Hussain Soomro)	H.Pump	1029	8.11	0.52	225	Nil	60	37	305	114	89	11.00	124	0.50	658	0.06	0.40	1.05	0	0	Safe
12	BDN-29	Madarassa, Darool Uloom Saeedia, Badin	Tap	655	8.32	0.00	130	10	36	12	140	86	83	6.00	72	0.49	419	0.01	0.27	5.05	8	3	Unsafe

Total No. of samples	12	12	12	12	6	12	12	12	12	12	12	12	12	12	10	12	12	12	12	12
%age of Samples Exceeding Permissible Limits		33%	25%					8%	8%				0%	17%	0%	0%	0%	58%	42%	92%
No. of Samples Exceeding Permissible Limits		4	3					1	1				0	2	0	0	0	7	5	11
Maximum Concentration	2960	9.01	56.00	310	15	152	96	775	597	312	23.00	375	4.93	1894	0.06	0.87	24.31	70	50	
Minimum Concentration	593	7.03	0.00	115	10	32	12	130	79	69	5.00	64	0.48	380	0.01	0.25	0.66	0	0	

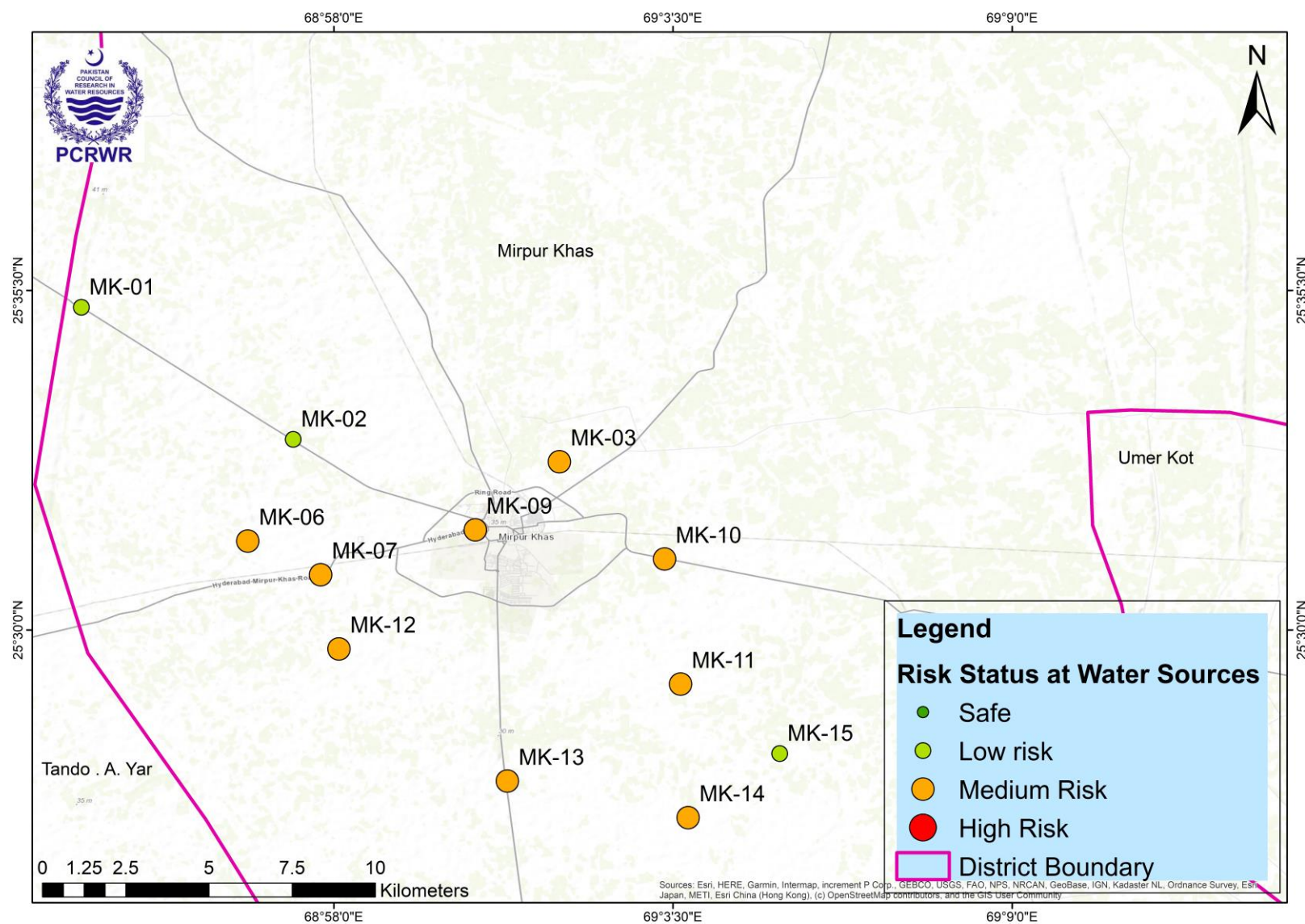


Water quality risk map of Badin

Detailed Water Quality Results of Mirpur Khās

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ / Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/ 100 ml)	(MPN/ 100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/ Unsafe
1	MK-01	Zafar Farm Sunji Mori (Tando Adam Road) Check post	H.Pump	1527	7.55	1.21	230	Nil	100	73	550	277	90	6.00	135	0.46	977	0.10	0.34	2.40	55	0	Unsafe
2	MK-02	MCH Turk Ali Mari	H.Pump	740	7.67	77.00	240	Nil	52	34	270	61	42	2.20	40	0.28	474	0.33	0.69	0.12	19	0	Unsafe
3	MK-03	Bagoo Kolhi Village P.P	T.Well	1817	7.54	1.00	400	Nil	100	85	600	211	134	3.80	202	0.72	1163	0.06	0.88	0.70	10	0	Unsafe
4	MK-06	Jamia Masjid Bhitaro	H.Pump	7520	7.37	7.00	350	Nil	328	287	2000	1836	783	14.50	785	0.44	4813	2.54	1.15	1.50	80	40	Unsafe
5	MK-07	Noman Town Hyderabad Road Mirpur Khas.	H.Pump	5350	7.77	1.70	550	Nil	160	148	1010	1008	750	6.20	667	0.58	3424	0.08	0.91	0.12	30	0	Unsafe
6	MK-09	Jamia Maka Masjid Ring Road Mirpur Khas.	Tap	15410	7.41	1.31	400	Nil	700	546	4000	3871	1658	22.70	1765	2.43	9862	0.11	1.18	0.60	26	0	Unsafe
7	MK-10	Haji Sodho Khan Hotel MPK Sugar Mills.	H.Pump	2440	7.68	1.00	340	Nil	128	100	730	263	215	6.20	485	0.34	1562	0.07	0.41	0.80	27	0	Unsafe
8	MK-11	Syed Haji Shah Village P.P	H.Pump	1883	7.36	167.00	370	Nil	92	90	600	256	145	7.80	200	0.24	1205	0.11	0.35	2.00	31	0	Unsafe
9	MK-12	Goth Pirbhu Lal Jivan Das Village P.P.	H.Pump	2410	7.63	0.99	400	Nil	112	97	180	243	233	5.00	439	1.36	1542	0.12	0.78	0.12	65	3	Unsafe
10	MK-13	Jam Laghari Village P.P	H.Pump	4030	7.41	1.60	350	Nil	280	179	1440	755	250	9.20	573	0.92	2579	0.11	1.5	0.12	6	0	Unsafe
11	MK-14	Jamia Masjid Haji Khan Laghari.	H.Pump	33500	7.18	18.00	400	Nil	1200	972	7000	10105	4405	68.00	2013	0.34	21400	0.15	0.69	0.12	1	0	Unsafe
12	MK-15	Haji Muhammad Ali Halepoto Village P.P	H.Pump	843	7.63	7.00	250	Nil	48	39	280	71	58	5.20	68	0.65	539	0.06	0.18	0.12	17	0	Unsafe

Total No. of samples	12	12	12	12	0	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12	12
%age of Samples Exceeding Permissible Limits		0%	42%							75%	67%					0%	75%	17%	0%	0%	100%	17%	100%
No. of Samples Exceeding Permissible Limits		0	5							9	8					0	9	2	0	0	12	2	12
Maximum Concentration		33500	7.77	167.00	550	0	1200	972.00	7000	10105	4405	68.00	2013	2.43	21400	2.54	1.50	2.40		80	40		
Minimum Concentration		740	7.18	0.99	230	0	48	34.00	180	61	42	2.20	40	0.24	474	0.06	0.18	0.12		1	0		

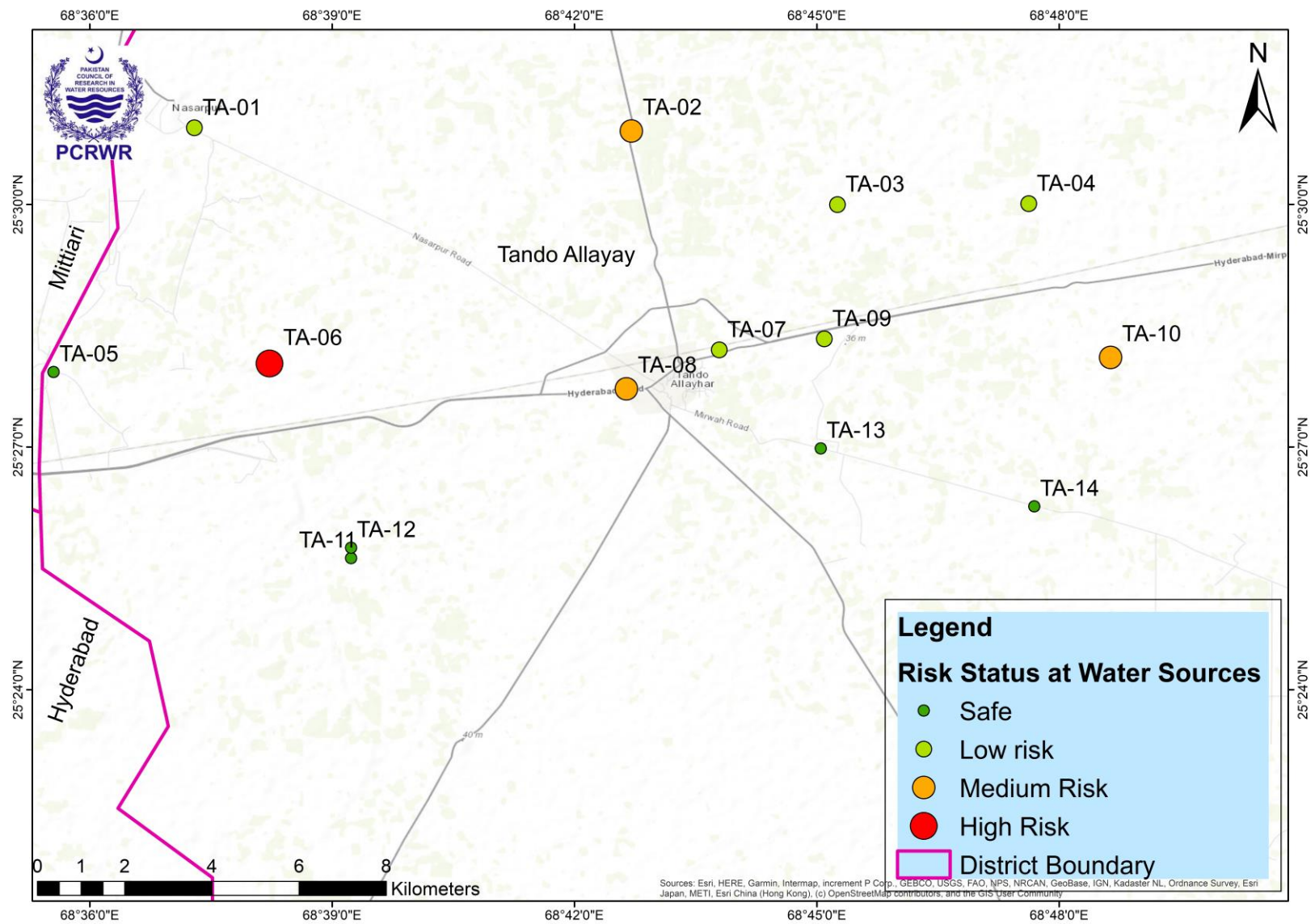


Water quality risk map of Mirpur Khas

Detailed Water Quality Results of Tando Allahyar

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	TA-01-GW	WaterRHC Nasrpur	Rooter	457	7.80	1.00	150	Nil	36	19	170	37	24	2.00	25	0.34	292	0.14	0.1	30.00	23	0	Unsafe
2	TA-02-GW	Near Middle School Shahpur Rizvia.	H.Pump	1660	7.86	3.00	300	Nil	72	58	420	174	179	6.50	271	0.25	1062	0.27	0.92	1.10	34	0	Unsafe
3	TA-03-GW	Jamia Masjid Hanfia Rizvia Ghulam Hussain Lund.	H.Pump	1695	7.61	3.00	320	Nil	74	59	430	132	184	5.00	325	0.81	1085	0.30	0.33	4.20	0	0	Unsafe
4	TA-04-GW	BHU Ditto Kalro.	H.Pump	1732	8.06	0.50	470	Nil	68	44	350	113	230	3.30	225	0.31	1108	0.23	1.3	0.90	0	0	Unsafe
5	TA-05-GW	Tajpur Village Government Dispensary	Well	1537	7.66	0.96	330	Nil	80	73	500	181	113	7.50	172	1.44	984	0.16	0.21	0.90	0	0	Safe
6	TA-06-GW	Jamia Masjid Qaim Khan Lashari.	H.Pump	1585	7.48	2.00	350	Nil	88	75	530	132	111	7.60	242	0.40	1014	0.14	0.28	26.50	4	0	Unsafe
7	TA-07-GW	WSS Tando Allahyar City.	T.Well	568	7.89	1.00	170	Nil	40	24	200	51	35	3.00	40	0.59	364	0.16	0.55	7.20	24	0	Unsafe
8	TA-08-GW	Civil Hospital Tando Allahyar.	H.Pump	883	7.83	1.01	270	Nil	50	38	280	60	70	3.00	82	0.66	565	0.17	0.13	18.00	0	0	Safe
9	TA-09-GW	Guru Rama Bai Sahab Mandir, Karo Patel.	H.Pump	1738	7.64	2.05	320	Nil	96	87	600	239	115	5.20	200	1.06	1112	0.17	0.4	0.80	0	0	Unsafe
10	TA-10-GW	Masjid & Madrsa Babul Rahmat Haji Usman Laghari.	H.Pump	921	7.89	1.10	270	Nil	56	39	300	86	70	3.00	65	0.65	589	0.23	0.39	18.50	2	0	Unsafe
11	TA-11-GW	Dhingano Bozdar, PPHI Hospital.	Rooter	788	8.17	1.20	230	Nil	50	30	250	69	62	1.80	60	3.34	504	0.19	0.27	0.90	0	0	Safe
12	TA-12-GW	Bilal Masjid, Village Bachal Kolachi.	H.Pump	646	7.94	0.56	200	Nil	40	24	200	47	50	5.50	53	0.41	413	0.18	0.07	1.80	0	0	Safe
13	TA-13-GW	Isa Madani Masjid Yousif Farm, Bhatiai Hotel Pak Sanghaar Road.	H.Pump	852	7.72	1.35	260	Nil	56	32	270	59	68	3.00	77	1.34	545	0.07	0.62	0.50	0	0	Safe
14	TA-14-GW	Near Govt. Dispensary (PPHI) Pak Sanghaar.	H.Pump	1146	7.77	2.00	300	Nil	60	36	300	79	120	4.00	152	1.81	733	0.07	0.36	0.60	0	0	Safe

Total No. of samples	14	14	14	14	0	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	
%age of Samples Exceeding Permissible Limits			0%					14%	0%					0%	36%	0%	0%	0%	36%	0%	57%
No. of Samples Exceeding Permissible Limits		0	0					2	0					0	5	0	0	0	5	0	8
Maximum Concentration	1738	8.17	3.00	470	0	96	87.00	600	239	230	7.60	325	3.34	1112	0.30	1.30	30.00	34	0		
Minimum Concentration	457	7.48	0.50	150	0	36	19.00	170	37	24	1.80	25	0.25	292	0.07	0.07	0.50	0	0		



Detailed Water Quality Results of Shaheed Benazirabad

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	Na	K	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	SB-01-GW	GBPS Village Azim shah	H.Pump	757	7.58	11.00	240	0	64	39	320	42	23	2.10	75	0.20	484	1.58	0.58	0.12	70	0	Unsafe
2	SB-02-GW	Jamiya Masjid at Village Haji Ghulam Shah	Tap	819	7.56	0.85	210	0	60	44	330	63	33	2.10	105	0.71	524	0.10	0.5	0.12	80	0	Unsafe
3	SB-03-GW	hospital road near solarized RO Plant Bucheri	H.Pump	2250	7.64	1.00	380	0	80	73	500	333	279	3.50	260	1.01	1440	0.17	0.46	0.12	56	0	Unsafe
4	SB-04-GW	Village Chakar Khan Shahani	H.Pump	707	8.01	0.56	190	0	40	32	230	60	59	3.60	75	0.33	452	0.32	0.89	2.60	78	0	Unsafe
5	SB-05-GW	ibrahim chandio Shop stop, Village Jafar Chutto.	H.Pump	1047	7.58	1.00	250	0	64	51	370	83	65	2.50	148	0.76	670	0.36	0.66	0.12	100	0	Unsafe
6	SB-06-GW	House of Mr. Shair Muhammad Bahrani, Village Sahib Khan Bahrani	H.Pump	1642	7.85	1.00	300	0	60	36	250	181	50	2.70	253	0.59	1051	2.44	0.87	3.10	64	0	Unsafe
7	SB-08-GW	Hotel Abdul Hakim Khaskheli, Village Mir Jan Muhammad Khaskheli	H.Pump	1098	7.81	0.99	320	0	48	39	280	97	119	2.10	87	1.97	703	0.26	0.41	0.12	70	0	Unsafe
8	SB-09-GW	Main Pangria Mori Stop, Village Ahmed Shah.	H.Pump	2003	7.91	1.80	380	0	40	36	250	231	339	2.60	280	0.33	1282	0.16	0.9	5.10	105	0	Unsafe
9	SB-10-GW	GBPS Village Ghulam Ali Chandio	H.Pump	2103	7.70	1.00	250	0	100	73	550	273	224	3.50	395	0.23	1346	0.12	0.05	5.30	80	0	Unsafe
10	SB-13-GW	Otaq BaShir Ahmed Chandio Village Masti Khan Chandio	H.Pump	2760	7.77	2.00	200	0	90	79	550	401	372	3.30	385	0.64	1766	0.48	0.89	2.80	50	0	Unsafe
11	SB-14-GW	Basic health center (PPHI) Village Haji Allah Bux Mangsi	H.Pump	531	8.03	1.20	150	0	32	27	190	52	31	2.90	40	0.18	340	0.31	0.24	2.40	88	1	Unsafe
12	SB-15-GW	Otaq Village Sulemen Lakho	H.Pump	1555	7.90	0.77	300	0	58	50	300	139	189	3.00	268	0.39	995	0.30	0.35	2.40	100	2	Unsafe
13	SB-18-GW	Basic health center (PPHI) Village Haji Allah Bux Mangsi	H.Pump	2850	7.72	1.00	330	0	100	85	600	389	369	5.40	529	1.52	1824	0.49	0.73	0.12	45	0	Unsafe
Total No. of samples				13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13
%age of Samples Exceeding Permissible Limits						8%					23%	31%				0%	46%	54%	0%	0%	100%	15%	100%
No. of Samples Exceeding Permissible Limits					0	1					3	4				0	6	7	0	0	13	2	13
Maximum Concentration				2850	8.03	11.00	380	0	100	85.00	600	401	372	5.40	529	1.97	1824	2.44	0.90	5.30	105	2	
Minimum Concentration				531	7.56	0.56	150	0	32	27.00	190	42	23	2.10	40	0.18	340	0.10	0.05	<0.13	45	0	

Detailed Water Quality Results of Muzaffarabad

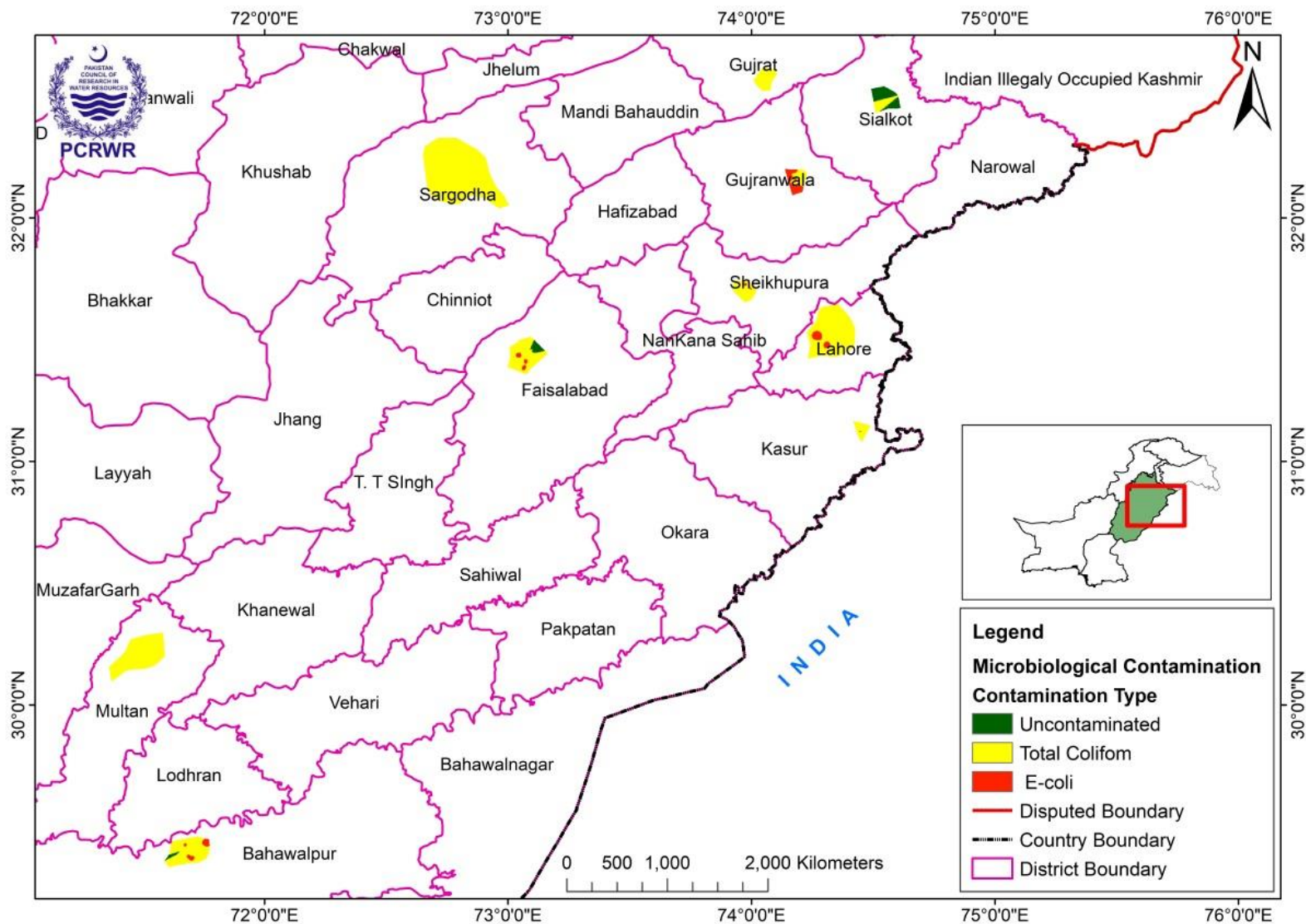
Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	K	Na	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	Safe/ Unsafe
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	
1	MZD-01	Tariqabad	Spring Water	1029	7.40	0.6	310	0	52	51	340	55	1.90	9	16	4.18	565	0.02	0.2	0.12	80	35	Unsafe
2	MZD-02	Bachon School Zero Point	Spring Water	755	7.50	9.8	350	0	28	58	310	24	1.00	5	15	0.75	415	0.02	0.15	0.12	70	30	Unsafe
3	MZD-03	Old Secretriare	F. Plant	178	7.60	0.0	60	0	40	7.29	130	12	0.90	5	14	0.66	97	0.02	0.25	0.12	0	0	Safe
4	MZD-04	Darbar Spring near river side	Spring Water	750	7.50	0.4	300	0	20	72	350	24	1.30	6	15	0.78	412	0.02	0.31	0.12	80	30	Unsafe
5	MZD-05	Upper Chatter Water Supply	W. Supply	133	7.70	12.2	40	0	36	7	120	10	0.80	4	14	0.71	73	0.02	0.04	0.12	0	0	Unsafe
6	MZD-06	Assembly AJK Filtration Plant	F. Plant	140	7.40	0.0	50	0	28	3	80	12	1.60	9	18	3.36	77	0.02	0.4	0.12	0	0	Safe
7	MZD-07	Read College Gojra	W. Supply	504	7.60	1.5	200	0	56	24	240	24	1.60	8	16	2.55	277	0.02	0.21	0.12	30	16	Unsafe
8	MZD-08	Chela Bandi	Spring Water	717	7.30	1.3	220	0	28	72	370	19	0.80	11	13	0.72	394	0.02	0.15	0.12	33	0	Unsafe
9	MZD-09	CHM Hospital MZD	W. Supply	134	7.60	1.3	50	0	40	51	310	10	0.70	15	26	8	73	0.02	0.19	0.12	0	0	Safe
10	MZD-10	Domail Chatter	Spring Water	809	7.10	0.0	310	0	112	29	400	24	0.80	4	16	0.73	444	0.02	0.05	0.12	80	38	Unsafe
Total No. of samples				10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
%age of Samples Exceeding Permissible Limits						20%					0%	0%				0%	0%	0%	0%	0%	60%	50%	70%
No. of Samples Exceeding Permissible Limits					0	2					0	0				0	0	0	0	0	6	5	7
Maximum Concentration				1029	7.70	12.2	350	0	112	72	500	250	1.90	15	26	10.00	1000	0.30	1.50	50.00	80	38	
Minimum Concentration				133	7.10	0.0	40.00	0	20	3	80	10	0.70	4.00	13	0.66	73	0.02	0.04	0.12	0	0	

Annexure-29

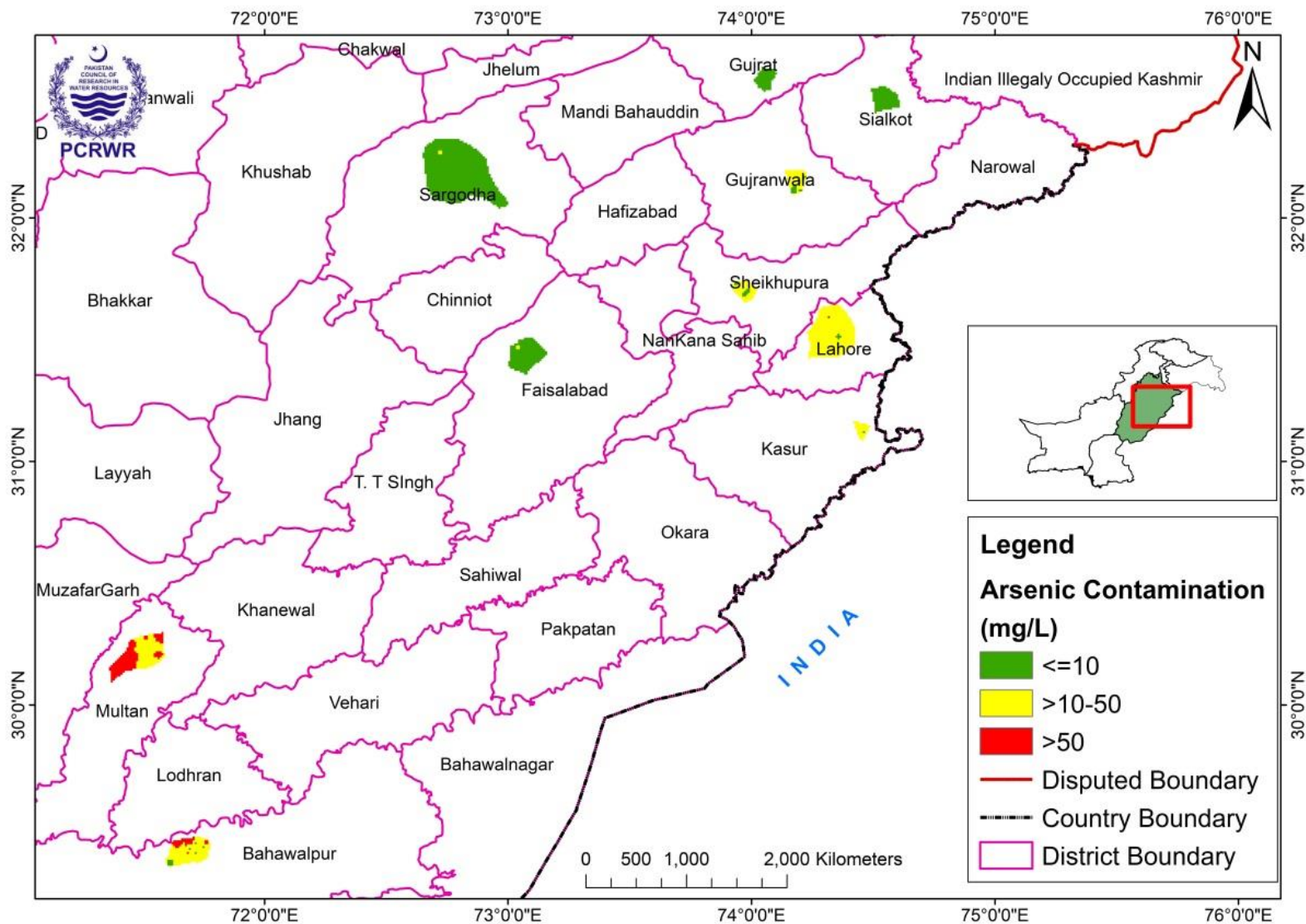
Detailed Water Quality Results of Gilgit

Sr. #	Sample Code	Location	Source	E.C	pH	Turbidity	HCO ₃ /Alk	CO ₃	Ca	Mg	Hard	Cl	K	Na	SO ₄	NO ₃ (N)	TDS	Fe	F	As	Total Coliforms	E. coli	Remarks
Units			-	μS/cm	-	NTU	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	mg/l	μg/l	(MPN/100 ml)	(MPN/100 ml)	
Maximum Permissible Limits NDWQS			-	NGVS	6.50-8.50	5.00	NGVS	NGVS	NGVS	NGVS	500	250	NGVS	NGVS	NGVS	10.00	1000	0.30	1.50	50.00	0	0	Safe/Unsafe
1	GLT-01	Sakardu River Hurnush	Surface Water	305	7.60	960	50	0	36	10	130	1.9	4	8.4	85	4.1	168	0.02	0.18	0.89	44	10	Unsafe
2	GLT-02	Indus River Bunjee Bridge	Surface Water	277	7.00	803	50	0	28	12	120	1.9	3	6.4	80	3.1	152	0.02	0.08	0.76	38	12	Unsafe
3	GLT-03	Water Tank Jutial, Cantt.	W. Supply	64	7.00	2	20	0	8	2	30	1.9	2	1.3	14	0.3	35	0.02	0.02	0.12	4	0	Unsafe
4	GLT-04	Tap Water Khumar Chowk	W. Supply	65	7.20	3	20	0	8	2	30	1.9	2	1.1	15	0.3	36	0.02	0.03	0.12	8	0	Unsafe
5	GLT-05	Barmas water Tank	W. Supply	58	7.30	4	25	0	6	2	25	1.9	2	0.5	0.27	1.1	32	0.02	0.03	0.12	12	0	Unsafe
6	GLT-06	New Polo Ground Gilgit	W. Supply	58	7.30	1	25	0	6	2	25	1.9	2	0.4	0.27	1.0	32	0.02	0.03	0.12	8	0	Unsafe
7	GLT-07	Jugloat Goor Village on Karakoram Highway	W. Supply	447	8.00	1	100	0	48	17	190	1.9	8	6.4	110	1.4	246	0.02	0.22	0.29	10	0	Unsafe
8	GLT-08	Karakorum international university Gilgit	Surface Water	150	8.10	91	40	0	16	7	70	1.9	2	1.9	30	1.4	83	0.02	0.03	0.12	16	4	Unsafe
9	GLT-09	Gilgit River City	Surface Water	145	7.70	82	40	0	12	10	70	1.9	2	1.8	28	1.3	80	0.02	0.03	0.12	40	14	Unsafe
10	GLT-10	Hunza River Doynor Bridge	Surface Water	196	8.10	426	30	0	16	11	85	1.9	6	2	55	4.6	108	0.02	0.03	0.12	42	12	Unsafe
Total No. of samples				10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
%age of Samples Exceeding Permissible Limits						50%					0%	0%			0%	0%	0%	0%	0%	0%	100%	50%	100%
No. of Samples Exceeding Permissible Limits					0	5					0	0			0	0	0	0	0	0	10	5	10
Maximum Concentration				447	8.10	960	100	0	48	17	190	1.9	8	8.4	110	4.6	246	0.02	0.22	0.89	44	14	
Minimum Concentration				58	7.00	1	20	0	6	2	25	1.9	2	0.4	0.27	0.3	32	0.02	0.02	0.12	4	0	

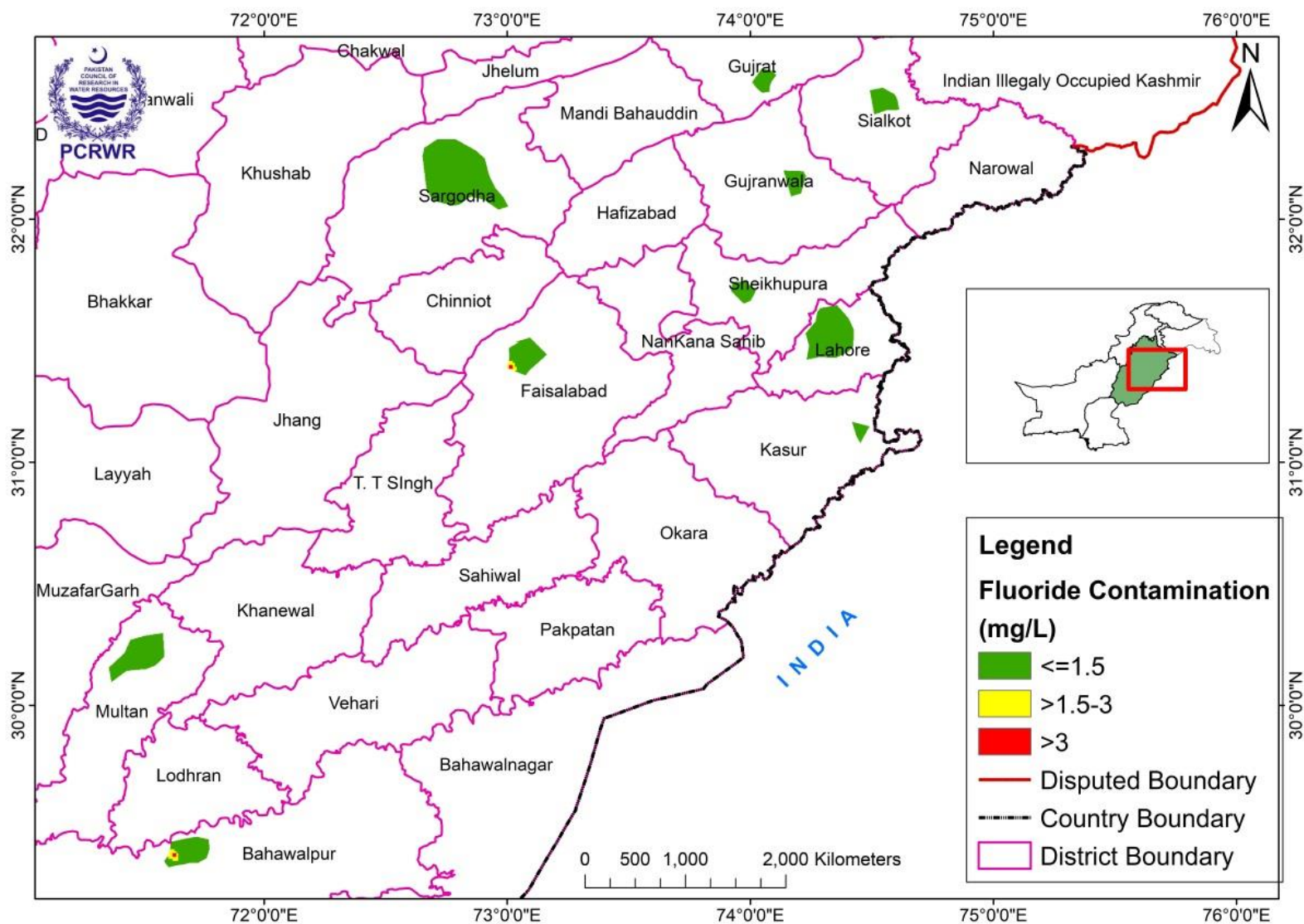
Hotspot of Microbial Contamination in Punjab



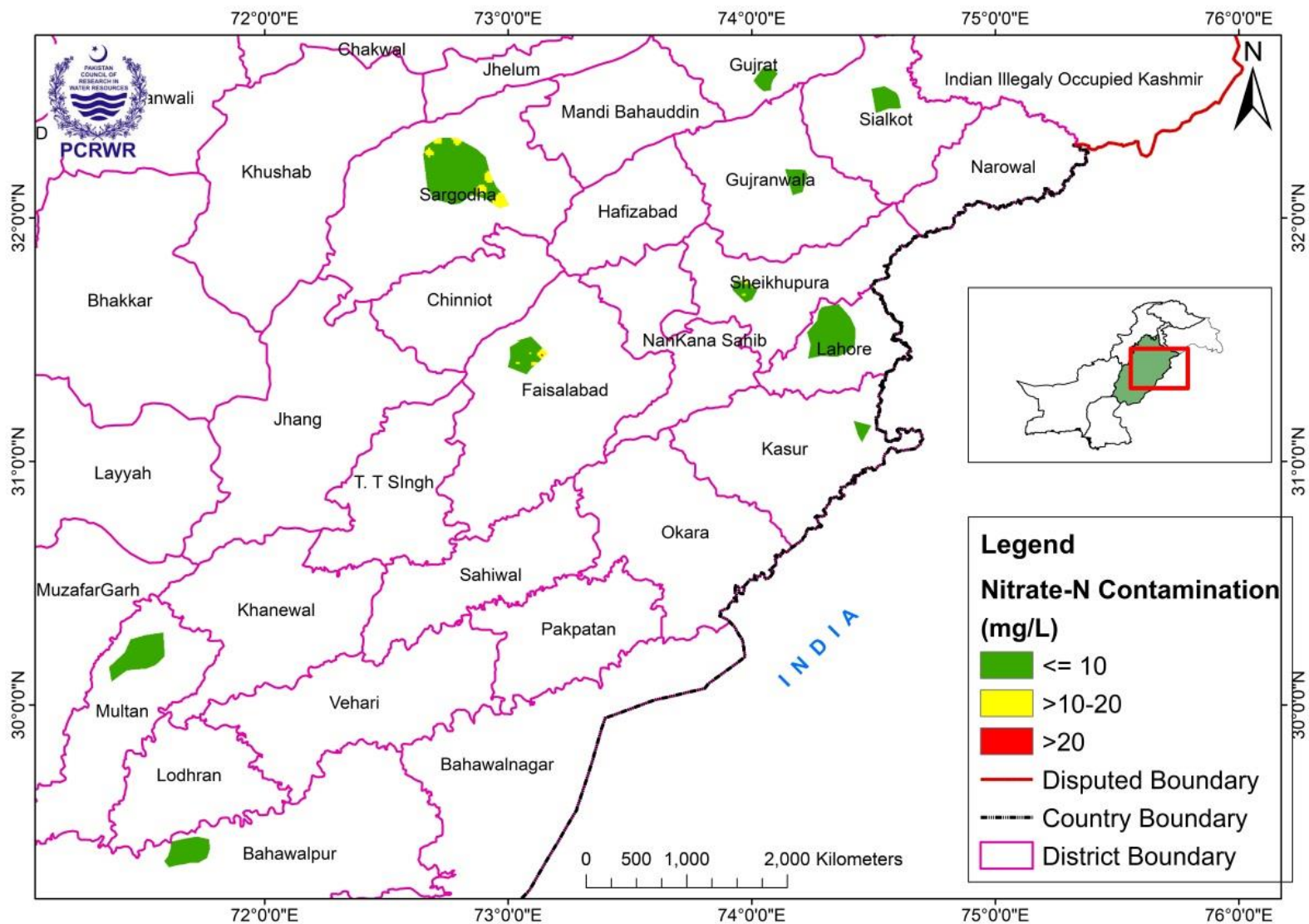
Hotspot of Arsenic Contamination in Punjab



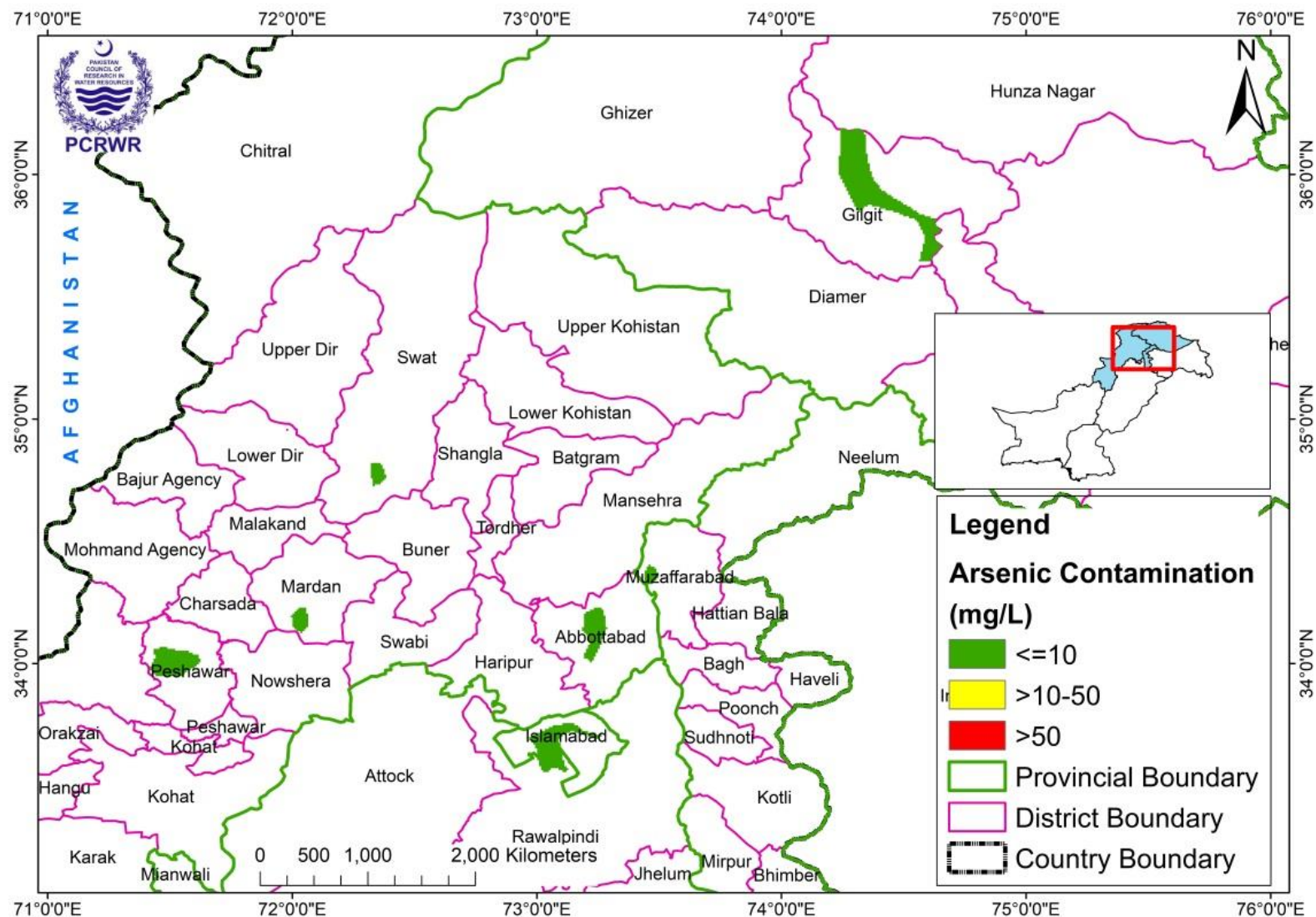
Hotspot of Fluoride Contamination in Punjab



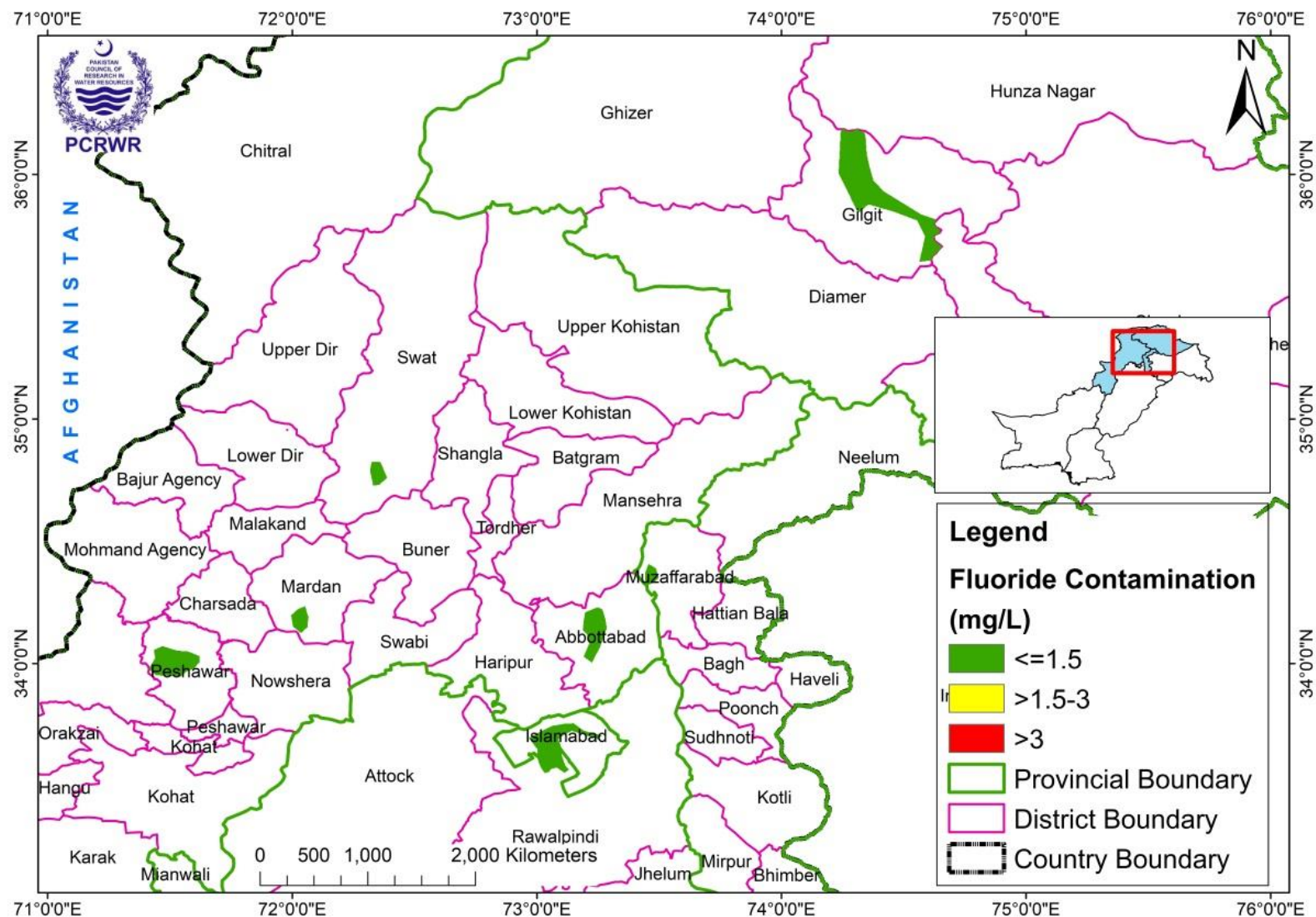
Hotspot of Nitrate Contamination in Punjab



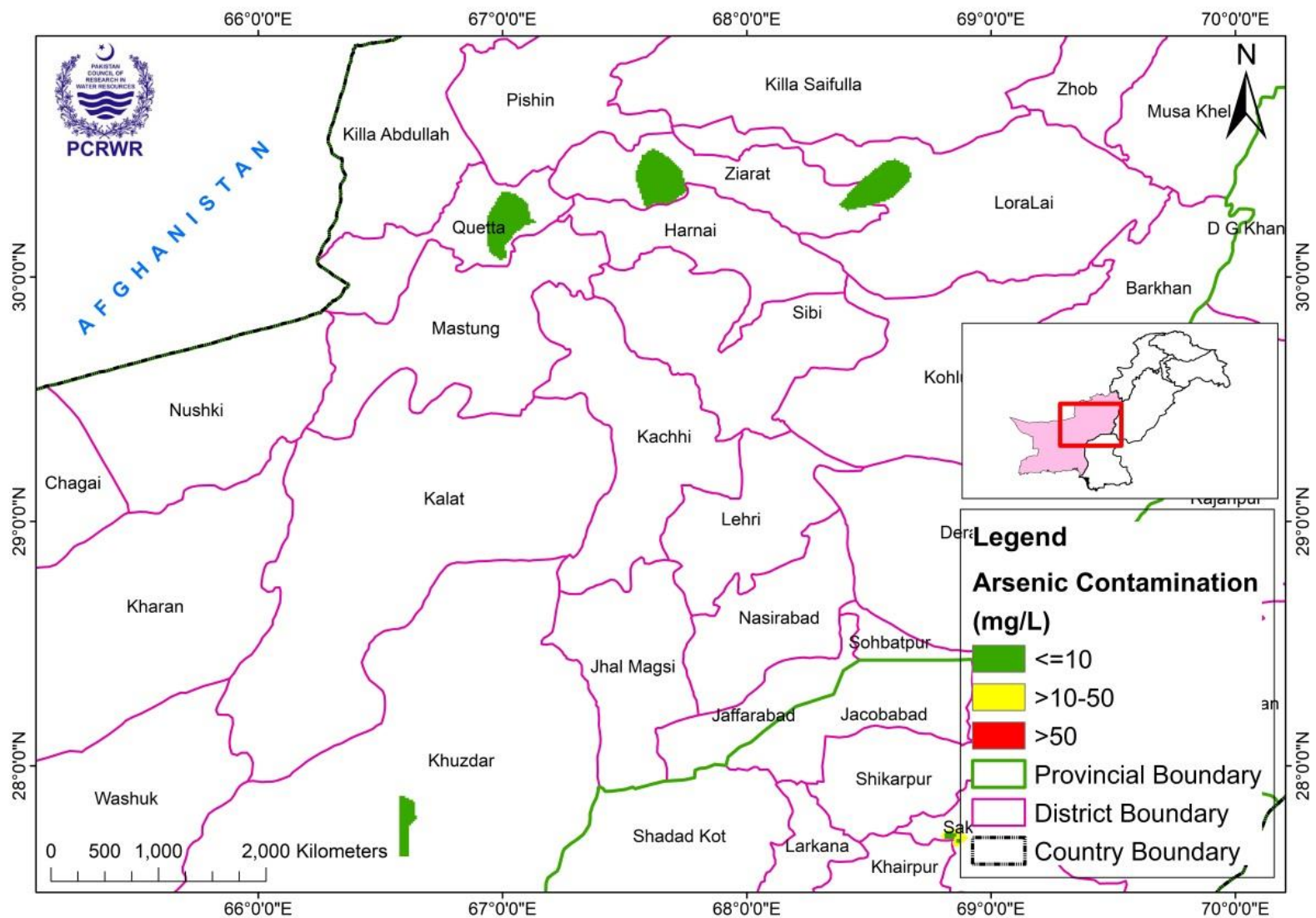
Hotspot of Arsenic Contamination in KPK



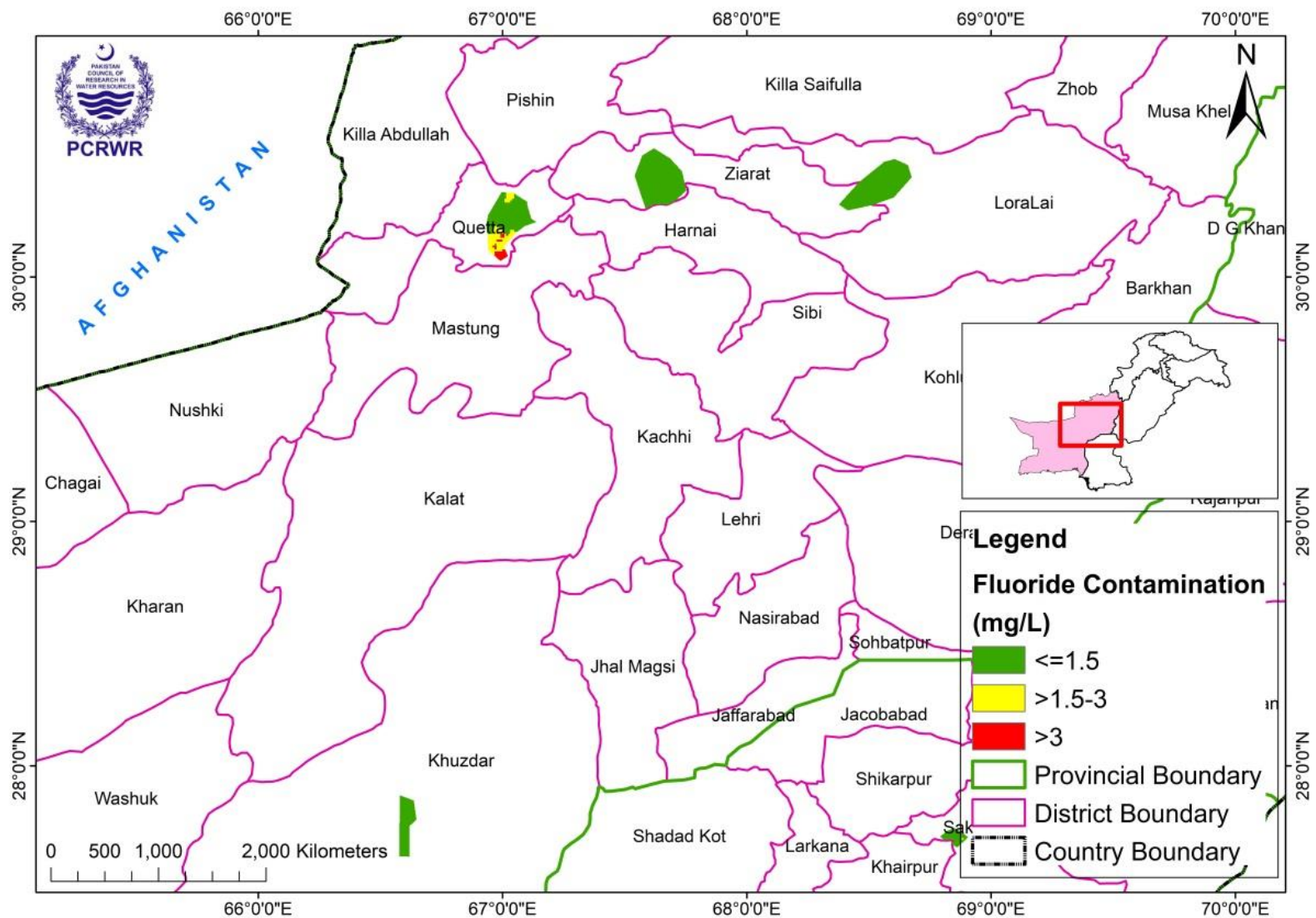
Hotspot of Fluoride Contamination in KPK



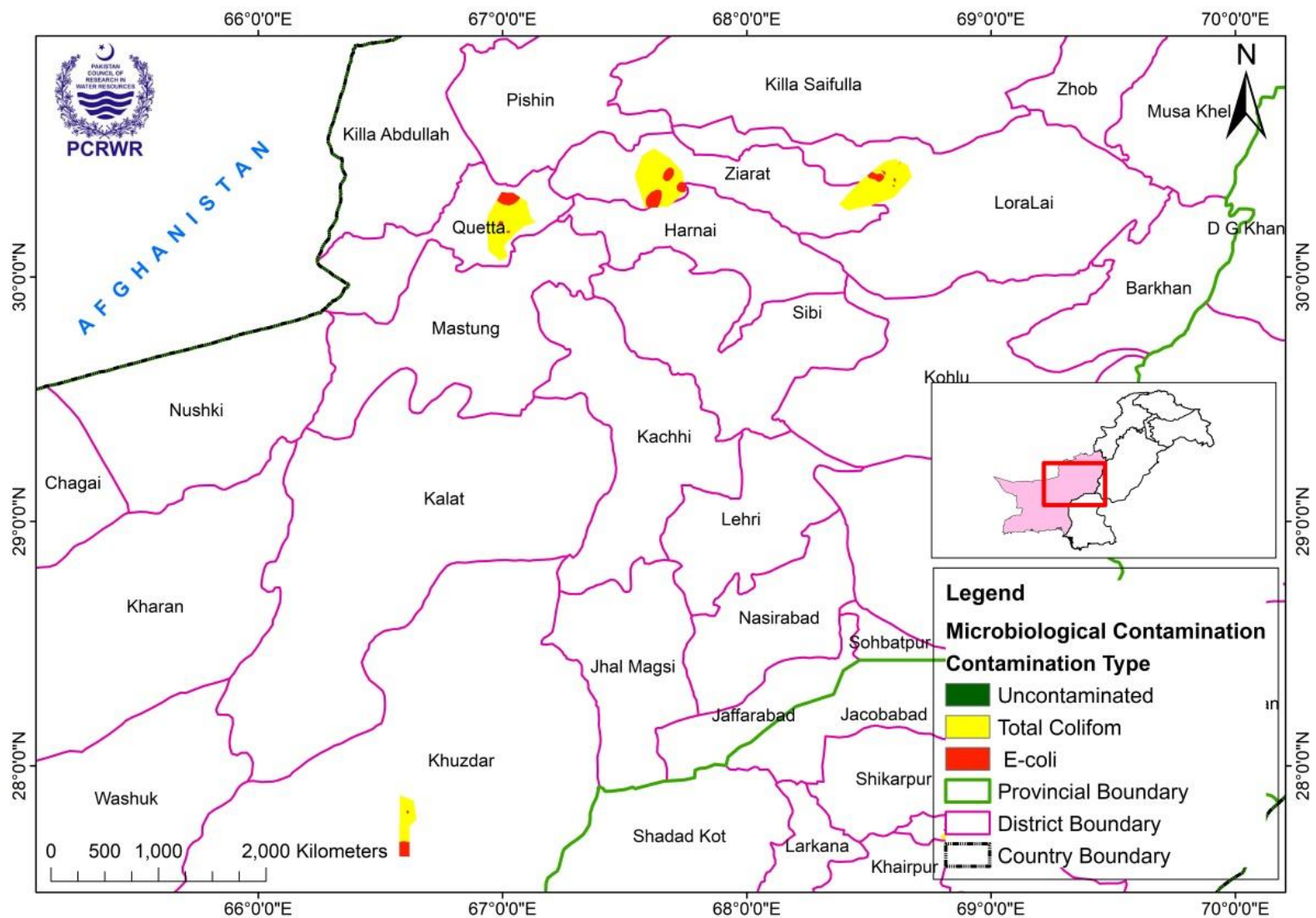
Hotspot of Arsenic Contamination in Balochistan



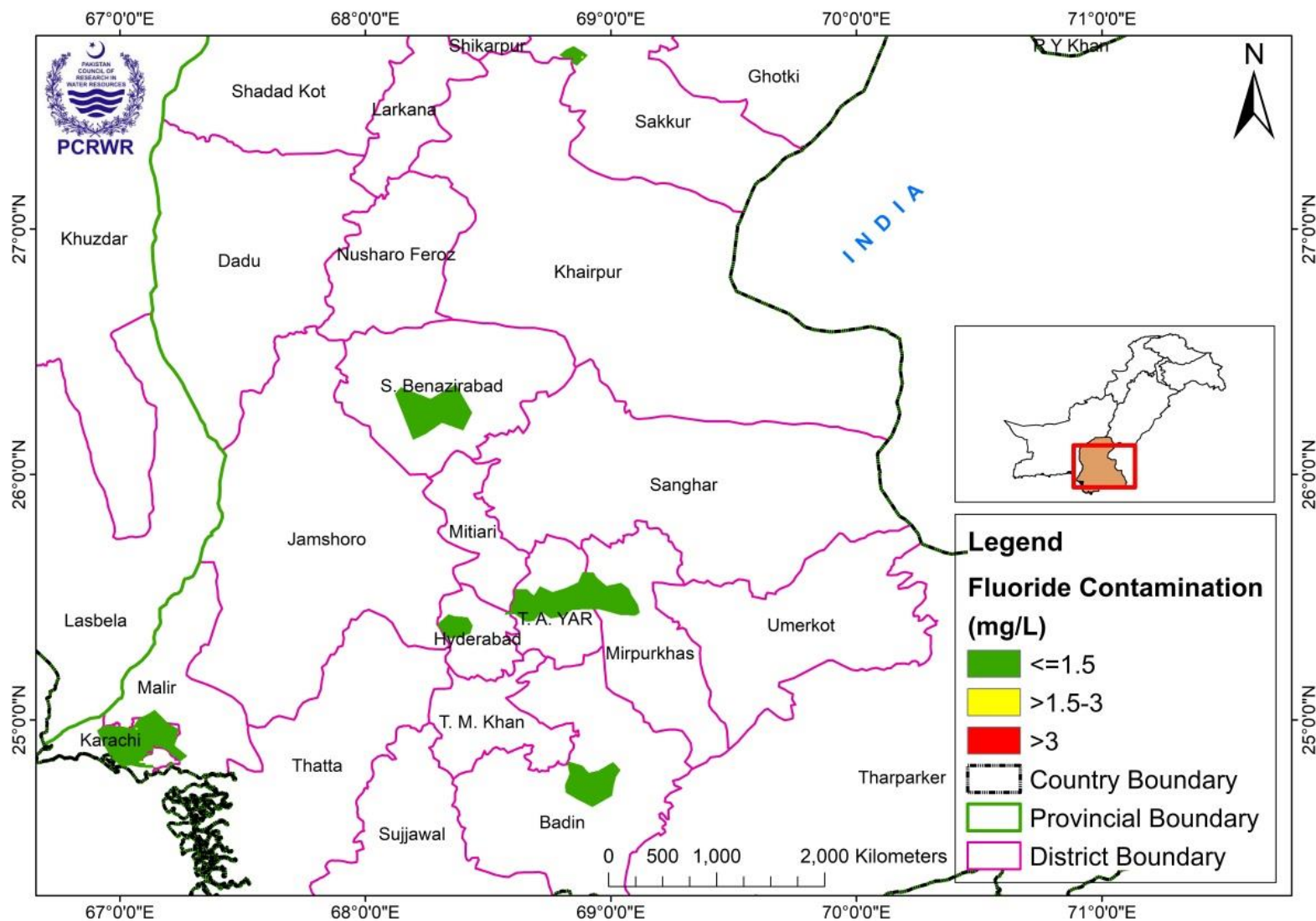
Hotspot of Fluoride Contamination in Balochistan



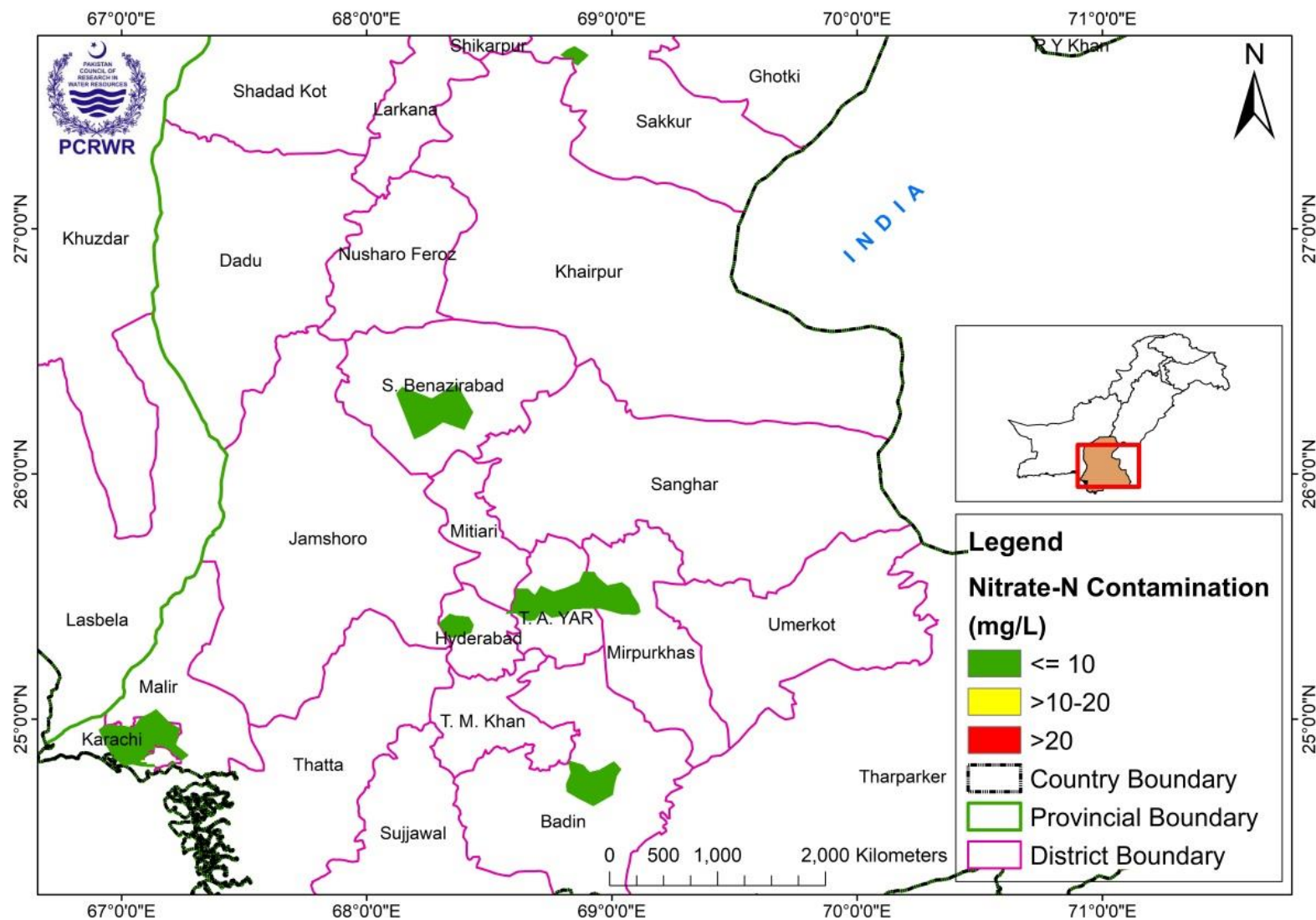
Hotspot of Microbial Contamination in Balochistan



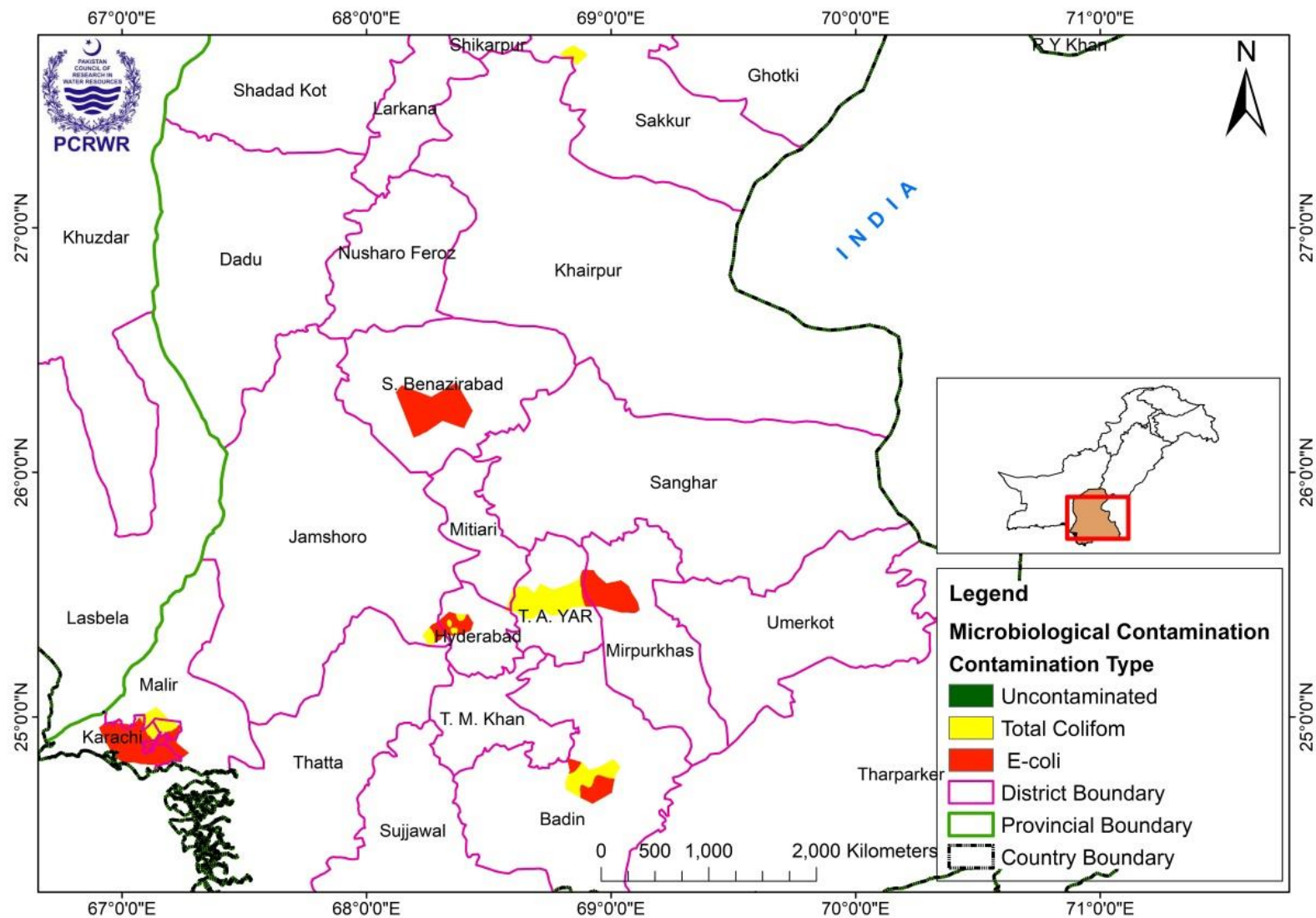
Hotspot of Fluoride Contamination in Sindh



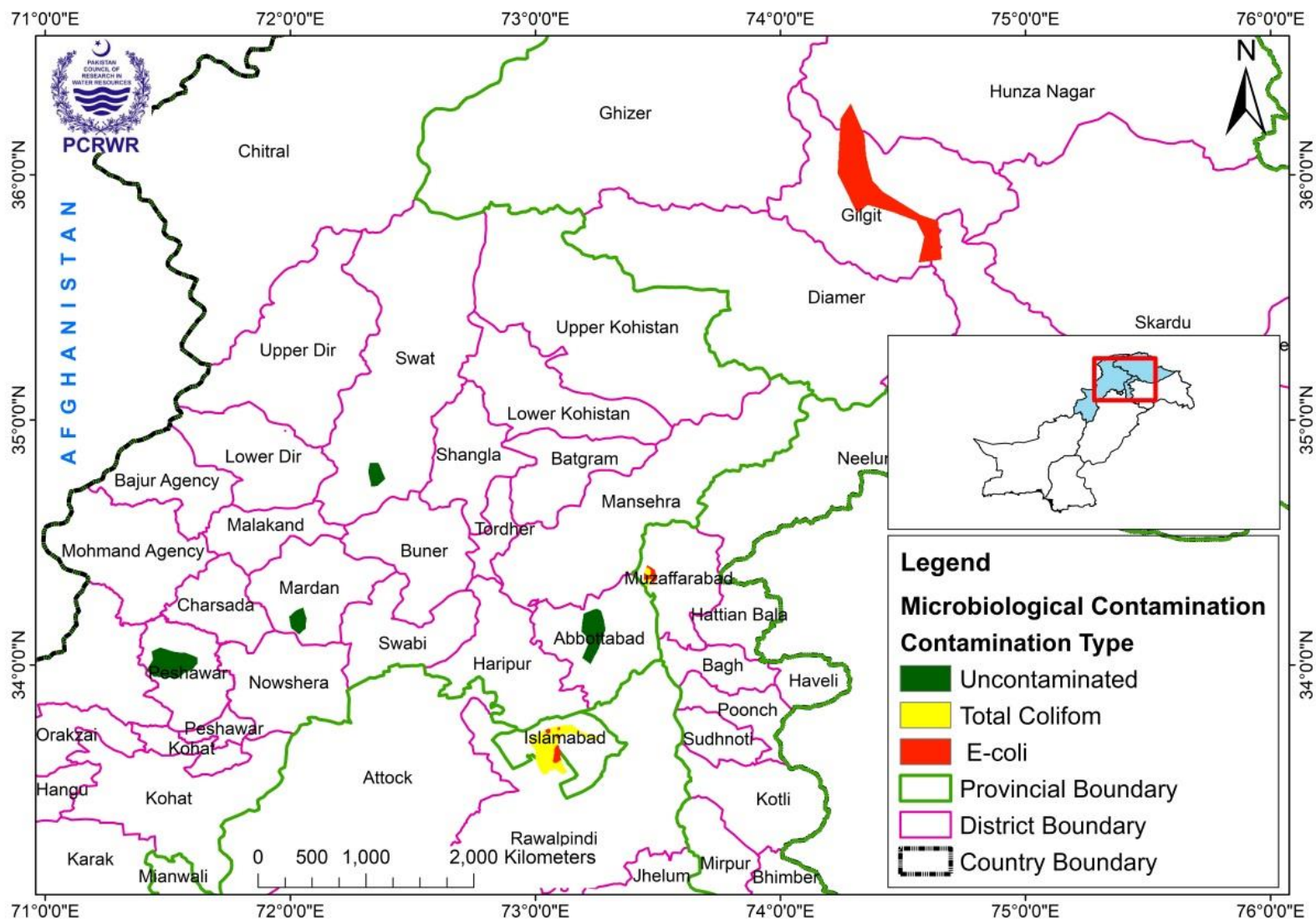
Hotspot of Nitrate Contamination in Sindh



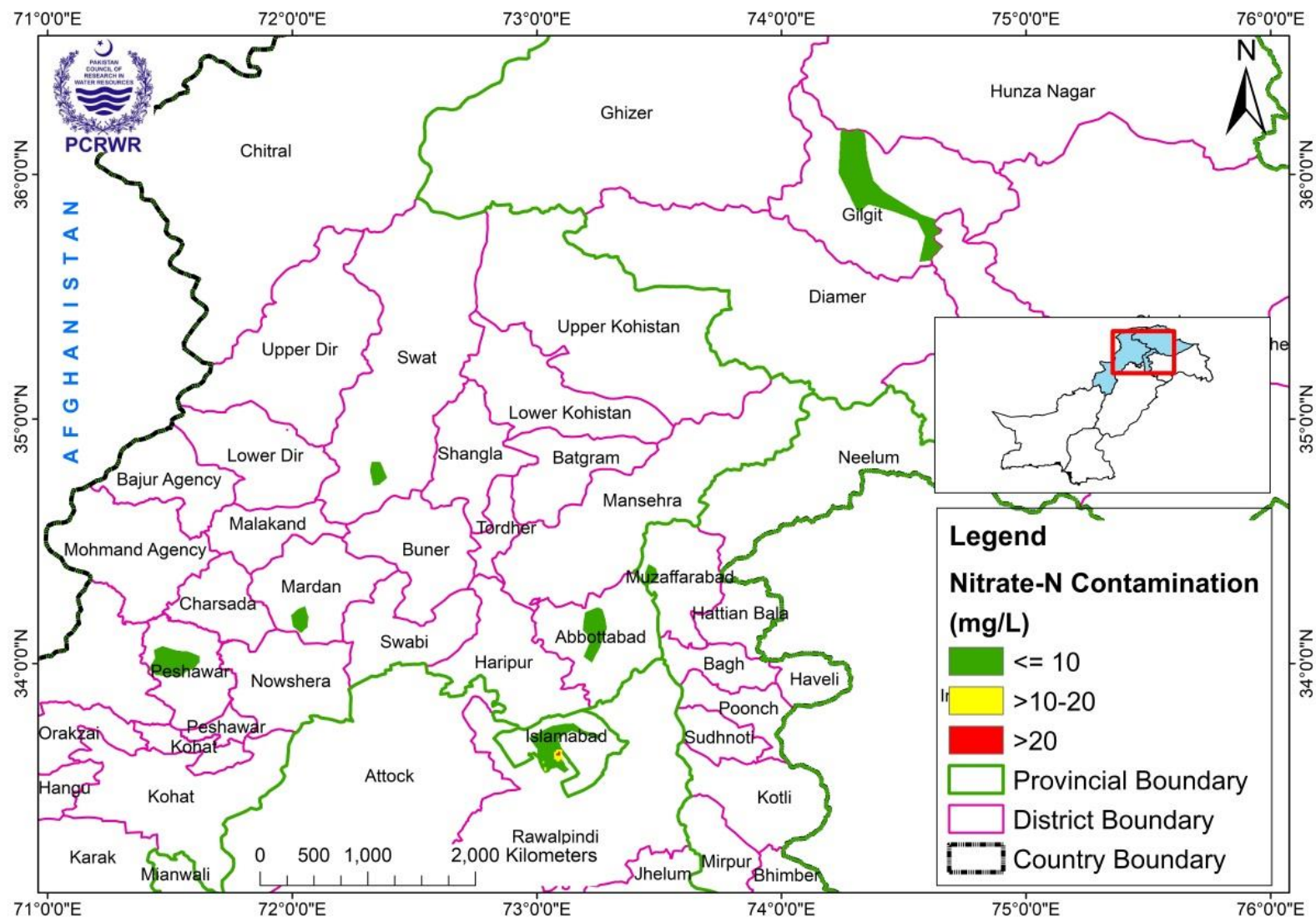
Hotspot of Microbial Contamination in Sindh



Hotspot of Microbial Contamination in Gilgit-Baltistan

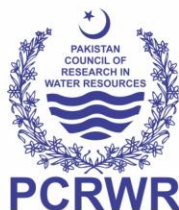


Hotspot of Nitrate Contamination in Gilgit-Baltistan



About PCRWR

PCRWR is an apex body of the Ministry of Science and Technology and is mandated to conduct, organize, coordinate and promote research on all aspects of water resources including irrigation (surface and groundwater), drainage, soil reclamation, drinking water and wastewater. It has eight regional offices located at different agro-ecological zones and each centre conducts research on water-related issues of the respective zones. These Regional Offices are located at Lahore, Bahawalpur, Tandojam, Quetta, Peshawar, Karachi, Gilgit and Muzaffarabad. Besides these eight Regional Offices, PCRWR has a setup of 24 water quality testing and research laboratories in major cities of the country. This includes ISO-17025 accredited National Water Quality Laboratory having its own Laboratory Information Management System (LIMS). PCRWR has all types of infrastructure such as soil and water testing laboratories, groundwater assessment equipment, research farms to conduct and disseminate the research. It is the only organization in Pakistan that owns drainage type lysimeters in Lahore, Tandojam, Quetta and Peshawar. PCRWR has done considerable work on crop water requirements, tile drainage, soil reclamation, on-farm water management technologies, rainwater harvesting, artificial recharge, groundwater assessment and management, skimming wells, drinking water, and indigenous development of water testing and treatment tools, salinity and moisture sensors. To help in developing the capacity of in-service professionals and fresh graduates, PCRWR has also a well-equipped National Capacity Building Institute in Islamabad to provide short and long term trainings on all aspects of water.



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